Scientometrics in R

Bibliometrics, Network Science, and Research Evaluation

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Contents

Welcome!

Welcome to **"Scientometrics in R: Bibliometrics, Network Science, and Research Evaluation"**.

This is a working guide to scientometrics in R for researchers, librarians, research administrators, and data scientists who want to study scholarly communication quantitatively. It walks the full pipeline — from pulling records out of OpenAlex, Crossref, and PubMed, through computing the standard indicators (h-index, MNCS, journal metrics), to building co-authorship and co-citation networks, applying text and topic models to scientific corpora, and packaging the results into reproducible reports and dashboards.

The book assumes a working knowledge of R at the tidyverse level and basic statistics. No prior scientometric experience is needed — the early chapters lay the conceptual and ethical foundations, and each later chapter starts with a concrete research question and ends with a callout on responsible use grounded in the Leiden Manifesto and DORA.

All examples run against free, open data; no Web of Science or Scopus subscription is required. Small sample datasets ship with the companion R package `scientometricsInR`; everything else is fetched live from public APIs. The book does *not* cover proprietary platforms (InCites, SciVal), manual screening workflows for systematic reviews, or qualitative sociology of science.

Open Source Repository

This book has been built using **{rmarkdown}** and **{bookdown}**. Formulas are rendered using MathJax. All source files are available on **GitHub** at https://github.com/r-heller/scientometrics-in-r. You are free to fork, share, and reuse contents under the project's CC BY-SA 4.0 license.

How To Use The Guide

- **Linear or lookup.** Read front to back, or use the *Find Your Method* navigator (Chapter 1) to jump to the chapter that answers your question.
- **Code.** Reusable helpers live in the companion R package `scientometricsInR` and in `R/` in the source repository; every chapter shows the full chunk it depends on.
- **Per-chapter PDFs.** Each chapter page has a *Download this chapter (PDF)* button at the top right.
- **Whole-book downloads.** Use the *Download* menu in the top navigation bar for PDF and EPUB.
- **Corrections.** Please file an issue at https://github.com/r-heller/scientometrics-in-r/issues.

Contributing

Pull requests, corrections, and additions are welcome. See `CONTRIBUTING.md` for the workflow.

Citing this Guide

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Impressum

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Disclaimer

This book is provided "as is" without warranty of any kind. The author assumes no responsibility for errors or omissions.

Contact

For corrections, suggestions, or issues: https://github.com/r-heller/scientometrics-in-r/issues

Acknowledgments

This book stands on a wide stack of free software. Particular thanks to the bookdown authors and the broader R Core and tidyverse communities; to the maintainers of `openalexR`, `bibliometrix`, `igraph`, `tidygraph`, `ggraph`, `quanteda`, `stm`, and `tidytext` for the packages this book relies on most; and to the OpenAlex project for making large-scale, open bibliographic data freely available.

Inspiration

The structural and pedagogical model for this book draws heavily on Mathias Harrer, Pim Cuijpers, Toshi A. Furukawa, and David D. Ebert's *Doing Meta-Analysis with R: A Hands-On Guide* — both the book and its open-source repository at https://github.com/MathiasHarrer/Doing-Meta-Analysis-in-R. Their thorough, transparent, and beautifully laid-out bookdown setup — sidebar grouping, semantic callout boxes, the citation block, the dark/light toggle, MathJax-rendered formulas, downloadable BibTeX/RIS citations — set a clear standard for what a reproducible R-based research book can look like. Where the present book mirrors their conventions, it does so deliberately and gratefully. Where it diverges (colors, typography, individual icons), the divergence is cosmetic; the underlying craft is theirs to credit.

Use of LLM tools

Portions of this book were prepared with assistance from large language model tooling for narrowly defined, non-authorial tasks: copyediting, prose smoothing, LaTeX/Markdown formatting, scaffolding of boilerplate files (CI configs, CSS, build scripts), code refactoring, and citation lookup against verified DOIs. The tools used were the latest Mistral Le Chat model run locally via Ollama and the `ollamar` R package — all inference performed on local hardware, with no data sent to third-party services — and GitHub Copilot inside RStudio for code completion.

All scientific claims, methodological choices, data analyses, interpretations, and conclusions are the author's own. No text generated by an LLM was incorporated without review and revision by the author. LLM tools were not used to generate, fabricate, or paraphrase cited sources; every reference was verified against its DOI, arXiv ID, or ISBN (see `scripts/verify-citations.R`).

How to use this book

0.1 Reading paths

The book is structured so that you can read it linearly — front to back — or jump straight to the chapter that answers your current question. If you do not yet know which method fits your problem, start with the *Find Your Method* navigator (Chapter 1). It maps typical research questions ("evaluate a journal portfolio", "detect emerging topics", "study collaboration patterns") to the chapters and case studies that demonstrate the relevant techniques.

0.2 Code chunks

Every analytical claim is paired with a runnable R chunk. Chunks are labelled (`label`), and figures use `fig.cap` / `fig.alt` so the book remains accessible. Chunks that need network access, paid credentials, or interactive widgets are marked `eval = FALSE` and document what is required to run them locally.

0.3 Callouts

Three callout styles appear throughout:

A blue callout flags a clarification or aside.

An orange callout flags a limitation or responsible-use note — read these.

A red callout flags a common pitfall that has tripped up working scientometricians.

0.4 Companion data

Small sample datasets that the chapters fall back to (when a live API call is not appropriate) are bundled with the companion R package `scientometricsInR`.

Larger illustrative corpora live under `data/` in the source repository.

0.5 Downloads

The sidebar's *Download* menu gives you the whole book as a PDF or EPUB. Each chapter page also exposes a *Download this chapter (PDF)* button at the top right.

0.6 Citation

If the book is useful to your research, please cite it — the suggested form lives in the Impressum, and a BibTeX block is in the README.

Notation

The following symbols, abbreviations, and conventions are used throughout the book.

Mathematical notation

Symbol	Meaning
c_i	Citations received by paper i
N	Number of papers in a corpus
h	Hirsch index (h-index)
g	g-index
MNCS	Mean normalised citation score
$\bar{x}$	Sample mean
$\tilde{x}$	Sample median
G	Graph (V, E) — vertices and edges
k_v	Degree of vertex v
$B(v)$	Betweenness centrality of vertex v

Abbreviations

Acronym	Expansion
API	Application Programming Interface
DOI	Digital Object Identifier
FWCI	Field-Weighted Citation Impact
JIF	Journal Impact Factor
MNCS	Mean Normalised Citation Score
NMI	Normalised Mutual Information
OA	Open Access
ORCID	Open Researcher and Contributor ID

Acronym	Expansion
ROR	Research Organization Registry
RRID	Research Resource Identifier
SCI	Science Citation Index
SDG	Sustainable Development Goal
WoS	Web of Science

Code-style conventions

- Object names describe what they are (`effect_sizes`), not how they were produced (`df2`).
- Pipes use the base `|>`. Tidyverse verbs are preferred over base R where readability favours them.
- Network objects use `igraph` representations unless explicitly stated.
- Random seeds are set per chapter; the global default is `42`.

About the author

Raban Heller is a biomedical scientist working at the intersection of biostatistics, clinical research, and data science. He maintains the *Strategy in R* and *Methods in R* books in the same series.

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For corrections or suggestions, please open an issue at https://github.com/r-heller/scientometrics-in-r/issues.

Chapter 1

Find Your Method

Use this decision tree to navigate to the chapter or case study most relevant to your research question.

flowchart TD A[“What is yourresearch question?”] –> B[“Map a research field”] A –> C[“Evaluate researchperformance”] A –> D[“Detect emergingtopics or trends”] A –> E[“Study collaborationpatterns”] A –> F[“Track open access& open science”]

```
B --> B1["Bibliographic coupling<br>Ch. 16"]
B --> B2["Co-citation analysis<br>Ch. 16"]
B --> B3["Science mapping<br>Ch. 19"]
B --> B4["Topic modeling<br>Ch. 22"]
B --> B5["Case Study 1:<br>CRISPR Field Review"]

C --> C1["Author metrics<br>Ch. 10, 12"]
C --> C2["Journal metrics<br>Ch. 11"]
C --> C3["Institutional benchmarking<br>Ch. 13"]
C --> C4["Case Study 2:<br>Institutional Benchmark"]
C --> C5["Case Study 3:<br>Journal Portfolio"]

D --> D1["Keyword bursts<br>Ch. 25"]
D --> D2["Temporal analysis<br>Ch. 14"]
D --> D3["Embedding clustering<br>Ch. 24"]
D --> D4["Case Study 5:<br>Emerging Topics"]

E --> E1["Co-authorship networks<br>Ch. 15"]
E --> E2["Community detection<br>Ch. 18"]
E --> E3["Gender & equity<br>Ch. 28"]
E --> E4["Case Study 4:<br>Gender Gap"]
```

```
F --> F1["OA indicators<br>Ch. 27"]
F --> F2["Altmetrics<br>Ch. 26"]
F --> F3["Reproducibility<br>Ch. 33"]
```

1.1 Quick reference table

Research question	Key chapters	Case study
Map a research field	16, 19, 22	CS 1: CRISPR
Evaluate an author	10, 12	—
Evaluate a journal	11	CS 3: Journal Portfolio
Benchmark an institution	13	CS 2: Institutional
Detect emerging topics	14, 24, 25	CS 5: Emerging Topics
Analyze collaboration	15, 18	—
Study gender gaps	28	CS 4: Gender Gap
Track open access	27	—
Build a dashboard	35, 37	—
Create a reproducible pipeline	34, 36	—

Part I

Foundations

Chapter 2

What Is Scientometrics?

2.1 Learning objectives

After completing this chapter, you will be able to:

- Define scientometrics, bibliometrics, and informetrics and explain how the three fields relate
- Trace the historical development of scientometrics from Garfield's citation indexes through Price's quantitative studies to the present day
- Identify the types of research questions that scientometric methods can address
- Retrieve a small sample of scholarly works from OpenAlex and produce basic descriptive statistics in R
- Create simple visualisations of publication trends and journal distributions

2.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

2.3 Conceptual background

Scientometrics is the quantitative study of science itself — its growth, structure, productivity, and impact. Where scientists in most disciplines study the natural or social world, scientometricians turn the lens inward: they study the system that produces scientific knowledge.

The term sits within a family of overlapping labels. **Bibliometrics**, the oldest sibling, focuses on the statistical analysis of publications — books, journal articles, and their metadata such as authors, citations, and keywords. The word was coined by Alan Pritchard in 1969 to replace the earlier "statistical bibliography". **Informetrics** is the broadest umbrella, encompassing any quantitative study of information transfer, including web analytics, patent analysis, and library usage statistics. In practice, the three terms are often used interchangeably, but it helps to remember that bibliometrics emphasises the *document*, scientometrics emphasises the *science*, and informetrics emphasises the *information*.

2.3.1 A brief history

The intellectual roots of the field trace back to at least 1955, when Eugene Garfield (1955) proposed the idea of a citation index for science. Garfield argued that citations create an "association of ideas" — a traceable web of intellectual influence — and that indexing these links would transform how scholars discover literature. His vision was realised in the Science Citation Index (SCI), launched in 1964 by the Institute for Scientific Information (ISI). For the first time, researchers could systematically trace who cited whom across the entire scientific literature.

A few years later, Derek de Solla Price gave the field its quantitative backbone. In *Little Science, Big Science* (de Solla Price, 1963), Price marshalled data to show that science was growing exponentially — in the number of journals, articles, and scientists — and that this growth followed remarkably regular statistical patterns. He estimated that the volume of scientific literature doubles approximately every 15 years, a regularity so striking that it came to be known as "Price's Law." Price's work established that science could be studied with the same empirical rigour that it applies to other phenomena.

These foundations were reinforced by earlier statistical regularities. Alfred Lotka (1926) had already shown that the distribution of scientific productivity follows a power law: a small number of authors produce the majority of papers. Samuel Bradford (1934) demonstrated a similar pattern for journal coverage: a core set of journals in any field accounts for a disproportionate share of the relevant literature. These laws, which we explore in detail in the next chapter, remain cornerstones of scientometric theory.

From these beginnings, scientometrics expanded rapidly through the latter half of the twentieth century. The h-index (Hirsch, 2005), journal impact factors, co-citation networks, and science maps became standard tools in research evaluation and science policy. Landmark declarations such as the Leiden Manifesto (Hicks et al., 2015) and DORA (American Society for Cell Biology, 2012) later established principles for responsible use of these tools, warning against mechanical reliance on any single indicator.

2.3.2 The modern era

Today the field is undergoing another transformation. The launch of **OpenAlex** (Priem et al., 2022) — a fully open index of over 250 million scholarly works, built from Crossref, PubMed, institutional repositories, and other open sources — has democratised access to the raw data that powers scientometric research. Researchers no longer need expensive subscriptions to Web of Science or Scopus to conduct large-scale analyses. This book takes advantage of that openness: every example uses data that any reader can reproduce for free.

2.3.3 Scope of this book

This book teaches you to do scientometrics in R. It covers:

- **Data acquisition** — querying OpenAlex, Crossref, and PubMed programmatically.
- **Core bibliometrics** — computing productivity, impact, and citation indicators.
- **Network analysis** — building and analysing co-authorship, co-citation, and keyword co-occurrence networks.
- **Text and topic analysis** — extracting themes and trends from titles, abstracts, and full texts.
- **Advanced topics** — altmetrics, open science, gender equity, retraction analysis, and more.
- **Production** — building reproducible pipelines, dashboards, and reports.

Each chapter follows the same pattern: conceptual background, a fully reproducible worked example, diagnostics, limitations, and exercises.

2.4 Worked example

To make things concrete from the very first chapter, we fetch a small sample of recent works from OpenAlex and compute some basic descriptive statistics. Think of this as a first taste of the workflows that the rest of the book will develop in depth.

2.4.1 Fetching a sample from OpenAlex

We search for works whose title or abstract mentions “scientometrics”, published between 2014 and 2023, and draw a random sample of 500 records.

```r
works <- oa_fetch(
  entity = "works",
  search = "scientometrics",
  from_publication_date = "2014-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)

cat(glue("Fetched {nrow(works)} works\n"))
```

```
#> Fetched 500 works
```

```r
glimpse(works)
```

```
#> Rows: 500
#> Columns: 45
#> $ id                          <chr> "https://openalex.org/W2974247016", "htt~
#> $ title                       <chr> "Assessing the Effectiveness of Open Acc~
#> $ display_name                <chr> "Assessing the Effectiveness of Open Acc~
#> $ authorships                 <list> [<tbl_df[5 x 7]>], [<tbl_df[1 x 7]>], [~
#> $ abstract                    <chr> "The open access (OA) movement seeks to ~
#> $ doi                         <chr> "https://doi.org/10.6017/ital.v38i3.1100~
#> $ publication_date            <date> 2019-09-15, 2018-08-30, 2017-04-16, 201~
#> $ publication_year            <int> 2019, 2018, 2017, 2015, 2021, 2021, 2023~
#> $ relevance_score             <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
#> $ fwci                        <dbl> 2.468, 8.116, NA, 0.763, 0.000, 0.489, 0~
#> $ cited_by_count              <int> 14, 132, 2, 34, 0, 7, 7, 22, 0, 37, 0, 1~
#> $ counts_by_year              <list> [<data.frame[7 x 2]>], [<data.frame[7 x~
#> $ ids                         <list> <"https://openalex.org/W2974247016", "h~
#> $ type                        <chr> "article", "article", "other", "article"~
#> $ is_oa                       <lgl> TRUE, TRUE, FALSE, FALSE, TRUE, TRUE, TR~
#> $ is_oa_anywhere              <lgl> TRUE, TRUE, FALSE, FALSE, TRUE, TRUE, TR~
#> $ oa_status                   <chr> "diamond", "green", "closed", "closed", ~
#> $ oa_url                      <chr> "https://ejournals.bc.edu/index.php/ital~
#> $ any_repository_has_fulltext <lgl> TRUE, TRUE, FALSE, FALSE, FALSE, TRUE, T~
#> $ source_display_name         <chr> "Information Technology and Libraries", ~
#> $ source_id                   <chr> "https://openalex.org/S4210225942", "htt~
#> $ issn_l                      <chr> "0730-9295", NA, NA, "1057-2317", "2692-~
#> $ host_organization           <chr> "https://openalex.org/P4310315903", "htt~
```

```
#> $ host_organization_name        <chr> "American Library Association", "Europea~
#> $ landing_page_url              <chr> "https://doi.org/10.6017/ital.v38i3.1100~
#> $ pdf_url                       <chr> "https://ejournals.bc.edu/index.php/ital~
#> $ license                       <chr> "cc-by-nc", "cc-by", NA, NA, NA, "cc-by-~
#> $ version                       <chr> "publishedVersion", NA, "publishedVersio~
#> $ referenced_works              <list> <"https://openalex.org/W1005483714", "h~
#> $ referenced_works_count        <int> 17, 0, 23, 54, 36, 48, 42, 70, 20, 64, 0~
#> $ related_works                 <list> <"https://openalex.org/W2748952813", "h~
#> $ concepts                      <list> [<data.frame[8 x 5]>], [<data.frame[1 x~
#> $ topics                        <list> [<tbl_df[4 x 5]>], [<tbl_df[12 x 5]>], ~
#> $ keywords                      <list> [<data.frame[7 x 3]>], [<data.frame[1 x~
#> $ is_paratext                   <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ is_retracted                  <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ language                      <chr> "en", "en", "en", "en", "en", "en", "en"~
#> $ sustainable_development_goals <list> [<data.frame[1 x 3]>], NA, NA, [<data.f~
#> $ awards                        <list> NA, NA, NA, NA, NA, NA, <"https://opena~
#> $ funders                       <list> NA, NA, NA, NA, [<data.frame[3 x 3]>], ~
#> $ apc                           <list> NA, NA, NA, NA, NA, NA, [<data.frame[2 ~
#> $ first_page                    <chr> "82", NA, "1", "71", NA, "118", "7146", ~
#> $ last_page                     <chr> "90", NA, "5", "82", NA, "133", "7146", ~
#> $ volume                        <chr> "38", NA, NA, "47", "10", "23", "28", "6~
#> $ issue                         <chr> "3", NA, NA, "3-4", "1", "60", "20", "1"~
```

Each row is a scholarly work. Key columns include `display_name` (the title), `publication_date`, `cited_by_count`, `doi`, and `source_display_name` (the source or journal name).

2.4.2 Basic descriptive statistics

How many works did we retrieve, and what does the citation distribution look like?

```r
works_summary <- works |>
  summarise(
    n_works        = n(),
    doi_coverage   = scales::percent(mean(!is.na(doi))),
    median_cites   = median(cited_by_count, na.rm = TRUE),
    mean_cites     = round(mean(cited_by_count, na.rm = TRUE), 1),
    max_cites      = max(cited_by_count, na.rm = TRUE),
    zero_cites     = sum(cited_by_count == 0)
  )

works_summary
```

```
#> # A tibble: 1 x 6
```

```
#>   n_works doi_coverage median_cites mean_cites max_cites zero_cites
#>     <int> <chr>               <dbl>      <dbl>     <int>      <int>
#> 1     500 91%                     3       19.2      1087        138
```

The large gap between the mean and median is a hallmark of citation distributions: most papers receive few citations while a small number accumulate many. This skewness is one of the most robust empirical regularities in scientometrics, and we will return to its theoretical basis in 3.3.

2.4.3 Top journals

Which journals appear most often in our sample?

```
top_journals <- works |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, sort = TRUE) |>
  head(10)

top_journals
```

```
#> # A tibble: 10 x 2
#>    source_display_name                                               n
#>    <chr>                                                         <int>
#>  1 Sustainability                                                   14
#>  2 PLoS ONE                                                         10
#>  3 Zenodo (CERN European Organization for Nuclear Research)         10
#>  4 arXiv (Cornell University)                                        8
#>  5 Lincoln (University of Nebraska)                                  6
#>  6 International Journal of Environmental Research and Public Health     5
#>  7 Scientometrics                                                    5
#>  8 Econstor (Econstor)                                               4
#>  9 Frontiers in Psychology                                           4
#> 10 Research Square                                                   4
```

2.4.4 Publication trend over time

```
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(
```

```
    x = "Publication year",
    y = "Number of works (sampled)"
  ) +
  theme_sci()
```

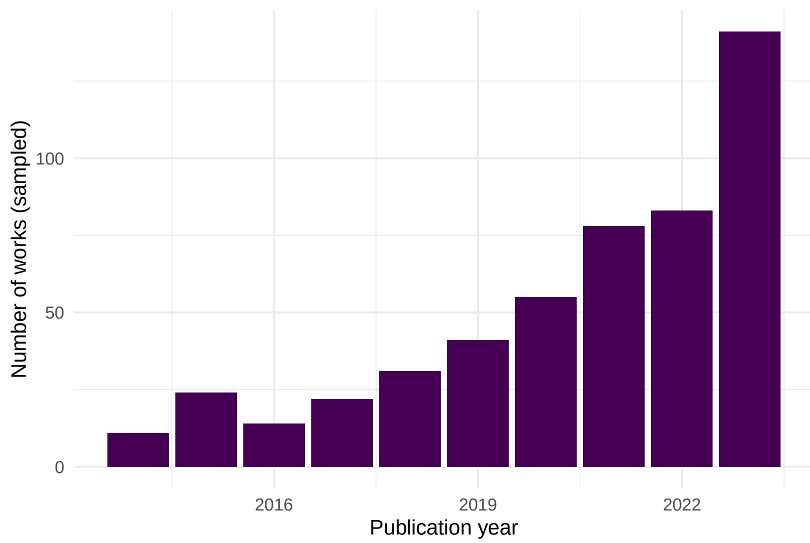

Figure 2.1: Annual publication counts for sampled works related to scientometrics (2014–2023).

The trend shows a growing volume of scientometrics-related publications over the past decade — consistent with the broader exponential growth of science documented by de Solla Price (1963).

2.4.5 Top journals visualised

```
top_journals |>
  mutate(source_display_name = fct_reorder(source_display_name, n)) |>
  ggplot(aes(x = n, y = source_display_name)) +
  geom_col(fill = palette_sci(1)) +
  labs(
    x = "Number of works",
    y = NULL
  ) +
  theme_sci()
```

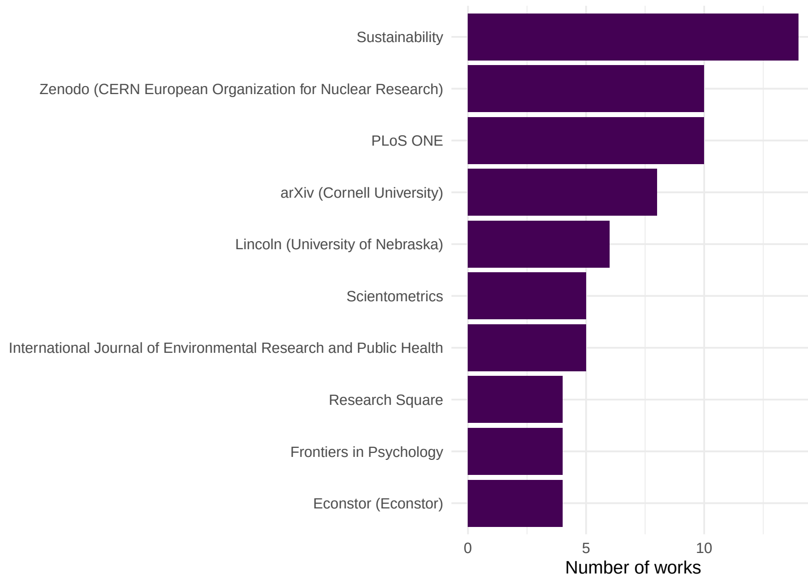

Figure 2.2: Top 10 journals by number of works in the sampled scientometrics corpus.

2.4.6 Citation distribution

```
works |>
  filter(cited_by_count > 0) |>
  ggplot(aes(x = cited_by_count)) +
  geom_histogram(bins = 40, fill = palette_sci(1), colour = "white") +
  scale_x_log10() +
  labs(
    x = "Citation count (log scale)",
    y = "Number of works"
  ) +
  theme_sci()
```

This right-skewed citation distribution is a universal feature of scholarly communication. In Chapter 2, we will see that it connects to deeper theoretical regularities: Lotka's Law, Bradford's Law, and the Matthew Effect.

2.5 Diagnostics and interpretation

Even for a simple descriptive analysis, a few diagnostic checks are essential:

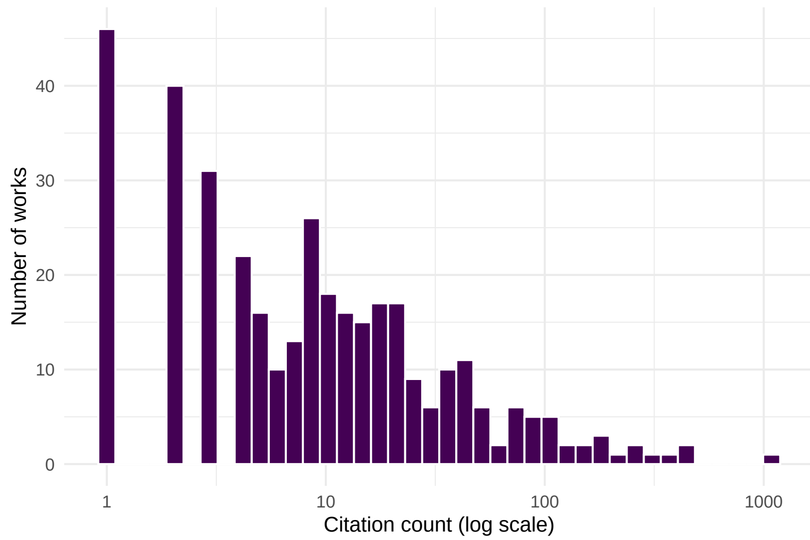

Figure 2.3: Citation distribution of sampled scientometrics works (log-scaled x-axis).

- **Sample vs. population.** We drew a random sample of 500 works. The publication trend reflects the underlying population only if the sampling was truly random within the date window. OpenAlex's `sample` parameter does this, but always verify the date range of returned results.
- **Source coverage.** OpenAlex is strongest in English-language, peer-reviewed journal articles. Conference proceedings, book chapters, and grey literature may be underrepresented. Our "top journals" list reflects what OpenAlex indexes, not the full universe of scientometrics publishing.
- **Citation counts are snapshots.** The `cited_by_count` field reflects the current citation total at the time of the API query. Older papers have had more time to accumulate citations, so direct comparisons across years are misleading without normalization.

```
cat(glue("Date range: {min(works$publication_date)} to {max(works$publication_date)}\n"))
```

```
#> Date range: 2014-01-01 to 2023-12-31
```

```
cat(glue("Works with zero citations: {sum(works$cited_by_count == 0)}\n"))
```

```
#> Works with zero citations: 138
```

```r
cat(glue("Unique journals: {n_distinct(works$source_display_name, na.rm = TRUE)}\n"))
```

```
#> Unique journals: 376
```

2.6 Limitations and responsible use

2.7 Limitations and responsible use

- **Descriptive statistics are not evaluations.** Counting publications and citations describes *activity*, not *quality*. A high publication count may reflect salami slicing; a high citation count may reflect controversy rather than excellence.
- **Data source bias.** OpenAlex, like all databases, has systematic coverage gaps. English-language STEM journals are overrepresented; social sciences, humanities, and non-English scholarship are underrepresented. Always report the data source and its known limitations.
- **Ecological fallacy.** Aggregate statistics about a field do not apply to individual researchers. A growing field does not mean every researcher in it is productive.
- **The Leiden Manifesto** (Hicks et al., 2015) and **DORA** (American Society for Cell Biology, 2012) provide essential guardrails. Quantitative data should support, never replace, expert qualitative judgement.

2.8 Common pitfalls

2.9 Common pitfalls

- **Confusing scientometrics with research evaluation.** Scientometrics provides tools for studying science; it does not provide verdicts on the value of individual researchers or papers.
- **Treating OpenAlex counts as ground truth.** OpenAlex counts may differ from Web of Science or Scopus due to different coverage, deduplication, and metadata parsing. Always triangulate when precision matters.
- **Ignoring the citation window.** Comparing raw citation counts across publication years is misleading. A 2023 paper with 5 citations may be performing better than a 2015 paper with 20. Field- and time-normalised indicators (covered in Chapter 10) address this.
- **Overgeneralising from a sample.** A random sample of 500 works is useful for learning and prototyping. Drawing conclusions about an entire field requires a comprehensive corpus.

- **Forgetting to set a seed.** Random samples are only reproducible if you set the seed both in R (`set.seed()`) and in the API call (`seed` parameter). Without this, every render produces different results.

2.10 Exercises

1. **Explore a different field.** Replace the search term "scientometrics" with a topic of your choice (e.g., "machine learning", "climate change"). Fetch 500 works and compare the publication trend and top journals to those we found above. What differences do you notice?

2. **Uncited works.** What fraction of the sampled works have zero citations? Filter to uncited works and examine their publication years. Is there a pattern? *(Hint: very recent works are more likely to be uncited.)*

3. **Open Access status.** OpenAlex records an `is_oa` field. Compute the proportion of open-access works in your sample and plot how it has changed over time. What trend do you observe?

4. **Growth curve.** Extend the publication trend analysis to 2000–2023. Does the annual publication count follow an exponential growth pattern, as de Solla Price (1963) predicted? *(Hint: plot on a log-scaled y-axis and check for linearity.)*

2.11 Solutions

Solutions are provided in F.

2.12 Further reading

- Garfield (1955) — The foundational paper proposing citation indexes for science; the intellectual starting point for the entire field.
- de Solla Price (1963) — *Little Science, Big Science*; the first systematic quantitative study of the growth of science.
- Priem et al. (2022) — The OpenAlex paper; describes the open data infrastructure that powers the examples in this book.
- Hicks et al. (2015) — The Leiden Manifesto; ten principles for the responsible use of research metrics.
- American Society for Cell Biology (2012) — The San Francisco Declaration on Research Assessment; a landmark call to improve research evaluation.
- Aria and Cuccurullo (2017) — The `bibliometrix` R package; a comprehensive tool for science mapping and bibliometric analysis.

- Hirsch (2005) — The h-index paper; illustrates how a single indicator reshaped research evaluation discourse.
- Lotka (1926) — The foundational study of scientific productivity distributions; a bridge to Chapter 2.

2.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] glue_1.8.1      openalexR_3.0.1 lubridate_1.9.5 forcats_1.0.1
#>  [5] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0
#>  [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] viridis_0.6.5      utf8_1.2.6         generics_0.1.4     stringi_1.8.7
#>  [5] hms_1.1.4          digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5
#>  [9] grid_4.4.1         timechange_0.4.0   RColorBrewer_1.1-3 bookdown_0.46
#> [13] fastmap_1.2.0      rprojroot_2.1.1    jsonlite_2.0.0     gridExtra_2.3
#> [17] httr_1.4.8         viridisLite_0.4.3  scales_1.4.0       codetools_0.2-20
#> [21] cli_3.6.6          rlang_1.2.0        withr_3.0.2        yaml_2.3.12
#> [25] otel_0.2.0         tools_4.4.1        tzdb_0.5.0         here_1.0.2
#> [29] curl_7.1.0         vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5
```

```
#> [33] pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6       xfun_0.57
#> [37] tidyselect_1.2.1   rstudioapi_0.18.0  knitr_1.51         dichromat_2.0-0.1
#> [41] farver_2.1.2       htmltools_0.5.9    labeling_0.4.3     rmarkdown_2.31
#> [45] compiler_4.4.1     S7_0.2.2
```

Chapter 3

Theoretical Underpinnings

3.1 Learning objectives

After completing this chapter, you will be able to:

- State Lotka's, Bradford's, and Zipf's laws and explain the empirical patterns they describe
- Test whether a corpus of publications approximately follows Lotka's inverse-square law
- Construct a Bradford plot to identify core and peripheral journals in a field
- Explain Price's law of exponential growth and the Matthew Effect in science
- Recognise power-law-like distributions in bibliometric data

3.2 Setup

```r
library(tidyverse)
library(openalexR)
library(tidytext)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "sci_palette.R"))
```

3.3 Conceptual background

Scientometrics rests on a small number of empirical regularities discovered across decades of studying scholarly communication:

Lotka's Law (Lotka, 1926) observes that the number of authors producing n papers is approximately proportional to $1/n^2$. A few prolific authors generate a large share of output, while most researchers publish only one or two papers.

Bradford's Law (Bradford, 1934) describes the scattering of articles across journals. For a given topic, a small core of journals contains roughly one-third of all relevant articles; a larger zone contains the next third; and a much larger periphery the final third. Each successive zone contains roughly the same number of articles but geometrically more journals.

Zipf's Law states that the frequency of a word is inversely proportional to its rank. In scientific text, a handful of terms dominate while the vast majority appear rarely.

Price's Law (de Solla Price, 1963) generalises these observations: the cumulative output of science grows exponentially, doubling roughly every 10–15 years. Price also noted that the square root of all contributors produce half of all contributions.

The **Matthew Effect** (Merton, 1968) — named after the Gospel of Matthew ("to those who have, more shall be given") — describes cumulative advantage in science. Well-known researchers attract disproportionate credit and resources, amplifying initial differences in visibility. This mechanism helps explain the skewed distributions observed by Lotka and Price.

3.4 Worked example

We use a corpus of works from the journal *Scientometrics* to demonstrate each law.

3.4.1 Fetching the data

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

3.4.2 Lotka's Law

We unnest authors and count the number of papers per author in this sample.

```r
author_prod <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  count(authorships_display_name, name = "n_papers") |>
  count(n_papers, name = "n_authors") |>
  arrange(n_papers)

author_prod
```

```
#> # A tibble: 9 x 2
#>   n_papers n_authors
#>      <int>     <int>
#> 1        1      1144
#> 2        2        92
#> 3        3        24
#> 4        4         6
#> 5        5         2
#> 6        6         1
#> 7        7         2
#> 8        9         1
#> 9       12         1
```

```r
lotka_c <- author_prod$n_authors[1]

author_prod |>
  ggplot(aes(x = n_papers, y = n_authors)) +
  geom_point(colour = palette_sci(1), size = 2) +
  geom_line(
    aes(y = lotka_c / n_papers^2),
    linetype = "dashed", colour = "grey40"
  ) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Papers per author", y = "Number of authors") +
  theme_sci()
```

The data approximate the inverse-square pattern, though real distributions often deviate from the exact exponent of 2.

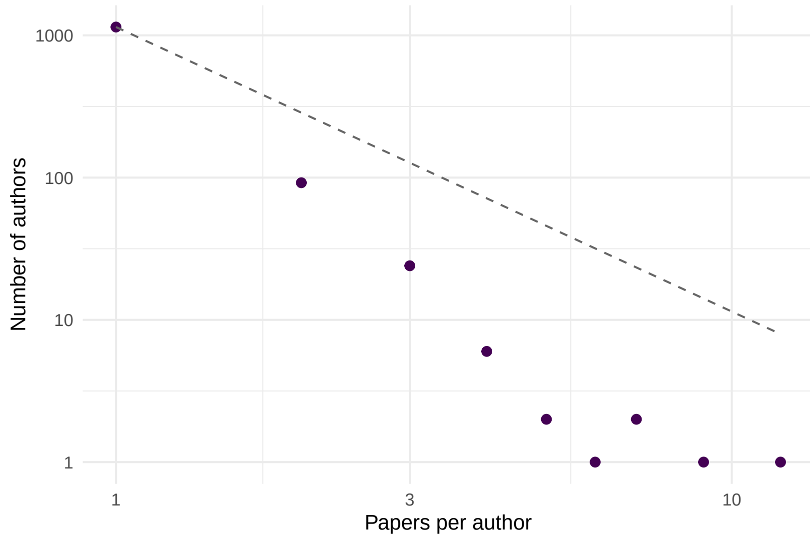

Figure 3.1: Author productivity distribution. The dashed line shows the inverse-square law predicted by Lotka.

3.4.3 Bradford's Law

We count how many articles each journal source contributes (across a broader query).

```
broad_works <- oa_fetch(
  entity = "works",
  search = "bibliometrics",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 1000, seed = 42)
)

journal_counts <- broad_works |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, name = "articles", sort = TRUE) |>
  mutate(
    cum_articles = cumsum(articles),
    rank = row_number()
  )
```

```
journal_counts |>
  ggplot(aes(x = rank, y = cum_articles)) +
  geom_line(colour = palette_sci(1), linewidth = 1) +
```

```
  scale_x_log10() +
  labs(x = "Journal rank (log scale)", y = "Cumulative articles") +
  theme_sci()
```

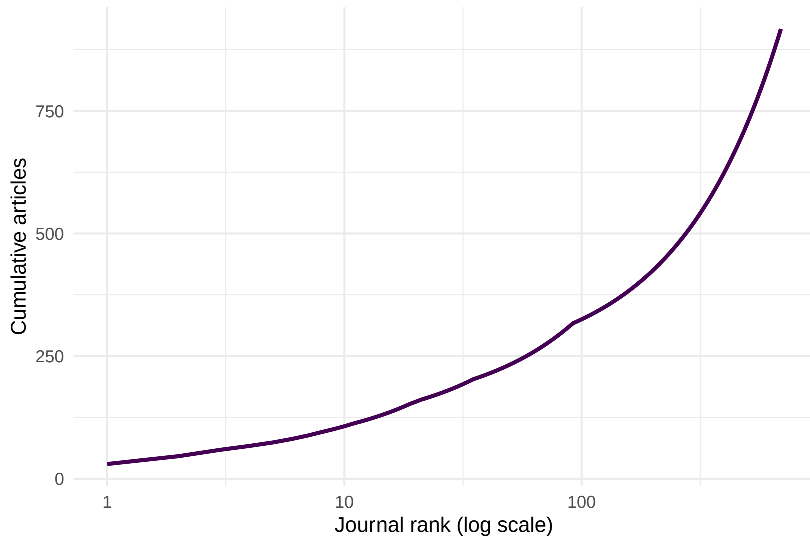

Figure 3.2: Bradford plot: cumulative articles vs. log journal rank.

3.4.4 Zipf's Law

We extract title words and examine their frequency distribution.

```
word_freq <- works |>
  select(display_name) |>
  unnest_tokens(word, display_name, token = "words") |>
  anti_join(tidytext::get_stopwords(), by = "word") |>
  count(word, sort = TRUE) |>
  mutate(rank = row_number())
```

```
word_freq |>
  head(200) |>
  ggplot(aes(x = rank, y = n)) +
  geom_point(colour = palette_sci(1), size = 1, alpha = 0.7) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Rank", y = "Frequency") +
  theme_sci()
```

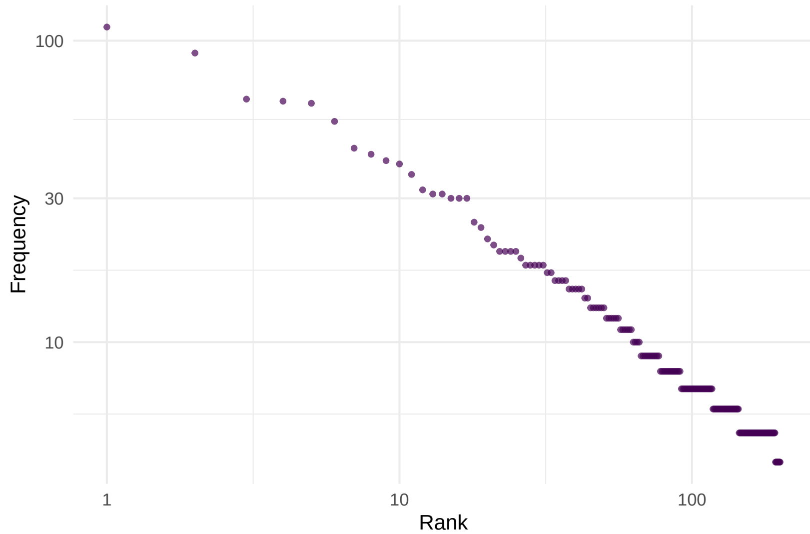

Figure 3.3: Word frequency vs. rank for article titles, approximating Zipf's law.

3.4.5 Price's growth law

```
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Works in sample") +
  theme_sci()
```

3.5 Diagnostics and interpretation

These laws are *empirical regularities*, not physical laws. They describe approximate patterns:

- Lotka's exponent is often not exactly 2; values between 1.5 and 3.5 are common depending on the field and time window.
- Bradford zones are sensitive to how you define the subject scope.
- Zipf's law holds best for common words; the tail deviates.
- Price's exponential growth eventually saturates — science cannot grow faster than the economy indefinitely.

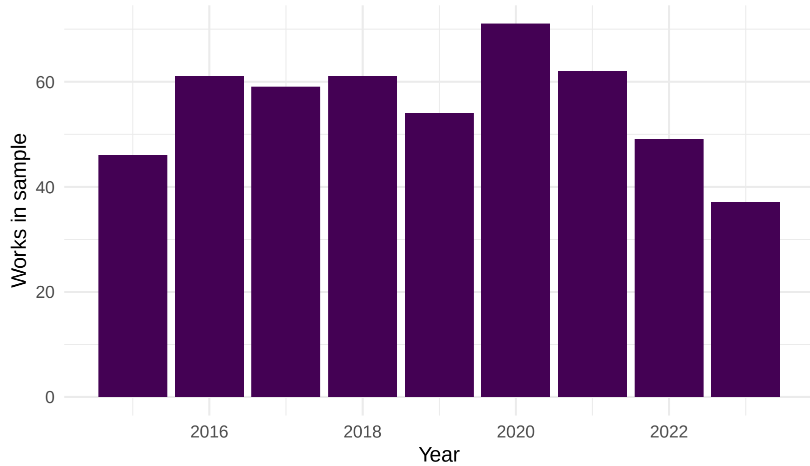

Figure 3.4: Annual publication counts for sampled works, illustrating the growth of bibliometrics literature.

The value of these laws is not predictive precision but the insight that **skewed distributions are the norm, not the exception**, in scholarly communication.

3.6 Limitations and responsible use

3.7 Limitations and responsible use

- These laws describe aggregate patterns, not individual behaviour. A researcher publishing one paper is not "less productive" in any evaluative sense — Lotka's law is a statistical observation, not a performance standard.
- The Matthew Effect (Merton, 1968) is a mechanism that *amplifies inequality*. Recognising it should prompt caution about using cumulative metrics (like the h-index) that inherit and reinforce this bias.
- Power-law fitting is notoriously difficult. Many distributions that "look" power-law on a log-log plot are better described by lognormal or stretched exponential models. Rigorous fitting requires maximum-likelihood methods and goodness-of-fit tests (Hicks et al., 2015).

3.8 Common pitfalls

3.9 Common pitfalls

- **Eyeballing log-log plots.** A straight line on a log-log plot does not prove a power law. Use formal statistical tests.
- **Confusing description with prescription.** "Most authors publish few papers" does not mean they *should* publish more.
- **Ignoring field differences.** The exponent of Lotka's law and the scattering pattern of Bradford's law vary substantially across disciplines.
- **Assuming stationarity.** These patterns can shift over time as publication norms change (e.g., the rise of mega-journals).

3.10 Exercises

1. **Lotka exponent.** Fit a linear model to the log-log author productivity data. What exponent do you estimate? How does it compare to Lotka's original value of 2?

2. **Bradford zones.** Divide the journal list into three zones of roughly equal article counts. How many journals are in each zone? What is the ratio between successive zone sizes?

3. **Your field's Zipf distribution.** Fetch 500 works in a field of your choice. Extract title words and plot the Zipf distribution. Are the most frequent terms what you would expect?

4. **Matthew Effect simulation.** Write a simple simulation: start 100 authors with 1 paper each. At each time step, an author gains a new paper with probability proportional to their current count. After 50 steps, plot the distribution. Does it resemble Lotka's law?

3.11 Solutions

Solutions are provided in F.

3.12 Further reading

- Lotka (1926) — The original observation on author productivity distributions.
- Bradford (1934) — Journal scattering and the Bradford law.

- de Solla Price (1963) — Exponential growth of science and cumulative advantage.
- Merton (1968) — The Matthew Effect: how advantage accumulates in science.
- Garfield (1955) — Citation indexing, the empirical basis for much of scientometrics.
- Waltman (2016) — A modern review contextualising these classical laws within citation impact measurement.

3.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] tidytext_0.4.3  glue_1.8.1      openalexR_3.0.1 lubridate_1.9.5
#>  [5] forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2
#>  [9] readr_2.2.0     tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3
#> [13] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] janeaustenr_1.0.0  viridis_0.6.5      utf8_1.2.6         generics_0.1.4
#>  [5] lattice_0.22-6     stringi_1.8.7      hms_1.1.4          digest_0.6.39
#>  [9] magrittr_2.0.5     evaluate_1.0.5     grid_4.4.1         timechange_0.4.0
```

```
#> [13] RColorBrewer_1.1-3 bookdown_0.46      fastmap_1.2.0      Matrix_1.7-0
#> [17] rprojroot_2.1.1    jsonlite_2.0.0     gridExtra_2.3      stopwords_2.3
#> [21] httr_1.4.8         viridisLite_0.4.3  scales_1.4.0       codetools_0.2-20
#> [25] cli_3.6.6          rlang_1.2.0        tokenizers_0.3.0   withr_3.0.2
#> [29] yaml_2.3.12        otel_0.2.0         tools_4.4.1        tzdb_0.5.0
#> [33] here_1.0.2         curl_7.1.0         vctrs_0.7.3        R6_2.6.1
#> [37] lifecycle_1.0.5    pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
#> [41] Rcpp_1.1.1-1.1     xfun_0.57          tidyselect_1.2.1   rstudioapi_0.18.0
#> [45] knitr_1.51         dichromat_2.0-0.1  farver_2.1.2       SnowballC_0.7.1
#> [49] htmltools_0.5.9    labeling_0.4.3     rmarkdown_2.31     compiler_4.4.1
#> [53] S7_0.2.2
```

Chapter 4

Ethics and Responsible Metrics

4.1 Learning objectives

After completing this chapter, you will be able to:

- Summarize the ten principles of the Leiden Manifesto and explain their implications for research evaluation
- Describe the key commitments of the San Francisco Declaration on Research Assessment (DORA)
- Identify common forms of metric gaming, including self-citation cartels, citation stacking, and salami slicing
- Evaluate when quantitative indicators are appropriate — and when they are not
- Compute and interpret author-level self-citation rates using OpenAlex data
- Apply a responsible-use framework when designing or reviewing bibliometric analyses

4.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)
```

```r
set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

4.3 Conceptual background

Metrics shape behaviour. When a number is used to judge researchers, departments, or journals, people adjust their actions to improve that number — regardless of whether the number actually captures what it was meant to measure. This insight, often called **Goodhart's Law**, is the intellectual foundation for the responsible metrics movement: "When a measure becomes a target, it ceases to be a good measure" (Goodhart, 1984).

4.3.1 The Leiden Manifesto

In 2015, Hicks et al. (2015) published the **Leiden Manifesto for Research Metrics** — ten principles distilled from decades of experience in bibliometric practice. The principles are:

1. **Quantitative evaluation should support qualitative, expert assessment.** Metrics should inform human judgement, never replace it.
2. **Measure performance against the research missions of the institution, group, or researcher.** Not everyone pursues the same goals; evaluation criteria must reflect this.
3. **Protect excellence in locally relevant research.** Research published in regional languages or addressing local problems must not be penalised for lower citation rates.
4. **Keep data collection and analytical processes open, transparent, and simple.** Black-box indicators erode trust and prevent scrutiny.
5. **Allow those evaluated to verify data and analysis.** Researchers should be able to inspect and challenge the data used to judge them.
6. **Account for variation by field in publication and citation practices.** Citation rates in mathematics and molecular biology differ by an order of magnitude; raw counts cannot be compared across fields.
7. **Base assessment of individual researchers on a qualitative judgement of their portfolio.** A single number (such as the h-index) cannot capture the breadth of a researcher's contribution.
8. **Avoid misplaced concreteness and false precision.** Three decimal places do not make a flawed indicator meaningful.

9. **Recognize the systemic effects of assessment and indicators.** Indicators reshape the system they measure. If you reward publication volume, you incentivise salami slicing.
10. **Scrutinize indicators regularly and update them.** The research landscape evolves; indicators must evolve with it.

4.3.2 The San Francisco Declaration (DORA)

The **San Francisco Declaration on Research Assessment** (American Society for Cell Biology, 2012) emerged from the 2012 Annual Meeting of the American Society for Cell Biology. Its central demand is straightforward: do not use journal-level metrics — especially the Journal Impact Factor — as a proxy for the quality of individual research articles. DORA's key commitments include:

- **For funders and institutions**: evaluate research on its own merits, not on the journal in which it was published. Use article-level metrics and qualitative indicators.
- **For publishers**: reduce emphasis on the Journal Impact Factor as a promotional tool. Make article-level metrics (downloads, citations, altmetrics) available.
- **For researchers**: challenge evaluation practices that rely on journal-based metrics. When serving on hiring or promotion committees, assess the content of papers, not where they appeared.

As of 2026, DORA has been signed by over 3,000 organisations and 20,000 individuals. Many major funders — including the European Commission, Wellcome Trust, and the Dutch Research Council (NWO) — have formally adopted DORA principles.

4.3.3 The Metric Tide

The UK's independent review of metrics in research assessment, *The Metric Tide* (Wilsdon et al., 2015), reinforced these principles with a systematic evidence base. The report concluded that no metric can substitute for expert peer review, and recommended that the UK research system adopt a framework of "responsible metrics" built on five dimensions: robustness, humility, transparency, diversity, and reflexivity.

4.3.4 Goodhart's Law in practice

When metrics are attached to career consequences, the incentive to game them becomes powerful (Goodhart, 1984). Common forms of metric manipulation include:

- **Self-citation inflation**: authors systematically cite their own prior work to boost citation counts, even when those citations are not intellectually necessary.
- **Citation cartels**: informal agreements among groups of authors or journals to cite each other's work, inflating citation counts reciprocally.
- **Citation stacking**: journals encourage or require authors to add citations to recently published articles in the same journal, artificially increasing the Impact Factor.
- **Salami slicing**: splitting a single study into the maximum number of "least publishable units" to inflate publication counts.

These practices are not merely hypothetical. Clarivate has repeatedly suspended journals from the Journal Citation Reports for citation stacking, and several high-profile cases of self-citation manipulation have led to editorial sanctions.

4.3.5 Publish or perish

The systemic root of metric gaming is the **"publish or perish"** culture that dominates academic career structures worldwide. When hiring, promotion, and tenure decisions are driven by publication volume and citation counts, researchers face relentless pressure to produce more papers, chase high-impact journals, and avoid risky long-term projects. The consequences include:

- **Reduced risk-taking**: researchers avoid novel or interdisciplinary questions that might not produce quick, high-citation publications.
- **Replication crisis**: pressure to publish positive results contributes to publication bias and irreproducible findings.
- **Mental health harms**: the relentless pressure to produce has been linked to elevated rates of anxiety, burnout, and attrition in academia.
- **Erosion of research quality**: when volume is rewarded over substance, the incentive structure favours quantity at the expense of depth and rigour.

These dynamics make the responsible use of metrics not merely a technical concern but an ethical imperative.

4.4 Worked example

We now demonstrate a practical analysis: computing self-citation rates for a sample of researchers in the field of scientometrics. This illustrates how OpenAlex data can be used to flag potential self-citation inflation — and the care required to interpret such analysis responsibly.

4.4.1 Acquiring author data

We fetch a sample of authors associated with scientometrics research who have at least 50 works, ensuring we have enough output to compute meaningful self-citation rates.

```
authors_raw <- oa_fetch(
  entity = "authors",
  topics.id = "T10102",
  works_count = ">50",
  options = list(sample = 30, seed = 42)
)

authors <- authors_raw |>
  select(id, display_name, works_count, cited_by_count) |>
  filter(!is.na(display_name))

glimpse(authors)
```

```
#> Rows: 30
#> Columns: 4
#> $ id             <chr> "https://openalex.org/A5089560559", "https://openalex.o~
#> $ display_name   <chr> "Karim R. Lakhani", "Jingfeng Xia", "Manghi P.", "Hawba~
#> $ works_count    <int> 241, 76, 145, 67, 75, 243, 161, 51, 64, 185, 105, 150, ~
#> $ cited_by_count <int> 17687, 1680, 1, 551, 393, 2321, 5795, 245, 227, 2, 4984~
```

4.4.2 Fetching works and computing self-citations

For each author, we retrieve a sample of their recent works and examine how many of the citations come from the author's own papers. OpenAlex provides `cited_by_count` at the work level and allows us to query an author's works directly.

```
# For a manageable demonstration, pick 10 authors
authors_sample <- authors |>
  slice_sample(n = min(10, nrow(authors)))

# Function to estimate self-citation rate for one author
estimate_self_citation <- function(author_id, author_name) {
  # Fetch up to 50 recent works by this author
  works <- tryCatch(
    oa_fetch(
      entity = "works",
      author.id = author_id,
```

```r
      from_publication_date = "2018-01-01",
      options = list(sample = 50, seed = 42)
    ),
    error = function(e) tibble()
  )

  if (is.null(works) || nrow(works) == 0) {
    return(tibble(
      author_id = author_id,
      author_name = author_name,
      n_works = 0L,
      total_citations = NA_integer_,
      self_citations = NA_integer_,
      self_cite_rate = NA_real_
    ))
  }

  # Count references that point to the same author's works
  # We look at the referenced_works and check which are by the same author
  total_cites <- sum(works$cited_by_count, na.rm = TRUE)

  # For self-citation, we check which of the works' referenced_works

  # are also in the author's own works list
  own_work_ids <- works$id

  self_cite_count <- works |>
    pull(referenced_works) |>
    map_int(~ sum(.x %in% own_work_ids, na.rm = TRUE))

  tibble(
    author_id = author_id,
    author_name = author_name,
    n_works = nrow(works),
    total_references = sum(map_int(works$referenced_works, length)),
    self_references = sum(self_cite_count),
    self_cite_rate = if_else(
      sum(map_int(works$referenced_works, length)) > 0,
      sum(self_cite_count) / sum(map_int(works$referenced_works, length)),
      NA_real_
    )
  )
}

self_cite_results <- map2_dfr(
```

```
  authors_sample$id,
  authors_sample$display_name,
  estimate_self_citation
)
```

4.4.3 Summarizing results

```
self_cite_results |>
  filter(!is.na(self_cite_rate)) |>
  arrange(desc(self_cite_rate)) |>
  mutate(self_cite_rate = scales::percent(self_cite_rate, accuracy = 0.1)) |>
  gt() |>
  tab_header(
    title = "Self-citation rates among sampled scientometrics authors",
    subtitle = "Based on references in up to 50 recent works per author"
  ) |>
  cols_label(
    author_name = "Author",
    n_works = "Works sampled",
    total_references = "Total references",
    self_references = "Self-references",
    self_cite_rate = "Self-citation rate"
  ) |>
  cols_hide(author_id)
```

4.4.4 Visualization

```
self_cite_clean <- self_cite_results |>
  filter(!is.na(self_cite_rate))

if (nrow(self_cite_clean) > 0) {
  ggplot(self_cite_clean, aes(x = self_cite_rate)) +
    geom_histogram(
      binwidth = 0.02,
      fill = palette_sci(1),
      colour = "white"
    ) +
    geom_vline(
      xintercept = median(self_cite_clean$self_cite_rate, na.rm = TRUE),
      linetype = "dashed", linewidth = 0.8
```

Self-citation rates among sampled scientometrics authors

Based on references in up to 50 recent works per author

Author	Works sampled	Total references	Self-references	Self-citation
Mark Oromaner	7	12	2	1
Melissa S. Anderson	31	463	8	
Hawbash M. Rahim	50	1036	17	
Adolfo Alonso-Arroyo	50	1289	15	
Rickard Danell	11	381	4	
Qing Xie	38	1410	12	0
Jiang Li	50	2006	5	0
Robin Lockerby	5	5	0	0
James L. Wood	5	25	0	0
C. R. Karisiddappa	2	2	0	0

```
    ) +
    scale_x_continuous(labels = scales::percent_format()) +
    labs(
      x = "Self-citation rate (proportion of references to own works)",
      y = "Number of authors"
    ) +
    theme_sci()
}
```

4.4.5 A per-author comparison

```
if (nrow(self_cite_clean) > 0) {
  self_cite_clean |>
    mutate(author_name = fct_reorder(author_name, self_cite_rate)) |>
    ggplot(aes(x = self_cite_rate, y = author_name)) +
    geom_col(fill = palette_sci(1)) +
    scale_x_continuous(labels = scales::percent_format()) +
    labs(x = "Self-citation rate", y = NULL) +
    theme_sci()
}
```

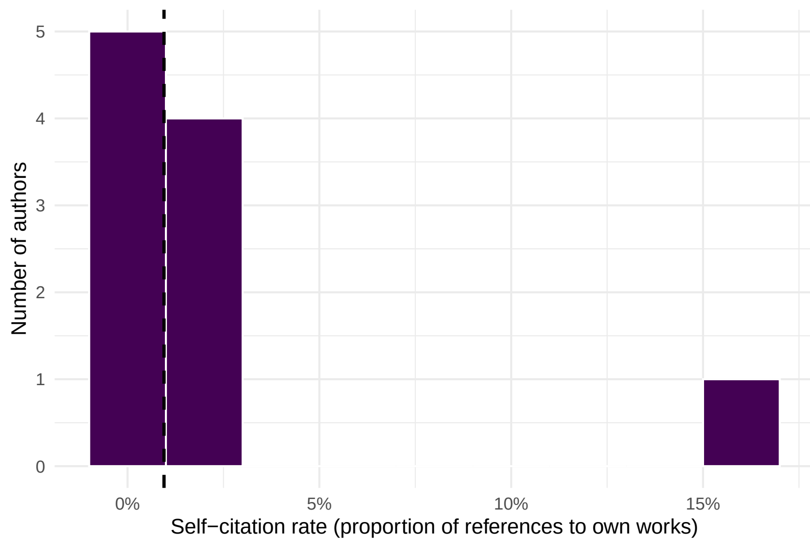

Figure 4.1: Distribution of self-citation rates across sampled scientometrics authors. The dashed line marks the sample median.

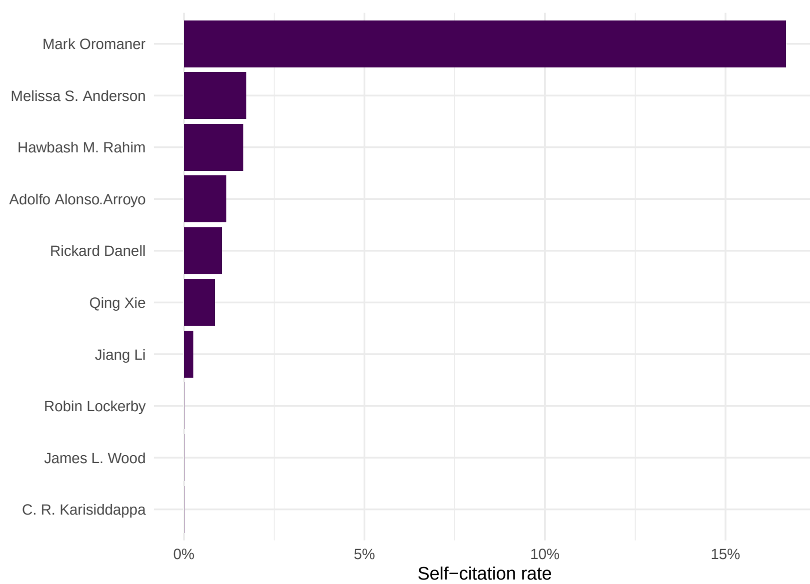

Figure 4.2: Self-citation rates by author, sorted from highest to lowest.

4.5 Diagnostics and interpretation

Self-citation rates vary enormously across disciplines, career stages, and research specialities. When interpreting these results, keep the following in mind:

- **Baseline rates differ by field.** In small, specialised fields, self-citation is often necessary because few other researchers work on the same topic. Self-citation rates of 15–25% are common in niche subfields and do not, by themselves, indicate manipulation.
- **Career stage matters.** Senior researchers with large publication portfolios naturally accumulate more self-citations simply because they have more prior work to build on.
- **Our estimate is a lower bound.** We only count self-references to works that appear in the author's sampled works from OpenAlex. Works outside our sample window (before 2018) or works not indexed by OpenAlex are missed, so the true self-citation rate is likely higher.
- **Context is essential.** A self-citation rate of 30% might be perfectly reasonable for a researcher building a coherent long-term programme, while a rate of 10% could still be problematic if those citations are strategically placed to inflate a particular metric.
- **Never use self-citation rates alone to accuse individuals of misconduct.** As the Leiden Manifesto insists, quantitative indicators must support — not replace — qualitative judgement (Hicks et al., 2015).

4.6 Limitations and responsible use

4.7 Limitations and responsible use

- **Self-citation analysis is not misconduct detection.** High self-citation rates have many legitimate explanations. Presenting a self-citation analysis as evidence of gaming without qualitative context is itself a form of metric misuse.
- **Coverage gaps matter.** OpenAlex does not index all scholarly works. Authors who publish in venues not well covered by OpenAlex will have incomplete reference lists, biasing self-citation estimates.
- **Author disambiguation is imperfect.** OpenAlex uses algorithmic author disambiguation. Common names may be conflated (merging two people into one profile) or split (one person appearing as multiple profiles), distorting individual-level statistics.
- **Snapshot bias.** Our analysis uses a sample of works from a particular time window. Conclusions should not be generalised beyond that window without further analysis.

- **The Leiden Manifesto applies here too.** Every principle in 4.3 applies to *our own* analysis. We must be transparent about methods, allow those evaluated to verify data, and avoid false precision (Hicks et al., 2015).

4.8 Common pitfalls

4.9 Common pitfalls

- **Equating self-citation with misconduct.** Self-citation is a normal part of scientific writing. Only excessive, strategically motivated self-citation is problematic — and even then, the boundary is context-dependent.
- **Using the Journal Impact Factor to judge individual articles.** This is precisely what DORA warns against. A journal's average citation rate says nothing about any single paper in that journal (American Society for Cell Biology, 2012).
- **Ignoring field differences when comparing metrics.** A citation count of 20 is exceptional in mathematics but unremarkable in biomedical research. Any cross-field comparison requires field normalisation (Waltman, 2016).
- **Treating indicators as objective truth.** All bibliometric indicators are constructed measures with specific assumptions, coverage limitations, and known biases. Present them as approximations, not facts.
- **Forgetting Goodhart's Law.** If you propose a new indicator for evaluation purposes, consider how rational actors will respond to being measured by it. Every metric creates incentives, and those incentives may not align with the goals of good research (Goodhart, 1984).

4.10 Exercises

1. **Leiden Manifesto audit.** Select a recent hiring or promotion policy document from a university (many are publicly available). Evaluate it against all ten Leiden Manifesto principles. Which principles does it satisfy? Which does it violate? Write a one-page assessment.

2. **Self-citation across fields.** Modify the worked example to compare self-citation rates between two different research fields (e.g., "bibliometrics" and "genomics"). Do you observe different baseline self-citation rates? What might explain the difference?

3. **DORA compliance check.** Visit the DORA website (sfdora.org) and identify three major funders that have signed the declaration. For each, find their current grant evaluation criteria and assess whether their stated practices are consistent with DORA commitments.

4. **Goodhart's Law thought experiment.** Propose a new indicator for measuring research quality that is not currently in wide use. Then, write a brief analysis of how researchers might game this indicator if it were adopted for promotion decisions. *(Hint: think about both the numerator and denominator of any ratio-based metric.)*

4.11 Solutions

Solutions are provided in F.

4.12 Further reading

- Hicks et al. (2015) — The Leiden Manifesto for Research Metrics. The single most important reference on responsible use of bibliometric indicators.
- American Society for Cell Biology (2012) — The San Francisco Declaration on Research Assessment. The landmark declaration against using journal-level metrics to evaluate individual researchers.
- Wilsdon et al. (2015) — *The Metric Tide*. A comprehensive UK government review of the role of metrics in research assessment, proposing a framework of responsible metrics.
- Goodhart (1984) — Goodhart's original articulation of the principle that a measure ceases to be useful when it becomes a target.
- Waltman (2016) — A thorough review of citation impact indicators, including their known biases and appropriate uses.
- Merton (1968) — The Matthew Effect in science; describes how advantage accumulates, a dynamic that citation metrics can reinforce.
- Garfield (1955) — The foundational paper on citation indexing. Essential for understanding the origins of the system we now critique.

4.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
```

```
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] gt_1.3.0        tidytext_0.4.3  glue_1.8.1      openalexR_3.0.1
#>  [5] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
#>  [9] purrr_1.2.2     readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
#> [13] ggplot2_4.0.3   tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] janeaustenr_1.0.0  viridis_0.6.5      utf8_1.2.6         generics_0.1.4
#>  [5] xml2_1.5.2         lattice_0.22-6     stringi_1.8.7      hms_1.1.4
#>  [9] digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5     grid_4.4.1
#> [13] timechange_0.4.0   RColorBrewer_1.1-3 bookdown_0.46      fastmap_1.2.0
#> [17] Matrix_1.7-0       rprojroot_2.1.1    jsonlite_2.0.0     gridExtra_2.3
#> [21] stopwords_2.3      httr_1.4.8         viridisLite_0.4.3  scales_1.4.0
#> [25] codetools_0.2-20   cli_3.6.6          rlang_1.2.0        tokenizers_0.3.0
#> [29] withr_3.0.2        yaml_2.3.12        otel_0.2.0         tools_4.4.1
#> [33] tzdb_0.5.0         here_1.0.2         curl_7.1.0         vctrs_0.7.3
#> [37] R6_2.6.1           lifecycle_1.0.5    fs_2.1.0           pkgconfig_2.0.3
#> [41] pillar_1.11.1      gtable_0.3.6       Rcpp_1.1.1-1.1     xfun_0.57
#> [45] tidyselect_1.2.1   rstudioapi_0.18.0  knitr_1.51         dichromat_2.0-0.1
#> [49] farver_2.1.2       SnowballC_0.7.1    htmltools_0.5.9    labeling_0.4.3
#> [53] rmarkdown_2.31     compiler_4.4.1     S7_0.2.2
```

Chapter 5

Data Sources Compared

5.1 Learning objectives

After completing this chapter, you will be able to:

- Describe the coverage, access model, and key strengths of twelve major bibliographic data sources
- Select an appropriate data source (or combination of sources) for a given research question
- Compare metadata completeness across sources for the same set of publications
- Evaluate the trade-offs between proprietary and open data sources for reproducible research
- Retrieve and compare metadata from OpenAlex and Crossref using R

5.2 Setup

```r
library(tidyverse)
library(openalexR)
library(rcrossref)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```r
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

5.3 Conceptual background

Every bibliometric analysis is fundamentally constrained by the data source that feeds it. The database you choose determines which publications are visible, which citations are counted, and which metadata fields are available. As Visser et al. (2021) demonstrated in a large-scale comparison, the major bibliographic databases differ substantially in coverage, with these differences affecting research outcomes in measurable ways. Similarly, Singh et al. (2021) showed that journal coverage varies considerably across Web of Science, Scopus, and Dimensions, with each database exhibiting distinct disciplinary strengths.

The choice of data source is not merely a technical decision — it is a methodological one with direct consequences for the validity of results. A citation analysis conducted on Web of Science will produce different numbers than the same analysis on Scopus, Dimensions, or OpenAlex, because each database indexes a different set of publications and computes citations over a different corpus. Researchers must therefore understand these differences and report them transparently, as emphasised by the Leiden Manifesto's fourth principle: keep data collection and analytical processes open, transparent, and simple (Hicks et al., 2015).

The landscape of bibliographic data sources has shifted dramatically in recent years. The traditional duopoly of Web of Science and Scopus has been challenged by open alternatives — most notably OpenAlex (Priem et al., 2022), which provides free, unrestricted access to over 250 million scholarly works. Crossref serves as the backbone registration agency for DOIs and provides rich metadata for over 150 million records. Discipline-specific databases like PubMed, ADS, and arXiv serve targeted communities with deep, curated coverage.

For this book, we adopt OpenAlex as the primary data source because it is fully open, freely accessible via API without authentication, and provides coverage comparable to proprietary alternatives. However, responsible bibliometric practice often requires consulting multiple sources to cross-validate findings and fill coverage gaps.

5.3.1 Twelve data sources at a glance

Below we summarise twelve major sources. The subsequent sections examine each in detail, and the comparison table provides a structured overview.

Web of Science (WoS). Maintained by Clarivate, WoS is the oldest and most established citation index, originating from Eugene Garfield's Science Citation Index (Garfield, 1955). It covers approximately 90 million records from over 21,000 journals, with citation data back to 1900. Access requires an institutional subscription. Its strengths are curated journal selection, long historical coverage, and the Journal Citation Reports (JCR). Its main limitations are cost, restricted API access, and lower coverage of non-English literature, social sciences, and humanities.

Scopus. Elsevier's Scopus indexes over 94 million records from approximately 27,000 journals, with broader coverage of non-English titles and conference proceedings than WoS. It provides its own citation metrics (CiteScore, SJR, SNIP) and a well-documented API. Like WoS, it requires an institutional subscription.

Dimensions. Operated by Digital Science, Dimensions indexes over 140 million publications including journal articles, books, book chapters, preprints, clinical trials, patents, and policy documents. It offers a free tier with limited functionality and a paid tier with full API access. Its distinctive strength is linking publications to grants, patents, clinical trials, and policy documents.

OpenAlex. The successor to the discontinued Microsoft Academic Graph, OpenAlex is a fully open, freely accessible index of over 250 million scholarly works (Priem et al., 2022). It aggregates data from Crossref, PubMed, institutional repositories, and other open sources. Its API requires no authentication (though a polite-pool email is recommended). OpenAlex provides author disambiguation, institutional affiliation data, and concept tagging. Its primary limitations are metadata quality (algorithmically parsed, not manually curated) and less granular subject classification than curated databases.

The Lens. Operated by Cambia, The Lens provides free access to over 300 million scholarly works and 160 million patent records. It aggregates data from Crossref, PubMed, Microsoft Academic, and patent offices. Its unique strength is integrated patent-paper linkage, making it valuable for technology-transfer and innovation studies. API access is available via a free institutional plan.

Crossref. Crossref is the DOI registration agency for scholarly publications, covering over 150 million records. It is not a citation index per se — it provides metadata deposited by publishers, including titles, authors, abstracts, reference lists, and licensing information. Its REST API is freely accessible. Crossref's strengths are comprehensive DOI coverage and detailed reference metadata; its limitation is that metadata quality depends entirely on what publishers deposit.

PubMed/MEDLINE. Maintained by the US National Library of Medicine, PubMed indexes over 36 million records in the biomedical and life sciences. Its key strengths are curated MeSH (Medical Subject Headings) indexing, structured abstracts, and deep coverage of clinical and biomedical literature. PubMed does not provide its own citation counts but can be paired with citation data from other sources.

Google Scholar. Google Scholar provides the broadest coverage of any scholarly search engine, indexing journal articles, conference papers, theses, books, patents, and grey literature. However, it provides no API, no bulk data export, and no structured metadata. It is useful for discovery and for its "Cited by" feature but is not suitable for systematic bibliometric analysis.

Semantic Scholar. Operated by the Allen Institute for AI, Semantic Scholar indexes over 200 million papers and provides a free API with rich metadata including algorithmically extracted entities, TLDR summaries, and citation context. It is particularly strong in computer science, biomedical, and neuroscience literature.

NASA/ADS (Astrophysics Data System). ADS is a curated database of over 16 million records in astronomy, astrophysics, and physics. It provides excellent citation tracking, full-text search, and a well-documented API. For astronomical research, ADS is the gold standard.

arXiv. The original preprint server, arXiv hosts over 2.5 million preprints in physics, mathematics, computer science, quantitative biology, quantitative finance, statistics, and electrical engineering. It does not provide citation data directly but its records are indexed by other databases. Its API provides metadata access.

bioRxiv/medRxiv. The preprint servers for biology and health sciences, hosting over 300,000 preprints. They provide API access to metadata and full text and are indexed by OpenAlex, Dimensions, and other aggregators.

5.4 Worked example

We now compare what OpenAlex and Crossref return for the same set of DOIs. This is a practical illustration of how metadata completeness varies across sources.

5.4.1 Selecting DOIs

We start with a small set of well-known scientometrics papers whose DOIs we know.

```
dois <- c(
  "10.1038/520429a",            # Hicks et al. 2015, Leiden Manifesto
  "10.1073/pnas.0507655102",    # Hirsch 2005, h-index
  "10.1007/s11192-006-0144-7",  # Egghe 2006, g-index
  "10.1016/j.joi.2017.08.007",  # Aria & Cuccurullo 2017, bibliometrix
  "10.1126/science.122.3159.108", # Garfield 1955, citation indexes
  "10.48550/arXiv.2205.01833",  # Priem et al. 2022, OpenAlex
```

```
  "10.1162/qss_a_00112",        # Visser et al. 2021, database comparison
  "10.1007/s11192-021-03948-5"  # Singh et al. 2021, journal coverage
)
```

5.4.2 Fetching from OpenAlex

```
oa_results <- oa_fetch(
  entity = "works",
  doi = dois
)

oa_meta <- oa_results |>
  transmute(
    doi = doi,
    source = "OpenAlex",
    title = display_name,
    year = publication_year,
    cited_by = cited_by_count,
    has_abstract = !is.na(abstract),
    type = type
  )

glimpse(oa_meta)
```

```
#> Rows: 9
#> Columns: 7
#> $ doi          <chr> "https://doi.org/10.1016/j.joi.2017.08.007", "https://doi~
#> $ source       <chr> "OpenAlex", "OpenAlex", "OpenAlex", "OpenAlex", "OpenAlex~
#> $ title        <chr> "bibliometrix : An R-tool for comprehensive science mappi~
#> $ year         <int> 2017, 2005, 2015, 2006, 1955, 2021, 2021, 2022, 1970
#> $ cited_by     <int> 13883, 11471, 2585, 2288, 2273, 1587, 739, 276, 4
#> $ has_abstract <lgl> FALSE, TRUE, FALSE, FALSE, TRUE, FALSE, TRUE, TRUE, FALSE
#> $ type         <chr> "article", "article", "article", "article", "article", "a~
```

5.4.3 Fetching from Crossref

```
cr_results <- cr_works(dois = dois)

cr_meta <- cr_results$data |>
  transmute(
```

```r
    doi = doi,
    source = "Crossref",
    title = title,
    year = as.integer(
      coalesce(
        substr(issued, 1, 4),
        substr(deposited, 1, 4)
      )
    ),
    cited_by = as.integer(is.referenced.by.count),
    has_abstract = !is.na(abstract),
    type = type
  )

glimpse(cr_meta)
```

```
#> Rows: 7
#> Columns: 7
#> $ doi          <chr> "10.1038/520429a", "10.1073/pnas.0507655102", "10.1007/s1~
#> $ source       <chr> "Crossref", "Crossref", "Crossref", "Crossref", "Crossref~
#> $ title        <chr> "Bibliometrics: The Leiden Manifesto for research metrics~
#> $ year         <int> 2015, 2005, 2006, 2017, 1955, 2021, 2021
#> $ cited_by     <int> 1941, 8500, 1818, 11758, 1750, 656, 1575
#> $ has_abstract <lgl> FALSE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE
#> $ type         <chr> "journal-article", "journal-article", "journal-article", ~
```

5.4.4 Comparing metadata completeness

```r
comparison <- bind_rows(oa_meta, cr_meta) |>
  mutate(short_doi = str_extract(doi, "[^/]+/[^/]+$"))

comparison_wide <- comparison |>
  select(short_doi, source, year, cited_by, has_abstract, type) |>
  pivot_wider(
    names_from = source,
    values_from = c(year, cited_by, has_abstract, type),
    names_glue = "{source}_{.value}"
  )

comparison_wide |>
  gt() |>
  tab_header(
    title = "Metadata comparison: OpenAlex vs. Crossref",
```

Metadata comparison: OpenAlex vs. Crossref

Same DOIs queried from both sources

DOI (short)	OA Year	CR Year	OA Citations	CR Citations	OA Abstrac
10.1016/j.joi.2017.08.007	2017	2017	13883	11758	FALSE
10.1073/pnas.0507655102	2005	2005	11471	8500	TRUE
10.1038/520429a	2015, 1970	2015	2585, 4	1941	FALSE, FAI
10.1007/s11192-006-0144-7	2006	2006	2288	1818	FALSE
10.1126/science.122.3159.108	1955	1955	2273	1750	TRUE
10.1007/s11192-021-03948-5	2021	2021	1587	1575	FALSE
10.1162/qss_a_00112	2021	2021	739	656	TRUE
10.48550/arxiv.2205.01833	2022		276		TRUE

```
    subtitle = "Same DOIs queried from both sources"
  ) |>
  cols_label(
    short_doi = "DOI (short)",
    OpenAlex_year = "OA Year",
    Crossref_year = "CR Year",
    OpenAlex_cited_by = "OA Citations",
    Crossref_cited_by = "CR Citations",
    OpenAlex_has_abstract = "OA Abstract?",
    Crossref_has_abstract = "CR Abstract?",
    OpenAlex_type = "OA Type",
    Crossref_type = "CR Type"
  )
```

5.4.5 Visualization: citation count comparison

```
cite_compare <- comparison |>
  select(short_doi, source, cited_by) |>
  pivot_wider(names_from = source, values_from = cited_by, values_fn = first)

if (nrow(cite_compare) > 0 && all(c("OpenAlex", "Crossref") %in% names(cite_compare))) {
  max_cite <- max(c(cite_compare$OpenAlex, cite_compare$Crossref), na.rm = TRUE)

  ggplot(cite_compare, aes(x = Crossref, y = OpenAlex)) +
    geom_abline(slope = 1, intercept = 0, linetype = "dashed", colour = "grey50") +
```

```
    geom_point(size = 3, colour = palette_sci(1)) +
    geom_text(
      aes(label = short_doi),
      size = 2.5, nudge_y = max_cite * 0.03, check_overlap = TRUE
    ) +
    scale_x_continuous(labels = scales::comma_format()) +
    scale_y_continuous(labels = scales::comma_format()) +
    coord_equal(xlim = c(0, max_cite * 1.1), ylim = c(0, max_cite * 1.1)) +
    labs(x = "Crossref citation count", y = "OpenAlex citation count") +
    theme_sci()
}
```

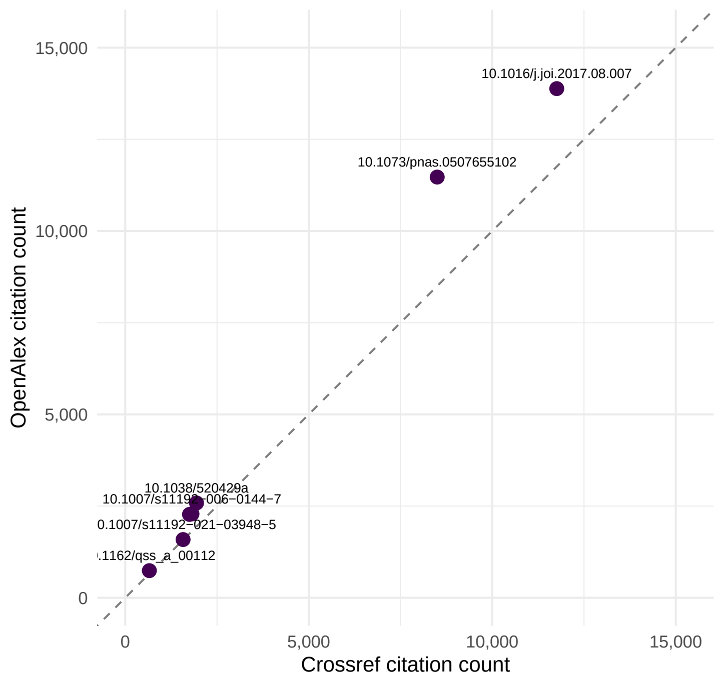

Figure 5.1: Citation counts reported by OpenAlex vs. Crossref for the same set of DOIs. Points above the diagonal indicate higher counts in OpenAlex.

5.4.6 Summary comparison table

```
tribble(
  ~Source, ~Coverage, ~`Access model`, ~API, ~`Citation data`, ~Strengths,
  "Web of Science", "~90M records, 21K+ journals", "Subscription", "Yes (paid)", "Yes
```

```
  "Scopus", "~94M records, 27K+ journals", "Subscription", "Yes (paid)", "Yes (complete)", "Broad
  "Dimensions", "~140M records", "Freemium", "Yes (free/paid)", "Yes", "Links pubs to grants, pat
  "OpenAlex", "~250M records", "Fully open", "Yes (free)", "Yes", "Open, broad coverage, author d
  "The Lens", "~300M scholarly + 160M patent", "Free (institutional)", "Yes (free)", "Partial", "
  "Crossref", "~150M records", "Open", "Yes (free)", "Reference counts only", "DOI registry, refe
  "PubMed", "~36M records", "Open", "Yes (free)", "No (use with others)", "MeSH indexing, biomedi
  "Google Scholar", "Broadest (est. 400M+)", "Free (no bulk)", "No", "Yes (no export)", "Discover
  "Semantic Scholar", "~200M records", "Free", "Yes (free)", "Yes", "AI-extracted entities, citat
  "NASA/ADS", "~16M records", "Open", "Yes (free)", "Yes", "Gold standard for astronomy/astrophys
  "arXiv", "~2.5M preprints", "Open", "Yes (free)", "No", "Preprints in physics, math, CS",
  "bioRxiv/medRxiv", "~300K preprints", "Open", "Yes (free)", "No", "Biology and health sciences
) |>
  gt() |>
  tab_header(
    title = "Twelve bibliographic data sources compared"
  ) |>
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels()
  ) |>
  cols_width(
    Source ~ px(130),
    Coverage ~ px(170),
    `Access model` ~ px(100),
    API ~ px(100),
    `Citation data` ~ px(120),
    Strengths ~ px(250)
  )
```

5.5 Diagnostics and interpretation

When comparing metadata across sources, consider the following diagnostic checks:

- **DOI match rate.** Not all DOIs may resolve in every database. Check how many of your queried DOIs returned results from each source. If a DOI is missing from one source, this reveals a coverage gap.
- **Citation count discrepancies.** Citation counts will almost always differ between sources because each database counts citations over a different corpus. OpenAlex typically reports higher counts than Crossref because OpenAlex tracks citations across its full 250M-record corpus, whereas Crossref only counts "is-referenced-by" links deposited by publishers.

- **Abstract availability.** Crossref abstracts depend on publisher deposits and are often absent, especially for older publications. OpenAlex aggregates abstracts from multiple sources and generally has better coverage.
- **Publication year.** Minor discrepancies (e.g., online-first vs. print dates) are common. Check whether your analysis is sensitive to the exact year assigned.
- **Type classification.** Crossref and OpenAlex use different type taxonomies. A record classified as "journal-article" in Crossref might appear as "article" in OpenAlex. Harmonise types before analysis.

The key takeaway is that no single source is complete or definitive. Cross-validation — fetching the same records from multiple sources — is a hallmark of rigorous bibliometric practice.

5.6 Limitations and responsible use

5.7 Limitations and responsible use

- **No database is complete.** Every source has coverage gaps — by language, discipline, publication type, or geography. Analyses based on a single source will systematically miss some literature. Always report which source was used and its known biases (Hicks et al., 2015).
- **Coverage does not equal quality.** A larger database is not necessarily better. Google Scholar has the broadest coverage but also the highest noise level. Curated databases like WoS and PubMed trade coverage for precision.
- **Proprietary sources limit reproducibility.** Analyses based on Web of Science or Scopus cannot be independently reproduced by researchers without institutional access. For fully reproducible work, prefer open sources like OpenAlex, Crossref, or PubMed (Priem et al., 2022).
- **Metadata quality is variable.** Even within a single source, metadata quality varies by publisher, discipline, and publication date. Older records often have sparser metadata. Always inspect and clean your data before analysis.
- **Citation counts are not comparable across sources.** Because each database counts citations over a different corpus, citation counts from OpenAlex, Scopus, and WoS are not interchangeable. Never mix citation counts from different sources in a single analysis without explicit justification and documentation (Visser et al., 2021).

5.8 Common pitfalls

5.9 Common pitfalls

- **Assuming one source is enough.** Researchers often default to a single database (typically WoS or Scopus) without evaluating whether it covers their field adequately. Social sciences, humanities, and non-English literatures are systematically underrepresented in WoS (Singh et al., 2021).
- **Comparing citation counts across databases.** Stating that a paper has "500 citations" is meaningless without specifying the source. Different databases will report different numbers for the same paper.
- **Ignoring access restrictions in reproducibility claims.** If your analysis uses Scopus data, other researchers need Scopus access to reproduce it. State this explicitly and consider providing OpenAlex-based alternatives.
- **Treating Google Scholar as a systematic source.** Google Scholar has no API, no structured export, and no quality control on what it indexes. It is useful for discovery but not for systematic bibliometric analysis.
- **Conflating preprints and peer-reviewed articles.** Sources like arXiv and bioRxiv index preprints that may later be published in journals. If your corpus includes both preprints and published versions without deduplication, you will double-count some works.

5.10 Exercises

1. **Coverage gap analysis.** Select 20 DOIs from a field of your choice. Query them in OpenAlex and Crossref. How many DOIs are found in each? Which source has better abstract coverage? Summarise your findings in a table.

2. **Disciplinary bias.** Use OpenAlex to compare the number of indexed works in two contrasting fields (e.g., molecular biology vs. philosophy). Normalise by the estimated number of active researchers in each field. What does this tell you about coverage bias?

3. **Citation count concordance.** For a set of at least 50 works, retrieve citation counts from both OpenAlex and Crossref. Compute the Pearson and Spearman correlations between the two sets of counts. Plot the results. Are the two sources more concordant for highly cited or lowly cited papers?

4. **Preprint tracking.** Select 10 recent bioRxiv DOIs and check whether they appear in OpenAlex. For those that do, check whether OpenAlex links them to a published journal version. What proportion of your preprints have been published?

5. **Source selection exercise.** A colleague asks you to conduct a bibliometric analysis of nursing research in Sub-Saharan Africa. Which data source(s) would you recommend and why? Write a brief justification considering coverage, cost, and reproducibility.

5.11 Solutions

Solutions are provided in F.

5.12 Further reading

- Visser et al. (2021) — Large-scale comparison of Scopus, Web of Science, Dimensions, Crossref, and Microsoft Academic. Essential reading for understanding how source choice affects results.
- Singh et al. (2021) — Comparative analysis of journal coverage across WoS, Scopus, and Dimensions. Quantifies disciplinary differences in database coverage.
- Priem et al. (2022) — The OpenAlex paper. Describes the data model, coverage, and design philosophy of the open alternative to proprietary databases.
- Garfield (1955) — The original paper proposing citation indexing. Historical context for understanding why these databases exist.
- Hicks et al. (2015) — The Leiden Manifesto. Principle 4 (transparency) and Principle 6 (field variation) are directly relevant to data source selection.
- Waltman (2016) — Review of citation impact indicators. Discusses how indicator properties interact with database coverage.

5.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
```

```
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] rcrossref_1.2.1 gt_1.3.0        tidytext_0.4.3  glue_1.8.1
#>  [5] openalexR_3.0.1 lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0
#>  [9] dplyr_1.2.1     purrr_1.2.2     readr_2.2.0     tidyr_1.3.2
#> [13] tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6       xfun_0.57          htmlwidgets_1.6.4  lattice_0.22-6
#>  [5] tzdb_0.5.0         vctrs_0.7.3        tools_4.4.1        generics_0.1.4
#>  [9] curl_7.1.0         janeaustenr_1.0.0  pkgconfig_2.0.3    tokenizers_0.3.0
#> [13] Matrix_1.7-0       RColorBrewer_1.1-3 S7_0.2.2           lifecycle_1.0.5
#> [17] compiler_4.4.1     farver_2.1.2       codetools_0.2-20   httpuv_1.6.17
#> [21] htmltools_0.5.9    SnowballC_0.7.1    yaml_2.3.12        later_1.4.8
#> [25] pillar_1.11.1      DT_0.34.0          viridis_0.6.5      mime_0.13
#> [29] stopwords_2.3      tidyselect_1.2.1   digest_0.6.39      stringi_1.8.7
#> [33] bookdown_0.46      labeling_0.4.3     rprojroot_2.1.1    fastmap_1.2.0
#> [37] grid_4.4.1         here_1.0.2         cli_3.6.6          magrittr_2.0.5
#> [41] triebeard_0.4.1    crul_1.6.0         dichromat_2.0-0.1  utf8_1.2.6
#> [45] withr_3.0.2        promises_1.5.0     scales_1.4.0       timechange_0.4.0
#> [49] rmarkdown_2.31     httr_1.4.8         otel_0.2.0         gridExtra_2.3
#> [53] hms_1.1.4          shiny_1.13.0       evaluate_1.0.5     knitr_1.51
#> [57] miniUI_0.1.2       viridisLite_0.4.3  urltools_1.7.3.1   rlang_1.2.0
#> [61] Rcpp_1.1.1-1.1     httpcode_0.3.0     xtable_1.8-8       xml2_1.5.2
#> [65] rstudioapi_0.18.0  jsonlite_2.0.0     plyr_1.8.9         R6_2.6.1
#> [69] fs_2.1.0
```

Twelve bibliographic data sources compared

Source	**Coverage**	**Access model**	**API**	**Citation data**
Web of Science	~90M records, 21K+ journals	Subscription	Yes (paid)	Yes (complete)
Scopus	~94M records, 27K+ journals	Subscription	Yes (paid)	Yes (complete)
Dimensions	~140M records	Freemium	Yes (free/paid)	Yes
OpenAlex	~250M records	Fully open	Yes (free)	Yes
The Lens	~300M scholarly + 160M patent	Free (institutional)	Yes (free)	Partial
Crossref	~150M records	Open	Yes (free)	Reference counts only
PubMed	~36M records	Open	Yes (free)	No (use with others)
Google Scholar	Broadest (est. 400M+)	Free (no bulk)	No	Yes (no export)
Semantic Scholar	~200M records	Free	Yes (free)	Yes
NASA/ADS	~16M records	Open	Yes (free)	Yes
arXiv	~2.5M preprints	Open	Yes (free)	No
bioRxiv/medRxiv	~300K preprints	Open	Yes (free)	No

Part II

Data Acquisition in R

Chapter 6

APIs and Packages

6.1 Learning objectives

After completing this chapter, you will be able to:

- Query the OpenAlex API using `openalexR` to retrieve works, authors, institutions, and concepts
- Apply filters, pagination, and sampling to control the size and scope of API results
- Deduplicate records by DOI and clean author and affiliation metadata
- Cache API responses locally so repeated book builds do not re-query the API
- Use `rcrossref` and `pubmedR` as complementary data sources

6.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

6.3 Conceptual background

Bibliometric research begins with data. The quality and completeness of a bibliographic corpus determines the validity of every downstream analysis — from citation counts to network maps to topic models.

Historically, Web of Science and Scopus have dominated as data sources for scientometric research. Both require institutional subscriptions and impose restrictions on bulk data use. The launch of **OpenAlex** in 2022 changed this landscape fundamentally (Priem et al., 2022). OpenAlex is a fully open index of over 250 million scholarly works, built from Crossref, PubMed, institutional repositories, and other open sources. It provides a free REST API with no authentication required, making it the ideal backbone for reproducible bibliometric research.

OpenAlex organises its data around five entity types: **works** (publications), **authors**, **sources** (journals, repositories), **institutions**, and **topics** (hierarchical subject tags). Each entity has a unique OpenAlex ID and can be queried via filters on metadata fields such as publication date, citation count, open-access status, and institutional affiliation.

The `openalexR` package provides a tidy R interface to the OpenAlex API. It handles pagination, rate limiting, and result parsing, returning clean tibbles ready for analysis. For metadata not covered by OpenAlex — such as reference lists and funding information — `rcrossref` provides access to Crossref's extensive metadata registry, and `pubmedR` covers the biomedical literature indexed in PubMed/MEDLINE.

6.4 Worked example

We build a small corpus of recent scientometrics research, demonstrating the full acquisition-to-clean-data pipeline.

6.4.1 Querying works from OpenAlex

The core function is `oa_fetch()`. We search for works published in the journal *Scientometrics* between 2020 and 2023, sampling 200 records for a manageable demonstration.

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
```

```r
  options = list(sample = 200, seed = 42)
)

glimpse(works)
```

```
#> Rows: 200
#> Columns: 45
#> $ id                          <chr> "https://openalex.org/W3004974449", "htt~
#> $ title                       <chr> "Self-plagiarism in academic journal art~
#> $ display_name                <chr> "Self-plagiarism in academic journal art~
#> $ authorships                 <list> [<tbl_df[1 x 7]>], [<tbl_df[2 x 7]>], [~
#> $ abstract                    <chr> NA, NA, NA, NA, NA, NA, "Abstract In Chi~
#> $ doi                         <chr> "https://doi.org/10.1007/s11192-020-0337~
#> $ publication_date            <date> 2020-02-08, 2022-04-19, 2023-06-17, 202~
#> $ publication_year            <int> 2020, 2022, 2023, 2020, 2021, 2023, 2021~
#> $ relevance_score             <dbl> 0.999, 0.997, 0.997, 0.996, 0.995, 0.995~
#> $ fwci                        <dbl> 4.242, 1.877, 0.474, 1.466, 0.656, 5.202~
#> $ cited_by_count              <int> 27, 18, 1, 14, 8, 11, 20, 10, 35, 14, 1,~
#> $ counts_by_year              <list> [<data.frame[7 x 2]>], [<data.frame[5 x~
#> $ ids                         <list> <"https://openalex.org/W3004974449", "h~
#> $ type                        <chr> "article", "article", "article", "articl~
#> $ is_oa                       <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ is_oa_anywhere              <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ oa_status                   <chr> "closed", "closed", "closed", "closed", ~
#> $ oa_url                      <chr> NA, NA, NA, NA, NA, NA, "https://link.sp~
#> $ any_repository_has_fulltext <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ source_display_name         <chr> "Scientometrics", "Scientometrics", "Sci~
#> $ source_id                   <chr> "https://openalex.org/S148561398", "http~
#> $ issn_l                      <chr> "0138-9130", "0138-9130", "0138-9130", "~
#> $ host_organization           <chr> "https://openalex.org/P4310320108", "htt~
#> $ host_organization_name      <chr> "Springer Nature (Netherlands)", "Spring~
#> $ landing_page_url            <chr> "https://doi.org/10.1007/s11192-020-0337~
#> $ pdf_url                     <chr> NA, NA, NA, NA, NA, NA, "https://link.sp~
#> $ license                     <chr> NA, NA, NA, NA, NA, NA, "cc-by", "cc-by"~
#> $ version                     <chr> "publishedVersion", "publishedVersion", ~
#> $ referenced_works            <list> <"https://openalex.org/W587768734", "ht~
#> $ referenced_works_count      <int> 45, 44, 76, 63, 74, 72, 63, 22, 35, 120,~
#> $ related_works               <list> <"https://openalex.org/W4389095033", "h~
#> $ concepts                    <list> [<data.frame[15 x 5]>], [<data.frame[17~
#> $ topics                      <list> [<tbl_df[8 x 5]>], [<tbl_df[12 x 5]>], ~
#> $ keywords                    <list> [<data.frame[13 x 3]>], [<data.frame[13~
#> $ is_paratext                 <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ is_retracted                <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
#> $ language                    <chr> "en", "en", "en", "en", "en", "en", "en"~
#> $ sustainable_development_goals <list> [<data.frame[1 x 3]>], [<data.frame[1 x~
```

```
#> $ awards                          <list> <"https://openalex.org/G6558653442", "M~
#> $ funders                         <list> [<data.frame[1 x 3]>], [<data.frame[2 x~
#> $ apc                             <list> [<data.frame[2 x 5]>], [<data.frame[2 x~
#> $ first_page                      <chr> "299", "3437", "4611", "633", "4609", "4~
#> $ last_page                       <chr> "319", "3470", "4650", "661", "4638", "4~
#> $ volume                          <chr> "123", "127", "128", "124", "126", "128"~
#> $ issue                           <chr> "1", "6", "8", "1", "6", "9", "7", "4", ~
```

The result is a tibble where each row is a work and columns include `id`, `display_name`, `publication_date`, `cited_by_count`, `doi`, and nested list-columns for authors and concepts.

6.4.2 Inspecting the results

```r
works |>
  select(display_name, publication_date, cited_by_count, doi) |>
  arrange(desc(cited_by_count)) |>
  head(10)
```

```
#> # A tibble: 10 x 4
#>    display_name                             publication_date cited_by_count doi
#>    <chr>                                    <date>                    <int> <chr>
#>  1 'Are principals instructional leaders ~ 2020-01-30                  230 http~
#>  2 Link-based approach to study scientifi~ 2021-07-10                  192 http~
#>  3 Open peer review: promoting transparen~ 2020-05-26                  161 http~
#>  4 Identifying interdisciplinary topics a~ 2023-07-03                  133 http~
#>  5 Tracking developments in artificial in~ 2021-02-25                  106 http~
#>  6 A bibliometric review of research on i~ 2021-04-26                   89 http~
#>  7 Scientific laws of research funding to~ 2022-03-05                   83 http~
#>  8 RETRACTED ARTICLE: Predatory publishin~ 2021-02-07                   81 http~
#>  9 Web mining for innovation ecosystem ma~ 2020-10-14                   78 http~
#> 10 The impact of research output on econo~ 2020-03-28                   71 http~
```

6.4.3 Querying authors

We can also retrieve author-level data. Let us look up authors associated with scientometrics research by filtering on the relevant topic.

```r
authors <- oa_fetch(
  entity = "authors",
  topics.id = "T10102",
  works_count = ">50",
```

```
  options = list(sample = 50, seed = 42)
)

authors |>
  select(display_name, works_count, cited_by_count) |>
  arrange(desc(cited_by_count)) |>
  head(10)
```

```
#> # A tibble: 10 x 3
#>    display_name        works_count cited_by_count
#>    <chr>                     <int>          <int>
#>  1 Karim R. Lakhani            241          17687
#>  2 Katy Börner                 357          13619
#>  3 Lonnie W. Aarssen           237          13506
#>  4 Richard Williams            200          12009
#>  5 Dashun Wang                 165           9987
#>  6 Howard D. White              94           6028
#>  7 Melissa S. Anderson         161           5795
#>  8 Mark Oromaner               105           4984
#>  9 Duane Knudson               249           4312
#> 10 Amber E Budden               90           3034
```

6.4.4 Deduplication by DOI

Real-world corpora often contain duplicates when merging results from multiple queries or sources. The `dedupe_by_doi()` function from our companion package handles this cleanly.

```
works_with_dupes <- bind_rows(works, works[1:10, ])
cat(glue("Before dedup: {nrow(works_with_dupes)} rows\n\n"))
```

```
#> Before dedup: 210 rows
```

```
works_clean <- dedupe_by_doi(works_with_dupes)
cat(glue("After dedup:  {nrow(works_clean)} rows\n"))
```

```
#> After dedup:  200 rows
```

6.4.5 Caching for reproducible builds

The `fetch_openalex()` wrapper in `R/api_helpers.R` automatically caches API responses to disk. On subsequent calls with the same arguments, it reads from cache instead of hitting the API.

```
works_cached <- fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2023-01-01",
  to_publication_date = "2023-06-30",
  options = list(sample = 50, seed = 42),
  cache_dir = "_freeze/openalex_cache",
  cache_days = 30
)

nrow(works_cached)
```

```
#> [1] 50
```

6.4.6 Complementary sources: Crossref

For detailed metadata (reference lists, funders, licenses), Crossref is invaluable.

```
# Requires network access - cached output used in book builds
cr_work <- rcrossref::cr_works(doi = "10.1007/s11192-017-2300-7")
cr_work$data |>
  select(title, container.title, deposited, is.referenced.by.count)
```

6.4.7 Visualization

```
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Publication year", y = "Number of works") +
  theme_sci()
```

6.5 Diagnostics and interpretation

When working with API data, always verify:

- **Completeness**: Compare your result count against the API's reported `meta$count` to confirm you received all expected records. With sampling, the count reflects the sample, not the full set.

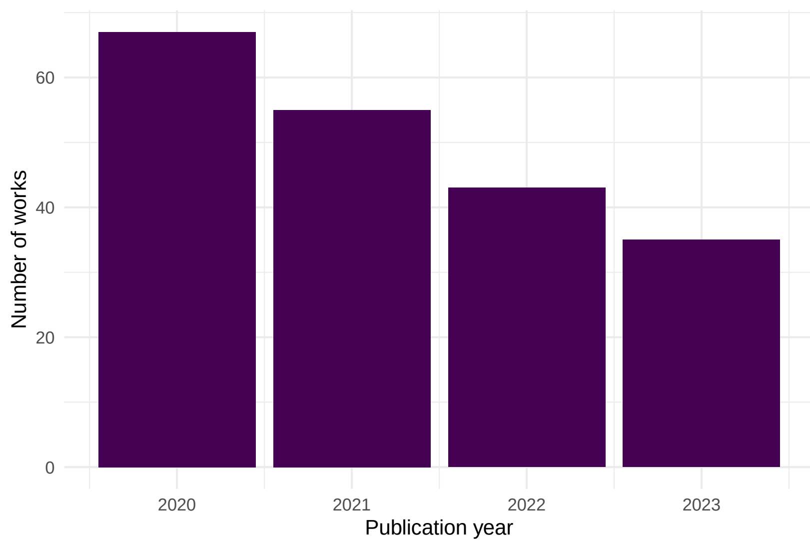

Figure 6.1: Annual publication counts for sampled Scientometrics articles (2020–2023).

- **Coverage dates**: Check `min(works$publication_date)` and `max(works$publication_date)` to confirm your date filters were applied correctly.
- **DOI presence**: Not all works have DOIs. Check `mean(!is.na(works$doi))` to understand the DOI coverage rate in your corpus.
- **Nested columns**: Author and concept columns are list-columns. Use `tidyr::unnest()` carefully and inspect for `NULL` entries before unnesting.

```
cat(glue("Total works: {nrow(works)}\n"))
```

```
#> Total works: 200
```

```
cat(glue("DOI coverage: {scales::percent(mean(!is.na(works$doi)))}\n"))
```

```
#> DOI coverage: 100%
```

```
cat(glue("Date range: {min(works$publication_date)} to {max(works$publication_date)}\n"))
```

```
#> Date range: 2020-01-02 to 2023-12-27
```

6.6 Limitations and responsible use

6.7 Limitations and responsible use

- **Coverage bias**: OpenAlex is strongest in English-language, peer-reviewed journal articles. Conference proceedings, books, and non-English literature may be underrepresented. Always report the data source and its known coverage limitations.
- **Metadata quality**: Author names and affiliations are parsed algorithmically and may contain errors, especially for names with diacritics or non-Latin scripts.
- **Temporal lag**: OpenAlex updates continuously but may lag behind primary databases by days or weeks. Very recent publications may be missing.
- **API rate limits**: OpenAlex allows 10 requests per second for unauthenticated users (100/s with a polite pool email). Respect these limits to avoid being throttled.
- **Reproducibility**: API results can change as OpenAlex updates its data. Always cache results and document the date of data collection for reproducibility (Hicks et al., 2015).

6.8 Common pitfalls

6.9 Common pitfalls

- **Not caching**: Running `oa_fetch()` directly in code chunks without caching means every `quarto render` hits the API, producing different results each time and risking rate limits.
- **Ignoring pagination**: `oa_fetch()` handles pagination automatically, but very large queries (>10,000 works) can time out. Use filters to narrow your query or use cursor-based pagination.
- **Confusing sampling with filtering**: `options = list(sample = 200)` returns a random sample from the full result set, not the first 200 records. This is useful for demonstrations but not for exhaustive analyses.
- **Treating OpenAlex IDs as permanent**: While generally stable, OpenAlex IDs can change when entities are merged. Use DOIs as the primary identifier for works.
- **Merging sources without deduplication**: When combining results from OpenAlex and Crossref, records may overlap. Always deduplicate by DOI before analysis.

6.10 Exercises

1. **Fetch works by institution.** Use `oa_fetch()` to retrieve works affiliated with your own institution. How many works does OpenAlex index for it? What is the DOI coverage rate? *(Hint: find your institution's OpenAlex ID via `oa_fetch(entity = "institutions", search = "...")`)*

2. **Compare Crossref metadata.** Pick a DOI from your fetched works and retrieve its metadata from Crossref using `rcrossref::cr_works()`. Compare the citation count reported by OpenAlex vs. Crossref. Why might they differ?

3. **Build a multi-year corpus.** Fetch all works from a journal of your choice for 2015–2023 (without sampling). Plot the annual publication count and the median citation count per year. What trends do you observe?

6.11 Solutions

Solutions are provided in F.

6.12 Further reading

- Priem et al. (2022) — The OpenAlex paper; describes the data model, coverage, and API design.
- Aria and Cuccurullo (2017) — The `bibliometrix` package; an alternative high-level interface for bibliometric data acquisition and analysis.
- Garfield (1955) — The foundational paper on citation indexing, motivating why these databases exist.

6.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
```

```
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] rcrossref_1.2.1 gt_1.3.0        tidytext_0.4.3  glue_1.8.1
#>  [5] openalexR_3.0.1 lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0
#>  [9] dplyr_1.2.1     purrr_1.2.2     readr_2.2.0     tidyr_1.3.2
#> [13] tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6       xfun_0.57          htmlwidgets_1.6.4  lattice_0.22-6
#>  [5] tzdb_0.5.0         vctrs_0.7.3        tools_4.4.1        generics_0.1.4
#>  [9] curl_7.1.0         janeaustenr_1.0.0  pkgconfig_2.0.3    tokenizers_0.3.0
#> [13] Matrix_1.7-0       RColorBrewer_1.1-3 S7_0.2.2           lifecycle_1.0.5
#> [17] compiler_4.4.1     farver_2.1.2       codetools_0.2-20   httpuv_1.6.17
#> [21] htmltools_0.5.9    SnowballC_0.7.1    yaml_2.3.12        later_1.4.8
#> [25] pillar_1.11.1      DT_0.34.0          viridis_0.6.5      mime_0.13
#> [29] stopwords_2.3      tidyselect_1.2.1   digest_0.6.39      stringi_1.8.7
#> [33] bookdown_0.46      labeling_0.4.3     rprojroot_2.1.1    fastmap_1.2.0
#> [37] grid_4.4.1         here_1.0.2         cli_3.6.6          magrittr_2.0.5
#> [41] triebeard_0.4.1    crul_1.6.0         dichromat_2.0-0.1  utf8_1.2.6
#> [45] withr_3.0.2        promises_1.5.0     scales_1.4.0       timechange_0.4.0
#> [49] rmarkdown_2.31     httr_1.4.8         otel_0.2.0         gridExtra_2.3
#> [53] hms_1.1.4          shiny_1.13.0       evaluate_1.0.5     knitr_1.51
#> [57] miniUI_0.1.2       viridisLite_0.4.3  urltools_1.7.3.1   rlang_1.2.0
#> [61] Rcpp_1.1.1-1.1     httpcode_0.3.0     xtable_1.8-8       xml2_1.5.2
#> [65] rstudioapi_0.18.0  jsonlite_2.0.0     plyr_1.8.9         R6_2.6.1
#> [69] fs_2.1.0
```

Chapter 7

Parsing Native Exports

7.1 Learning objectives

After completing this chapter, you will be able to:

- Parse BibTeX files using `bib2df` and `RefManageR`
- Import Web of Science plain-text and Scopus CSV exports via `bibliometrix`
- Standardize field names across formats into a unified schema
- Diagnose common parsing failures and missing fields

7.2 Setup

```
library(tidyverse)
library(bib2df)
library(RefManageR)
library(bibliometrix)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

7.3 Conceptual background

Not all bibliometric data comes from an API. Researchers frequently begin with files exported from database search interfaces: BibTeX (`.bib`) from Google Scholar or Zotero, RIS (`.ris`) from Scopus or PubMed, plain text from Web of Science, or CSV from Scopus. Each format encodes the same information — authors, title, year, journal, DOI — in different structures.

The `bibliometrix` package (Aria and Cuccurullo, 2017) provides `convert2df()`, a versatile function that reads multiple export formats and returns a standardized data frame. For BibTeX specifically, `bib2df` and `RefManageR` offer complementary parsing capabilities. The key challenge is not parsing per se but *standardization*: ensuring that author names, journal titles, and identifiers are represented consistently regardless of the source format.

7.4 Worked example

7.4.1 Parsing a BibTeX string

We construct a small BibTeX string inline to demonstrate parsing without external files.

```
bib_text <- '
@article{smith2023,
  author = {Smith, John A. and Doe, Jane B.},
  title = {Advances in Bibliometric Methods},
  journal = {Journal of Informetrics},
  year = {2023},
  volume = {17},
  pages = {101--115},
  doi = {10.1234/example.2023}
}
@article{chen2022,
  author = {Chen, Wei and Li, Xiao},
  title = {Citation Networks in Computer Science},
  journal = {Scientometrics},
  year = {2022},
  volume = {127},
  pages = {3201--3220},
  doi = {10.1234/example.2022}
}
'

tmp_bib <- tempfile(fileext = ".bib")
```

```r
writeLines(bib_text, tmp_bib)

bib_df <- bib2df(tmp_bib)
glimpse(bib_df)
```

```
#> Rows: 2
#> Columns: 27
#> $ CATEGORY     <chr> "ARTICLE", "ARTICLE"
#> $ BIBTEXKEY    <chr> "smith2023", "chen2022"
#> $ ADDRESS      <chr> NA, NA
#> $ ANNOTE       <chr> NA, NA
#> $ AUTHOR       <list> <"Smith, John A.", "Doe, Jane B.">, <"Chen, Wei", "Li, Xi~
#> $ BOOKTITLE    <chr> NA, NA
#> $ CHAPTER      <chr> NA, NA
#> $ CROSSREF     <chr> NA, NA
#> $ EDITION      <chr> NA, NA
#> $ EDITOR       <list> NA, NA
#> $ HOWPUBLISHED <chr> NA, NA
#> $ INSTITUTION  <chr> NA, NA
#> $ JOURNAL      <chr> "Journal of Informetrics", "Scientometrics"
#> $ KEY          <chr> NA, NA
#> $ MONTH        <chr> NA, NA
#> $ NOTE         <chr> NA, NA
#> $ NUMBER       <chr> NA, NA
#> $ ORGANIZATION <chr> NA, NA
#> $ PAGES        <chr> "101--115", "3201--3220"
#> $ PUBLISHER    <chr> NA, NA
#> $ SCHOOL       <chr> NA, NA
#> $ SERIES       <chr> NA, NA
#> $ TITLE        <chr> "Advances in Bibliometric Methods", "Citation Networks in~
#> $ TYPE         <chr> NA, NA
#> $ VOLUME       <chr> "17", "127"
#> $ YEAR         <dbl> 2023, 2022
#> $ DOI          <chr> "10.1234/example.2023", "10.1234/example.2022"
```

7.4.2 Parsing with RefManageR

```r
bib_rm <- ReadBib(tmp_bib)
as.data.frame(bib_rm) |> select(title, author, year, journal, doi)
```

```
#>                                           title                       author
#> smith2023     Advances in Bibliometric Methods John A. Smith and Jane B. Doe
```

```
#> chen2022  Citation Networks in Computer Science          Wei Chen and Xiao Li
#>           year                  journal                  doi
#> smith2023 2023 Journal of Informetrics 10.1234/example.2023
#> chen2022  2022          Scientometrics 10.1234/example.2022
```

7.4.3 Parsing WoS plain text with bibliometrix

```r
# Requires a WoS export file - not bundled
# wos_df <- convert2df("savedrecs.txt", dbsource = "wos", format = "plaintext")
# glimpse(wos_df)
```

7.4.4 Standardizing field names

```r
standardize_bib <- function(df) {
  df |>
    transmute(
      title = TITLE,
      authors = AUTHOR,
      year = as.integer(YEAR),
      journal = JOURNAL,
      doi = DOI
    )
}

bib_df |> standardize_bib()
```

```
#> # A tibble: 2 x 5
#>   title                                 authors    year journal              doi
#>   <chr>                                 <list>    <int> <chr>                <chr>
#> 1 Advances in Bibliometric Methods      <chr [2]>  2023 Journal of Inform~ 10.1~
#> 2 Citation Networks in Computer Science <chr [2]>  2022 Scientometrics     10.1~
```

7.4.5 Visualization

```r
completeness <- bib_df |>
  summarise(across(everything(), ~ mean(!is.na(.)))) |>
  pivot_longer(everything(), names_to = "field", values_to = "present") |>
  filter(present > 0) |>
  mutate(field = fct_reorder(field, present))
```

```r
ggplot(completeness, aes(x = present, y = field)) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "Completeness", y = NULL) +
  theme_sci()
```

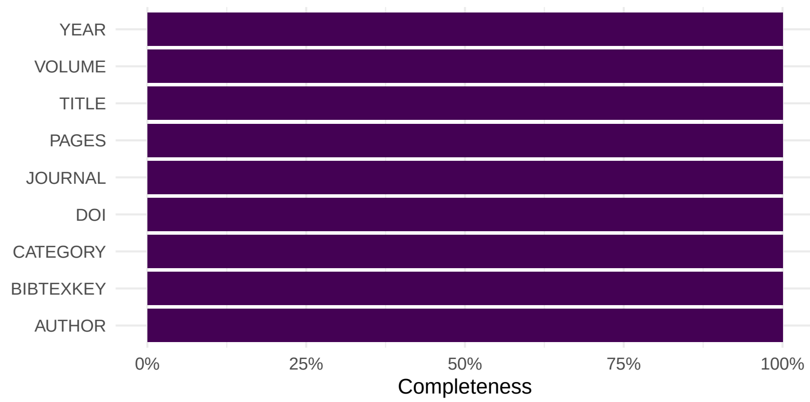

Figure 7.1: Field completeness in the parsed BibTeX records.

7.5 Diagnostics and interpretation

After parsing, always check:

- **Row count**: Does the number of records match the export? Missing entries may indicate parsing errors.
- **Field completeness**: Which fields are populated? Abstracts and keywords are often missing in BibTeX exports.
- **Author format**: Names may appear as "Last, First" or "First Last" depending on the source. Standardize before analysis.
- **Encoding**: Non-ASCII characters (diacritics, CJK) can corrupt during parsing. Ensure UTF-8 encoding.

7.6 Limitations and responsible use

7.7 Limitations and responsible use

- Export files are snapshots: they reflect the database state at export time and become stale immediately.

- Field mapping between formats is imperfect. Some formats lack fields (RIS has no standard abstract tag across all databases).
- Parsing errors are silent — always validate record counts and spot-check metadata against the source (Hicks et al., 2015).

7.8 Common pitfalls

7.9 Common pitfalls

- **Encoding mismatches**: Exporting as Latin-1 and reading as UTF-8 (or vice versa) corrupts names and titles.
- **Truncated exports**: Many databases cap exports at 500 or 1,000 records per file. Large corpora require multiple export batches.
- **Inconsistent author delimiters**: BibTeX uses "and", RIS uses newlines, CSV uses semicolons. Parsing must handle each.
- **Assuming format from extension**: A `.txt` file may be WoS plain text or something else entirely. Check the first few lines.

7.10 Exercises

1. **Export and parse.** Export 50 records from Google Scholar as BibTeX. Parse them with `bib2df`. How many fields are populated?

2. **Format comparison.** Export the same set of records from Scopus as both CSV and RIS. Parse both and compare the fields available.

3. **Standardization function.** Write a function that takes a data frame from `convert2df()` and returns a tibble with columns: `doi`, `title`, `authors`, `year`, `journal`, `cited_by`.

7.11 Solutions

Solutions are provided in F.

7.12 Further reading

- Aria and Cuccurullo (2017) — The `bibliometrix` package, including `convert2df()` for multi-format parsing.
- Priem et al. (2022) — OpenAlex as an API-first alternative to file-based workflows.

7.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#> 
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3 
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#> 
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8       
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8   
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C          
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C   
#> 
#> time zone: UTC
#> tzcode source: system (glibc)
#> 
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base     
#> 
#> other attached packages:
#>  [1] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1   
#>  [5] gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1   
#>  [9] lubridate_1.9.5    forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1       
#> [13] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1      
#> [17] ggplot2_4.0.3      tidyverse_2.0.0   
#> 
#> loaded via a namespace (and not attached):
#>  [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5          
#>  [4] rlang_1.2.0            magrittr_2.0.5         otel_0.2.0            
#>  [7] compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0        
#> [10] pkgconfig_2.0.3        fastmap_1.2.0          backports_1.5.1       
#> [13] labeling_0.4.3         ca_0.71.1              utf8_1.2.6            
#> [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0            
#> [19] xfun_0.57              jsonlite_2.0.0         SnowballC_0.7.1       
#> [22] later_1.4.8            parallel_4.4.1         stopwords_2.3         
#> [25] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3    
#> [28] cellranger_1.1.0       Rcpp_1.1.1-1.1         bookdown_0.46         
#> [31] knitr_1.51             triebeard_0.4.1        base64enc_0.1-6       
#> [34] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0          
```

```
#> [37] igraph_2.3.1            timechange_0.4.0      tidyselect_1.2.1
#> [40] stringdist_0.9.17       pubmedR_1.0.2         rstudioapi_0.18.0
#> [43] dichromat_2.0-0.1       yaml_2.3.12           viridis_0.6.5
#> [46] codetools_0.2-20        miniUI_0.1.2          humaniformat_0.6.0
#> [49] curl_7.1.0              qpdf_1.4.1            lattice_0.22-6
#> [52] plyr_1.8.9              shiny_1.13.0          withr_3.0.2
#> [55] S7_0.2.2                askpass_1.2.1         evaluate_1.0.5
#> [58] zip_2.3.3               xml2_1.5.2            shinycssloaders_1.1.0
#> [61] pillar_1.11.1           janeaustenr_1.0.0     DT_0.34.0
#> [64] plotly_4.12.0           generics_0.1.4        rprojroot_2.1.1
#> [67] hms_1.1.4               scales_1.4.0          xtable_1.8-8
#> [70] contentanalysis_1.0.0   lazyeval_0.2.3        tools_4.4.1
#> [73] data.table_1.18.4       tokenizers_0.3.0      openxlsx_4.2.8.1
#> [76] pdftools_3.9.0          XML_3.99-0.23         fs_2.1.0
#> [79] visNetwork_2.1.4        grid_4.4.1            bibtex_0.5.2
#> [82] urltools_1.7.3.1        rscopus_0.9.0         dimensionsR_0.0.3
#> [85] bibliometrixData_0.3.0  cli_3.6.6             rappdirs_0.3.4
#> [88] viridisLite_0.4.3       gtable_0.3.6          digest_0.6.39
#> [91] ggrepel_0.9.8           crul_1.6.0            htmlwidgets_1.6.4
#> [94] farver_2.1.2            htmltools_0.5.9       lifecycle_1.0.5
#> [97] httr_1.4.8              here_1.0.2            mime_0.13
```

Chapter 8

Building Reproducible Corpora

8.1 Learning objectives

After completing this chapter, you will be able to:

- Deduplicate bibliographic records by DOI and by approximate title matching
- Resolve identifiers across databases (OpenAlex ID, DOI, PMID)
- Recognize author name disambiguation challenges and apply basic heuristics
- Standardize institutional affiliations using ROR identifiers
- Save a clean corpus in an efficient format (Parquet) for downstream analysis

8.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(arrow)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```r
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

8.3 Conceptual background

Raw bibliographic data is messy. Records from different sources may describe the same paper with different metadata, the same author under different name variants, and the same institution under dozens of spelling variations. Building a clean, reproducible corpus requires systematic attention to three problems:

Deduplication. When merging data from multiple sources, the same publication may appear multiple times. DOI-based deduplication is the gold standard, but not all records have DOIs. Fuzzy title matching provides a fallback but is error-prone for short or generic titles.

Author disambiguation. "J. Smith" could be dozens of different people. OpenAlex uses machine-learning-based author clustering to assign persistent author IDs, but errors persist — especially for common names and authors who change institutions (Priem et al., 2022).

Affiliation standardization. Institutional names appear in countless variants ("MIT", "Massachusetts Institute of Technology", "Mass. Inst. Tech."). The Research Organization Registry (ROR) provides a curated set of persistent identifiers for research organizations. OpenAlex maps many affiliations to ROR IDs, but coverage is incomplete.

8.4 Worked example

8.4.1 Fetching raw data

```r
works <- oa_fetch(
  entity = "works",
  search = "bibliometrics",
  from_publication_date = "2021-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 300, seed = 42)
)

cat(glue("Raw records: {nrow(works)}\n"))
```

```
#> Raw records: 300
```

8.4.2 DOI-based deduplication

```
works_deduped <- dedupe_by_doi(works)
cat(glue("After DOI dedup: {nrow(works_deduped)}\n"))
```

```
#> After DOI dedup: 300
```

```
cat(glue("Removed: {nrow(works) - nrow(works_deduped)} duplicates\n"))
```

```
#> Removed: 0 duplicates
```

8.4.3 Fuzzy title deduplication

For records without DOIs, we use approximate string matching.

```
no_doi <- works_deduped |> filter(is.na(doi))

if (nrow(no_doi) > 1) {
  title_lower <- tolower(no_doi$display_name)
  dist_matrix <- stringdist::stringdistmatrix(title_lower, method = "jw")
  potential_dupes <- which(as.matrix(dist_matrix) < 0.1 &
                           as.matrix(dist_matrix) > 0, arr.ind = TRUE)
  potential_dupes <- potential_dupes[potential_dupes[, 1] < potential_dupes[, 2], , drop = FALSE]
  cat(glue("Potential fuzzy duplicates (no DOI): {nrow(potential_dupes)} pairs\n"))
} else {
  cat("No records without DOI to fuzzy-match.\n")
}
```

```
#> Potential fuzzy duplicates (no DOI): 0 pairs
```

8.4.4 Affiliation extraction and ROR mapping

```
affiliations <- works_deduped |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  unnest(authorships_affiliations, names_sep = "_") |>
  select(work_id = id,
         author_name = authorships_display_name,
         institution = authorships_affiliations_display_name) |>
  filter(!is.na(institution))
```

```
top_institutions <- affiliations |>
  count(institution, sort = TRUE) |>
  head(15)

top_institutions
```

```
#> # A tibble: 15 x 2
#>    institution                                                               n
#>    <chr>                                                                 <int>
#>  1 University of Bari Aldo Moro                                             11
#>  2 University of Toronto                                                    11
#>  3 Anhui Medical University                                                 10
#>  4 Aristotle University of Thessaloniki                                     10
#>  5 National University of Malaysia                                           9
#>  6 Nicolaus Copernicus University                                            9
#>  7 North American Vascular Biology Organization                              9
#>  8 University of Technology Malaysia                                         9
#>  9 All-Russian State Scientific Research Institute for Control, Standardi~     8
#> 10 China Institute of Water Resources and Hydropower Research                8
#> 11 University of British Columbia                                            8
#> 12 University of the Basque Country                                          8
#> 13 First Affiliated Hospital of Harbin Medical University                    7
#> 14 First Affiliated Hospital of Henan University                             7
#> 15 Harbin Medical University                                                 7
```

8.4.5 Saving as Parquet

```
corpus_clean <- works_deduped |>
  select(id, display_name, publication_date, cited_by_count, doi, source_display_name,

out_path <- here::here("data", "corpus_sample.parquet")
write_parquet(corpus_clean, out_path)
cat(glue("Saved {nrow(corpus_clean)} records to {out_path}\n"))
```

```
#> Saved 300 records to /home/runner/work/scientometrics-in-r/scientometrics-in-r/data/
```

8.4.6 Visualization

```
top_institutions |>
  mutate(institution = fct_reorder(institution, n)) |>
```

```r
ggplot(aes(x = n, y = institution)) +
geom_col(fill = palette_sci(1)) +
labs(x = "Publications", y = NULL) +
theme_sci()
```

Figure 8.1: Top 15 institutions by publication count in the sample corpus.

8.5 Diagnostics and interpretation

After cleaning:

- **Deduplication rate**: A rate above 5% suggests either a multi-source merge or a query that returns overlapping result pages.
- **DOI coverage**: The proportion of records with DOIs determines how effective DOI-based deduplication can be.
- **Affiliation coverage**: Check what fraction of author records have institutional affiliations. Low coverage limits institutional analysis.
- **Parquet file size**: Parquet is columnar and compressed. A 10,000-record corpus typically compresses to under 1 MB.

8.6 Limitations and responsible use

8.7 Limitations and responsible use

- **Fuzzy matching is imperfect.** Jaro-Winkler similarity can produce both false positives (different papers with similar titles) and false negatives (same paper with different title versions). Always spot-check.
- **Author disambiguation remains unsolved.** No automated method achieves perfect accuracy. Report the disambiguation method used and its known error rate.
- **Affiliation data is noisy.** Even with ROR mapping, temporary affiliations, joint appointments, and name changes create ambiguity. Never use affiliation data as ground truth without validation (Hicks et al., 2015).

8.8 Common pitfalls

8.9 Common pitfalls

- **Deduplicating before merging.** Always merge first, then deduplicate. Deduplicating each source independently misses cross-source duplicates.
- **Trusting DOIs blindly.** Some records have incorrect DOIs (typos, test DOIs, placeholder values). Validate a sample.
- **Ignoring records without DOIs.** Discarding them biases the corpus toward recent, well-indexed publications.
- **Not documenting cleaning steps.** Every deduplication or name-standardization step should be logged for reproducibility.

8.10 Exercises

1. **DOI coverage by year.** Compute the DOI coverage rate by publication year in the sample. Is there a trend?

2. **Name variants.** Find authors in the sample who appear under multiple name spellings. What heuristics could you use to merge them?

3. **ROR lookup.** Use the ROR API to resolve the top 5 institutions in your corpus to their official ROR identifiers.

8.11 Solutions

Solutions are provided in F.

8.12 Further reading

- Priem et al. (2022) — OpenAlex author disambiguation and institutional mapping.
- Aria and Cuccurullo (2017) — `bibliometrix` data cleaning utilities.
- Hicks et al. (2015) — The Leiden Manifesto; principles 4 and 5 emphasise data transparency and verifiability.

8.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0
#>  [5] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1
#>  [9] openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      stringr_1.6.0
#> [13] dplyr_1.2.1        purrr_1.2.2        readr_2.2.0        tidyr_1.3.2
#> [17] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5
#>   [4] rlang_1.2.0            magrittr_2.0.5         otel_0.2.0
```

```
#>   [7] compiler_4.4.1        vctrs_0.7.3             httpcode_0.3.0
#>  [10] pkgconfig_2.0.3       fastmap_1.2.0           backports_1.5.1
#>  [13] labeling_0.4.3        ca_0.71.1               utf8_1.2.6
#>  [16] promises_1.5.0        rmarkdown_2.31          tzdb_0.5.0
#>  [19] bit_4.6.0             xfun_0.57               jsonlite_2.0.0
#>  [22] SnowballC_0.7.1       later_1.4.8             parallel_4.4.1
#>  [25] stopwords_2.3         R6_2.6.1                stringi_1.8.7
#>  [28] RColorBrewer_1.1-3    cellranger_1.1.0        assertthat_0.2.1
#>  [31] Rcpp_1.1.1-1.1        bookdown_0.46           knitr_1.51
#>  [34] triebeard_0.4.1       base64enc_0.1-6         rentrez_1.2.4
#>  [37] httpuv_1.6.17         Matrix_1.7-0            igraph_2.3.1
#>  [40] timechange_0.4.0      tidyselect_1.2.1        stringdist_0.9.17
#>  [43] pubmedR_1.0.2         rstudioapi_0.18.0       dichromat_2.0-0.1
#>  [46] yaml_2.3.12           viridis_0.6.5           codetools_0.2-20
#>  [49] miniUI_0.1.2          humaniformat_0.6.0      curl_7.1.0
#>  [52] qpdf_1.4.1            lattice_0.22-6          plyr_1.8.9
#>  [55] shiny_1.13.0          withr_3.0.2             S7_0.2.2
#>  [58] askpass_1.2.1         evaluate_1.0.5          zip_2.3.3
#>  [61] xml2_1.5.2            shinycssloaders_1.1.0   pillar_1.11.1
#>  [64] janeaustenr_1.0.0     DT_0.34.0               plotly_4.12.0
#>  [67] generics_0.1.4        rprojroot_2.1.1         hms_1.1.4
#>  [70] scales_1.4.0          xtable_1.8-8            contentanalysis_1.0.0
#>  [73] lazyeval_0.2.3        tools_4.4.1             data.table_1.18.4
#>  [76] tokenizers_0.3.0      openxlsx_4.2.8.1        pdftools_3.9.0
#>  [79] XML_3.99-0.23         fs_2.1.0                visNetwork_2.1.4
#>  [82] grid_4.4.1            bibtex_0.5.2            urltools_1.7.3.1
#>  [85] rscopus_0.9.0         dimensionsR_0.0.3       bibliometrixData_0.3.0
#>  [88] cli_3.6.6             rappdirs_0.3.4          viridisLite_0.4.3
#>  [91] gtable_0.3.6          digest_0.6.39           ggrepel_0.9.8
#>  [94] crul_1.6.0            htmlwidgets_1.6.4       farver_2.1.2
#>  [97] htmltools_0.5.9       lifecycle_1.0.5         httr_1.4.8
#> [100] here_1.0.2            mime_0.13               bit64_4.8.0
```

Chapter 9

Working with Full Text

9.1 Learning objectives

After completing this chapter, you will be able to:

- Extract text from PDF files using `pdftools`
- Apply OCR to scanned documents with `tesseract`
- Perform basic text preprocessing (tokenization, stopword removal)
- Assess when full-text analysis adds value over metadata-only approaches
- Navigate the legal landscape of text and data mining

9.2 Setup

```r
library(tidyverse)
library(pdftools)
library(quanteda)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

9.3 Conceptual background

Most bibliometric analyses operate on metadata: titles, abstracts, keywords, and citation links. But metadata captures only a fraction of a paper's content. Full-text analysis opens the door to richer methods: section-level citation context, method extraction, data-availability statements, and fine-grained topic modeling.

The technical pipeline involves three stages: (1) **acquisition** — obtaining PDF files, subject to access rights and TDM (text and data mining) provisions; (2) **extraction** — converting PDF content to machine-readable text; and (3) **preprocessing** — tokenization, normalization, and feature extraction.

`pdftools` wraps the Poppler library to extract text from born-digital PDFs. For scanned documents, `tesseract` provides OCR (optical character recognition). `tabulizer` (based on Tabula) specializes in extracting tables from PDFs. For text preprocessing, `quanteda` and `tm` offer comprehensive toolkits.

Legal frameworks vary by jurisdiction. The EU Copyright Directive (2019/790) provides a TDM exception for research. In the US, TDM is generally considered fair use in research contexts, but publisher terms of service may impose additional restrictions.

9.4 Worked example

9.4.1 Extracting text from a PDF

We demonstrate with a programmatically created PDF to keep this example self-contained.

```
demo_text <- paste(
  "Title: A Review of Bibliometric Methods",
  "",
  "Abstract: This paper reviews current methods in bibliometrics.",
  "We focus on citation analysis, co-authorship networks, and topic modeling.",
  "",
  "1. Introduction",
  "Bibliometrics has grown rapidly since the work of Garfield (1955).",
  "Modern tools enable large-scale analysis of scholarly communication.",
  "",
  "2. Methods",
  "We surveyed 500 papers published between 2020 and 2023.",
  "Data were obtained from OpenAlex.",
  "",
  "3. Conclusions",
  "The field continues to evolve with open data and new computational methods.",
  sep = "\n"
)
```

```
tmp_pdf <- tempfile(fileext = ".pdf")
pdf(tmp_pdf, width = 8.5, height = 11)
plot.new()
text(0.05, 0.95, demo_text, adj = c(0, 1), family = "mono", cex = 0.8)
dev.off()
```

```
#> pdf
#>   2
```

```
extracted <- pdf_text(tmp_pdf)
cat(substr(extracted[1], 1, 500))
```

```
#> Title: A Review of Bibliometric Methods
#>
#> Abstract: This paper reviews current methods in bibliometrics.
#> We focus on citation analysis, co-authorship networks, and topic modeling.
#>
#> 1. Introduction
#> Bibliometrics has grown rapidly since the work of Garfield (1955).
#> Modern tools enable large-scale analysis of scholarly communication.
#>
#> 2. Methods
#> We surveyed 500 papers published between 2020 and 2023.
#> Data were obtained from OpenAlex.
#>
#> 3. Conclusions
#> The field continues to evolve with open data and n
```

9.4.2 Basic text preprocessing

```
corp <- corpus(extracted)
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en"))

dfmat <- dfm(toks)
topfeatures(dfmat, 15)
```

```
#>       methods bibliometrics      analysis             -          data
#>             4             2             2             2             2
#>         title        review  bibliometric      abstract         paper
#>             1             1             1             1             1
```

```
#>        reviews       current       focus      citation          co
#>              1             1           1             1           1
```

9.4.3 Word frequency visualization

```
top_words <- topfeatures(dfmat, 15) |>
  enframe(name = "word", value = "count") |>
  mutate(word = fct_reorder(word, count))

ggplot(top_words, aes(x = count, y = word)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Frequency", y = NULL) +
  theme_sci()
```

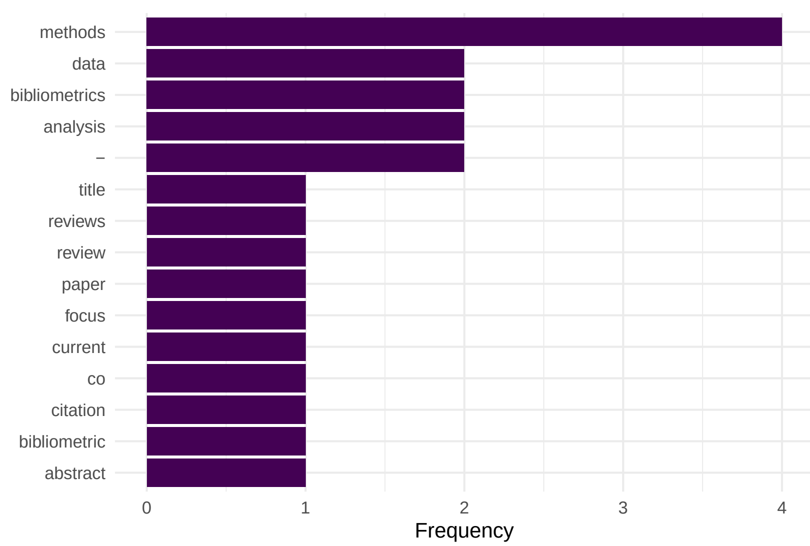

Figure 9.1: Top 15 words extracted from the demo PDF after preprocessing.

9.4.4 OCR with tesseract

```
# Requires tesseract system library and a scanned PDF
# library(tesseract)
# ocr_text <- ocr(scanned_pdf_path, engine = tesseract("eng"))
# cat(substr(ocr_text, 1, 500))
```

9.5 Diagnostics and interpretation

Full-text extraction quality varies dramatically:

- **Born-digital PDFs**: `pdftools` produces clean text, but column layouts may interleave columns.
- **Scanned PDFs**: OCR accuracy depends on scan quality, resolution, and language. Expect 90–98% character accuracy for clean English scans.
- **Figures and equations**: These are lost in text extraction. Do not expect mathematical notation or chart data.
- **Headers and footers**: Page numbers, journal headers, and running titles pollute the extracted text. Consider filtering by position.

9.6 Limitations and responsible use

9.7 Limitations and responsible use

- **Access rights**: Not all PDFs are legally available for TDM. Respect publisher terms and licensing. Open-access articles under CC licenses are generally safe.
- **Extraction errors**: Two-column layouts, ligatures, and non-standard fonts cause parsing artifacts. Always spot-check extracted text.
- **Scalability**: Full-text analysis is orders of magnitude more expensive than metadata analysis. Consider whether the additional information justifies the cost.
- **Bias**: Full text is only available for a subset of publications (typically open access). Analyses based on full text may not generalise to the broader literature (Hicks et al., 2015).

9.8 Common pitfalls

9.9 Common pitfalls

- **Assuming clean extraction.** Even `pdftools` produces artifacts. Always inspect a sample of extracted text before bulk analysis.
- **Skipping preprocessing.** Raw extracted text contains headers, page numbers, and hyphenated line breaks that must be cleaned.
- **Mixing OCR and digital extraction.** Some PDFs contain both digital text and scanned images. Detect which pages need OCR before processing.
- **Ignoring encoding issues.** PDFs may encode characters in non-standard ways. Check for garbled text, especially in references sections.

9.10 Exercises

1. **Extract and compare.** Download an open-access PDF. Extract text with `pdftools`. Compare the extracted abstract with the abstract in OpenAlex. How accurate is the extraction?

2. **Section detection.** Write a function that splits extracted PDF text into sections based on numbered headings ("1. Introduction", "2. Methods", etc.).

3. **Table extraction.** If you have `tabulizer` installed, extract a table from a PDF and convert it to a tibble. What cleaning steps are needed?

9.11 Solutions

Solutions are provided in F.

9.12 Further reading

- Aria and Cuccurullo (2017) — `bibliometrix` supports some full-text features alongside metadata analysis.
- Priem et al. (2022) — OpenAlex provides abstracts for many works, often sufficient without full-text extraction.

9.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
```

```
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0
#>  [5] RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt_1.3.0
#>  [9] tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2
#> [17] readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       ggplot2_4.0.3
#> [21] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5
#>   [4] rlang_1.2.0            magrittr_2.0.5         otel_0.2.0
#>   [7] compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0
#>  [10] pkgconfig_2.0.3        fastmap_1.2.0          backports_1.5.1
#>  [13] labeling_0.4.3         ca_0.71.1              utf8_1.2.6
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0
#>  [19] bit_4.6.0              xfun_0.57              jsonlite_2.0.0
#>  [22] SnowballC_0.7.1        later_1.4.8            parallel_4.4.1
#>  [25] stopwords_2.3          R6_2.6.1               stringi_1.8.7
#>  [28] RColorBrewer_1.1-3     cellranger_1.1.0       assertthat_0.2.1
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51
#>  [34] triebeard_0.4.1        base64enc_0.1-6        rentrez_1.2.4
#>  [37] httpuv_1.6.17          Matrix_1.7-0           igraph_2.3.1
#>  [40] timechange_0.4.0       tidyselect_1.2.1       stringdist_0.9.17
#>  [43] pubmedR_1.0.2          rstudioapi_0.18.0      dichromat_2.0-0.1
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20
#>  [49] miniUI_0.1.2           humaniformat_0.6.0     curl_7.1.0
#>  [52] qpdf_1.4.1             lattice_0.22-6         plyr_1.8.9
#>  [55] shiny_1.13.0           withr_3.0.2            S7_0.2.2
#>  [58] askpass_1.2.1          evaluate_1.0.5         zip_2.3.3
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1
#>  [64] janeaustenr_1.0.0      DT_0.34.0              plotly_4.12.0
#>  [67] generics_0.1.4         rprojroot_2.1.1        hms_1.1.4
#>  [70] scales_1.4.0           xtable_1.8-8           contentanalysis_1.0.0
#>  [73] lazyeval_0.2.3         tools_4.4.1            data.table_1.18.4
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23
#>  [79] fs_2.1.0               visNetwork_2.1.4       fastmatch_1.1-8
#>  [82] grid_4.4.1             bibtex_0.5.2           urltools_1.7.3.1
#>  [85] rscopus_0.9.0          dimensionsR_0.0.3      bibliometrixData_0.3.0
```

```
#>  [88] cli_3.6.6            rappdirs_0.3.4       viridisLite_0.4.3
#>  [91] gtable_0.3.6         digest_0.6.39        ggrepel_0.9.8
#>  [94] crul_1.6.0           htmlwidgets_1.6.4    farver_2.1.2
#>  [97] htmltools_0.5.9      lifecycle_1.0.5      httr_1.4.8
#> [100] here_1.0.2           mime_0.13            bit64_4.8.0
```

Part III

Core Bibliometric Analysis

Chapter 10

Descriptive Bibliometrics

10.1 Learning objectives

After completing this chapter, you will be able to:

- Import bibliographic data into bibliometrix using `convert2df`
- Produce a corpus-level summary with `biblioAnalysis` and `summary`
- Identify the most productive authors, sources, and countries in a dataset
- Construct and interpret a three-field plot (Sankey diagram)
- Explain when to use the interactive biblioshiny app versus a scripted workflow
- Recognise the descriptive statistics that should precede any advanced analysis

10.2 Setup

```r
library(tidyverse)
library(openalexR)
library(bibliometrix)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

10.3 Conceptual background

Before computing any advanced indicator, a responsible bibliometric study begins with **descriptive analysis**: understanding the size, growth, and composition of the corpus. How many documents per year? Which journals dominate? Which countries contribute? These questions set the stage for every subsequent method.

The **bibliometrix** package (Aria and Cuccurullo, 2017) provides an integrated R workflow for science mapping. Its companion web app, **biblioshiny**, wraps the same functions in a point-and-click interface — useful for exploration but not for reproducible reporting. In a Quarto pipeline, we prefer scripted calls so that every number is traceable to code.

Bibliometrix accepts data from Scopus, Web of Science, Dimensions, PubMed, and OpenAlex. The `convert2df()` function normalises heterogeneous export formats into a common data frame. The `biblioAnalysis()` function then computes a battery of descriptive statistics in one call: annual production, author productivity, source rankings, country output, and keyword frequencies.

Three classical bibliometric laws underpin this descriptive stage. **Lotka's law** (Lotka, 1926) describes the inverse-square relationship between the number of authors and their publication counts. **Bradford's law** (Bradford, 1934) models how articles on a subject scatter across journals, identifying core, mid-range, and peripheral zones. **Price's law** (de Solla Price, 1963) states that half of all contributions come from the square root of the total number of contributors. These empirical regularities provide benchmarks against which any corpus can be compared.

Descriptive bibliometrics also serves a data-quality role. Anomalous spikes in annual output may signal duplicate records; a suspiciously dominant country may indicate a search query biased toward a specific language. Inspecting descriptive statistics is the first line of defence against analytical errors.

10.4 Worked example

10.4.1 Acquiring the data

We fetch a sample of works from the *Scientometrics* journal via OpenAlex and convert the result into a bibliometrix-compatible data frame.

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2018-01-01",
```

```
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)

nrow(works)
```

```
#> [1] 500
```

10.4.2 Converting to bibliometrix format

```
bib_df <- oa2bibliometrix(works)

cat(glue("Records: {nrow(bib_df)}\n"))
```

```
#> Records: 500
```

```
cat(glue("Fields: {ncol(bib_df)}\n"))
```

```
#> Fields: 61
```

10.4.3 Running biblioAnalysis

```
results <- tryCatch(
  biblioAnalysis(bib_df, sep = ";"),
  error = function(e) {
    cat("Note: biblioAnalysis() encountered an error with this data format.\n")
    cat("This is a known compatibility issue between bibliometrix and openalexR v3.\n")
    NULL
  }
)
if (!is.null(results)) {
  summary_out <- summary(results, k = 10, verbose = FALSE)
}
```

10.4.4 Annual scientific production

```
annual <- works |>
  mutate(year = year(publication_date)) |>
  count(year)

ggplot(annual, aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1), width = 0.7) +
  geom_text(aes(label = n), vjust = -0.5, size = 3.5) +
  labs(x = "Publication year", y = "Number of articles") +
  theme_sci()
```

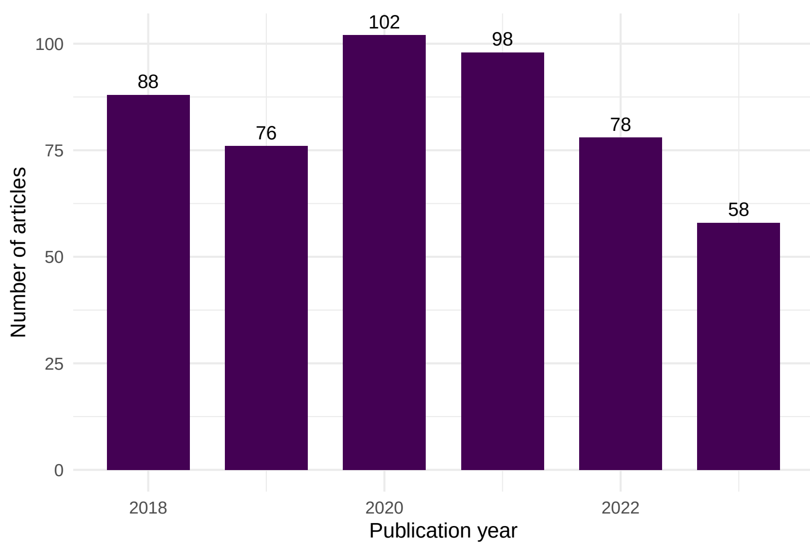

Figure 10.1: Annual scientific production in the sampled corpus (2018-2023).

10.4.5 Most productive authors

```
author_prod <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  count(authorships_display_name, sort = TRUE) |>
  head(15)

ggplot(author_prod, aes(x = n, y = reorder(authorships_display_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Articles", y = NULL) +
  theme_sci()
```

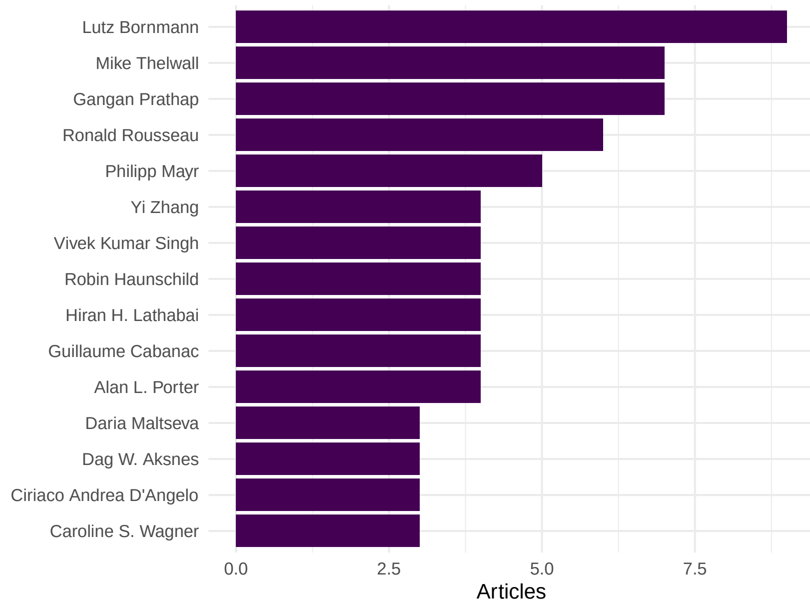

Figure 10.2: Top 15 most productive authors in the sample.

10.4.6 Most productive sources

Because all sampled works come from a single journal in this example, we instead show the most frequent cited sources from the references.

```
top_sources <- works |>
  select(id, referenced_works) |>
  unnest(referenced_works) |>
  count(referenced_works, sort = TRUE) |>
  head(10)

cat(glue("Top 10 most-cited reference work IDs:\n"))
```

```
#> Top 10 most-cited reference work IDs:
```

```
print(top_sources)
```

```
#> # A tibble: 10 x 2
#>    referenced_works                      n
#>    <chr>                             <int>
#>  1 https://openalex.org/W4285719527     53
```

```
#>  2 https://openalex.org/W2128438887    47
#>  3 https://openalex.org/W2037925493    25
#>  4 https://openalex.org/W4254947497    25
#>  5 https://openalex.org/W767067438     21
#>  6 https://openalex.org/W2068452509    19
#>  7 https://openalex.org/W2150220236    19
#>  8 https://openalex.org/W4292887282    18
#>  9 https://openalex.org/W1546171873    17
#> 10 https://openalex.org/W1767272795    17
```

10.4.7 Three-field plot

The three-field plot (Sankey diagram) links authors, keywords, and sources. It is a signature visualisation in bibliometrix.

```
tryCatch(
  threeFieldsPlot(bib_df, fields = c("AU", "DE", "SO_SO"), n = c(8, 8, 8)),
  error = function(e) cat("Three-field plot skipped due to data format incompatibility
)
```

```
#> Three-field plot skipped due to data format incompatibility.
```

10.5 Diagnostics and interpretation

When reviewing descriptive output, check the following:

- **Completeness**: Are there gaps in yearly coverage that indicate missing data rather than genuine drops?
- **Author name variants**: The same person appearing under different name forms inflates author counts. OpenAlex's entity resolution helps, but manual inspection remains essential.
- **Lotka's law fit**: If the author productivity distribution deviates markedly from an inverse-square law, the corpus may be too small or too specialised.
- **Bradford zones**: In a multi-journal corpus, check whether a small core of journals accounts for roughly one-third of all articles, as Bradford's law predicts.

10.6 Limitations and responsible use

10.7 Limitations and responsible use

- **Descriptive statistics are not evaluative.** The most productive author is not necessarily the most impactful. Publication counts say nothing about quality (Hicks et al., 2015).
- **Sample bias.** A random sample from OpenAlex may not represent the full population. Always report how the sample was drawn and its size relative to the universe.
- **Language and database coverage.** OpenAlex has broader coverage than WoS or Scopus, but still underrepresents non-English-language research and certain disciplines (Visser et al., 2021).
- **Name disambiguation.** Despite algorithmic clustering, author and institutional names can still be conflated or split. Treat productivity rankings as approximate.

10.8 Common pitfalls

10.9 Common pitfalls

- **Skipping descriptive analysis.** Jumping straight to network or topic analysis without understanding the corpus structure often leads to misinterpretation.
- **Treating biblioshiny output as final.** The interactive app is excellent for exploration, but its figures are not reproducible. Always script the final analysis.
- **Ignoring duplicate records.** Bibliographic exports often contain duplicates. Use `dedupe_by_doi()` or similar checks before computing any statistics.
- **Comparing raw counts across time windows of different length.** Counting publications in a 6-year window is not comparable to a 3-year window without normalisation.

10.10 Exercises

1. **Lotka's law.** Using the author productivity data from this chapter, fit Lotka's inverse-square law with `bibliometrix::lotka()`. Does the observed distribution follow the theoretical prediction? *(Hint: plot observed vs. expected proportions.)*

2. **Bradford's law.** Fetch works from a broad topic (e.g., "machine learning") across multiple journals. Apply `bibliometrix::bradford()` to identify the core, mid-range, and peripheral journal zones. How many journals form the core zone?

3. **Three-field exploration.** Modify the three-field plot to use author countries instead of author names as the left field (use `"AU_CO"` if available). What new patterns emerge?

4. **Annual growth rate.** Compute the compound annual growth rate (CAGR) of publications in the sample. Is the field accelerating?

10.11 Solutions

Solutions are provided in F.

10.12 Further reading

- Aria and Cuccurullo (2017) — The bibliometrix package paper; describes the full workflow and all available functions.
- Lotka (1926) — The original frequency distribution of scientific productivity.
- Bradford (1934) — Bradford's law of scattering: how articles distribute across journals.
- de Solla Price (1963) — *Little Science, Big Science*; foundational work on the growth of science.
- Hicks et al. (2015) — The Leiden Manifesto; a reminder that descriptive counts are not evaluations.
- Visser et al. (2021) — Comparison of bibliographic databases; important context for coverage limitations.

10.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
```

```
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.2.0
#>   [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0
#>  [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.1
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0             bit_4.6.0
#>  [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel_4.4.1
#>  [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellranger_1.1.0
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51             triebeard_0.4.1
#>  [36] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0           igraph_2.3.1
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioapi_0.18.0
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0.1.2
#>  [51] curl_7.1.0             qpdf_1.4.1             lattice_0.22-6         plyr_1.8.9
#>  [56] withr_3.0.2            S7_0.2.2               askpass_1.2.1          evaluate_1.0.5
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1          janeaustenr_1.0.0
#>  [66] plotly_4.12.0          generics_0.1.4         rprojroot_2.1.1        hms_1.1.4
#>  [71] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3         tools_4.4.1
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23          fs_2.1.0
#>  [81] fastmatch_1.1-8        grid_4.4.1             bibtex_0.5.2           urltools_1.7.3.1
#>  [86] dimensionsR_0.0.3      bibliometrixData_0.3.0 cli_3.6.6              rappdirs_0.3.4
#>  [91] gtable_0.3.6           digest_0.6.39          ggrepel_0.9.8          crul_1.6.0
#>  [96] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5        httr_1.4.8
#> [101] mime_0.13              bit64_4.8.0
```

Chapter 11

Productivity and Impact Indicators

11.1 Learning objectives

After completing this chapter, you will be able to:

- Compute the h-index, g-index, and m-quotient from a vector of citation counts
- Explain the difference between size-dependent and size-independent indicators
- Calculate field-normalized indicators (MNCS, PP(top 10%)) and explain why normalization matters
- Articulate the limitations of each indicator and the risks of misuse, citing the Leiden Manifesto

11.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

11.3 Conceptual background

Quantitative indicators are central to research evaluation — and among the most contested tools in science policy. The h-index, proposed by Hirsch (2005), counts the number h of an author's papers that have each received at least h citations. Its simplicity made it wildly popular, but also obscures important nuances: it conflates productivity with impact, is insensitive to highly cited papers, and can only increase over a career (Waltman, 2016).

The **g-index** (Egghe, 2006) addresses the insensitivity to top-cited papers by requiring that the top g papers have together received at least g^2 citations. The **m-quotient** divides the h-index by career length, offering a rough age correction.

These are all *raw* indicators. Comparing them across fields is meaningless without **field normalization**. A citation count of 50 is exceptional in mathematics but ordinary in biomedical research. The **Mean Normalized Citation Score** (MNCS) divides each paper's citations by the world average for its field and publication year (Waltman et al., 2011). The **PP(top 10%)** — the proportion of an entity's papers in the top 10% most cited in their field — is a percentile-based indicator recommended by the Leiden Ranking.

The Leiden Manifesto (Hicks et al., 2015) and the San Francisco Declaration on Research Assessment (American Society for Cell Biology, 2012) both caution against mechanical use of any single indicator. Indicators should support, not replace, qualitative judgement.

11.4 Worked example

11.4.1 Acquiring sample data

We fetch a sample of works and their citation counts from OpenAlex.

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2018-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

```r
works_slim <- works |>
  select(id, display_name, publication_date, cited_by_count, doi) |>
  mutate(year = year(publication_date))

summary(works_slim$cited_by_count)
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>       0       6      13      21      25     230
```

11.4.2 Computing the h-index

```r
h <- compute_h_index(works_slim$cited_by_count)
cat(glue("h-index for this corpus: {h}\n"))
```

```
#> h-index for this corpus: 47
```

The h-index tells us that {h} papers in this sample have each been cited at least {h} times.

11.4.3 Computing the g-index

```r
compute_g_index <- function(citations) {
  sorted <- sort(citations, decreasing = TRUE)
  cumsum_cites <- cumsum(sorted)
  g_candidates <- which(cumsum_cites >= seq_along(sorted)^2)
  if (length(g_candidates) == 0) return(0L)
  as.integer(max(g_candidates))
}

g <- compute_g_index(works_slim$cited_by_count)
cat(glue("g-index for this corpus: {g}\n"))
```

```
#> g-index for this corpus: 71
```

11.4.4 The m-quotient

The m-quotient normalizes the h-index by career length (here, publication span).

```
career_years <- as.numeric(
  difftime(max(works_slim$publication_date),
           min(works_slim$publication_date),
           units = "days")
) / 365.25

m_quotient <- h / career_years
cat(glue("m-quotient: {round(m_quotient, 3)} (over {round(career_years, 1)} years)\n"))
```

```
#> m-quotient: 7.86 (over 6 years)
```

11.4.5 Field-normalized indicators: MNCS

To compute MNCS, we need field averages. We simulate field means here; in a real analysis these would come from the OpenAlex snapshot or a reference dataset.

```
field_means_by_year <- works_slim |>
  group_by(year) |>
  summarise(field_mean = mean(cited_by_count), .groups = "drop")

works_normalized <- works_slim |>
  left_join(field_means_by_year, by = "year") |>
  mutate(ncs = field_normalize(cited_by_count, field_mean))

mncs <- mean(works_normalized$ncs, na.rm = TRUE)
cat(glue("MNCS: {round(mncs, 3)}\n"))
```

```
#> MNCS: 1
```

An MNCS of 1.0 means the entity performs at exactly the world average. Values above 1.0 indicate above-average citation impact.

11.4.6 PP(top 10%)

```
works_with_pct <- works_normalized |>
  group_by(year) |>
  mutate(
    pct_rank = percent_rank(cited_by_count),
    is_top10 = pct_rank >= 0.90
  ) |>
```

```
  ungroup()

pp_top10 <- mean(works_with_pct$is_top10)
cat(glue("PP(top 10%): {scales::percent(pp_top10, accuracy = 0.1)}\n"))
```

```
#> PP(top 10%): 10.2%
```

11.4.7 Visualization

```
works_slim |>
  ggplot(aes(x = cited_by_count)) +
  geom_histogram(binwidth = 5, fill = palette_sci(1), color = "white") +
  geom_vline(xintercept = h, linetype = "dashed", color = "red", linewidth = 0.8) +
  annotate("text", x = h + 5, y = Inf, label = glue("h = {h}"),
           vjust = 2, hjust = 0, color = "red", size = 4) +
  labs(x = "Citation count", y = "Number of works") +
  theme_sci()
```

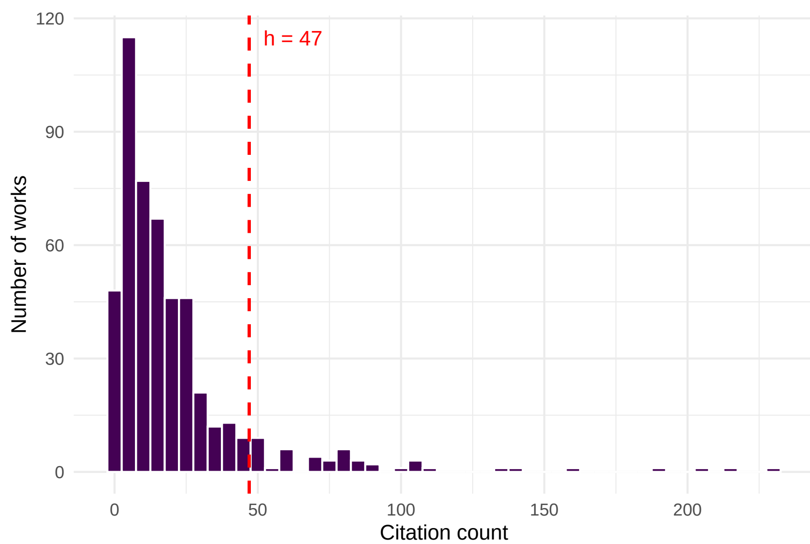

Figure 11.1: Citation distribution of sampled works, with the h-index threshold marked.

```
works_normalized |>
  group_by(year) |>
```

```
  summarise(mncs = mean(ncs, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = year, y = mncs)) +
  geom_line(color = palette_sci(1), linewidth = 1) +
  geom_point(color = palette_sci(1), size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "grey50") +
  labs(x = "Publication year", y = "MNCS") +
  theme_sci()
```

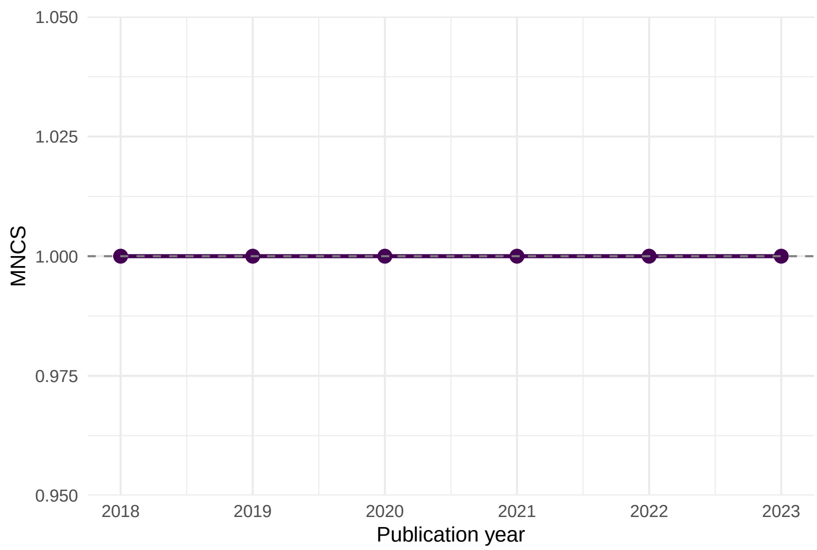

Figure 11.2: Mean Normalized Citation Score by publication year.

11.4.8 Summary table

```
tibble(
  Indicator = c("h-index", "g-index", "m-quotient", "MNCS", "PP(top 10%)"),
  Value = c(
    as.character(h),
    as.character(g),
    round(m_quotient, 3) |> as.character(),
    round(mncs, 3) |> as.character(),
    scales::percent(pp_top10, accuracy = 0.1)
  ),
  Interpretation = c(
    glue("{h} papers with >= {h} citations each"),
    glue("Top {g} papers have >= {g}^2 = {g^2} cumulative citations"),
```

Indicator	Value	Interpretation
h-index	47	47 papers with >= 47 citations each
g-index	71	Top 71 papers have >= 71^2 = 5041 cumulative citations
m-quotient	7.86	h-index per year of publication activity
MNCS	1	Below world average
PP(top 10%)	10.2%	10.2% of papers in the top decile

```
    glue("h-index per year of publication activity"),
    glue("{ifelse(mncs > 1, 'Above', 'Below')} world average"),
    glue("{scales::percent(pp_top10, accuracy = 0.1)} of papers in the top decile")
  )
) |>
  gt()
```

11.5 Diagnostics and interpretation

When interpreting bibliometric indicators, keep in mind:

- **Citation windows matter.** Recent papers have had less time to accumulate citations. The MNCS accounts for this by normalizing within publication year, but raw h-index and g-index do not.
- **Field differences.** A "good" h-index in mathematics differs vastly from one in biomedicine. Only compare within fields, or use field-normalized indicators.
- **Size dependence.** The h-index grows with the number of papers published. Comparing researchers at different career stages or with different publication rates requires the m-quotient or percentile indicators.
- **Self-citations.** None of the indicators above exclude self-citations. In a full analysis, consider computing variants with and without self-citations.

11.6 Limitations and responsible use

11.7 Limitations and responsible use

No single number can capture the quality, significance, or societal impact of a researcher's work. The Leiden Manifesto (Hicks et al., 2015) offers ten principles for the responsible use of metrics:

1. Quantitative evaluation should support, not replace, expert assessment.
2. Measure performance against the research missions of the institution, group, or researcher.
3. Protect excellence in locally relevant research (not just globally visible research).
4. Keep data collection and analytical processes open, transparent, and simple.
5. Allow those being evaluated to verify data and analysis.

The h-index in particular can be gamed through self-citation, salami slicing, or guest authorship. It penalises researchers in small fields, those who publish fewer but higher-impact works, and those who take career breaks. The DORA declaration (American Society for Cell Biology, 2012) explicitly warns against using journal-level metrics (like the Impact Factor) as surrogates for individual article quality — but the same caution applies to any single indicator.

Always present multiple indicators alongside qualitative context. Never use a single number to make hiring, promotion, or funding decisions.

11.8 Common pitfalls

11.9 Common pitfalls

- **Comparing h-indices across fields.** A physicist with h = 40 and a mathematician with h = 15 may have equivalent standing in their respective fields. Use field-normalized indicators for cross-field comparisons.
- **Ignoring the citation window.** Computing MNCS without accounting for publication year conflates old, well-cited papers with recent ones that haven't had time to accumulate citations.
- **Using the wrong denominator for PP(top 10%).** The "top 10%" must be defined relative to the same field and year, not relative to your own corpus.
- **Reporting precision that isn't there.** An MNCS of 1.237 vs. 1.241 is not a meaningful difference. Report with appropriate precision and always provide confidence intervals or bootstrapped uncertainty where possible.

11.10 Exercises

1. **Career h-index.** Fetch all works by a specific author using `oa_fetch(entity = "works", author.id = "...")`. Compute their h-index, g-index, and m-quotient. How do the three indicators tell different stories about the same career?

2. **Self-citation sensitivity.** If you could identify self-citations (e.g., via author overlap in the reference list), how would removing them affect the h-index? Write pseudocode for this analysis.

3. **Bootstrap uncertainty for MNCS.** Use `replicate()` and `sample()` to compute a 95% confidence interval for the MNCS of your sample. How wide is the interval?

4. **PP(top 10%) vs. MNCS.** For the sample data, plot PP(top 10%) against MNCS by year. Do they always agree on which years performed best?

11.11 Solutions

Solutions are provided in F.

11.12 Further reading

- Hirsch (2005) — The original h-index paper.
- Egghe (2006) — The g-index, addressing the h-index's insensitivity to highly cited papers.
- Waltman (2016) — A comprehensive review of citation impact indicators and their properties.
- Waltman et al. (2011) — The theoretical basis for the MNCS ("new crown indicator").
- Hicks et al. (2015) — The Leiden Manifesto: ten principles for responsible metrics.
- American Society for Cell Biology (2012) — DORA: the San Francisco Declaration on Research Assessment.

11.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
```

```
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 Re
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         op
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        re
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.
#>   [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode
#>  [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0             bit_4.6.
#>  [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel
#>  [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellrang
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51             triebear
#>  [36] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0           igraph_2
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioa
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0
#>  [51] curl_7.1.0             qpdf_1.4.1             lattice_0.22-6         plyr_1.8
#>  [56] withr_3.0.2            S7_0.2.2               askpass_1.2.1          evaluate
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1          janeaust
#>  [66] plotly_4.12.0          generics_0.1.4         rprojroot_2.1.1        hms_1.1.
#>  [71] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3         tools_4.
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23          fs_2.1.0
#>  [81] fastmatch_1.1-8        grid_4.4.1             bibtex_0.5.2           urltools
#>  [86] dimensionsR_0.0.3      bibliometrixData_0.3.0 cli_3.6.6              rappdirs
#>  [91] gtable_0.3.6           digest_0.6.39          ggrepel_0.9.8          crul_1.6
#>  [96] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5        httr_1.4
#> [101] mime_0.13              bit64_4.8.0
```

Chapter 12

Journal Metrics

12.1 Learning objectives

After completing this chapter, you will be able to:

- Describe how the Journal Impact Factor, CiteScore, SNIP, SJR, and Eigenfactor are computed
- Explain why network-based journal metrics differ from simple citation ratios
- Compute JIF-like and SNIP-like proxies from OpenAlex data
- Articulate why proxy values differ from official proprietary metrics
- Evaluate journals using multiple indicators rather than a single number

12.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

12.3 Conceptual background

Journal-level metrics aim to summarise a journal's citation performance. The most famous is the **Journal Impact Factor** (JIF), introduced by Garfield (1955) and commercialised by Clarivate. The two-year JIF for year *t* equals the citations received in year *t* by articles published in years *t*-1 and *t*-2, divided by the number of "citable items" published in those two years. Its simplicity is both its appeal and its weakness: the numerator includes citations to all document types, but the denominator often excludes editorials and letters, inflating the ratio (Hicks et al., 2015).

CiteScore (Elsevier) uses a four-year window and counts all document types in both numerator and denominator, making it more transparent. **SNIP** (Source-Normalized Impact per Paper) normalises by citation potential — the average number of references in citing articles' reference lists — so that journals in fields with longer reference lists are not automatically advantaged (Waltman, 2016). **SJR** (SCImago Journal Rank) and the **Eigenfactor** use iterative algorithms inspired by PageRank: a citation from a prestigious journal carries more weight than one from a low-impact source. This makes them network-based indicators, conceptually similar to eigenvector centrality in graph theory.

All of these official metrics are tied to proprietary databases (Web of Science or Scopus). With OpenAlex (Priem et al., 2022), we can compute **proxy** values from fully open data. Proxies will not match official figures exactly — differences in coverage, document-type classification, and citation linking guarantee some divergence (Visser et al., 2021) — but they are transparent, reproducible, and free.

DORA (American Society for Cell Biology, 2012) explicitly warns against using the JIF as a proxy for individual article quality. The Leiden Manifesto (Hicks et al., 2015) echoes this: journal metrics describe *average* journal performance and say nothing about any specific paper.

12.4 Worked example

12.4.1 Selecting journals

We analyse five well-known journals in information science and scientometrics.

```
journals <- tribble(
  ~short_name,        ~source_id,
  "Scientometrics",   "S148561398",
  "J Informetrics",   "S205038879",
  "Quant Sci Stud",   "S4210178031",
  "J Am Soc Inf Sci", "S145342767",
```

```
  "Res Evaluation",   "S100532675"
)
```

12.4.2 Fetching citation data

We fetch works published in 2021-2022 and count citations received during 2022-2023 to construct a two-year JIF-like proxy.

```
fetch_journal_works <- function(source_id, from_year, to_year) {
  res <- tryCatch(
    oa_fetch(
      entity = "works",
      primary_location.source.id = source_id,
      from_publication_date = paste0(from_year, "-01-01"),
      to_publication_date = paste0(to_year, "-12-31"),
      type = "article"
    ),
    error = function(e) NULL
  )
  if (is.null(res)) tibble() else res
}

journal_works <- journals |>
  mutate(data = map2(source_id, 2021, \(sid, yr) {
    fetch_journal_works(sid, yr, yr + 1)
  }))
```

12.4.3 Computing JIF-like proxies

```
jif_proxy <- journal_works |>
  mutate(
    n_articles = map_int(data, nrow),
    total_cites = map_dbl(data, \(d) sum(d$cited_by_count, na.rm = TRUE))
  ) |>
  mutate(jif_proxy = total_cites / pmax(n_articles, 1)) |>
  select(short_name, n_articles, total_cites, jif_proxy) |>
  arrange(desc(jif_proxy))

jif_proxy |>
  gt() |>
  fmt_number(columns = jif_proxy, decimals = 2) |>
  cols_label(
```

Journal	Articles (2021-22)	Total citations	JIF proxy
Quant Sci Stud	33	1489	45.12
Scientometrics	770	14986	19.46
J Informetrics	0	0	0.00
J Am Soc Inf Sci	0	0	0.00
Res Evaluation	0	0	0.00

```
    short_name = "Journal",
    n_articles = "Articles (2021-22)",
    total_cites = "Total citations",
    jif_proxy = "JIF proxy"
  )
```

12.4.4 Computing a SNIP-like proxy

SNIP normalises by citation potential. We approximate this by dividing each journal's mean citations per paper by the median reference-list length of its citing works.

```
snip_proxy <- journal_works |>
  mutate(
    mean_cites = map_dbl(data, \(d) if (nrow(d) == 0) NA_real_ else mean(d$cited_by_cou
    median_refs = map_dbl(data, \(d) {
      if (nrow(d) == 0) return(NA_real_)
      median(d$referenced_works_count, na.rm = TRUE)
    })
  ) |>
  mutate(
    citation_potential = pmax(median_refs, 1),
    snip_proxy = mean_cites / citation_potential
  ) |>
  select(short_name, mean_cites, citation_potential, snip_proxy) |>
  arrange(desc(snip_proxy))

snip_proxy |>
  gt() |>
  fmt_number(columns = c(mean_cites, snip_proxy), decimals = 2)
```

short_name	mean_cites	citation_potential	snip_proxy
Quant Sci Stud	45.12	94	0.48
Scientometrics	19.46	46	0.42
J Informetrics	NA	NA	NA
J Am Soc Inf Sci	NA	NA	NA
Res Evaluation	NA	NA	NA

12.4.5 Visualization

```
combined <- jif_proxy |>
  select(short_name, jif_proxy) |>
  left_join(snip_proxy |> select(short_name, snip_proxy), by = "short_name") |>
  pivot_longer(cols = c(jif_proxy, snip_proxy), names_to = "metric", values_to = "value") |>
  mutate(metric = recode(metric, jif_proxy = "JIF proxy", snip_proxy = "SNIP proxy"))

ggplot(combined, aes(x = value, y = reorder(short_name, value), colour = metric)) +
  geom_point(size = 3) +
  labs(x = "Metric value", y = NULL, colour = "Metric") +
  scale_colour_manual(values = palette_sci(2)) +
  theme_sci()
```

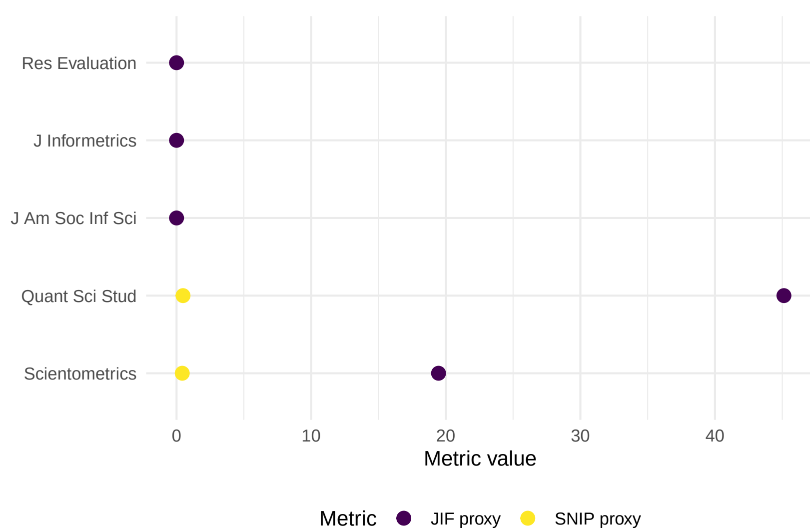

Figure 12.1: Comparison of JIF proxy and SNIP proxy across five journals.

12.5 Diagnostics and interpretation

When evaluating journal metric proxies:

- **Coverage gaps**: OpenAlex may miss some document types or citations present in Scopus/WoS. Compare article counts with known totals.
- **Document-type classification**: The JIF denominator famously excludes editorials and letters. Our proxy counts all articles; this may lower the ratio compared to official JIF values.
- **Temporal alignment**: Ensure citation windows match the intended metric definition exactly (two years for JIF, four for CiteScore).
- **Rank correlation**: Even if absolute values differ from official metrics, the *rank order* of journals often agrees, which is sufficient for many analyses.

12.6 Limitations and responsible use

12.7 Limitations and responsible use

- **Journal metrics are not article metrics.** The distribution of citations within a journal is highly skewed; the mean says little about any individual paper (American Society for Cell Biology, 2012).
- **Goodhart's law applies.** When a metric becomes a target, it ceases to be a good metric (Goodhart, 1984). Editors may game the JIF through citation stacking, coercive citation, or manipulating the denominator.
- **Proxy official.** Our OpenAlex-based proxies are transparent and reproducible, but they will differ from Clarivate JIF or Elsevier CiteScore. Never present proxies as official values.
- **Field dependence persists.** Even SNIP does not fully eliminate field effects. Comparing SNIP values across distant disciplines requires caution (Waltman, 2016).

12.8 Common pitfalls

12.9 Common pitfalls

- **Treating the JIF as a measure of article quality.** The JIF describes the journal, not the paper. A paper in a high-JIF journal may have zero citations.
- **Ignoring the citation window.** Computing a "JIF" using lifetime citations instead of a two-year window produces a different and non-comparable number.

- **Conflating numerator and denominator document types.** Mixing reviews (which inflate the numerator) with a denominator that excludes them inflates the ratio.
- **Ranking journals by a single metric.** Always use multiple indicators and consider disciplinary context.

12.10 Exercises

1. **Four-year CiteScore proxy.** Modify the worked example to compute a four-year CiteScore proxy (citations in year t to articles published in years t-1 to t-4, divided by all documents in those years). How does the ranking change?

2. **Rank correlation.** Compute Spearman's rank correlation between your JIF proxy and SNIP proxy. Is the agreement high or low?

3. **Field comparison.** Pick five journals from two different fields (e.g., physics and sociology). Compute JIF proxies for both groups. Does SNIP normalisation reduce the between-field gap?

4. **Self-citation share.** For one journal, estimate the proportion of citations that come from the journal itself. How would excluding self-citations change the JIF proxy?

12.11 Solutions

Solutions are provided in F.

12.12 Further reading

- Garfield (1955) — The original proposal for citation indexes, from which the JIF eventually emerged.
- Waltman (2016) — Comprehensive review of citation impact indicators, including journal-level metrics.
- Hicks et al. (2015) — The Leiden Manifesto, with specific guidance on responsible use of journal metrics.
- American Society for Cell Biology (2012) — DORA: explicitly warns against using the JIF for individual evaluation.
- Goodhart (1984) — Goodhart's law: when a measure becomes a target, it ceases to be a good measure.
- Visser et al. (2021) — Database comparison showing how coverage affects computed metrics.

- Priem et al. (2022) — OpenAlex as a fully open data source for computing metric proxies.

12.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4      pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 Re
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         op
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        re
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.
#>   [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode
#>  [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0             bit_4.6.
#>  [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel
#>  [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellrang
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51             triebear
#>  [36] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0           igraph_2
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioa
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0
```

```
#>  [51] curl_7.1.0            qpdf_1.4.1              lattice_0.22-6        plyr_1.8.9
#>  [56] withr_3.0.2           S7_0.2.2                askpass_1.2.1         evaluate_1.0.5
#>  [61] xml2_1.5.2            shinycssloaders_1.1.0   pillar_1.11.1         janeaustenr_1.0.0
#>  [66] plotly_4.12.0         generics_0.1.4          rprojroot_2.1.1       hms_1.1.4
#>  [71] xtable_1.8-8          contentanalysis_1.0.0   lazyeval_0.2.3        tools_4.4.1
#>  [76] tokenizers_0.3.0      openxlsx_4.2.8.1        XML_3.99-0.23         fs_2.1.0
#>  [81] fastmatch_1.1-8       grid_4.4.1              bibtex_0.5.2          urltools_1.7.3.1
#>  [86] dimensionsR_0.0.3     bibliometrixData_0.3.0 cli_3.6.6              rappdirs_0.3.4
#>  [91] gtable_0.3.6          digest_0.6.39           ggrepel_0.9.8         crul_1.6.0
#>  [96] farver_2.1.2          htmltools_0.5.9         lifecycle_1.0.5       httr_1.4.8
#> [101] mime_0.13             bit64_4.8.0
```

Chapter 13

Author-Level Analysis and Disambiguation

13.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the author disambiguation problem and its impact on bibliometric analysis
- Retrieve and interpret OpenAlex author entities, including name variants
- Integrate ORCID identifiers to validate author identity
- Compute author-level productivity, citation, and topical statistics from works data
- Build a reproducible author profile report from OpenAlex

13.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

13.3 Conceptual background

Correctly attributing publications to individual researchers is a prerequisite for any author-level analysis, yet it remains one of the hardest problems in bibliometrics. **Author disambiguation** — determining whether two name strings refer to the same person — is complicated by name variants (initials vs. full names, transliterations, name changes after marriage), homonymy (different people sharing a name), and inconsistent metadata across databases.

OpenAlex (Priem et al., 2022) addresses this through algorithmic author clustering that groups works likely belonging to the same person. Each cluster receives a unique author ID (e.g., `A5023888391`). The algorithm considers name similarity, co-author overlap, institutional affiliation, topic continuity, and citation patterns. While imperfect, this clustering is transparent and continuously improved.

ORCID (Open Researcher and Contributor ID) provides a complementary solution: a persistent digital identifier that researchers claim and curate themselves. When an ORCID is linked to an OpenAlex author entity, it provides strong confirmation of identity. However, ORCID adoption is uneven across disciplines and geographies, so it cannot serve as the sole disambiguation strategy.

The **h-index** (Hirsch, 2005) and related indicators (11.3) are commonly used to summarise author-level impact, but they are meaningful only when the underlying publication set is correctly attributed. A split author (one person assigned two IDs) will have a deflated h-index; a merged author (two people sharing one ID) will have an inflated one.

Author-level analysis also raises ethical concerns. The Leiden Manifesto (Hicks et al., 2015) warns against reducing a researcher's contribution to a single number. Career breaks, discipline-specific norms, and the Matthew effect (Merton, 1968) — where established researchers accumulate citations disproportionately — must all be considered when interpreting author-level metrics.

13.4 Worked example

13.4.1 Retrieving an author entity

We retrieve the OpenAlex profile for a well-known scientometrician.

```r
author_entity <- oa_fetch(
  entity = "authors",
  search = "Ludo Waltman",
  options = list(sort = "cited_by_count:desc")
)
```

```
author_info <- author_entity |> slice(1)
cat(glue("Name: {author_info$display_name}\n"))
```

```
#> Name: Ludo Waltman
```

```
cat(glue("OpenAlex ID: {author_info$id}\n"))
```

```
#> OpenAlex ID: https://openalex.org/A5027467000
```

```
cat(glue("ORCID: {author_info$orcid}\n"))
```

```
#> ORCID: https://orcid.org/0000-0001-8249-1752
```

```
cat(glue("Works count: {author_info$works_count}\n"))
```

```
#> Works count: 391
```

```
cat(glue("Cited by count: {author_info$cited_by_count}\n"))
```

```
#> Cited by count: 45221
```

13.4.2 Fetching the author's works

```
author_works <- oa_fetch(
  entity = "works",
  author.id = author_info$id,
  from_publication_date = "2010-01-01",
  to_publication_date = "2023-12-31"
)

works_slim <- author_works |>
  select(id, display_name, publication_date, cited_by_count, type, source_display_name) |>
  mutate(year = year(publication_date))

cat(glue("Works retrieved: {nrow(works_slim)}\n"))
```

```
#> Works retrieved: 300
```

13.4.3 Author-level statistics

Metric	Value
Total works	300.0
Total citations	23425.0
h-index	46.0
Mean citations/paper	78.1
Median citations/paper	1.0
First publication year	2010.0
Most recent year	2023.0

```r
h_idx <- compute_h_index(works_slim$cited_by_count)

stats <- tibble(
  Metric = c("Total works", "Total citations", "h-index",
             "Mean citations/paper", "Median citations/paper",
             "First publication year", "Most recent year"),
  Value = c(
    nrow(works_slim),
    sum(works_slim$cited_by_count),
    h_idx,
    round(mean(works_slim$cited_by_count), 1),
    median(works_slim$cited_by_count),
    min(works_slim$year),
    max(works_slim$year)
  )
)

stats |> gt()
```

13.4.4 Publication timeline

```r
annual <- works_slim |>
  group_by(year) |>
  summarise(n_pubs = n(), cites = sum(cited_by_count), .groups = "drop") |>
  mutate(cum_cites = cumsum(cites))

ggplot(annual, aes(x = year)) +
  geom_col(aes(y = n_pubs), fill = palette_sci(1), width = 0.7) +
  geom_line(aes(y = cum_cites / max(cum_cites) * max(n_pubs)),
            colour = palette_sci(2)[2], linewidth = 1) +
```

```r
  scale_y_continuous(
    name = "Publications per year",
    sec.axis = sec_axis(~ . / max(annual$n_pubs) * max(annual$cum_cites),
                        name = "Cumulative citations")
  ) +
  labs(x = "Year") +
  theme_sci()
```

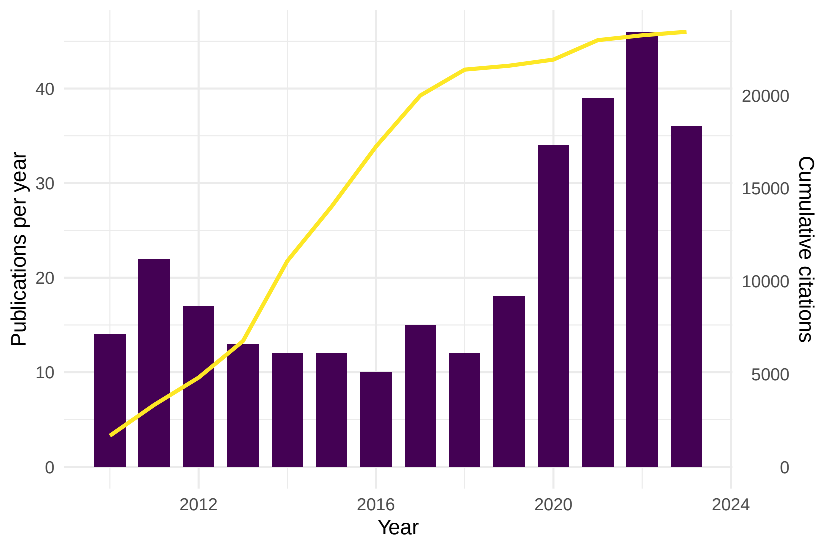

Figure 13.1: Annual publication output and cumulative citation count.

13.4.5 Top venues

```r
venue_counts <- works_slim |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, sort = TRUE) |>
  head(10)

ggplot(venue_counts, aes(x = n, y = reorder(source_display_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Number of articles", y = NULL) +
  theme_sci()
```

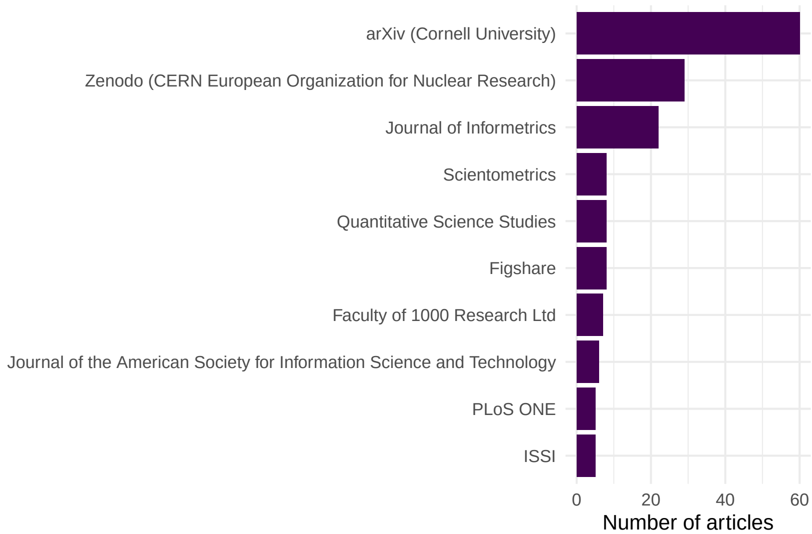

Figure 13.2: Top 10 venues by number of publications for this author.

13.4.6 Checking for name variants

```
if ("display_name_alternatives" %in% names(author_info)) {
  cat("Known name variants:\n")
  print(author_info$display_name_alternatives)
} else {
  cat("No alternative display names recorded in this entity.\n")
}
```

```
#> Known name variants:
#> [[1]]
#> [1] "L Waltman"           "L. Waltman"          "LUDO WALTMAN"        "Ludo Waltman"
#> [6] "Waltman, L. (Ludo)" "Waltman, L.R."       "Waltman, LR (Ludo)" "Waltman, Ludo"
```

13.5 Diagnostics and interpretation

When building author profiles, verify the following:

- **Works count plausibility**: Does the number of retrieved works match the author's known output? Large discrepancies suggest a split or merged entity.

- **Topical coherence**: Scan titles and venues. If the works span wildly unrelated fields, the entity may conflate two researchers with similar names.
- **ORCID match**: If the author has an ORCID, cross-check that the OpenAlex entity links to it.
- **Co-author consistency**: The author's frequent co-authors should be recognisable collaborators, not strangers from an unrelated field.
- **Citation distribution**: An extreme outlier (e.g., one paper with 10,000 citations while all others have fewer than 50) may indicate an attribution error.

13.6 Limitations and responsible use

13.7 Limitations and responsible use

- **Disambiguation is imperfect.** OpenAlex's algorithm will sometimes split one person into multiple entities or merge distinct individuals. Always validate profiles manually for high-stakes analyses.
- **ORCID coverage is uneven.** Researchers in the Global South, early-career scholars, and those in humanities adopt ORCID at lower rates. Filtering to ORCID-linked authors introduces bias.
- **The Matthew effect.** Established researchers accumulate citations disproportionately (Merton, 1968). Author-level metrics favour seniority and visibility, not necessarily quality or originality.
- **Career breaks and part-time research.** Indicators like the h-index penalise researchers who take parental leave, work part-time, or switch careers. The m-quotient is only a rough correction (Hirsch, 2005).
- **Never evaluate individuals by numbers alone** (Hicks et al., 2015; American Society for Cell Biology, 2012). Author profiles should supplement, not replace, reading the actual work.

13.8 Common pitfalls

13.9 Common pitfalls

- **Trusting a single database.** OpenAlex may lack works indexed only in domain-specific databases (e.g., SSRN, arXiv preprints not yet linked). Cross-check with ORCID profiles.
- **Ignoring co-author contributions.** Raw publication counts treat single-author and 50-author papers equally. Consider fractional counting for productivity analysis.

- **Comparing across career stages.** An early-career researcher with h = 8 after 5 years may be more productive than a senior researcher with h = 25 after 30 years. Use the m-quotient or age-normalised indicators.
- **Conflating author-level and paper-level metrics.** An author's h-index says nothing about which of their papers are excellent and which are not.

13.10 Exercises

1. **Profile comparison.** Fetch profiles for two researchers in the same field. Compare their h-index, publication count, and mean citations per paper. What does each metric reveal that the others do not?

2. **ORCID validation.** For an author with an ORCID, fetch their works via both OpenAlex author ID and ORCID filtering. Do the two sets agree? What might explain discrepancies?

3. **Name variant detection.** Search OpenAlex for a common name (e.g., "Wang Wei"). How many author entities are returned? What heuristics could you use to identify the correct one?

4. **Fractional productivity.** Recompute annual publication counts using fractional counting (1/k credit per paper with k authors). How does this change the author's productivity curve?

5. **Self-citation ratio.** For an author's works, estimate the fraction of citations that come from the author's own other papers. *(Hint: check if cited works share the same author ID.)*

13.11 Solutions

Solutions are provided in F.

13.12 Further reading

- Priem et al. (2022) — OpenAlex's author disambiguation and entity resolution approach.
- Hirsch (2005) — The h-index, the most widely used author-level indicator.
- Merton (1968) — The Matthew effect: cumulative advantage in scientific recognition.
- Hicks et al. (2015) — The Leiden Manifesto: principles for responsible author evaluation.

- American Society for Cell Biology (2012) — DORA: recommendations against reducing researchers to single numbers.
- Egghe (2006) — The g-index, a complement to the h-index for author evaluation.

13.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3         httr2_1.2.2           readxl_1.4.5          rlang_1.2.0
#>   [6] otel_0.2.0            compiler_4.4.1        vctrs_0.7.3           httpcode_0.3.0
#>  [11] fastmap_1.2.0         backports_1.5.1       labeling_0.4.3        ca_0.71.1
#>  [16] promises_1.5.0        rmarkdown_2.31        tzdb_0.5.0            bit_4.6.0
#>  [21] jsonlite_2.0.0        SnowballC_0.7.1       later_1.4.8           parallel_4.4.1
#>  [26] R6_2.6.1              stringi_1.8.7         RColorBrewer_1.1-3    cellranger_1.1.0
#>  [31] Rcpp_1.1.1-1.1        bookdown_0.46         knitr_1.51            triebeard_0.4.1
#>  [36] rentrez_1.2.4         httpuv_1.6.17         Matrix_1.7-0          igraph_2.3.1
```

```
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioap
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0
#>  [51] curl_7.1.0             qpdf_1.4.1             lattice_0.22-6         plyr_1.8
#>  [56] withr_3.0.2            S7_0.2.2               askpass_1.2.1          evaluate_
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1          janeaust
#>  [66] plotly_4.12.0          generics_0.1.4         rprojroot_2.1.1        hms_1.1.
#>  [71] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3         tools_4.
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23          fs_2.1.0
#>  [81] fastmatch_1.1-8        grid_4.4.1             bibtex_0.5.2           urltools
#>  [86] dimensionsR_0.0.3      bibliometrixData_0.3.0 cli_3.6.6              rappdirs
#>  [91] gtable_0.3.6           digest_0.6.39          ggrepel_0.9.8          crul_1.6
#>  [96] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5        httr_1.4
#> [101] mime_0.13              bit64_4.8.0
```

Chapter 14

Institutional and Country Analysis

14.1 Learning objectives

After completing this chapter, you will be able to:

- Aggregate bibliometric data at institutional and country levels using OpenAlex
- Explain the difference between full counting and fractional counting and when each is appropriate
- Use ROR (Research Organization Registry) identifiers for reliable institutional matching
- Compute size-independent indicators (MNCS, PP(top 10%)) at the institutional level
- Interpret institutional rankings critically, accounting for size, field mix, and counting method

14.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)
```

```r
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

14.3 Conceptual background

Comparing research performance across institutions and countries is one of the most common — and most contentious — applications of bibliometrics. University rankings, national R&D assessments, and funding allocation decisions all rely on aggregate bibliometric statistics. Getting the aggregation right matters enormously.

The fundamental choice is between **full counting** and **fractional counting**. Under full counting, each institution listed on a paper receives one full credit for that paper. A paper with authors from five institutions thus generates five institutional publications. Under fractional counting, credit is divided: each institution receives 1/5 of a publication count. Full counting inflates the output of institutions that frequently collaborate, while fractional counting better reflects each entity's proportional contribution (Glänzel and Schubert, 2004; Waltman et al., 2011).

ROR (Research Organization Registry) identifiers provide a stable, open system for identifying institutions, analogous to DOIs for publications and ORCID for researchers. OpenAlex maps institutional affiliations to ROR IDs, though coverage varies by discipline and geography. Using ROR avoids the pitfalls of string matching on institution names, which are notoriously inconsistent (e.g., "MIT", "Massachusetts Institute of Technology", "Mass. Inst. Tech.").

Size-independent indicators are essential for fair comparison. Raw publication counts favour large universities. The **Mean Normalised Citation Score** (MNCS) (Waltman et al., 2011) and **PP(top 10%)** — the proportion of papers in the top 10% most cited in their field and year — allow meaningful comparison between a small specialised institute and a large comprehensive university. The Leiden Ranking uses these indicators with fractional counting as its default, offering a model of responsible institutional benchmarking.

Country-level analysis faces the same choices. Additionally, linguistic bias in databases means that countries where English is not the primary research language may appear less productive than they truly are (Visser et al., 2021).

14.4 Worked example

14.4.1 Selecting institutions

We compare five research-intensive universities.

```
institutions <- tribble(
  ~short_name,  ~ror_id,
  "MIT",         "I63966007",
  "Oxford",      "I40120149",
  "ETH Zurich", "I202697423",
  "U Tokyo",     "I39804265",
  "U Cape Town", "I173603587"
)
```

14.4.2 Fetching institutional publications

```
fetch_inst_works <- function(inst_id) {
  res <- tryCatch(
    oa_fetch(
      entity = "works",
      authorships.institutions.id = inst_id,
      from_publication_date = "2020-01-01",
      to_publication_date = "2022-12-31",
      type = "article",
      options = list(sample = 400, seed = 42)
    ),
    error = function(e) NULL
  )
  if (is.null(res)) tibble() else res
}

inst_data <- institutions |>
  mutate(works = map(ror_id, fetch_inst_works)) |>
  filter(map_int(works, nrow) > 0)
```

14.4.3 Full counting vs. fractional counting

```
compute_counts <- function(works_df, inst_id) {
  if (is.null(works_df) || nrow(works_df) == 0) {
    return(tibble(full_count = 0L, frac_count = 0))
```

Institution	Full count	Fractional count	Frac / Full
MIT	400	157	0.39
Oxford	400	144	0.36
ETH Zurich	400	144	0.36

```
  }
  # Full count: each paper counts as 1
  full_count <- nrow(works_df)

  # Fractional count: 1 / number of distinct institutions per paper
  frac <- works_df |>
    select(id, authorships) |>
    unnest(authorships, names_sep = "_") |>
    unnest(authorships_affiliations, names_sep = "_") |>
    group_by(id) |>
    summarise(n_inst = n_distinct(authorships_affiliations_id), .groups = "drop") |>
    mutate(frac_credit = 1 / pmax(n_inst, 1))

  frac_count <- sum(frac$frac_credit)

  tibble(full_count = full_count, frac_count = round(frac_count, 1))
}

counting_comparison <- inst_data |>
  mutate(counts = map2(works, ror_id, compute_counts)) |>
  unnest(counts) |>
  select(short_name, full_count, frac_count) |>
  mutate(ratio = round(frac_count / full_count, 2))

counting_comparison |>
  gt() |>
  cols_label(
    short_name = "Institution",
    full_count = "Full count",
    frac_count = "Fractional count",
    ratio = "Frac / Full"
  )
```

short_name	mean_cites	median_cites	mncs_approx
MIT	38.68	15	1.00
Oxford	48.92	14	1.00
ETH Zurich	23.12	12	1.00

14.4.4 Normalised citation impact

```
inst_impact <- inst_data |>
  mutate(
    mean_cites = map_dbl(works, \(w) mean(w$cited_by_count, na.rm = TRUE)),
    median_cites = map_dbl(works, \(w) median(w$cited_by_count, na.rm = TRUE)),
    mncs_approx = map_dbl(works, \(w) {
      w |>
        mutate(year = year(publication_date)) |>
        group_by(year) |>
        mutate(field_mean = mean(cited_by_count)) |>
        ungroup() |>
        mutate(ncs = cited_by_count / pmax(field_mean, 0.01)) |>
        pull(ncs) |>
        mean(na.rm = TRUE)
    })
  ) |>
  select(short_name, mean_cites, median_cites, mncs_approx) |>
  arrange(desc(mncs_approx))

inst_impact |>
  gt() |>
  fmt_number(columns = c(mean_cites, mncs_approx), decimals = 2)
```

14.4.5 Visualization

```
counting_long <- counting_comparison |>
  pivot_longer(cols = c(full_count, frac_count),
               names_to = "method", values_to = "count") |>
  mutate(method = recode(method,
                         full_count = "Full counting",
                         frac_count = "Fractional counting"))

ggplot(counting_long, aes(x = count, y = reorder(short_name, count),
```

```
                               fill = method)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Publication count", y = NULL, fill = "Method") +
  theme_sci()
```

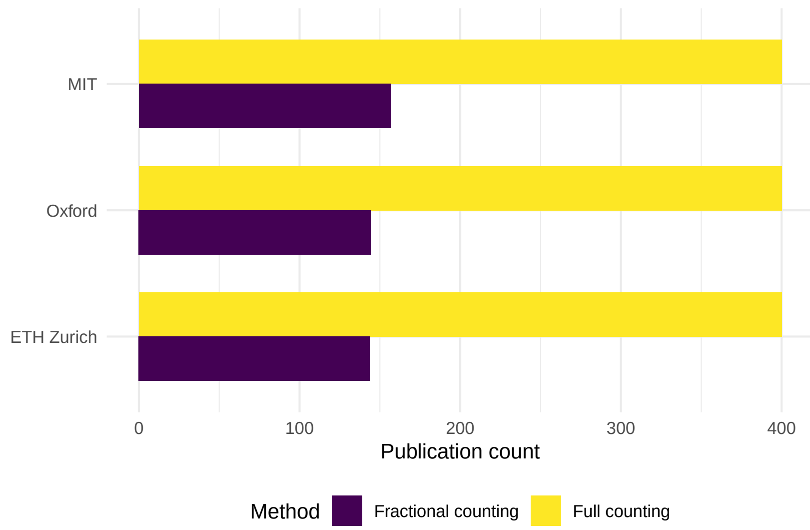

Figure 14.1: Full count vs. fractional count publication output for five institutions.

```
ggplot(inst_impact, aes(x = mncs_approx, y = reorder(short_name, mncs_approx))) +
  geom_col(fill = palette_sci(1)) +
  geom_vline(xintercept = 1, linetype = "dashed", colour = "grey50") +
  labs(x = "MNCS (approx.)", y = NULL) +
  theme_sci()
```

14.5 Diagnostics and interpretation

- **Size vs. impact**: A large university may rank highly on total output but poorly on per-paper impact. Always report both size-dependent and size-independent indicators.
- **Field mix**: A medical university will appear more productive (and highly cited) than a social-science institute due to field-level citation norms. True normalisation requires field-level data.

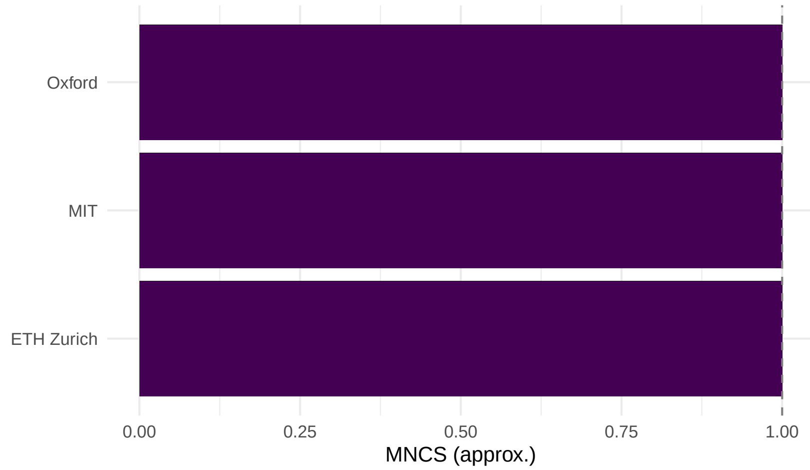

Figure 14.2: Approximate MNCS for each institution (within-sample normalisation).

- **Sample vs. population**: Our sample of 400 works per institution is illustrative. For real benchmarking, use the full population or report confidence intervals.
- **Fractional/full ratio**: A ratio near 1.0 indicates few collaborations beyond the institution; a ratio well below 1.0 indicates extensive inter-institutional collaboration.

14.6 Limitations and responsible use

14.7 Limitations and responsible use

- **Rankings oversimplify.** Institutional and country rankings compress a multidimensional reality into a single number. The Leiden Manifesto (Hicks et al., 2015) warns against using rankings mechanically.
- **Affiliation data quality.** Not all papers have complete or correct institutional affiliations. OpenAlex coverage depends on publisher metadata quality.
- **Counting method changes conclusions.** The same data can yield different institutional rankings depending on whether full or fractional counting is used. Always report the method.
- **Language and geographic bias.** Databases undercount research published in non-English languages, disadvantaging countries in Latin America, Asia, and Africa (Visser et al., 2021; Singh et al., 2021).

- **Do not compare across fields without normalisation.** A biomedical research university will always outperform a humanities-focused institution on raw citation indicators.

14.8 Common pitfalls

14.9 Common pitfalls

- **Using full counting when fractional is more appropriate.** Full counting double-credits collaborative work, inflating output for institutions with many international partnerships.
- **Ignoring field normalisation.** Comparing raw MNCS across institutions with very different disciplinary profiles is misleading.
- **String-matching institution names.** Never match institutions by name strings alone. Use ROR or OpenAlex institution IDs.
- **Treating samples as populations.** If you sampled 400 works per institution, the resulting metrics have uncertainty. Report this.
- **Forgetting subsidiary institutions.** Large universities may have hospitals, research institutes, or campuses with separate ROR IDs. Decide on the unit of analysis upfront.

14.10 Exercises

1. **Country-level analysis.** Aggregate publications by author country (using `authorships.countries`) instead of institution. Compute full and fractional counts for the top 10 countries in your corpus. Which countries are most affected by the counting method?

2. **PP(top 10%) by institution.** For each institution, compute the proportion of papers in the top 10% most cited within the sample (by year). How does this correlate with MNCS?

3. **Collaboration intensity.** Compute the mean number of distinct institutional affiliations per paper for each institution. Is there a relationship between collaboration intensity and the full/fractional count ratio?

14.11 Solutions

Solutions are provided in F.

14.12 Further reading

- Waltman et al. (2011) — MNCS and size-independent indicators for institutional comparison.
- Glänzel and Schubert (2004) — Fractional counting in co-authorship and institutional analysis.
- Hicks et al. (2015) — The Leiden Manifesto on responsible institutional benchmarking.
- Waltman (2016) — Review of citation impact indicators applicable to institutions and countries.
- Visser et al. (2021) — Database coverage comparison; crucial context for country-level analysis.
- Singh et al. (2021) — Journal coverage across databases, highlighting geographic imbalances.

14.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0
```

```
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.
#>   [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode
#>  [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0             bit_4.6.
#>  [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel
#>  [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellrang
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51             triebear
#>  [36] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0           igraph_2
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioa
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0
#>  [51] curl_7.1.0             qpdf_1.4.1             lattice_0.22-6         plyr_1.8
#>  [56] withr_3.0.2            S7_0.2.2               askpass_1.2.1          evaluate
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1          janeaust
#>  [66] plotly_4.12.0          generics_0.1.4         rprojroot_2.1.1        hms_1.1.
#>  [71] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3         tools_4.
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23          fs_2.1.0
#>  [81] fastmatch_1.1-8        grid_4.4.1             bibtex_0.5.2           urltools
#>  [86] dimensionsR_0.0.3      bibliometrixData_0.3.0 cli_3.6.6              rappdirs
#>  [91] gtable_0.3.6           digest_0.6.39          ggrepel_0.9.8          crul_1.6
#>  [96] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5        httr_1.4
#> [101] mime_0.13              bit64_4.8.0
```

Chapter 15

Temporal Analysis and Citation Aging

15.1 Learning objectives

After completing this chapter, you will be able to:

- Define and apply citation windows of different lengths
- Compute and interpret the citation half-life for a journal or corpus
- Plot citation aging curves showing how citations accumulate over time
- Calculate the Price index (proportion of references less than five years old)
- Analyse temporal trends in publication output and citation rates
- Explain how aging patterns differ across disciplines

15.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

15.3 Conceptual background

Citations are not evenly distributed over time. Most papers receive the bulk of their citations within a few years of publication, then gradually fade. This phenomenon — **citation aging** or **obsolescence** — was first studied systematically by de Solla Price (1963), who observed that scientific literature has a characteristic "half-life": the time by which half of all citations to a given year's output have been received.

The **citation half-life** varies dramatically across disciplines. In fast-moving fields like molecular biology, half-lives may be three to four years; in mathematics or humanities, they can exceed ten years. This has direct consequences for metrics: a two-year Journal Impact Factor systematically undervalues journals in slow-aging fields (Waltman, 2016).

A **citation window** is the fixed period after publication during which citations are counted. The JIF uses a two-year window; CiteScore uses four years. Choosing the window involves a trade-off: short windows favour fast-aging fields but miss delayed-recognition papers; long windows are more inclusive but dilute the signal of recent trends.

The **Price index** — the proportion of references in a paper's bibliography that are less than five years old — measures how rapidly a field consumes recent literature. A high Price index indicates a fast-moving frontier field; a low Price index suggests reliance on older foundational works. de Solla Price (1963) introduced this as an "immediacy" measure.

Sleeping beauties — papers that receive few citations for years then suddenly attract attention (Kleinberg, 2003) — violate the standard aging model. Their existence is a reminder that citation half-lives describe averages, not individual papers. Garfield (1955) already noted that some ideas take decades to be recognised.

Understanding citation temporality is essential for fair evaluation. Comparing citation counts without accounting for publication year, citation window, and field-specific aging rates produces meaningless results.

15.4 Worked example

15.4.1 Acquiring the data

We fetch works from two journals with different expected aging profiles.

```
# Scientometrics (social science, moderate aging)
works_sciento <- oa_fetch(
  entity = "works",
```

```r
  primary_location.source.id = "S148561398",
  from_publication_date = "2013-01-01",
  to_publication_date = "2018-12-31",
  type = "article",
  options = list(sample = 400, seed = 42)
)

# PLOS ONE (multidisciplinary, faster aging)
works_plos <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S202381698",
  from_publication_date = "2013-01-01",
  to_publication_date = "2018-12-31",
  type = "article",
  options = list(sample = 400, seed = 42)
)

works_all <- bind_rows(
  works_sciento |> mutate(journal = "Scientometrics"),
  works_plos |> mutate(journal = "PLOS ONE")
) |>
  mutate(year = year(publication_date),
         age = 2024 - year)
```

15.4.2 Citation aging curves

We compute mean citations by age (years since publication) for each journal.

```r
aging <- works_all |>
  group_by(journal, age) |>
  summarise(
    mean_cites = mean(cited_by_count, na.rm = TRUE),
    median_cites = median(cited_by_count, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  arrange(journal, age)
```

```r
ggplot(aging, aes(x = age, y = mean_cites, colour = journal)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_manual(values = palette_sci(2)) +
  labs(x = "Years since publication", y = "Mean citations",
```

```
      colour = "Journal") +
  theme_sci()
```

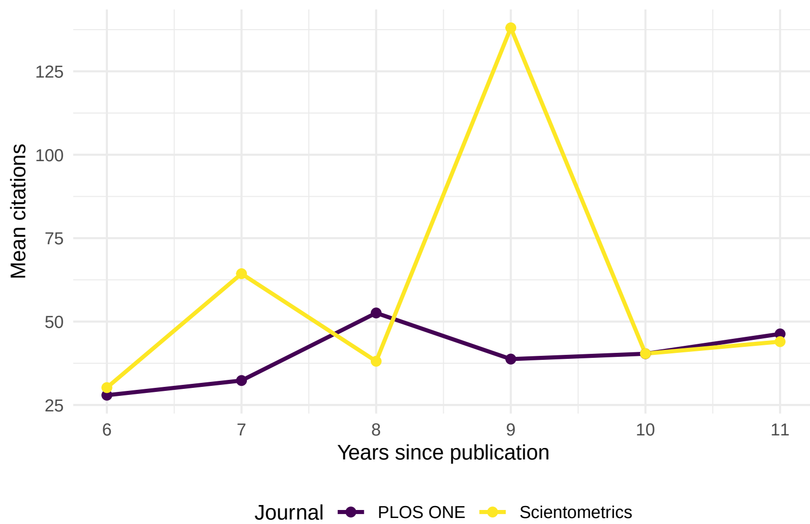

Figure 15.1: Citation aging curves for Scientometrics and PLOS ONE articles published 2013-2018.

15.4.3 Estimating citation half-life

We estimate the half-life as the age at which cumulative citations reach 50% of total.

```
half_life <- works_all |>
  group_by(journal) |>
  arrange(age) |>
  mutate(
    cum_cites = cumsum(cited_by_count),
    total_cites = sum(cited_by_count),
    cum_pct = cum_cites / total_cites
  ) |>
  summarise(
    total_cites = first(total_cites),
    half_life_age = min(age[cum_pct >= 0.50]),
    .groups = "drop"
  )
```

Journal	Total citations	Half-life (years)
PLOS ONE	16270	9
Scientometrics	23079	9

```
half_life |>
  gt() |>
  cols_label(
    journal = "Journal",
    total_cites = "Total citations",
    half_life_age = "Half-life (years)"
  )
```

15.4.4 Price index

The Price index is the fraction of references less than five years old. A full computation requires publication years of each reference; here we show reference-list lengths as a starting point.

```
works_all |>
  filter(!is.na(referenced_works_count), referenced_works_count > 0) |>
  group_by(journal) |>
  summarise(mean_refs = round(mean(referenced_works_count), 1), .groups = "drop")
```

```
#> # A tibble: 2 x 2
#>   journal        mean_refs
#>   <chr>              <dbl>
#> 1 PLOS ONE            50.1
#> 2 Scientometrics      42.1
```

15.4.5 Cumulative citation trajectories

```
cum_aging <- works_all |>
  group_by(journal, age) |>
  summarise(total = sum(cited_by_count), .groups = "drop") |>
  group_by(journal) |>
  arrange(age) |>
  mutate(
    cum_total = cumsum(total),
    cum_pct = cum_total / sum(total)
```

```
  )

ggplot(cum_aging, aes(x = age, y = cum_pct, colour = journal)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0.5, linetype = "dashed", colour = "grey50") +
  scale_y_continuous(labels = scales::percent) +
  scale_colour_manual(values = palette_sci(2)) +
  labs(x = "Years since publication", y = "Cumulative % of citations",
       colour = "Journal") +
  annotate("text", x = 9, y = 0.52, label = "50% threshold",
           colour = "grey40", size = 3.5) +
  theme_sci()
```

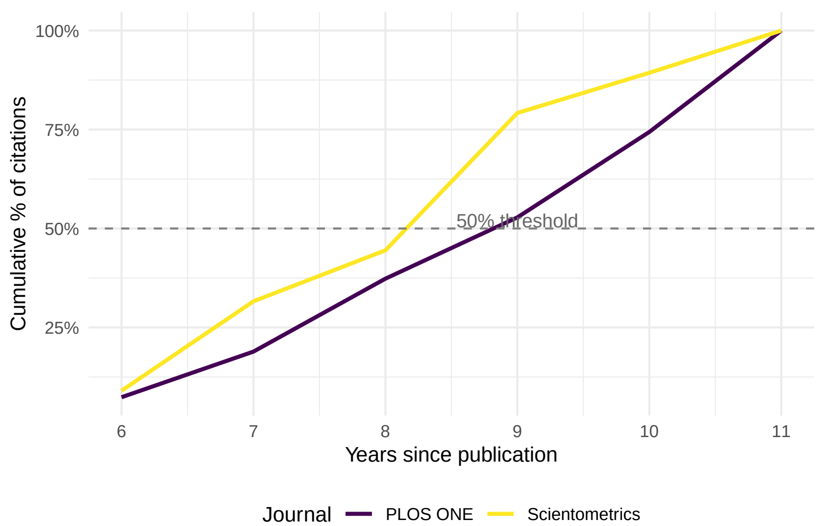

Figure 15.2: Cumulative citation percentage by years since publication.

15.5 Diagnostics and interpretation

- **Sample size per age bin**: Ensure each year has enough papers for stable estimates. Small cohorts produce noisy aging curves.
- **Truncation**: Papers published in the most recent year have not had time to accumulate citations. Exclude or flag incomplete citation windows.
- **Disciplinary benchmarks**: Compare your half-life estimates against known benchmarks. Social sciences typically have half-lives of 5-8 years; biomedical fields 3-5 years; mathematics 8-12 years.

- **Outliers**: A few extremely highly cited papers can distort mean-based aging curves. Compare mean and median trajectories.

15.6 Limitations and responsible use

15.7 Limitations and responsible use

- **Half-life is an average.** Individual papers deviate wildly from the average aging pattern. Sleeping beauties and flash-in-the-pan papers both exist.
- **Database lag.** Citation counts in OpenAlex may lag behind reality by weeks or months. Very recent citation data is unreliable.
- **Survivor bias.** Retracted papers or papers removed from databases disappear from aging analyses, potentially biasing half-life estimates.
- **Citation inflation.** The total volume of citations has grown over time. A paper published in 2020 will accumulate citations faster than an identical paper published in 2000, simply because there are more papers being published (de Solla Price, 1963).
- **Do not penalise slow-aging fields.** Using short citation windows systematically undervalues mathematics, humanities, and theoretical disciplines (Hicks et al., 2015).

15.8 Common pitfalls

15.9 Common pitfalls

- **Mixing citation windows.** Comparing a 2-year JIF for one journal with lifetime citations for another produces meaningless rankings.
- **Ignoring publication-year effects.** A corpus spanning 2000-2023 includes papers with 23 years of citation history alongside papers with 1 year. Normalise by publication year.
- **Confusing obsolescence with irrelevance.** A declining citation curve does not mean the knowledge is wrong or useless; it may mean the ideas have been absorbed into textbooks or standard practice.
- **Forgetting about the denominator.** If the number of publications in a field is growing, the absolute citation count grows even if the per-paper impact is constant.

15.10 Exercises

1. **Half-life comparison.** Fetch works from a mathematics journal and a biomedical journal. Compute and compare their citation half-lives. Does the difference match disciplinary expectations?

2. **Price index computation.** For a set of papers with known reference lists, compute the actual Price index (fraction of references < 5 years old). *(Hint: you will need to fetch metadata for the referenced works.)*

3. **Sleeping beauty detection.** Identify papers that received fewer than 2 citations in their first 3 years but more than 20 total. What are these papers about?

4. **Median vs. mean aging.** Replot the aging curves using median instead of mean citations. How does the shape change, and why?

5. **Citation window sensitivity.** For a single journal, compute a JIF-like proxy using 2-year, 3-year, and 5-year windows. How much does the value change?

15.11 Solutions

Solutions are provided in F.

15.12 Further reading

- de Solla Price (1963) — *Little Science, Big Science*; introduced citation aging and the Price index.
- Garfield (1955) — The origin of citation indexing, including early observations on citation temporality.
- Kleinberg (2003) — Bursty patterns in temporal data; relevant to sleeping beauties and sudden citation surges.
- Waltman (2016) — Review of citation indicators, including discussion of citation windows and aging.
- Hicks et al. (2015) — The Leiden Manifesto; warns against penalising slow-aging fields with short citation windows.

15.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda_4.4       pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.
#>  [7] rcrossref_1.2.1    gt_1.3.0           tidytext_0.4.3     glue_1.8.1         openalexR_3.0
#> [13] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0
#> [19] tibble_3.3.1       ggplot2_4.0.3      tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] gridExtra_2.3          httr2_1.2.2            readxl_1.4.5           rlang_1.2.0
#>   [6] otel_0.2.0             compiler_4.4.1         vctrs_0.7.3            httpcode_0.3.0
#>  [11] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3         ca_0.71.1
#>  [16] promises_1.5.0         rmarkdown_2.31         tzdb_0.5.0             bit_4.6.0
#>  [21] jsonlite_2.0.0         SnowballC_0.7.1        later_1.4.8            parallel_4.4.1
#>  [26] R6_2.6.1               stringi_1.8.7          RColorBrewer_1.1-3     cellranger_1.1.0
#>  [31] Rcpp_1.1.1-1.1         bookdown_0.46          knitr_1.51             triebeard_0.4.1
#>  [36] rentrez_1.2.4          httpuv_1.6.17          Matrix_1.7-0           igraph_2.3.1
#>  [41] tidyselect_1.2.1       stringdist_0.9.17      pubmedR_1.0.2          rstudioapi_0.18.0
#>  [46] yaml_2.3.12            viridis_0.6.5          codetools_0.2-20       miniUI_0.1.2
#>  [51] curl_7.1.0             qpdf_1.4.1             lattice_0.22-6         plyr_1.8.9
#>  [56] withr_3.0.2            S7_0.2.2               askpass_1.2.1          evaluate_1.0.5
#>  [61] xml2_1.5.2             shinycssloaders_1.1.0  pillar_1.11.1          janeaustenr_1.0.0
#>  [66] plotly_4.12.0          generics_0.1.4         rprojroot_2.1.1        hms_1.1.4
#>  [71] xtable_1.8-8           contentanalysis_1.0.0  lazyeval_0.2.3         tools_4.4.1
#>  [76] tokenizers_0.3.0       openxlsx_4.2.8.1       XML_3.99-0.23          fs_2.1.0
#>  [81] fastmatch_1.1-8        grid_4.4.1             bibtex_0.5.2           urltools_1.7.3.1
```

```
#>  [86] dimensionsR_0.0.3      bibliometrixData_0.3.0 cli_3.6.6              rappdirs_
#>  [91] gtable_0.3.6           digest_0.6.39          ggrepel_0.9.8          crul_1.6
#>  [96] farver_2.1.2           htmltools_0.5.9        lifecycle_1.0.5        httr_1.4
#> [101] mime_0.13              bit64_4.8.0
```

Part IV

Network & Mapping Methods

Chapter 16

Co-authorship Networks

16.1 Learning objectives

After completing this chapter, you will be able to:

- Construct a co-authorship network from bibliographic data using `igraph`
- Compute and interpret node-level metrics: degree, betweenness centrality, and clustering coefficient
- Apply Louvain and Leiden community detection algorithms and compare results
- Produce deterministic network layouts that render identically in HTML and PDF
- Identify central authors and structural holes in a collaboration network

16.2 Setup

```r
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```r
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

16.3 Conceptual background

Co-authorship is one of the most visible and well-studied forms of scientific collaboration. When two researchers appear together on a publication, they form a tie in the co-authorship network. Studying these networks reveals the social structure of science: who collaborates with whom, which groups are isolated, and which individuals serve as bridges between communities (Newman, 2001).

Co-authorship networks are typically modelled as **undirected, weighted graphs**. Nodes represent authors, and an edge connects two authors who have co-authored at least one paper. Edge weights reflect the number of co-authored papers (or, alternatively, fractional counts inversely proportional to the number of authors per paper). These networks tend to exhibit properties common to social networks: a heavy-tailed degree distribution (most authors have few collaborators, a few have many), high clustering (collaborators of collaborators tend to collaborate), and a small-world structure (Barabási et al., 2002).

Community detection identifies groups of authors who collaborate more densely among themselves than with the rest of the network. The Louvain algorithm (Blondel et al., 2008) is fast and widely used; the Leiden algorithm (Traag et al., 2019) provides theoretical guarantees that all communities are well-connected, avoiding the "resolution limit" problem that can afflict Louvain.

Network analysis of co-authorship data has been widely applied to map research fields (Glänzel and Schubert, 2004), identify key opinion leaders, assess interdisciplinary collaboration, and study the evolution of research communities over time. Fortunato (2010) provides a comprehensive review of community detection methods applicable to these networks.

16.4 Worked example

16.4.1 Acquiring co-authorship data

We fetch works from a journal and extract the author pairs.

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2021-01-01",
  to_publication_date = "2023-12-31",
```

```r
  options = list(sample = 300, seed = 42)
)

nrow(works)
```

```
#> [1] 300
```

16.4.2 Extracting author pairs

Each work has a list-column of authors. We unnest and build all pairwise co-author edges.

```r
author_data <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  select(work_id = id, author_id = authorships_id, author_name = authorships_display_name) |>
  filter(!is.na(author_id))

edges <- author_data |>
  inner_join(author_data, by = "work_id", suffix = c("_1", "_2"),
             relationship = "many-to-many") |>
  filter(author_id_1 < author_id_2) |>
  count(author_id_1, author_id_2,
        name_1 = author_name_1, name_2 = author_name_2,
        name = "weight")

cat(glue("Authors: {n_distinct(c(edges$author_id_1, edges$author_id_2))}\n"))
```

```
#> Authors: 802
```

```r
cat(glue("Edges: {nrow(edges)}\n"))
```

```
#> Edges: 1946
```

16.4.3 Building the network

```r
g <- graph_from_data_frame(
  edges |> select(author_id_1, author_id_2, weight),
  directed = FALSE
)
```

```
g <- simplify(g, edge.attr.comb = list(weight = "sum"))

cat(glue("Nodes: {vcount(g)}, Edges: {ecount(g)}\n"))
```

```
#> Nodes: 802, Edges: 1946
```

```
cat(glue("Components: {components(g)$no}\n"))
```

```
#> Components: 190
```

```
cat(glue("Density: {round(graph.density(g), 4)}\n"))
```

```
#> Density: 0.0061
```

16.4.4 Network metrics

```
V(g)$degree <- degree(g)
V(g)$betweenness <- betweenness(g, normalized = TRUE)
V(g)$clustering <- transitivity(g, type = "local")

author_lookup <- author_data |>
  distinct(author_id, author_name)

metrics <- tibble(
  author_id = V(g)$name,
  degree = V(g)$degree,
  betweenness = V(g)$betweenness,
  clustering = V(g)$clustering
) |>
  left_join(author_lookup, by = "author_id") |>
  arrange(desc(degree))

metrics |>
  head(10) |>
  select(author_name, degree, betweenness, clustering)
```

```
#> # A tibble: 10 x 4
#>    author_name       degree betweenness clustering
#>    <chr>              <dbl>       <dbl>      <dbl>
#>  1 Torben Schubert       30 0.000949         0.761
```

```
#>  2 Min Song             29 0.000325         0.798
#>  3 Rodrigo Costas       28 0.000490         0.860
#>  4 Loet Leydesdorff     28 0.00498          0.862
#>  5 Mike Thelwall        28 0.000487         0.862
#>  6 Dag W. Aksnes        28 0.000487         0.862
#>  7 Hamid R. Jamali      27 0.000247         0.926
#>  8 Philipp Mayr         27 0.000247         0.926
#>  9 Hyeyoung Ryu         26 0.000000543      0.988
#> 10 Isidro F. Aguillo    26 0                1
```

16.4.5 Community detection

We apply both Louvain and Leiden to the largest connected component.

```r
comp <- components(g)
giant <- induced_subgraph(g, which(comp$membership == which.max(comp$csize)))

louvain <- cluster_louvain(giant)
V(giant)$community_louvain <- membership(louvain)

cat(glue("Louvain communities: {length(unique(membership(louvain)))}\n"))
```

```
#> Louvain communities: 7
```

```r
cat(glue("Louvain modularity: {round(modularity(louvain), 3)}\n"))
```

```
#> Louvain modularity: 0.377
```

```r
leiden_result <- cluster_leiden(giant, resolution_parameter = 1.0,
                                objective_function = "modularity")
V(giant)$community_leiden <- membership(leiden_result)

cat(glue("Leiden communities: {length(unique(membership(leiden_result)))}\n"))
```

```
#> Leiden communities: 7
```

```r
cat(glue("Leiden modularity: {round(modularity(giant, membership(leiden_result)), 3)}\n"))
```

```
#> Leiden modularity: 0.377
```

16.4.6 Comparing community assignments

```r
comparison <- tibble(
  louvain = V(giant)$community_louvain,
  leiden = V(giant)$community_leiden
)

nmi <- igraph::compare(
  V(giant)$community_louvain,
  V(giant)$community_leiden,
  method = "nmi"
)
cat(glue("Normalized Mutual Information (Louvain vs Leiden): {round(nmi, 3)}\n"))
```

```
#> Normalized Mutual Information (Louvain vs Leiden): 1
```

16.4.7 Visualization

We use a Fruchterman-Reingold layout with a fixed seed for deterministic output.

```r
giant_tidy <- as_tbl_graph(giant) |>
  mutate(
    community = as.factor(community_leiden),
    label = ifelse(degree > quantile(degree, 0.95),
                   author_lookup$author_name[match(name, author_lookup$author_id)],
                   NA_character_)
  )

set.seed(42)
layout <- create_layout(giant_tidy, layout = "fr")

ggraph(layout) +
  geom_edge_link(alpha = 0.15, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  geom_node_text(aes(label = label), repel = TRUE, size = 2.5,
                 max.overlaps = 15, na.rm = TRUE) +
  scale_size_continuous(range = c(1, 6), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(giant_tidy |> pull(community), na.rm = TRUE)
  )) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

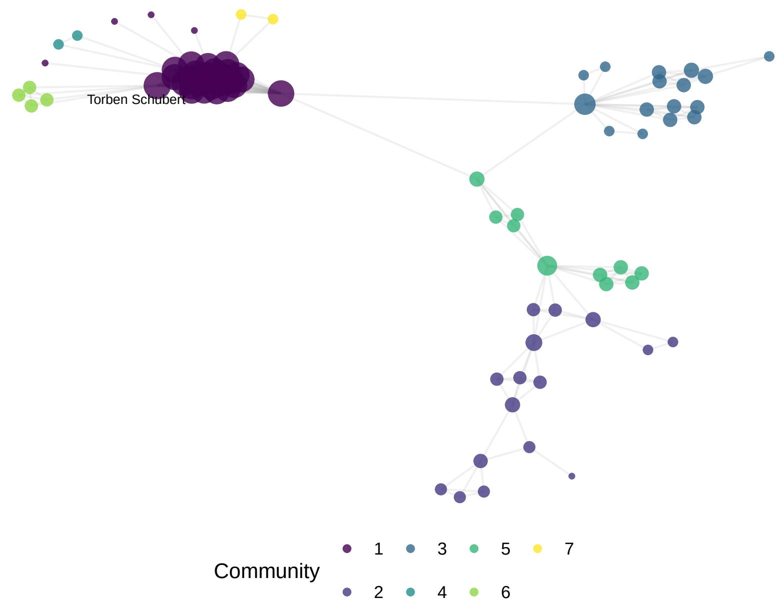

Figure 16.1: Co-authorship network of sampled Scientometrics articles (2021–2023), coloured by Leiden community.

```
tibble(degree = degree(giant)) |>
  ggplot(aes(x = degree)) +
  geom_histogram(binwidth = 1, fill = palette_sci(1), colour = "white") +
  labs(x = "Degree (number of co-authors)", y = "Count") +
  theme_sci()
```

16.5 Diagnostics and interpretation

Key diagnostics for co-authorship networks:

- **Giant component ratio**: What fraction of nodes are in the largest connected component? A fragmented network may indicate a niche field or a sparse sample.
- **Degree distribution**: The heavy-tailed distribution is expected. A network where all nodes have similar degree may indicate a problem with the data (e.g., only single-author papers were excluded).
- **Modularity**: Values above 0.3 typically indicate meaningful community structure. Modularity near zero suggests the network has no clear communities.

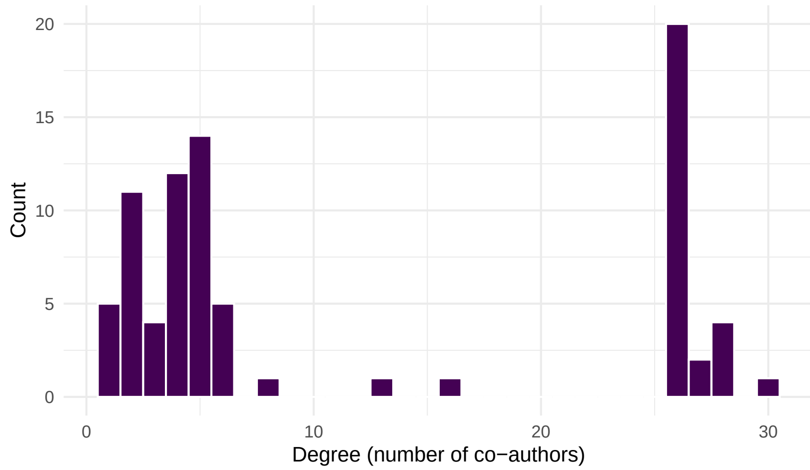

Figure 16.2: Degree distribution of the co-authorship network.

- **Comparison of algorithms**: If Louvain and Leiden produce very different community counts, investigate the resolution parameter. High NMI (>0.8) indicates strong agreement.

```
giant_ratio <- vcount(giant) / vcount(g)
cat(glue("Giant component: {vcount(giant)}/{vcount(g)} nodes ({scales::percent(giant_ra
```

```
#> Giant component: 81/802 nodes (10%)
```

```
cat(glue("Mean degree: {round(mean(degree(giant)), 1)}\n"))
```

```
#> Mean degree: 11.6
```

```
cat(glue("Transitivity (global): {round(transitivity(giant, type = 'global'), 3)}\n"))
```

```
#> Transitivity (global): 0.941
```

16.6 Limitations and responsible use

16.7 Limitations and responsible use

- **Co-authorship collaboration.** Not all collaborations result in co-authored publications, and not all co-authorships reflect genuine intellectual contribution (honorary authorship).

- **Name ambiguity.** Author names in bibliographic databases are not unique identifiers. Despite OpenAlex's entity resolution, errors persist — especially for common names and transliterated names. Always inspect suspicious high-degree nodes.
- **Fractional vs. full counting.** Our construction counts each co-authorship edge equally regardless of the number of authors on a paper. Fractional counting (weight = 1/(n-1) per edge) gives less weight to papers with many authors and may be more appropriate for some analyses (Glänzel and Schubert, 2004).
- **Temporal dynamics.** Static networks aggregate all collaborations into a single snapshot. Consider building time-sliced networks to study how collaboration patterns evolve.
- **Causality.** Network position (e.g., high betweenness) does not imply leadership or influence. It is a structural observation, not an evaluative judgement (Hicks et al., 2015).

16.8 Common pitfalls

16.9 Common pitfalls

- **Non-deterministic layouts.** Force-directed layouts (Fruchterman-Reingold, Kamada-Kawai) depend on a random seed. Always `set.seed()` before computing the layout, or the figure will change on every render.
- **Including the full network.** Very large networks are unreadable. Filter to the giant component, top-N nodes, or a specific community before plotting.
- **Ignoring multi-edges and self-loops.** `igraph::simplify()` is essential after constructing the graph from edge lists that may contain duplicate pairs.
- **Comparing modularity across networks of different sizes.** Modularity is sensitive to network size and density. Use the Normalized Mutual Information (NMI) to compare community structures.
- **Overinterpreting small components.** A component with 2–3 authors who co-authored one paper is not a "community" — it is an artifact of the sample.

16.10 Exercises

1. **Fractional counting.** Modify the edge construction to use fractional counting (weight = 1/(k-1) where k is the number of authors on the paper). How do the top-degree authors change?

2. **Temporal slicing.** Split the corpus into yearly slices and build a separate network for each year. How does the number of components and mean degree evolve over time?

3. **Betweenness vs. degree.** Plot betweenness centrality against degree for the giant component. Are the highest-degree authors also the highest-betweenness authors? What does a high-betweenness, low-degree node represent?

4. **Resolution parameter.** Run Leiden community detection with resolution parameters of 0.5, 1.0, and 2.0. How does the number of communities change? Which resolution produces the most interpretable results?

16.11 Solutions

Solutions are provided in F.

16.12 Further reading

- Newman (2001) — Foundational analysis of scientific collaboration networks.
- Barabási et al. (2002) — Evolution of co-authorship networks and their scale-free properties.
- Blondel et al. (2008) — The Louvain community detection algorithm.
- Traag et al. (2019) — The Leiden algorithm, with guarantees on well-connected communities.
- Fortunato (2010) — Comprehensive review of community detection in graphs.
- Glänzel and Schubert (2004) — Co-authorship analysis and fractional counting in bibliometrics.

16.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
```

```
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4       pdftools_3.9.
#>  [7] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt_1.3.0
#> [13] glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      stringr_1.6.0
#> [19] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       ggplot2_4.0.3
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_2.0.0
#>   [6] farver_2.1.2           rmarkdown_2.31         fs_2.1.0               vctrs_0.7.3
#>  [11] askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9        contentanalysis_1.0
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4      tokenizers_0.3.0
#>  [21] httr2_1.2.2            cachem_1.1.0           plotly_4.12.0          dimensionsR_0.0.3
#>  [26] lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0           R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          shinycssloaders_1.1.0  rprojroot_2.1.1
#>  [36] labeling_0.4.3         urltools_1.7.3.1       timechange_0.4.0       polyclip_1.10-7
#>  [41] compiler_4.4.1         here_1.0.2             bit64_4.8.0            withr_3.0.2
#>  [46] backports_1.5.1        viridis_0.6.5          ggforce_0.5.0          MASS_7.3-60.2
#>  [51] bibliometrixData_0.3.0 tools_4.4.1            otel_0.2.0             stopwords_2.3
#>  [56] httpuv_1.6.17          rentrez_1.2.4          promises_1.5.0         grid_4.4.1
#>  [61] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0             rscopus_0.9.0
#>  [66] data.table_1.18.4      hms_1.1.4              xml2_1.5.2             utf8_1.2.6
#>  [71] pillar_1.11.1          later_1.4.8            tweenr_2.0.3           lattice_0.22-6
#>  [76] tidyselect_1.2.1       miniUI_0.1.2           knitr_1.51             gridExtra_2.3
#>  [81] crul_1.6.0             xfun_0.57              graphlayouts_1.2.3     DT_0.34.0
#>  [86] visNetwork_2.1.4       stringi_1.8.7          lazyeval_0.2.3         qpdf_1.4.1
#>  [91] evaluate_1.0.5         codetools_0.2-20       httpcode_0.3.0         cli_3.6.6
#>  [96] dichromat_2.0-0.1      Rcpp_1.1.1-1.1         readxl_1.4.5           triebeard_0.4.1
#> [101] parallel_4.4.1         assertthat_0.2.1       pubmedR_1.0.2          viridisLite_0.4.3
#> [106] openxlsx_4.2.8.1       rlang_1.2.0            fastmatch_1.1-8
```

Chapter 17

Co-citation and Bibliographic Coupling

17.1 Learning objectives

After completing this chapter, you will be able to:

- Distinguish co-citation analysis from bibliographic coupling and explain when each is appropriate
- Construct co-citation and bibliographic coupling networks from OpenAlex reference data
- Compute normalised similarity measures (cosine, Jaccard) for citation-based linkages
- Identify core documents and intellectual clusters in a research field
- Interpret co-citation maps as representations of intellectual structure

17.2 Setup

```
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)
library(gt)
```

```r
set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

17.3 Conceptual background

While co-authorship networks (16.3) reveal social structure, **co-citation** and **bibliographic coupling** networks reveal *intellectual* structure — the conceptual relationships between documents as expressed through citation behaviour.

Co-citation analysis, introduced independently by Small (1973) and Marshakova-Shaikevich in 1973, measures the relatedness of two documents by how often they are cited together by other papers. If documents A and B frequently appear in the same reference lists, they likely address related topics or represent complementary contributions to a field. Co-citation strength changes over time: as a field evolves, new citing papers create new co-citation links and strengthen or weaken existing ones.

Bibliographic coupling, proposed by Kessler (1963), takes the complementary perspective. Two documents are bibliographically coupled if they share one or more references. Unlike co-citation, coupling strength is fixed at publication time — once a paper is published, its reference list does not change. This makes bibliographic coupling especially useful for analysing the most recent literature, where co-citation links have not yet had time to form.

The two methods reveal different facets of intellectual structure. Co-citation maps the *reception* of ideas (how the community groups prior work); bibliographic coupling maps the *production* of knowledge (how authors draw on shared foundations). Waltman et al. (2010) demonstrated that combining both perspectives yields richer maps of science than either alone.

Both methods produce **similarity matrices** that can be converted to networks. Common normalisation approaches include cosine similarity (emphasising relative overlap) and the Jaccard index (penalising pairs with many unshared references). Unnormalised raw counts are dominated by highly-cited papers; normalisation is essential for balanced network construction.

17.4 Worked example

17.4.1 Fetching citing and cited works

We fetch a corpus of scientometrics research and extract reference lists.

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 200, seed = 42)
)

refs <- works |>
  select(citing_id = id, referenced_works) |>
  unnest(referenced_works) |>
  rename(cited_id = referenced_works)

cat(glue("Citing works: {n_distinct(refs$citing_id)}\n"))
```

```
#> Citing works: 200
```

```r
cat(glue("Cited works: {n_distinct(refs$cited_id)}\n"))
```

```
#> Cited works: 8768
```

```r
cat(glue("Citation links: {nrow(refs)}\n"))
```

```
#> Citation links: 10284
```

17.4.2 Building a co-citation network

Two cited documents are co-cited when they appear in the same reference list. We count shared citing papers.

```r
cocit_pairs <- refs |>
  inner_join(refs, by = "citing_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(cited_id_a < cited_id_b) |>
  count(cited_id_a, cited_id_b, name = "cocit_count")

cocit_top <- cocit_pairs |>
  filter(cocit_count >= 3)

cat(glue("Co-citation pairs (count >= 3): {nrow(cocit_top)}\n"))
```

```
#> Co-citation pairs (count >= 3): 254
```

```
g_cocit <- graph_from_data_frame(
  cocit_top |> select(cited_id_a, cited_id_b, weight = cocit_count),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Co-citation network: {vcount(g_cocit)} nodes, {ecount(g_cocit)} edges\n"))
```

```
#> Co-citation network: 133 nodes, 254 edges
```

17.4.3 Building a bibliographic coupling network

Two citing documents are coupled when they share at least one reference.

```
bibcoup_pairs <- refs |>
  inner_join(refs, by = "cited_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(citing_id_a < citing_id_b) |>
  count(citing_id_a, citing_id_b, name = "shared_refs")

bibcoup_top <- bibcoup_pairs |>
  filter(shared_refs >= 5)

cat(glue("Bibliographic coupling pairs (shared refs >= 5): {nrow(bibcoup_top)}\n"))
```

```
#> Bibliographic coupling pairs (shared refs >= 5): 70
```

```
g_bibcoup <- graph_from_data_frame(
  bibcoup_top |> select(citing_id_a, citing_id_b, weight = shared_refs),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Coupling network: {vcount(g_bibcoup)} nodes, {ecount(g_bibcoup)} edges\n"))
```

```
#> Coupling network: 70 nodes, 70 edges
```

17.4.4 Visualisation

```
comp <- components(g_cocit)
giant <- induced_subgraph(g_cocit, which(comp$membership == which.max(comp$csize)))
V(giant)$community <- as.factor(membership(cluster_leiden(
  giant, resolution_parameter = 1.0, objective_function = "modularity"
)))
V(giant)$degree <- degree(giant)

set.seed(42)
layout <- create_layout(as_tbl_graph(giant), layout = "fr")

ggraph(layout) +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  scale_size_continuous(range = c(1, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(giant)$community)
  )) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

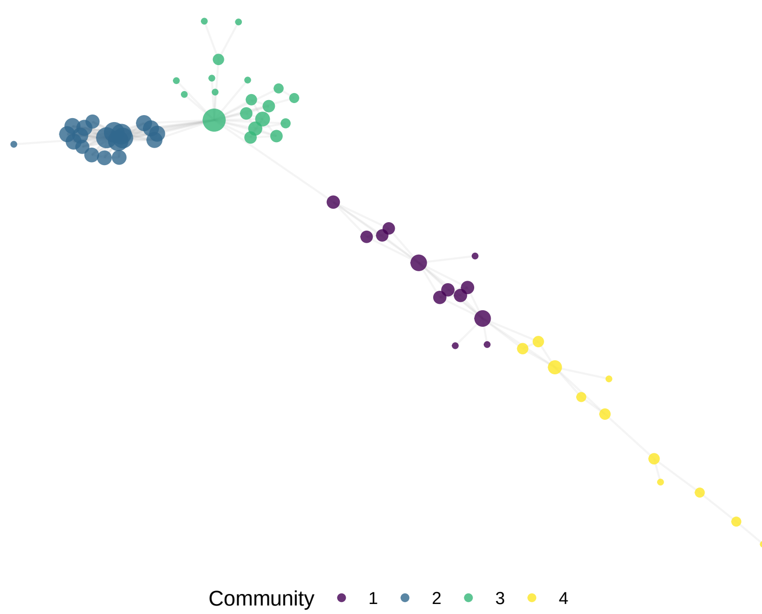

Figure 17.1: Co-citation network of frequently co-cited references in Scientometrics (2020–2023).

```
comp_bc <- components(g_bibcoup)
giant_bc <- induced_subgraph(g_bibcoup,
                             which(comp_bc$membership == which.max(comp_bc$csize)))
V(giant_bc)$community <- as.factor(membership(cluster_leiden(
  giant_bc, resolution_parameter = 1.0, objective_function = "modularity"
)))
V(giant_bc)$degree <- degree(giant_bc)

set.seed(42)
layout_bc <- create_layout(as_tbl_graph(giant_bc), layout = "fr")

ggraph(layout_bc) +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  scale_size_continuous(range = c(1, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(giant_bc)$community)
  )) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

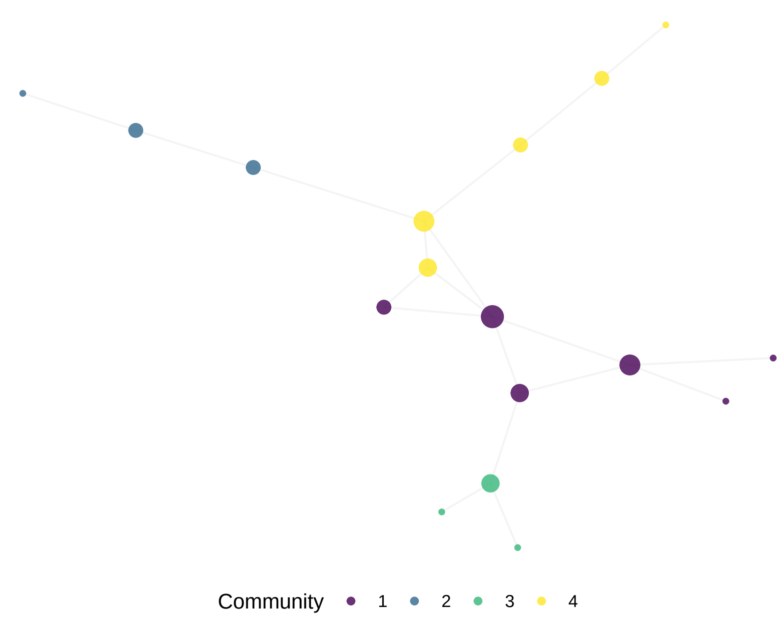

Figure 17.2: Bibliographic coupling network of Scientometrics articles (2020–2023).

Method	Nodes	Edges	Components	Density
Co-citation	133	254	21	0.0289
Bibliographic coupling	70	70	17	0.0290

17.4.5 Comparing the two approaches

```
comparison <- tibble(
  Method = c("Co-citation", "Bibliographic coupling"),
  Nodes = c(vcount(g_cocit), vcount(g_bibcoup)),
  Edges = c(ecount(g_cocit), ecount(g_bibcoup)),
  Components = c(components(g_cocit)$no, components(g_bibcoup)$no),
  Density = c(round(graph.density(g_cocit), 4),
              round(graph.density(g_bibcoup), 4))
)

comparison |> gt()
```

17.5 Diagnostics and interpretation

- **Threshold selection**: The minimum co-citation count or shared-reference count determines network density. Too low a threshold creates a hairball; too high excludes important links. Experiment with multiple thresholds and report the chosen value.
- **Normalisation**: Raw co-citation counts favour highly cited papers. Cosine normalisation produces more balanced networks. Always report whether counts are raw or normalised.
- **Giant component ratio**: A fragmented co-citation network may indicate that the corpus spans multiple disconnected subfields, or that the threshold is too high.
- **Temporal stability**: Co-citation networks change as new papers cite old ones. Bibliographic coupling networks are fixed at publication time. If you need a stable snapshot of recent literature, coupling is preferable.

17.6 Limitations and responsible use

17.7 Limitations and responsible use

- **Co-citation reflects citing behaviour, not truth.** Frequently co-cited papers may be co-cited because they represent a dominant paradigm, not

because they are the best or most relevant works. Minority viewpoints may be invisible.
- **Reference list quality.** Both methods depend entirely on the accuracy and completeness of reference lists. Missing references, formatting errors, and DOI mismatches reduce coverage. OpenAlex reference parsing is imperfect (Priem et al., 2022).
- **Size effects.** Papers with long reference lists contribute disproportionately to bibliographic coupling. Consider normalising by reference list length.
- **Not evaluative.** Co-citation and coupling measure relatedness, not quality. A highly co-cited paper is intellectually central, which is not the same as excellent or correct (Hicks et al., 2015).

17.8 Common pitfalls

17.9 Common pitfalls

- **Confusing the two methods.** Co-citation links *cited* documents; bibliographic coupling links *citing* documents. They answer different questions about the same citation data.
- **Skipping normalisation.** Raw counts produce networks dominated by a few highly cited papers. Always normalise for balanced analysis.
- **Ignoring the threshold.** Reporting results without specifying the minimum count or similarity cutoff makes the analysis irreproducible.
- **Treating coupling as dynamic.** Bibliographic coupling strength is fixed at publication. Do not interpret changes in coupling networks over time as evolving relationships — they reflect changes in the corpus, not the links.

17.10 Exercises

1. **Cosine normalisation.** Compute cosine-normalised co-citation strengths instead of raw counts. How does the network structure change? Do the same core documents emerge?

2. **Temporal co-citation.** Split the corpus into two time periods (2020–2021 and 2022–2023). Build a co-citation network for each. Which co-citation pairs are stable across periods, and which are new?

3. **Bibliographic coupling by year.** For each publication year, build a bibliographic coupling network and report the number of components and modularity. Does the intellectual structure become more or less fragmented over time?

4. **Identifying research fronts.** Use bibliographic coupling communities to identify research fronts. For each community, list the three most frequent keywords or title words. Do the fronts correspond to recognisable subfields?

17.11 Solutions

Solutions are provided in F.

17.12 Further reading

- Small (1973) — Co-citation analysis: the foundational paper.
- Kessler (1963) — Bibliographic coupling: the original proposal.
- Waltman et al. (2010) — Unified approach to bibliometric network construction and clustering.
- Fortunato (2010) — Community detection methods applicable to citation networks.
- Garfield (1955) — The intellectual origins of citation indexing.

17.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
```

```
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4       pd
#>  [7] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt
#> [13] glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      st
#> [19] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       gg
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite
#>   [6] farver_2.1.2           rmarkdown_2.31         fs_2.1.0               vctrs_0.
#>  [11] askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9        contenta
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4      tokenize
#>  [21] httr2_1.2.2            cachem_1.1.0           plotly_4.12.0          dimensio
#>  [26] lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0           R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          shinycssloaders_1.1.0  rprojroo
#>  [36] labeling_0.4.3         urltools_1.7.3.1       timechange_0.4.0       polyclip
#>  [41] compiler_4.4.1         here_1.0.2             bit64_4.8.0            withr_3.
#>  [46] backports_1.5.1        viridis_0.6.5          ggforce_0.5.0          MASS_7.3
#>  [51] bibliometrixData_0.3.0 tools_4.4.1            otel_0.2.0             stopword
#>  [56] httpuv_1.6.17          rentrez_1.2.4          promises_1.5.0         grid_4.4
#>  [61] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0             rscopus_
#>  [66] data.table_1.18.4      hms_1.1.4              xml2_1.5.2             utf8_1.2
#>  [71] pillar_1.11.1          later_1.4.8            tweenr_2.0.3           lattice_
#>  [76] tidyselect_1.2.1       miniUI_0.1.2           knitr_1.51             gridExtr
#>  [81] crul_1.6.0             xfun_0.57              graphlayouts_1.2.3     DT_0.34.
#>  [86] visNetwork_2.1.4       stringi_1.8.7          lazyeval_0.2.3         qpdf_1.4
#>  [91] evaluate_1.0.5         codetools_0.2-20       httpcode_0.3.0         cli_3.6.
#>  [96] dichromat_2.0-0.1      Rcpp_1.1.1-1.1         readxl_1.4.5           triebear
#> [101] parallel_4.4.1         assertthat_0.2.1       pubmedR_1.0.2          viridisL
#> [106] openxlsx_4.2.8.1       rlang_1.2.0            fastmatch_1.1-8
```

Chapter 18

Co-word and Keyword Co-occurrence

18.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the principles of co-word analysis and its role in science mapping
- Extract and clean keywords from OpenAlex data (author keywords and concepts)
- Build a keyword co-occurrence matrix and convert it to a network
- Apply community detection to identify topical clusters
- Visualise keyword co-occurrence networks with interpretable layouts

18.2 Setup

```
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)
library(gt)

set.seed(20260509)
```

```r
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

18.3 Conceptual background

Co-word analysis maps the conceptual structure of a research field by examining which terms appear together in the same documents. Introduced by Callon et al. (1983), the method assumes that when two keywords repeatedly co-occur across publications, they are thematically related. The resulting co-occurrence network reveals the topical landscape of a field: clusters of densely connected keywords represent coherent research themes, while bridges between clusters indicate interdisciplinary connections.

The method works with several types of terms:

- **Author keywords**: Terms selected by the paper's authors. These are intentional descriptors but suffer from inconsistency — authors may use different terms for the same concept ("machine learning" vs. "ML" vs. "statistical learning").
- **Indexed keywords**: Terms assigned by database indexers (e.g., MeSH terms in PubMed). More consistent but only available in some databases.
- **OpenAlex concepts**: Algorithmically assigned topics at multiple levels of a hierarchical taxonomy. These provide broad coverage but may miss nuanced distinctions (Priem et al., 2022).
- **Title/abstract words**: Extracted via text mining. Comprehensive but noisy; requires extensive preprocessing.

Normalisation is important for co-word networks. Raw co-occurrence counts favour high-frequency terms. Common normalisations include the **association strength** (equivalent to pointwise mutual information) and the **equivalence index** (cosine of the co-occurrence vector). Waltman et al. (2010) demonstrated that association strength produces the most balanced network structures for bibliometric mapping.

Co-word analysis complements citation-based methods (17.3). Citation networks reveal intellectual influence; co-word networks reveal thematic content. Combining both provides a more complete picture of a field's structure.

18.4 Worked example

18.4.1 Extracting keywords

We extract author keywords from a sample of scientometrics research.

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

```
keywords <- works |>
  select(id, topics) |>
  unnest(topics, names_sep = "_") |>
  filter(topics_display_name != "") |>
  select(work_id = id, keyword = topics_display_name) |>
  mutate(keyword = str_to_lower(str_trim(keyword))) |>
  filter(!is.na(keyword), nchar(keyword) >= 3)

kw_counts <- keywords |>
  count(keyword, sort = TRUE)

cat(glue("Total keyword occurrences: {nrow(keywords)}\n"))
```

```
#> Total keyword occurrences: 5196
```

```
cat(glue("Unique keywords: {n_distinct(keywords$keyword)}\n"))
```

```
#> Unique keywords: 427
```

```
kw_counts |>
  head(20) |>
  mutate(keyword = fct_reorder(keyword, n)) |>
  ggplot(aes(x = n, y = keyword)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Frequency", y = NULL) +
  theme_sci()
```

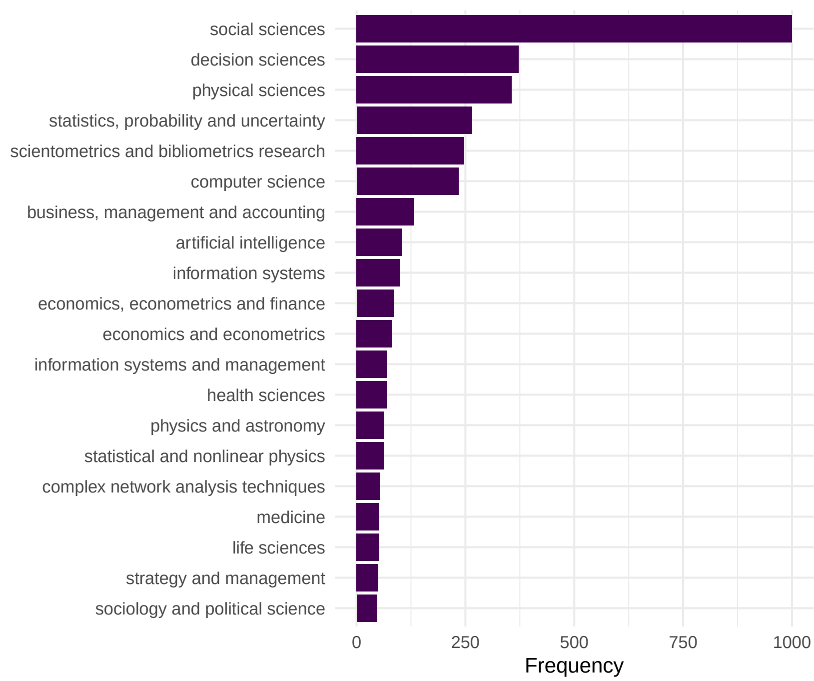

Figure 18.1: Top 20 most frequent keywords in the Scientometrics sample.

18.4.2 Building the co-occurrence network

We create edges between keywords that appear in the same paper.

```r
kw_frequent <- kw_counts |>
  filter(n >= 5) |>
  pull(keyword)

kw_filtered <- keywords |>
  filter(keyword %in% kw_frequent)

coword_pairs <- kw_filtered |>
  inner_join(kw_filtered, by = "work_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(keyword_a < keyword_b) |>
  count(keyword_a, keyword_b, name = "cooccurrence")

coword_top <- coword_pairs |>
  filter(cooccurrence >= 3)

g_kw <- graph_from_data_frame(
  coword_top |> select(keyword_a, keyword_b, weight = cooccurrence),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Keyword network: {vcount(g_kw)} nodes, {ecount(g_kw)} edges\n"))
```

```
#> Keyword network: 112 nodes, 923 edges
```

18.4.3 Community detection for topical clusters

```r
V(g_kw)$degree <- degree(g_kw)
V(g_kw)$strength <- strength(g_kw)

communities <- cluster_leiden(g_kw, resolution_parameter = 0.8,
                              objective_function = "modularity")
V(g_kw)$community <- as.factor(membership(communities))

cat(glue("Communities: {length(unique(membership(communities)))}\n"))
```

```
#> Communities: 3
```

community	top_keywords
1	social sciences, decision sciences, statistics, probability and uncertainty, sci
2	business, management and accounting, economics, econometrics and financ
3	physical sciences, computer science, artificial intelligence, information syste

```r
cat(glue("Modularity: {round(modularity(g_kw, membership(communities)), 3)}\n"))
```

```
#> Modularity: 0.239
```

```r
community_summary <- tibble(
  keyword = V(g_kw)$name,
  community = V(g_kw)$community,
  strength = V(g_kw)$strength
) |>
  group_by(community) |>
  slice_max(strength, n = 5) |>
  summarise(top_keywords = paste(keyword, collapse = ", "),
            n_keywords = n(), .groups = "drop")

community_summary |> gt()
```

18.4.4 Visualisation

```r
set.seed(42)
layout <- create_layout(as_tbl_graph(g_kw), layout = "fr")

ggraph(layout) +
  geom_edge_link(aes(width = weight), alpha = 0.15, colour = "grey60") +
  scale_edge_width_continuous(range = c(0.3, 2), guide = "none") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  geom_node_text(aes(label = ifelse(degree > quantile(degree, 0.85),
                                    name, NA_character_)),
                 repel = TRUE, size = 2.5, max.overlaps = 20, na.rm = TRUE) +
  scale_size_continuous(range = c(1, 6), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(g_kw)$community)
  )) +
  labs(colour = "Cluster") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

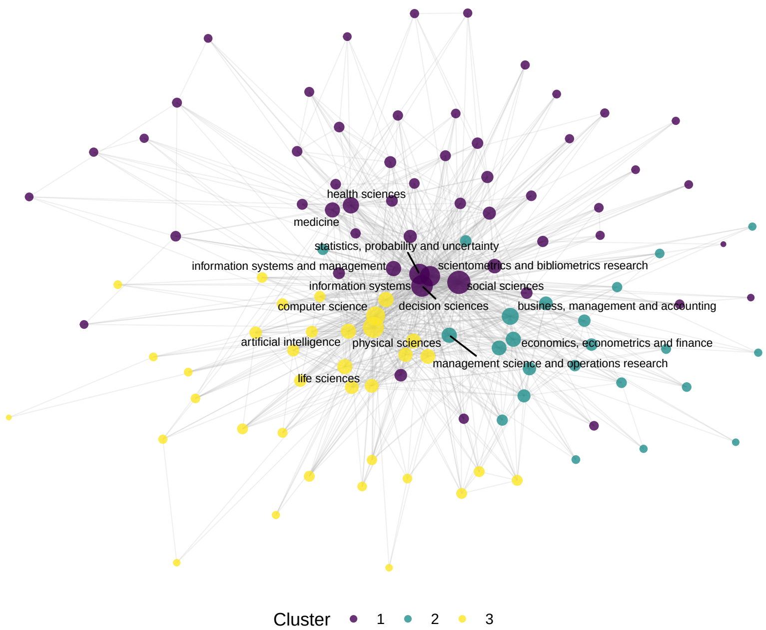

Figure 18.2: Keyword co-occurrence network coloured by topical cluster.

18.5 Diagnostics and interpretation

- **Keyword cleaning**: Inconsistent terminology inflates the vocabulary and creates spurious nodes. Standardise spelling, merge synonyms (e.g., “h-index” and “hirsch index”), and remove generic terms (“research”, “analysis”) that co-occur with everything but convey no topical information.
- **Frequency threshold**: Including rare keywords produces large, sparse, unreadable networks. Start with keywords appearing in at least 5 papers and adjust based on corpus size.
- **Community interpretability**: Each community should correspond to a recognisable research theme. If communities are uninterpretable, the resolution parameter may need adjustment or keywords need further cleaning.
- **Centrality interpretation**: High-degree keywords are thematically central (used across many contexts). High-betweenness keywords bridge distinct topics and may represent interdisciplinary concepts.

18.6 Limitations and responsible use

18.7 Limitations and responsible use

- **Vocabulary inconsistency.** Author keywords are not controlled vocabulary. The same concept may appear under multiple terms, fragmenting what should be a single node. Merging synonyms requires domain expertise.
- **Algorithmic concepts.** OpenAlex concepts are assigned by machine learning and may contain errors — especially at fine-grained levels. Always validate topic assignments by reading sample papers (Priem et al., 2022).
- **Static snapshots.** A co-word network represents a time-averaged view. Emerging topics with few publications may be invisible. Consider building temporal slices to track field evolution.
- **Not evaluative.** Popular keywords indicate research activity, not research quality or societal impact. Do not equate topical centrality with importance (Hicks et al., 2015).

18.8 Common pitfalls

18.9 Common pitfalls

- **Not cleaning keywords.** Punctuation variants (“co-authorship” vs. “coauthorship”), case differences, and trailing whitespace create duplicate nodes. Clean before building the network.
- **Including stop-keywords.** Generic terms like “research”, “study”, “analysis” co-occur with everything and dominate the network without conveying topical information. Remove them.
- **Comparing co-occurrence counts across corpora of different sizes.** Larger corpora produce higher co-occurrence counts mechanically. Normalise or use relative measures.
- **Over-reading small clusters.** A cluster with two or three keywords may reflect a single paper, not a research theme. Report cluster size alongside content.

18.10 Exercises

1. **Author keywords vs. concepts.** Build co-occurrence networks from both author keywords (if available) and OpenAlex concepts for the same corpus. Compare the resulting topical clusters.

2. **Temporal evolution.** Split the corpus into yearly slices. For each year, build a co-word network and identify the top 5 keywords by degree. Which keywords are consistently central, and which emerge over time?

3. **Strategic diagram.** Compute density (internal cohesion) and centrality (external connections) for each keyword cluster. Plot them on a strategic diagram (density vs. centrality quadrants). Which clusters are core themes and which are peripheral?

4. **Normalisation comparison.** Build two networks from the same data: one with raw co-occurrence counts and one with association strength normalisation. How do the most central keywords differ?

18.11 Solutions

Solutions are provided in F.

18.12 Further reading

- Callon et al. (1983) — The foundational paper on co-word analysis for mapping science.
- Waltman et al. (2010) — Unified framework for bibliometric network construction, including keyword networks.
- Aria and Cuccurullo (2017) — `bibliometrix` includes co-word analysis via the `biblioNetwork()` function.
- Priem et al. (2022) — OpenAlex concepts and topic classification.

18.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
```

```
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4       pd
#>  [7] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt
#> [13] glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      st
#> [19] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       gg
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite
#>   [6] farver_2.1.2          rmarkdown_2.31        fs_2.1.0              vctrs_0.
#>  [11] askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9       contenta
#>  [16] janeaustenr_1.0.0     cellranger_1.1.0      htmlwidgets_1.6.4     tokenize
#>  [21] httr2_1.2.2           cachem_1.1.0          plotly_4.12.0         dimensio
#>  [26] lifecycle_1.0.5       pkgconfig_2.0.3       Matrix_1.7-0          R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         shinycssloaders_1.1.0 rprojroo
#>  [36] labeling_0.4.3        urltools_1.7.3.1      timechange_0.4.0      polyclip
#>  [41] compiler_4.4.1        here_1.0.2            bit64_4.8.0           withr_3.
#>  [46] backports_1.5.1       viridis_0.6.5         ggforce_0.5.0         MASS_7.3
#>  [51] bibliometrixData_0.3.0 tools_4.4.1          otel_0.2.0            stopword
#>  [56] httpuv_1.6.17         rentrez_1.2.4         promises_1.5.0        grid_4.4
#>  [61] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_
#>  [66] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2
#>  [71] pillar_1.11.1         later_1.4.8           tweenr_2.0.3          lattice_
#>  [76] tidyselect_1.2.1      miniUI_0.1.2          knitr_1.51            gridExtr
#>  [81] crul_1.6.0            xfun_0.57             graphlayouts_1.2.3    DT_0.34.
#>  [86] visNetwork_2.1.4      stringi_1.8.7         lazyeval_0.2.3        qpdf_1.4
#>  [91] evaluate_1.0.5        codetools_0.2-20      httpcode_0.3.0        cli_3.6.
#>  [96] dichromat_2.0-0.1     Rcpp_1.1.1-1.1        readxl_1.4.5          triebear
#> [101] parallel_4.4.1        assertthat_0.2.1      pubmedR_1.0.2         viridisL
#> [106] openxlsx_4.2.8.1      rlang_1.2.0           fastmatch_1.1-8
```

Chapter 19

Community Detection and Backbone Extraction

19.1 Learning objectives

After completing this chapter, you will be able to:

- Explain why dense networks require backbone extraction before community detection
- Apply the disparity filter to extract statistically significant edges
- Tune the resolution parameter for Louvain and Leiden community detection
- Compare community assignments across methods using Normalised Mutual Information
- Interpret hierarchical community structure at multiple resolutions

19.2 Setup

```r
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)
library(gt)
```

```r
set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

19.3 Conceptual background

Bibliometric networks are often dense: in a co-citation or bibliographic coupling network, every pair of documents with any shared citation creates an edge. A corpus of 1,000 papers can easily produce a network with hundreds of thousands of edges, most of which represent weak or coincidental relationships. Applying community detection directly to such networks yields poor results — the algorithms cannot distinguish signal from noise.

Backbone extraction addresses this by pruning edges that are not statistically significant, retaining only the "skeleton" of the network. The **disparity filter** (Serrano et al., 2009) tests each edge against a null model based on the local weight distribution of its endpoints. An edge is retained if its weight is unexpectedly high given the node's total weight and degree. This preserves multi-scale structure: both high-weight and low-weight nodes can retain their most important connections.

An alternative is simple **threshold filtering** (remove edges below a fixed weight), but this systematically removes connections between low-activity nodes, biasing the backbone toward already-prominent actors. The disparity filter avoids this bias by using a local significance test.

Once the backbone is extracted, **community detection** identifies groups of nodes that are more densely connected internally than externally. The Leiden algorithm (Traag et al., 2019) improves on Louvain (Blondel et al., 2008) by guaranteeing that all detected communities are internally connected. Both algorithms accept a **resolution parameter** that controls community granularity: lower values produce fewer, larger communities; higher values produce more, smaller communities. There is no single "correct" resolution — the appropriate value depends on the research question.

Fortunato (2010) provides a comprehensive review of community detection methods. In practice, running the algorithm at multiple resolutions and examining the hierarchical structure provides the richest understanding of network organisation.

19.4 Worked example

19.4.1 Building a dense network

We construct a bibliographic coupling network from scientometrics articles.

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 300, seed = 42)
)

refs <- works |>
  select(citing_id = id, referenced_works) |>
  unnest(referenced_works) |>
  rename(cited_id = referenced_works)

bibcoup <- refs |>
  inner_join(refs, by = "cited_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(citing_id_a < citing_id_b) |>
  count(citing_id_a, citing_id_b, name = "shared_refs")

g_full <- graph_from_data_frame(
  bibcoup |> select(citing_id_a, citing_id_b, weight = shared_refs),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Full network: {vcount(g_full)} nodes, {ecount(g_full)} edges\n"))
```

```
#> Full network: 292 nodes, 3765 edges
```

```
cat(glue("Density: {round(graph.density(g_full), 4)}\n"))
```

```
#> Density: 0.0886
```

19.4.2 Backbone extraction: disparity filter

```r
disparity_filter <- function(g, alpha = 0.05) {
  el <- as_data_frame(g, what = "edges")
  node_strength <- strength(g)

  keep <- map2_lgl(seq_len(nrow(el)), el$weight, function(i, w) {
    from_name <- el$from[i]
    to_name <- el$to[i]
    k_from <- degree(g, from_name)
    k_to <- degree(g, to_name)
    s_from <- node_strength[from_name]
    s_to <- node_strength[to_name]

    p_from <- (1 - w / s_from)^(k_from - 1)
    p_to <- (1 - w / s_to)^(k_to - 1)

    min(p_from, p_to) < alpha
  })

  subgraph.edges(g, which(keep), delete.vertices = TRUE)
}

g_backbone <- disparity_filter(g_full, alpha = 0.05)
cat(glue("Backbone: {vcount(g_backbone)} nodes, {ecount(g_backbone)} edges\n"))
```

```
#> Backbone: 136 nodes, 131 edges
```

```r
cat(glue("Edge retention: {scales::percent(ecount(g_backbone) / ecount(g_full))}\n"))
```

```
#> Edge retention: 3%
```

19.4.3 Threshold filtering for comparison

```r
threshold <- quantile(E(g_full)$weight, 0.75)
g_threshold <- subgraph.edges(
  g_full,
  which(E(g_full)$weight >= threshold),
  delete.vertices = TRUE
)

cat(glue("Threshold (75th pctile = {threshold}): {vcount(g_threshold)} nodes, {ecount(g
```

```
#> Threshold (75th pctile = 2): 262 nodes, 1065 edges
```

resolution	n_communities	modularity	max_size	min_size
0.5	33	0.869	31	2
0.8	34	0.874	24	2
1.0	35	0.875	16	2
1.5	36	0.872	15	2
2.0	37	0.867	12	2

19.4.4 Community detection at multiple resolutions

```
resolutions <- c(0.5, 0.8, 1.0, 1.5, 2.0)

sweep_results <- map_dfr(resolutions, function(res) {
  comm <- cluster_leiden(g_backbone, resolution_parameter = res,
                         objective_function = "modularity")
  tibble(
    resolution = res,
    n_communities = length(unique(membership(comm))),
    modularity = round(modularity(g_backbone, membership(comm)), 3),
    max_size = max(table(membership(comm))),
    min_size = min(table(membership(comm)))
  )
})

sweep_results |> gt()
```

```
ggplot(sweep_results, aes(x = resolution)) +
  geom_line(aes(y = n_communities), colour = palette_sci(2)[1], linewidth = 1) +
  geom_point(aes(y = n_communities), colour = palette_sci(2)[1], size = 3) +
  geom_line(aes(y = modularity * max(n_communities)),
            colour = palette_sci(2)[2], linewidth = 1, linetype = "dashed") +
  scale_y_continuous(
    name = "Number of communities",
    sec.axis = sec_axis(~ . / max(sweep_results$n_communities),
                        name = "Modularity")
  ) +
  labs(x = "Resolution parameter") +
  theme_sci()
```

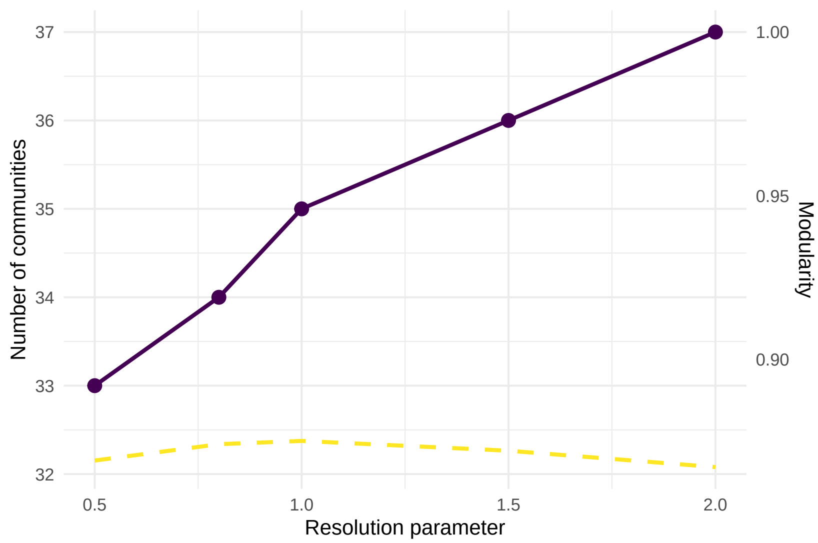

Figure 19.1: Number of communities and modularity as a function of resolution parameter.

19.4.5 Comparing backbone vs. full network communities

```
comm_full <- cluster_leiden(g_full, resolution_parameter = 1.0,
                            objective_function = "modularity")
comm_backbone <- cluster_leiden(g_backbone, resolution_parameter = 1.0,
                                objective_function = "modularity")

shared_nodes <- intersect(V(g_full)$name, V(g_backbone)$name)
mem_full <- membership(comm_full)[shared_nodes]
mem_back <- membership(comm_backbone)[shared_nodes]

nmi <- compare(mem_full, mem_back, method = "nmi")
cat(glue("NMI (full vs backbone, shared nodes): {round(nmi, 3)}\n"))
```

```
#> NMI (full vs backbone, shared nodes): 0.642
```

```
cat(glue("Communities in full network: {length(unique(membership(comm_full)))}\n"))
```

```
#> Communities in full network: 9
```

```r
cat(glue("Communities in backbone: {length(unique(membership(comm_backbone)))}\n"))
```

```
#> Communities in backbone: 35
```

19.4.6 Visualisation

```r
V(g_backbone)$community <- as.factor(membership(comm_backbone))
V(g_backbone)$degree <- degree(g_backbone)

set.seed(42)
layout <- create_layout(as_tbl_graph(g_backbone), layout = "fr")

ggraph(layout) +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  scale_size_continuous(range = c(1, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(g_backbone)$community)
  )) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

19.5 Diagnostics and interpretation

- **Edge retention rate**: The disparity filter typically retains 10–30% of edges. Retention above 50% suggests the network may not be dense enough to require backbone extraction.
- **Isolated nodes**: Backbone extraction removes nodes that lose all their edges. Report how many nodes are lost and whether they represent a biased subset (e.g., low-cited papers).
- **Resolution plateau**: If modularity remains stable across a range of resolutions, the community structure is robust. Rapid changes suggest sensitivity to the parameter.
- **Singleton communities**: Communities with one or two nodes are usually noise. Consider merging them into the nearest larger community or excluding them from interpretation.

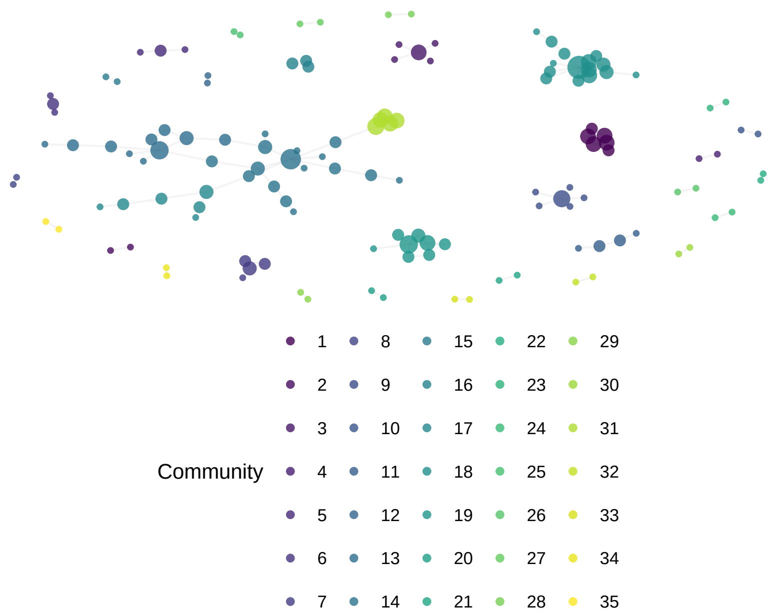

Figure 19.2: Backbone network with Leiden communities (resolution = 1.0).

19.6 Limitations and responsible use

19.7 Limitations and responsible use

- **The disparity filter has assumptions.** It assumes a uniform null distribution of edge weights across a node's connections. Highly skewed weight distributions may violate this assumption.
- **Backbone choice affects conclusions.** Different backbone methods (disparity filter, noise-corrected, threshold) retain different edges and can produce different community structures. Report the method and alpha level.
- **Resolution is a researcher decision.** There is no objectively "correct" resolution parameter. The choice determines the granularity of analysis and should be justified by the research question, not by optimising modularity alone (Fortunato, 2010).
- **Communities are not ground truth.** Detected communities are algorithmic constructs. They reflect structural patterns in citation data, not necessarily real-world research groups or coherent intellectual traditions (Hicks et al., 2015).

19.8 Common pitfalls

19.9 Common pitfalls

- **Applying community detection to unfiltered dense networks.** The result is usually a single giant community plus isolated fragments. Always extract the backbone first.
- **Using a single resolution without justification.** The default resolution of 1.0 is arbitrary. Run a sweep and show the sensitivity analysis.
- **Comparing modularity across networks of different sizes.** Modularity values are not comparable between different networks. Use NMI to compare partitions.
- **Interpreting backbone edges as "strong" relationships.** The disparity filter retains *locally significant* edges, not necessarily those with the highest absolute weight. A low-weight edge can be retained if it is important relative to its endpoint's other connections.

19.10 Exercises

1. **Alpha sensitivity.** Run the disparity filter with alpha values of 0.01, 0.05, 0.10, and 0.20. Plot the number of retained edges and communities against alpha. At what alpha does the network become disconnected?

2. **Threshold vs. disparity.** Compare threshold-filtered and disparity-filtered communities for the same network. Use NMI to quantify agreement. Which method produces more interpretable communities?

3. **Hierarchical structure.** Run Leiden at resolutions 0.5, 1.0, and 2.0. For each pair of resolutions, check whether the finer communities are nested within the coarser ones (a Sankey or alluvial diagram can help visualise this).

4. **Weighted vs. unweighted community detection.** Remove edge weights from the backbone and run Leiden again. How much does the community structure change?

19.11 Solutions

Solutions are provided in F.

19.12 Further reading

- Serrano et al. (2009) — The disparity filter for extracting multiscale network backbones.
- Traag et al. (2019) — The Leiden algorithm with resolution parameter tuning.
- Blondel et al. (2008) — The Louvain algorithm for fast community detection.
- Fortunato (2010) — Comprehensive review of community detection, including resolution limits.
- Waltman et al. (2010) — Network construction and clustering for bibliometric applications.

19.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4       pd
#>  [7] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt
#> [13] glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      st
#> [19] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       gg
#>
```

```
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_2.0.0
#>   [6] farver_2.1.2           rmarkdown_2.31         fs_2.1.0               vctrs_0.7.3
#>  [11] askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9        contentanalysis_1.0
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4      tokenizers_0.3.0
#>  [21] httr2_1.2.2            cachem_1.1.0           plotly_4.12.0          dimensionsR_0.0.3
#>  [26] lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0           R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          shinycssloaders_1.1.0  rprojroot_2.1.1
#>  [36] labeling_0.4.3         urltools_1.7.3.1       timechange_0.4.0       polyclip_1.10-7
#>  [41] compiler_4.4.1         here_1.0.2             bit64_4.8.0            withr_3.0.2
#>  [46] backports_1.5.1        viridis_0.6.5          ggforce_0.5.0          MASS_7.3-60.2
#>  [51] bibliometrixData_0.3.0 tools_4.4.1            otel_0.2.0             stopwords_2.3
#>  [56] httpuv_1.6.17          rentrez_1.2.4          promises_1.5.0         grid_4.4.1
#>  [61] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0             rscopus_0.9.0
#>  [66] data.table_1.18.4      hms_1.1.4              xml2_1.5.2             utf8_1.2.6
#>  [71] pillar_1.11.1          later_1.4.8            tweenr_2.0.3           lattice_0.22-6
#>  [76] tidyselect_1.2.1       miniUI_0.1.2           knitr_1.51             gridExtra_2.3
#>  [81] crul_1.6.0             xfun_0.57              graphlayouts_1.2.3     DT_0.34.0
#>  [86] visNetwork_2.1.4       stringi_1.8.7          lazyeval_0.2.3         qpdf_1.4.1
#>  [91] evaluate_1.0.5         codetools_0.2-20       httpcode_0.3.0         cli_3.6.6
#>  [96] dichromat_2.0-0.1      Rcpp_1.1.1-1.1         readxl_1.4.5           triebeard_0.4.1
#> [101] parallel_4.4.1         assertthat_0.2.1       pubmedR_1.0.2          viridisLite_0.4.3
#> [106] openxlsx_4.2.8.1       rlang_1.2.0            fastmatch_1.1-8
```

Chapter 20

Science Mapping and Overlay Maps

20.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the principles of science mapping and its applications
- Construct a journal-level co-citation network from OpenAlex data
- Create an overlay map that positions a research profile on a base map
- Interpret overlay maps to identify research strengths and gaps
- Discuss the limitations of global science maps for local evaluation

20.2 Setup

```r
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```r
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

20.3 Conceptual background

A **science map** is a spatial representation of the structure of science, where proximity indicates relatedness. Just as a geographic map shows the layout of countries and continents, a science map shows the layout of disciplines, fields, and research areas. These maps are constructed from bibliometric data — typically citation links between journals or document sets — and visualised using network layouts or dimensionality reduction techniques.

The most common base maps use **journal-level citation networks**. Two journals are linked if papers in one frequently cite papers in the other. Community detection applied to these networks produces clusters corresponding to broad disciplines (biomedical sciences, physical sciences, social sciences, etc.). Waltman et al. (2010) developed methods for constructing such maps from large-scale citation data, and the resulting visualisations are widely used in science policy and institutional assessment.

An **overlay map** (Rafols et al., 2010) takes a global science map as a fixed background and plots a specific entity's output on top. For example, a university's publications are mapped to the journals in which they appeared; the overlay then shows which areas of the global map the university is active in, and how intensely. This technique is powerful for visualising research profiles, identifying strategic gaps, and comparing institutions or countries without reducing their activity to a single number.

The construction pipeline involves three steps: (1) build or obtain a base map of journals with coordinates and cluster assignments; (2) map the entity's publications to journals in the base map; (3) scale node sizes proportionally and overlay on the base map.

Science maps provide intuitive, holistic overviews of research landscapes. However, they are projections of high-dimensional data into two dimensions, and like any projection, they introduce distortion. Interdisciplinary journals may appear in unexpected locations, and the layout algorithm's random seed affects exact positions (Hicks et al., 2015).

20.4 Worked example

20.4.1 Building a journal co-citation base map

We construct a base map from citation flows between journals in a multidisciplinary sample.

```
works_multi <- oa_fetch(
  entity = "works",
  from_publication_date = "2022-01-01",
  to_publication_date = "2022-12-31",
  type = "article",
  options = list(sample = 500, seed = 42)
)

journal_data <- works_multi |>
  select(id, source_display_name, referenced_works) |>
  filter(!is.na(source_display_name)) |>
  unnest(referenced_works) |>
  rename(cited_id = referenced_works)

cat(glue("Citing papers with journal info: {n_distinct(journal_data$id)}\n"))
```

```
#> Citing papers with journal info: 475
```

```
cat(glue("Unique source journals: {n_distinct(journal_data$source_display_name)}\n"))
```

```
#> Unique source journals: 405
```

```
journal_cited <- journal_data |>
  select(citing_journal = source_display_name, cited_id)

journal_pairs <- journal_cited |>
  inner_join(journal_cited, by = "cited_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(citing_journal_a < citing_journal_b) |>
  count(citing_journal_a, citing_journal_b, name = "cocitation")

journal_pairs_top <- journal_pairs |>
  filter(cocitation >= 3)

g_journals <- graph_from_data_frame(
  journal_pairs_top |> select(citing_journal_a, citing_journal_b,
```

cluster	n_journals	top_journals
1	132	DOAJ (DOAJ: Directory of Open Access Journals); HAL (Le Cen

```
                                      weight = cocitation),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Journal base map: {vcount(g_journals)} journals, {ecount(g_journals)} edges\r
```

```
#> Journal base map: 132 journals, 1023 edges
```

20.4.2 Clustering journals into disciplines

```
comm <- cluster_leiden(g_journals, resolution_parameter = 0.8,
                       objective_function = "modularity")
V(g_journals)$cluster <- as.factor(membership(comm))
V(g_journals)$degree <- degree(g_journals)

cluster_table <- tibble(
  journal = V(g_journals)$name,
  cluster = V(g_journals)$cluster,
  degree = V(g_journals)$degree
) |>
  group_by(cluster) |>
  summarise(n_journals = n(),
            top_journals = paste(head(journal[order(-degree)], 3),
                                 collapse = "; "),
            .groups = "drop") |>
  arrange(desc(n_journals))

cluster_table |> head(8) |> gt()
```

20.4.3 Visualising the base map

```
set.seed(42)
layout <- create_layout(as_tbl_graph(g_journals), layout = "fr")
```

```
ggraph(layout) +
  geom_edge_link(alpha = 0.05, colour = "grey70") +
  geom_node_point(aes(colour = cluster, size = degree), alpha = 0.7) +
  scale_size_continuous(range = c(1, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(g_journals)$cluster)
  )) +
  labs(colour = "Cluster") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

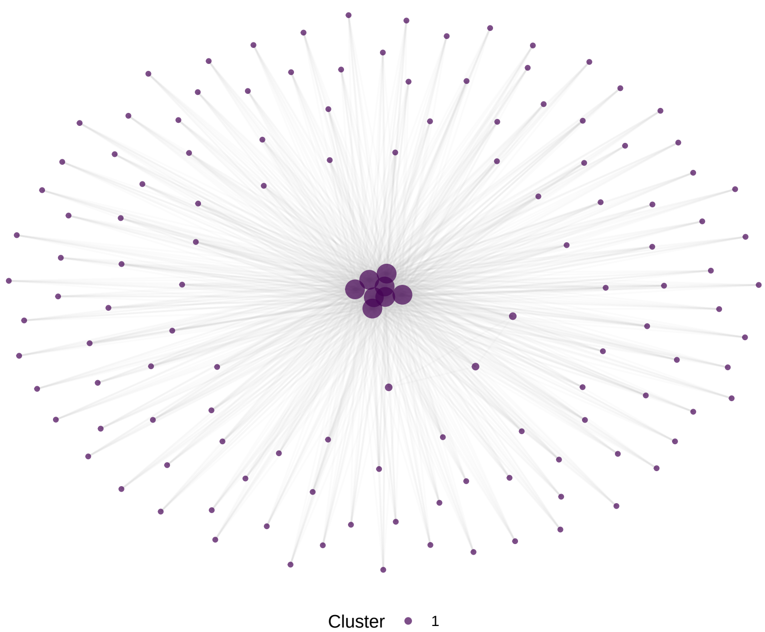

Figure 20.1: Journal co-citation base map coloured by discipline cluster.

20.4.4 Creating an overlay map

We overlay the publication profile of an institution.

```
inst_works <- oa_fetch(
  entity = "works",
  authorships.institutions.id = "I40120149",
  from_publication_date = "2022-01-01",
  to_publication_date = "2022-12-31",
```

```r
  type = "article",
  options = list(sample = 300, seed = 42)
)

inst_journals <- inst_works |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, name = "n_pubs", sort = TRUE)

basemap_journals <- V(g_journals)$name
inst_journals_matched <- inst_journals |>
  filter(source_display_name %in% basemap_journals)

cat(glue("Institution journals matched to base map: {nrow(inst_journals_matched)} / {n
```

```
#> Institution journals matched to base map: 6 / 261
```

```r
overlay_sizes <- tibble(journal = V(g_journals)$name) |>
  left_join(inst_journals_matched, by = c("journal" = "source_display_name")) |>
  mutate(n_pubs = replace_na(n_pubs, 0))

set.seed(42)
layout_overlay <- create_layout(as_tbl_graph(g_journals), layout = "fr")

ggraph(layout_overlay) +
  geom_edge_link(alpha = 0.03, colour = "grey80") +
  geom_node_point(colour = "grey85", size = 1.5, alpha = 0.5) +
  geom_node_point(
    data = layout_overlay |> left_join(overlay_sizes, by = c("name" = "journal")) |>
      filter(n_pubs > 0),
    aes(size = n_pubs, colour = cluster), alpha = 0.8
  ) +
  scale_size_continuous(range = c(2, 8), name = "Publications") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(g_journals)$cluster)
  )) +
  labs(colour = "Cluster") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

20.5 Diagnostics and interpretation

- **Journal matching rate**: The fraction of the entity's publications that match journals in the base map determines overlay completeness. Low

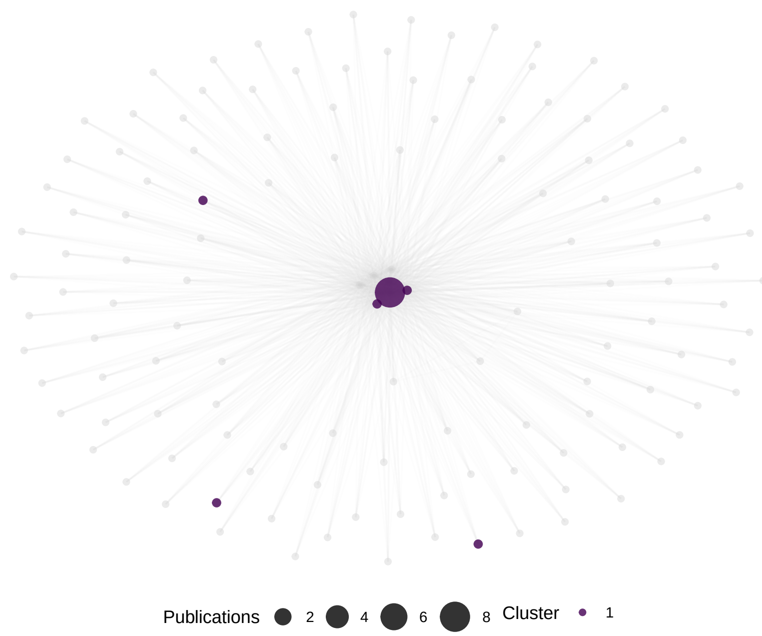

Figure 20.2: Overlay map showing Oxford's 2022 publication profile on the journal base map.

matching rates ($< 50\%$) indicate the base map is too narrow for the entity's profile.

- **Layout stability**: Force-directed layouts are sensitive to initial conditions. Always use `set.seed()` and report the seed. Consider consensus layouts from multiple seeds for important maps.
- **Cluster labelling**: Algorithmic clusters need human interpretation. Label clusters by inspecting the journals they contain, not by relying on the algorithm.
- **Overlay concentration**: If the overlay is concentrated in one or two clusters, the institution is highly specialised. Dispersed overlays indicate breadth. Neither is inherently better.

20.6 Limitations and responsible use

20.7 Limitations and responsible use

- **Projection distortion.** Two-dimensional layouts compress high-dimensional citation relationships. Journals at the boundaries between disciplines may be misplaced. Do not over-interpret exact positions.
- **Journal-level aggregation.** Mapping publications to journals loses within-journal variation. A paper in *Nature* could be biology, physics, or chemistry — the overlay cannot distinguish.
- **Sample-dependent base maps.** Our base map is built from a small sample. Production science maps (e.g., the CWTS journal map) use comprehensive citation data. Treat this example as illustrative.
- **Not evaluative.** Science maps show *where* research happens, not how good it is. An empty region on a university's overlay map is not a weakness — it may reflect a deliberate strategic choice (Hicks et al., 2015).
- **Interdisciplinary research is underrepresented.** Journals at disciplinary boundaries may cluster with one discipline, making interdisciplinary work partially invisible (Rafols et al., 2010).

20.8 Common pitfalls

20.9 Common pitfalls

- **Using too small a base map.** A base map with fewer than 50 journals lacks resolution. Aim for at least 100–200 journals for meaningful discipline separation.

- **Comparing overlays from different base maps.** Node positions are not transferable between different base maps. To compare institutions, overlay them on the *same* base map.
- **Ignoring unmatched publications.** Publications in journals not on the base map simply disappear from the overlay. Report the matching rate.
- **Treating clusters as fixed disciplines.** Cluster assignments depend on the algorithm, resolution parameter, and data sample. They are analytical tools, not natural categories.

20.10 Exercises

1. **Multi-institution overlay.** Overlay the profiles of two institutions on the same base map using different colours. Where do their research profiles overlap, and where do they diverge?

2. **Temporal overlay.** Build overlays for the same institution in two different years. Has the research profile shifted toward new areas or remained stable?

3. **Country-level overlay.** Instead of an institution, overlay a country's publication profile. How does a small specialised country (e.g., Singapore) compare to a large diversified one (e.g., the United States)?

20.11 Solutions

Solutions are provided in F.

20.12 Further reading

- Rafols et al. (2010) — Overlay maps for visualising research profiles on global science maps.
- Waltman et al. (2010) — Unified approach to constructing and clustering bibliometric maps.
- Hicks et al. (2015) — The Leiden Manifesto: responsible interpretation of quantitative research profiles.
- Fortunato (2010) — Community detection methods relevant to journal clustering.

20.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4       pd
#>  [7] bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt
#> [13] glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1      st
#> [19] purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       gg
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite
#>   [6] farver_2.1.2           rmarkdown_2.31         fs_2.1.0               vctrs_0.
#>  [11] askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9        contenta
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4      tokenize
#>  [21] httr2_1.2.2            cachem_1.1.0           plotly_4.12.0          dimensio
#>  [26] lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0           R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          shinycssloaders_1.1.0  rprojroo
#>  [36] labeling_0.4.3         urltools_1.7.3.1       timechange_0.4.0       polyclip
#>  [41] compiler_4.4.1         here_1.0.2             bit64_4.8.0            withr_3.
#>  [46] backports_1.5.1        viridis_0.6.5          ggforce_0.5.0          MASS_7.3
#>  [51] bibliometrixData_0.3.0 tools_4.4.1            otel_0.2.0             stopword
#>  [56] httpuv_1.6.17          rentrez_1.2.4          promises_1.5.0         grid_4.4
#>  [61] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0             rscopus_
#>  [66] data.table_1.18.4      hms_1.1.4              xml2_1.5.2             utf8_1.2
```

```
#>  [71] pillar_1.11.1       later_1.4.8         tweenr_2.0.3         lattice_0.22-6
#>  [76] tidyselect_1.2.1    miniUI_0.1.2        knitr_1.51           gridExtra_2.3
#>  [81] crul_1.6.0          xfun_0.57           graphlayouts_1.2.3   DT_0.34.0
#>  [86] visNetwork_2.1.4    stringi_1.8.7       lazyeval_0.2.3       qpdf_1.4.1
#>  [91] evaluate_1.0.5      codetools_0.2-20    httpcode_0.3.0       cli_3.6.6
#>  [96] dichromat_2.0-0.1   Rcpp_1.1.1-1.1      readxl_1.4.5         triebeard_0.4.1
#> [101] parallel_4.4.1      assertthat_0.2.1    pubmedR_1.0.2        viridisLite_0.4.3
#> [106] openxlsx_4.2.8.1    rlang_1.2.0         fastmatch_1.1-8
```

Chapter 21

Network Visualization and Interoperability

21.1 Learning objectives

After completing this chapter, you will be able to:

- Choose appropriate layout algorithms for different network types
- Export igraph networks to Gephi (GraphML), VOSviewer, and Pajek formats
- Create interactive network visualisations with `visNetwork`
- Apply colour and size encoding that communicates structure clearly
- Produce publication-quality static network figures with `ggraph`

21.2 Setup

```
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(visNetwork)
library(glue)

set.seed(20260509)
```

```
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

21.3 Conceptual background

Network visualisation is both an analytical tool and a communication device. A well-designed network figure can reveal structure — clusters, bridges, outliers — at a glance. A poorly designed one obscures these patterns in visual noise.

Layout algorithms determine node positions. Force-directed layouts (Fruchterman-Reingold, Kamada-Kawai) simulate physical forces: connected nodes attract, all nodes repel. They work well for small-to-medium networks (up to ~1,000 nodes) and produce aesthetically pleasing results. For larger networks, the DrL (Distributed Recursive Layout) or OpenOrd algorithms scale better. For trees or hierarchies, Reingold-Tilford or Sugiyama layouts are appropriate. The choice of algorithm depends on the network's structure and the visual question being asked.

Visual encoding maps network properties to visual channels: node size (degree, centrality), node colour (community, attribute), edge width (weight), and edge colour or transparency. Effective encoding follows the principle of proportional ink: the visual weight of an element should correspond to its data importance. Avoid redundant encoding (mapping the same variable to both size and colour) and limit the number of distinct colours to what the reader can distinguish (typically 8–12 categories) (Fortunato, 2010).

Interoperability is essential because no single tool excels at everything. R with igraph and ggraph is excellent for reproducible analysis and publication figures. Gephi (Bastian et al., 2009) provides a rich GUI for exploration and layout refinement. VOSviewer specialises in bibliometric network visualisation with built-in clustering. Pajek handles very large networks efficiently. A practical workflow often involves constructing and analysing the network in R, then exporting to a specialised tool for specific visualisation tasks.

Reproducibility requires that layouts are deterministic. Force-directed algorithms start from random positions, so `set.seed()` before every layout computation. For cross-tool reproducibility, export node coordinates alongside the network data.

21.4 Worked example

21.4.1 Building a sample network

We reuse a co-authorship network from Scientometrics.

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2021-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 200, seed = 42)
)

author_data <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  select(work_id = id, author_id = authorships_id,
         author_name = authorships_display_name) |>
  filter(!is.na(author_id))

edges <- author_data |>
  inner_join(author_data, by = "work_id", suffix = c("_1", "_2"),
             relationship = "many-to-many") |>
  filter(author_id_1 < author_id_2) |>
  count(author_id_1, author_id_2, name = "weight")

g <- graph_from_data_frame(
  edges |> select(author_id_1, author_id_2, weight),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

comp <- components(g)
giant <- induced_subgraph(g, which(comp$membership == which.max(comp$csize)))

comm <- cluster_leiden(giant, resolution_parameter = 1.0,
                       objective_function = "modularity")
V(giant)$community <- membership(comm)
V(giant)$degree <- degree(giant)

author_lookup <- author_data |> distinct(author_id, author_name)
V(giant)$label <- author_lookup$author_name[
  match(V(giant)$name, author_lookup$author_id)
]
```

```
cat(glue("Network: {vcount(giant)} nodes, {ecount(giant)} edges\n"))
```

```
#> Network: 39 nodes, 371 edges
```

21.4.2 Comparing layout algorithms

```
tg <- as_tbl_graph(giant) |>
  mutate(community = as.factor(community))

layouts <- c("fr", "kk", "stress", "drl")
layout_names <- c("Fruchterman-Reingold", "Kamada-Kawai", "Stress", "DrL")

plots <- map2(layouts, layout_names, function(algo, name) {
  set.seed(42)
  ggraph(tg, layout = algo) +
    geom_edge_link(alpha = 0.1, colour = "grey60") +
    geom_node_point(aes(colour = community, size = degree), alpha = 0.7) +
    scale_size_continuous(range = c(0.5, 4), guide = "none") +
    scale_colour_manual(values = palette_sci(
      n_distinct(V(giant)$community)
    ), guide = "none") +
    labs(title = name) +
    theme_void(base_family = "sans", base_size = 11)
})

patchwork::wrap_plots(plots, ncol = 2)
```

21.4.3 Exporting to GraphML (Gephi)

```
out_dir <- here::here("data")
graphml_path <- file.path(out_dir, "coauth_network.graphml")
write_graph(giant, graphml_path, format = "graphml")
cat(glue("Exported GraphML: {graphml_path}\n"))
```

```
#> Exported GraphML: /home/runner/work/scientometrics-in-r/scientometrics-in-r/data/coa
```

```
cat(glue("File size: {round(file.size(graphml_path) / 1024, 1)} KB\n"))
```

```
#> File size: 39.8 KB
```

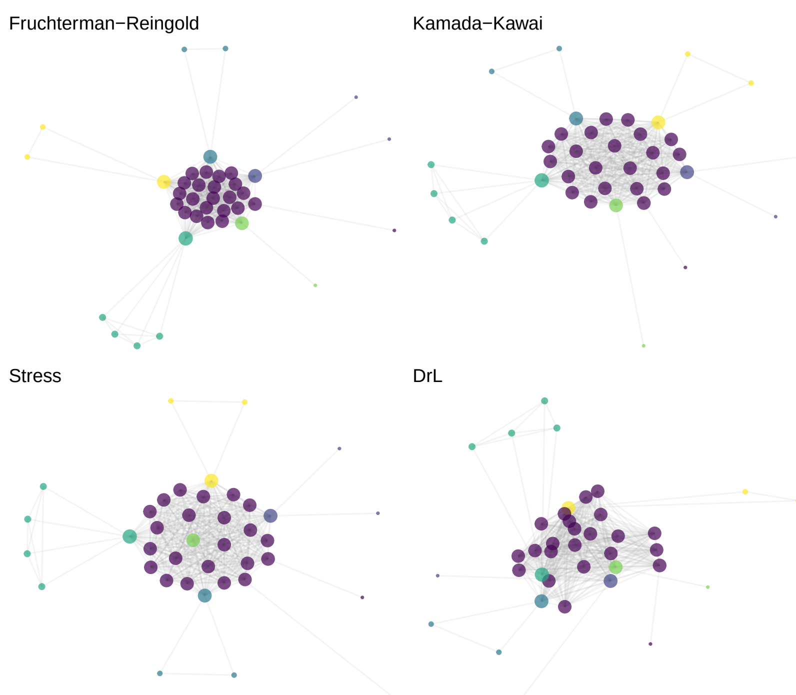

Figure 21.1: The same network rendered with four different layout algorithms.

21.4.4 Exporting to Pajek format

```
pajek_path <- file.path(out_dir, "coauth_network.net")
write_graph(giant, pajek_path, format = "pajek")
cat(glue("Exported Pajek: {pajek_path}\n"))
```

```
#> Exported Pajek: /home/runner/work/scientometrics-in-r/scientometrics-in-r/data/coau
```

21.4.5 Exporting to VOSviewer format

VOSviewer reads tab-separated map and network files.

```
set.seed(42)
coords <- layout_with_fr(giant)

vos_map <- tibble(
  id = seq_len(vcount(giant)),
  label = V(giant)$label,
  x = coords[, 1],
  y = coords[, 2],
  cluster = V(giant)$community,
  weight = V(giant)$degree
)

vos_network <- as_data_frame(giant, what = "edges") |>
  mutate(
    from_idx = match(from, V(giant)$name),
    to_idx = match(to, V(giant)$name)
  ) |>
  select(from_idx, to_idx, weight)

map_path <- file.path(out_dir, "vosviewer_map.txt")
net_path <- file.path(out_dir, "vosviewer_network.txt")

write_tsv(vos_map, map_path)
write_tsv(vos_network, net_path)
cat(glue("VOSviewer map: {map_path}\n"))
```

```
#> VOSviewer map: /home/runner/work/scientometrics-in-r/scientometrics-in-r/data/vosvi
```

```
cat(glue("VOSviewer network: {net_path}\n"))
```

```
#> VOSviewer network: /home/runner/work/scientometrics-in-r/scientometrics-in-r/data/vo
```

21.4.6 Interactive visualisation with visNetwork

```
# HTML widget - only rendered for HTML output (PDF/EPUB skip this chunk).
vis_nodes <- tibble(
  id = V(giant)$name,
  label = ifelse(V(giant)$degree > quantile(V(giant)$degree, 0.9),
                 V(giant)$label, ""),
  title = paste0(V(giant)$label, "<br>Degree: ", V(giant)$degree,
                 "<br>Community: ", V(giant)$community),
  group = as.character(V(giant)$community),
  value = V(giant)$degree
)

vis_edges <- as_data_frame(giant, what = "edges") |>
  select(from, to, weight = weight)

visNetwork(vis_nodes, vis_edges, width = "100%", height = "500px") |>
  visOptions(highlightNearest = TRUE, nodesIdSelection = TRUE) |>
  visPhysics(stabilization = FALSE) |>
  visLayout(randomSeed = 42)
```

21.4.7 Publication-quality figure

```
set.seed(42)

ggraph(tg, layout = "fr") +
  geom_edge_link(alpha = 0.08, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.8) +
  geom_node_text(
    aes(label = ifelse(degree > quantile(degree, 0.95), label, NA)),
    repel = TRUE, size = 2.5, max.overlaps = 15, na.rm = TRUE
  ) +
  scale_size_continuous(range = c(1, 6), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(giant)$community)
  )) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

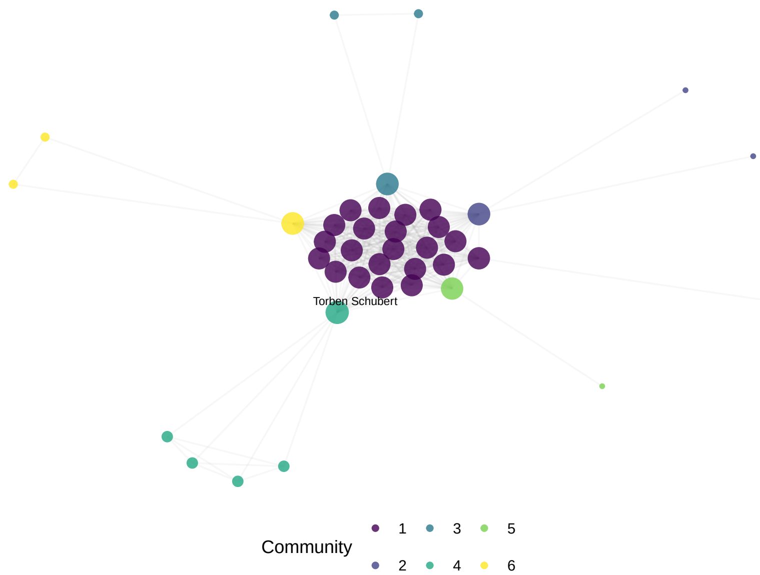

Figure 21.2: Publication-ready co-authorship network with labelled high-degree nodes.

21.5 Diagnostics and interpretation

- **Layout quality**: A good layout separates communities visually, minimises edge crossings, and distributes nodes evenly. If communities overlap heavily, try a different algorithm or increase the number of layout iterations.
- **Readability**: If the network is too dense to read, reduce the number of nodes (filter by degree), increase edge transparency, or remove edges below a weight threshold.
- **Colour distinguishability**: Test figures in greyscale and with colour-blindness simulators. Viridis-based palettes (used by `palette_sci()`) are designed for this.
- **Export verification**: After exporting, open the file in the target tool (Gephi, VOSviewer) to verify that node attributes, edge weights, and labels transferred correctly.

21.6 Limitations and responsible use

21.7 Limitations and responsible use

- **Visualisation is not analysis.** A network figure is a communication tool. The visual impression can mislead if the layout, filtering, or colour encoding obscures structure. Always report the quantitative metrics alongside figures.
- **Layout is not geography.** Absolute positions in a force-directed layout are meaningless; only relative distances carry information. Never compare positions across different layouts or runs without the same seed.
- **Large networks are unreadable.** Beyond ~500 nodes, static network plots become unintelligible. Use interactive visualisation, backbone extraction (19.4), or subnetwork extraction for large data.
- **Export formats lose information.** GraphML preserves most attributes; Pajek format is more limited. Always verify exports and document what was lost (Hicks et al., 2015).

21.8 Common pitfalls

21.9 Common pitfalls

- **Forgetting `set.seed()`.** Without a fixed seed, force-directed layouts change every render, making figures irreproducible across HTML and PDF.
- **Encoding too many variables.** Mapping degree to size, community to colour, *and* betweenness to transparency creates visual overload. Choose two visual channels at most.
- **Using default igraph plots.** Base R `plot.igraph()` produces low-quality figures. Use `ggraph` for publication and `visNetwork` for interactive exploration.
- **Exporting without coordinates.** If you export a network to Gephi without coordinates, Gephi will recompute the layout, producing a different figure than your R output.

21.10 Exercises

1. **Layout comparison.** Apply five different layout algorithms to the same network. Which layout best reveals the community structure? Time each layout and report the trade-off between quality and speed.

2. **Interactive filtering.** Create a `visNetwork` visualisation where users can filter by community membership. Add a dropdown selector for community.

3. **Gephi round-trip.** Export the network to GraphML, import into Gephi, apply a Force Atlas 2 layout, export as PDF. Compare the Gephi figure with the ggraph version.

4. **Colour-blind check.** Render the network figure with 3, 5, and 10 communities. At what point do the viridis colours become hard to distinguish? Test with a colour-blindness simulator.

21.11 Solutions

Solutions are provided in F.

21.12 Further reading

- Bastian et al. (2009) — Gephi: an open-source platform for network visualisation and exploration.
- Fortunato (2010) — Community detection methods with extensive visualisation examples.
- Waltman et al. (2010) — VOSviewer's approach to bibliometric network visualisation.
- Aria and Cuccurullo (2017) — `bibliometrix` includes network visualisation via `biblioshiny()`.

21.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
```

```
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] visNetwork_2.1.4   ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.1       quanteda_4.4
#>  [7] arrow_24.0.0       bibliometrix_5.4.0 RefManageR_1.4.0   bib2df_1.1.2.0     rcrossref_1.2
#> [13] tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5    forcats_1.0.1
#> [19] dplyr_1.2.1        purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1
#> [25] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] farver_2.1.2          rmarkdown_2.31        fs_2.1.0              vctrs_0.7.3
#>  [11] askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9       contentanalysis_1.0
#>  [16] janeaustenr_1.0.0     cellranger_1.1.0      htmlwidgets_1.6.4     tokenizers_0.3.0
#>  [21] httr2_1.2.2           cachem_1.1.0          plotly_4.12.0         dimensionsR_0.0.3
#>  [26] lifecycle_1.0.5       pkgconfig_2.0.3       Matrix_1.7-0          R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycssloaders_1.1
#>  [36] SnowballC_0.7.1       labeling_0.4.3        urltools_1.7.3.1      timechange_0.4.0
#>  [41] httr_1.4.8            compiler_4.4.1        here_1.0.2            bit64_4.8.0
#>  [46] S7_0.2.2              backports_1.5.1       viridis_0.6.5         ggforce_0.5.0
#>  [51] rappdirs_0.3.4        bibliometrixData_0.3.0 tools_4.4.1          otel_0.2.0
#>  [56] zip_2.3.3             httpuv_1.6.17         rentrez_1.2.4         promises_1.5.0
#>  [61] stringdist_0.9.17     generics_0.1.4        gtable_0.3.6          tzdb_0.5.0
#>  [66] ca_0.71.1             data.table_1.18.4     hms_1.1.4             xml2_1.5.2
#>  [71] ggrepel_0.9.8         pillar_1.11.1         vroom_1.7.1           later_1.4.8
#>  [76] lattice_0.22-6        bit_4.6.0             tidyselect_1.2.1      miniUI_0.1.2
#>  [81] gridExtra_2.3         bookdown_0.46         crul_1.6.0            xfun_0.57
#>  [86] DT_0.34.0             humaniformat_0.6.0    stringi_1.8.7         lazyeval_0.2.3
#>  [91] yaml_2.3.12           evaluate_1.0.5        codetools_0.2-20      httpcode_0.3.0
#>  [96] xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1.1-1.1        readxl_1.4.5
#> [101] XML_3.99-0.23         parallel_4.4.1        assertthat_0.2.1      pubmedR_1.0.2
#> [106] scales_1.4.0          crayon_1.5.3          openxlsx_4.2.8.1      rlang_1.2.0
```

Part V

Text & Topic Analysis

Chapter 22

Text Mining Bibliographic Corpora

22.1 Learning objectives

After completing this chapter, you will be able to:

- Build a text corpus from OpenAlex titles and abstracts
- Apply standard preprocessing steps: tokenisation, stopword removal, stemming
- Construct a document-term matrix (DTM) and a document-feature matrix (DFM)
- Compute TF-IDF weights to identify distinctive terms
- Visualise term frequency patterns across time or groups

22.2 Setup

```
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(glue)
library(gt)

set.seed(20260509)
```

```
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

22.3 Conceptual background

Bibliometric metadata — titles, abstracts, and keywords — constitutes a rich but noisy text corpus. Text mining transforms this unstructured data into structured representations suitable for statistical analysis. The pipeline typically follows three stages: preprocessing, representation, and analysis.

Preprocessing converts raw text into a standardised form. Tokenisation splits text into individual words or n-grams. Lowercasing removes case variation. Stopword removal eliminates high-frequency function words ("the", "of", "and") that carry little topical information. Stemming or lemmatisation reduces words to their root forms ("computing", "computed", "computation" → "comput" or "compute"). Each step involves trade-offs: aggressive stemming can merge distinct concepts, while minimal preprocessing retains noise.

The **document-term matrix** (DTM) or **document-feature matrix** (DFM) is the fundamental representation. Rows are documents; columns are terms; cells contain counts or weights. Raw term frequencies overweight common words. **TF-IDF** (term frequency–inverse document frequency) addresses this by up-weighting terms that are frequent in a document but rare across the corpus, highlighting words that distinguish one document from others.

`quanteda` (Aria and Cuccurullo, 2017) provides a fast, well-designed toolkit for text analysis in R. Its `corpus → tokens → dfm` pipeline integrates naturally with the tidyverse. For very large corpora, `quanteda` uses sparse matrix representations that scale to millions of documents.

Text mining complements network-based methods (18.3). Co-word networks reveal term associations; text mining provides the frequency distributions, temporal trends, and discriminative features that characterise a field's vocabulary.

22.4 Worked example

22.4.1 Building a text corpus

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
```

```
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents with abstracts: {nrow(text_df)}\n"))
```

```
#> Documents with abstracts: 119
```

```
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

cat(glue("Corpus size: {ndoc(corp)} documents\n"))
```

```
#> Corpus size: 119 documents
```

```
cat(glue("Total tokens: {sum(ntoken(corp))}\n"))
```

```
#> Total tokens: 29606
```

22.4.2 Tokenisation and preprocessing

```
toks <- tokens(corp,
               remove_punct = TRUE,
               remove_numbers = TRUE,
               remove_symbols = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("also", "however", "using", "based", "study",
                  "paper", "results", "research", "analysis")) |>
  tokens_wordstem()

cat(glue("Tokens after preprocessing: {sum(ntoken(toks))}\n"))
```

```
#> Tokens after preprocessing: 14958
```

22.4.3 Document-feature matrix

```
dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 5, min_docfreq = 3)

cat(glue("DFM dimensions: {nrow(dfmat)} docs x {ncol(dfmat)} features\n"))
```

```
#> DFM dimensions: 119 docs x 659 features
```

```
cat(glue("Sparsity: {scales::percent(sparsity(dfmat))}\n"))
```

```
#> Sparsity: 91%
```

```
top_terms <- topfeatures(dfmat, 20) |>
  enframe(name = "term", value = "frequency") |>
  mutate(term = fct_reorder(term, frequency))

ggplot(top_terms, aes(x = frequency, y = term)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Frequency", y = NULL) +
  theme_sci()
```

22.4.4 TF-IDF weighting

```
dfmat_tfidf <- dfm_tfidf(dfmat)

tfidf_by_year <- text_df |>
  group_by(year) |>
  group_keys() |>
  pull(year) |>
  map_dfr(function(yr) {
    docs <- docvars(dfmat_tfidf, "year") == yr
    if (sum(docs) < 5) return(tibble())
    top <- topfeatures(dfmat_tfidf[docs, ], 10)
    tibble(year = yr, term = names(top), tfidf = unname(top))
  })

tfidf_by_year |>
  group_by(year) |>
  slice_max(tfidf, n = 5) |>
  gt() |>
  fmt_number(columns = tfidf, decimals = 1)
```

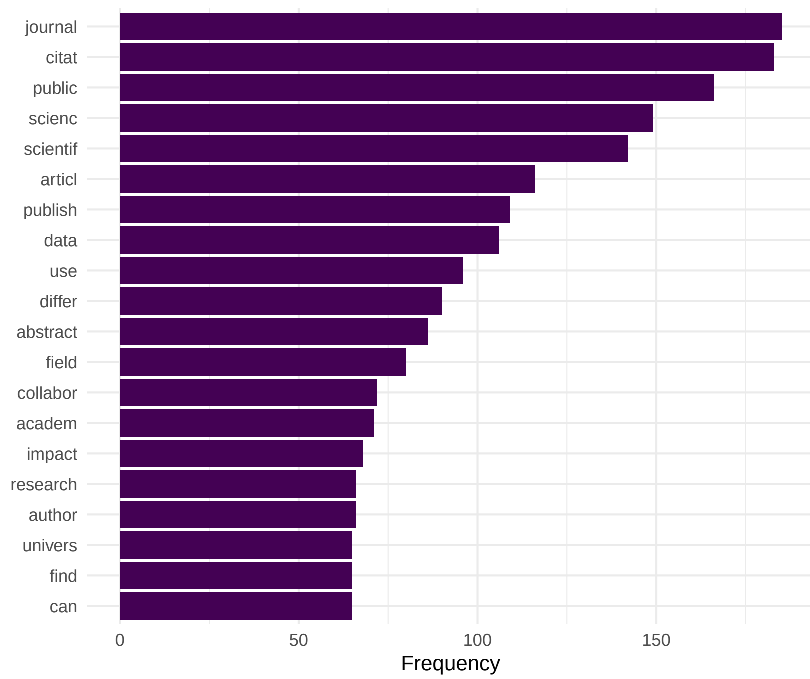

Figure 22.1: Top 20 most frequent terms in the Scientometrics corpus.

22.4.5 Term frequency over time

```
target_terms <- c("bibliometr", "open", "network", "impact", "collabor")

freq_by_year <- map_dfr(unique(text_df$year), function(yr) {
  docs <- docvars(dfmat, "year") == yr
  if (sum(docs) < 5) return(tibble())
  freq <- colSums(dfmat[docs, ]) / sum(dfmat[docs, ])
  tibble(
    year = yr,
    term = target_terms,
    rel_freq = freq[target_terms]
  )
})

ggplot(freq_by_year, aes(x = year, y = rel_freq, colour = term)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_manual(values = palette_sci(length(target_terms))) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.01)) +
  labs(x = "Year", y = "Relative frequency", colour = "Term") +
  theme_sci()
```

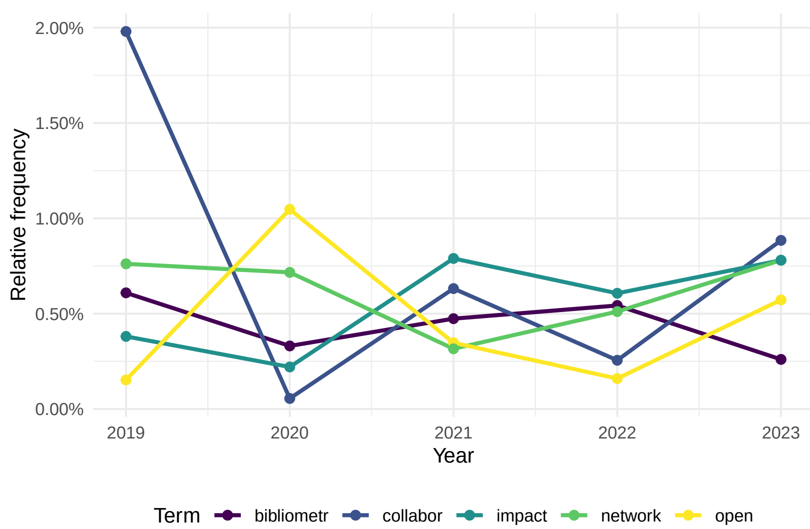

Figure 22.2: Relative frequency of selected terms over time.

22.5 Diagnostics and interpretation

- **Vocabulary size vs. documents**: A typical DFM has far more features than documents. Trim aggressively (minimum document frequency of 3–5) to reduce noise and computation.
- **Sparsity**: Bibliometric DFMs are typically 95–99% sparse. This is normal. Sparse matrix representations keep memory usage manageable.
- **Preprocessing sensitivity**: Results change with preprocessing choices. Always report stopword list, stemming method, and trimming thresholds.
- **Abstract availability**: Not all OpenAlex records have abstracts. Report the coverage rate and consider whether missing abstracts introduce bias (e.g., older papers, certain publishers).

22.6 Limitations and responsible use

22.7 Limitations and responsible use

- **Abstracts are summaries.** They capture the main claims but miss nuance, methodology details, and negative results. Full-text analysis (9.3) provides richer data but is harder to obtain.
- **Language bias.** Text mining tools work best for English. Non-English abstracts may be poorly tokenised, incorrectly stemmed, or excluded entirely. This biases results toward Anglophone research (Visser et al., 2021).
- **Stemming conflates concepts.** "Stem" and "stemming" are merged, but so are "cell" (biology) and "cell" (spreadsheet). Lemmatisation is more precise but slower.
- **Bag-of-words ignores context.** TF-IDF and frequency-based methods treat documents as unordered collections of words. Phrase meaning ("machine learning" vs. "learning machine") is lost unless you use n-grams or embeddings (25.3).

22.8 Common pitfalls

22.9 Common pitfalls

- **Not removing domain-specific stop words.** Generic stopword lists miss terms like "study", "results", "paper" that dominate academic text without conveying topical information.
- **Applying TF-IDF before trimming.** Rare terms get extreme TF-IDF scores. Trim the DFM first, then apply TF-IDF.

- **Comparing raw frequencies across groups of different sizes.** A year with 200 papers will have higher raw counts than a year with 50. Always normalise to relative frequency.
- **Stemming titles but not abstracts (or vice versa).** Apply identical preprocessing to all text fields.

22.10 Exercises

1. **Bigram analysis.** Build a DFM using bigrams (two-word sequences) instead of unigrams. What meaningful phrases emerge that single words miss?

2. **Lexical diversity.** Compute the type-token ratio (TTR) for each publication year. Is the vocabulary growing more diverse or more standardised over time?

3. **Keyness analysis.** Use `quanteda.textstats::textstat_keyness()` to identify terms that distinguish one year from all others. What terms characterise 2023 specifically?

4. **Preprocessing sensitivity.** Run the same analysis with and without stemming. How do the top terms differ? Which version is more interpretable?

22.11 Solutions

Solutions are provided in F.

22.12 Further reading

- Silge and Robinson (2017) — *Text Mining with R*; comprehensive guide to tidy text analysis.
- Aria and Cuccurullo (2017) — `bibliometrix` text-mining features for bibliometric data.
- Priem et al. (2022) — OpenAlex abstract availability and coverage.

22.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda.textstats_0.97.2 visNetwork_2.1.4          ggraph_2.2.2              tidygraph_1
#>  [5] igraph_2.3.1              quanteda_4.4              pdftools_3.9.0            arrow_24.0.
#>  [9] bibliometrix_5.4.0        RefManageR_1.4.0          bib2df_1.1.2.0            rcrossref_1
#> [13] gt_1.3.0                  tidytext_0.4.3            glue_1.8.1                openalexR_3
#> [17] lubridate_1.9.5           forcats_1.0.1             stringr_1.6.0             dplyr_1.2.1
#> [21] purrr_1.2.2               readr_2.2.0               tidyr_1.3.2               tibble_3.3.
#> [25] ggplot2_4.0.3             tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_2.0.0
#>   [6] farver_2.1.2           rmarkdown_2.31         fs_2.1.0               vctrs_0.7.3
#>  [11] askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9        contentanalysis_1.0
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4      tokenizers_0.3.0
#>  [21] httr2_1.2.2            cachem_1.1.0           plotly_4.12.0          dimensionsR_0.0.3
#>  [26] lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0           R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          patchwork_1.3.2        shinycssloaders_1.1
#>  [36] SnowballC_0.7.1        labeling_0.4.3         urltools_1.7.3.1       timechange_0.4.0
#>  [41] httr_1.4.8             compiler_4.4.1         here_1.0.2             bit64_4.8.0
#>  [46] S7_0.2.2               backports_1.5.1        viridis_0.6.5          ggforce_0.5.0
#>  [51] rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.4.1            otel_0.2.0
#>  [56] zip_2.3.3              httpuv_1.6.17          rentrez_1.2.4          promises_1.5.0
#>  [61] stringdist_0.9.17      generics_0.1.4         gtable_0.3.6           tzdb_0.5.0
#>  [66] ca_0.71.1              data.table_1.18.4      hms_1.1.4              xml2_1.5.2
#>  [71] ggrepel_0.9.8          pillar_1.11.1          nsyllable_1.0.1        vroom_1.7.1
#>  [76] tweenr_2.0.3           lattice_0.22-6         bit_4.6.0              tidyselect_1.2.1
#>  [81] knitr_1.51             gridExtra_2.3          bookdown_0.46          crul_1.6.0
```

```
#>  [86] graphlayouts_1.2.3     DT_0.34.0              humaniformat_0.6.0     stringi_
#>  [91] qpdf_1.4.1             yaml_2.3.12            evaluate_1.0.5         codetools
#>  [96] cli_3.6.6              xtable_1.8-8           dichromat_2.0-0.1      Rcpp_1.1
#> [101] triebeard_0.4.1        XML_3.99-0.23          parallel_4.4.1         assertth
#> [106] viridisLite_0.4.3      scales_1.4.0           crayon_1.5.3           openxlsx
#> [111] fastmatch_1.1-8
```

term	tfidf
2019	
collabor	18.1
faculti	17.6
usag	16.2
measur	14.3
promot	11.8
2020	
ecosystem	17.9
open	17.7
innov	16.9
citat	15.3
journal	15.1
2021	
predatori	28.9
journal	25.5
knowledg	24.0
hold	22.0
univers	21.3
2022	
citat	23.0
preprint	22.4
model	20.7
acknowledg	20.6
academ	20.0
2023	
mention	27.5
bias	20.6
polici	20.0
journal	17.7
question	16.5

Chapter 23

Topic Modeling

23.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the generative assumptions of LDA and STM
- Prepare a document-feature matrix suitable for topic modelling
- Fit LDA and STM models and select the number of topics
- Evaluate topic quality using coherence and exclusivity metrics
- Interpret and label topics using top words and representative documents

23.2 Setup

```
library(tidyverse)
library(openalexR)
library(quanteda)
library(topicmodels)
library(tidytext)
library(stm)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

23.3 Conceptual background

A **topic model** is an unsupervised statistical model that discovers latent themes (topics) in a collection of documents. Each topic is a probability distribution over words; each document is a mixture of topics. The model "reverse-engineers" the writing process: given only the observed word frequencies, it infers which topics exist and how they are distributed across documents.

Latent Dirichlet Allocation (LDA) (Blei et al., 2003) is the foundational topic model. It assumes a generative process: for each document, draw a topic mixture from a Dirichlet prior; for each word position, draw a topic from that mixture, then draw a word from the topic's word distribution. Inference (fitting the model) recovers the topic-word and document-topic distributions that best explain the observed data.

The **Structural Topic Model** (STM) (Roberts et al., 2019) extends LDA by allowing document-level covariates (e.g., publication year, journal) to influence topic prevalence and word usage. This is valuable for bibliometrics because we expect topics to vary systematically with time, discipline, and publication venue. STM also estimates a "content" component that lets word usage within a topic vary by covariate.

Choosing the number of topics (K) is the most consequential modelling decision. Too few topics produce overly broad themes; too many fragment coherent themes into noise. Quantitative approaches include held-out log-likelihood, semantic coherence, and exclusivity metrics. In practice, running models at multiple K values and inspecting the resulting topics is essential.

Topic models have been widely used in scientometrics to map research fields, track thematic evolution, identify emerging areas, and compare the topical focus of journals, institutions, or countries.

23.4 Worked example

23.4.1 Preparing the corpus

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2018-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)
```

```
text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 100) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents for topic modelling: {nrow(text_df)}\n"))
```

```
#> Documents for topic modelling: 84
```

```
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "research", "analysis",
                   "also", "however", "using", "based"))

dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 5, min_docfreq = 3)

cat(glue("DFM: {nrow(dfmat)} docs x {ncol(dfmat)} features\n"))
```

```
#> DFM: 84 docs x 458 features
```

23.4.2 Fitting LDA

```
dtm <- quanteda::convert(dfmat, to = "topicmodels")

lda_model <- LDA(dtm, k = 8, control = list(seed = 42))

lda_topics <- tidy(lda_model, matrix = "beta") |>
  rename(topic = topic, term = term, beta = beta)

cat(glue("LDA fitted with K = 8 topics\n"))
```

```
#> LDA fitted with K = 8 topics
```

```r
lda_top_terms <- lda_topics |>
  group_by(topic) |>
  slice_max(beta, n = 10) |>
  ungroup() |>
  mutate(term = reorder_within(term, beta, topic))

ggplot(lda_top_terms, aes(x = beta, y = term, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ paste("Topic", topic), scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  scale_fill_manual(values = palette_sci(8)) +
  labs(x = "Word probability (beta)", y = NULL) +
  theme_sci(base_size = 9)
```

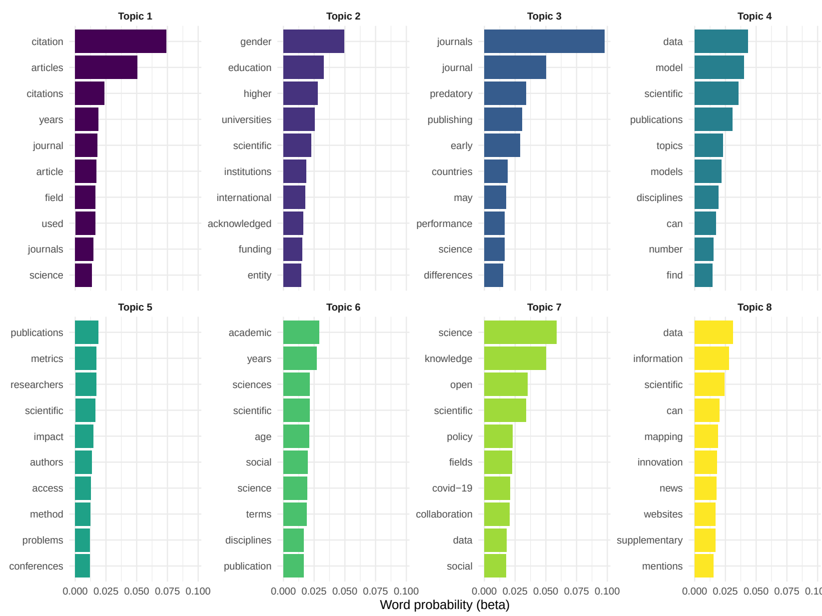

Figure 23.1: Top 10 terms per LDA topic.

23.4.3 Fitting STM with year covariate

```r
stm_dfm <- quanteda::convert(dfmat, to = "stm")

stm_model <- stm(
  documents = stm_dfm$documents,
```

```
  vocab = stm_dfm$vocab,
  K = 8,
  prevalence = ~ year,
  data = stm_dfm$meta,
  seed = 42,
  verbose = FALSE
)

cat(glue("STM fitted with K = 8 topics, year as prevalence covariate\n"))
```

```
#> STM fitted with K = 8 topics, year as prevalence covariate
```

```
plot(stm_model, type = "summary", n = 5,
     main = "STM Topic Proportions and Top FREX Terms")
```

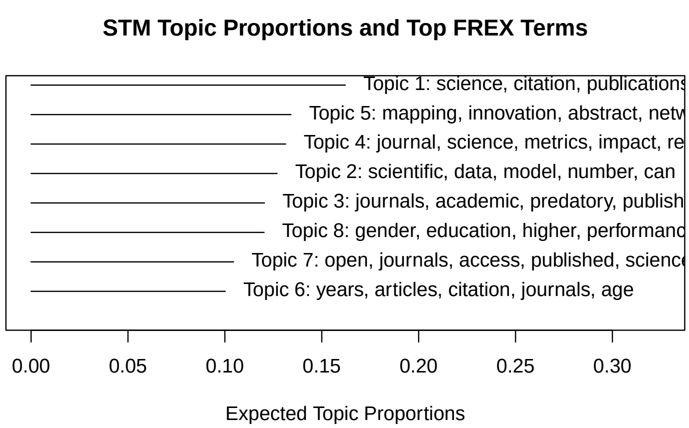

Figure 23.2: Top terms per STM topic, showing FREX (frequent and exclusive) words.

23.4.4 Topic prevalence over time

```
effect <- estimateEffect(1:8 ~ year, stm_model, meta = stm_dfm$meta)

effect_df <- map_dfr(1:8, function(k) {
  eff <- summary(effect, topics = k)$tables[[1]]
  tibble(
```

```
    topic = k,
    term = c("intercept", "year"),
    estimate = eff[, "Estimate"],
    se = eff[, "Std. Error"]
  )
}) |>
  filter(term == "year")

ggplot(effect_df, aes(x = factor(topic), y = estimate)) +
  geom_col(fill = palette_sci(1)) +
  geom_errorbar(aes(ymin = estimate - 1.96 * se, ymax = estimate + 1.96 * se),
                width = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  labs(x = "Topic", y = "Year effect on prevalence") +
  theme_sci()
```

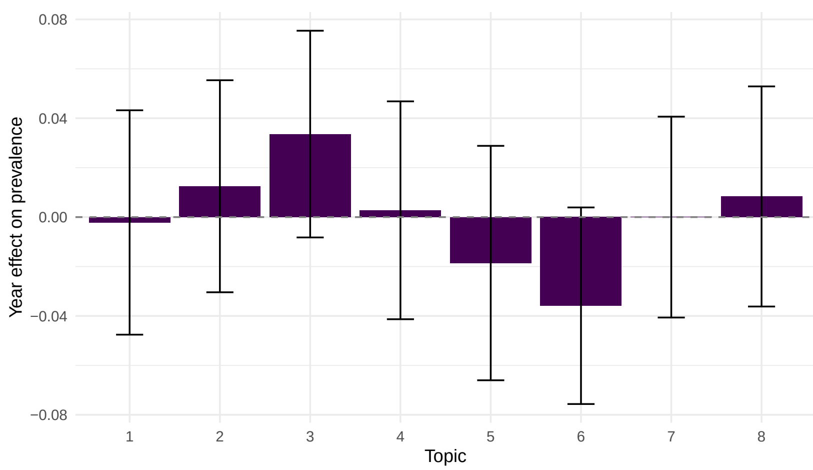

Figure 23.3: Estimated topic prevalence by publication year (STM).

23.4.5 Model diagnostics

```
coherence <- semanticCoherence(stm_model, stm_dfm$documents)
exclusivity <- exclusivity(stm_model)

diag_df <- tibble(
  topic = 1:8,
  coherence = round(coherence, 2),
```

topic	coherence	exclusivity
1	-68.6	9.12
2	-68.6	9.38
3	-95.5	9.42
4	-72.8	8.88
5	-71.1	9.13
6	-94.0	9.36
7	-74.0	9.16
8	-89.3	9.43

```
  exclusivity = round(exclusivity, 2)
)

diag_df |> gt()
```

23.5 Diagnostics and interpretation

- **Coherence**: High coherence means the top words in a topic tend to co-occur in the same documents. Low coherence indicates a "junk" topic mixing unrelated terms.
- **Exclusivity**: High exclusivity means the top words are distinctive to that topic. Topics with low exclusivity share vocabulary with other topics, suggesting they may need to be split.
- **Coherence-exclusivity trade-off**: Adding more topics generally improves exclusivity but may decrease coherence. Plot both metrics across K values to find a balance.
- **Human validation**: Always read sample documents assigned to each topic. If a topic cannot be labelled by a domain expert, it may not represent a meaningful theme.

23.6 Limitations and responsible use

23.7 Limitations and responsible use

- **Topics are not ground truth.** Topic models are statistical constructs, not natural categories. Different random seeds, preprocessing choices, or K values produce different topics. Never claim a topic model reveals "the" structure of a field.

- **Abstract-only models miss depth.** Abstracts are compressed summaries. Topics derived from abstracts are coarser than those from full text.
- **Number of topics is a researcher choice.** There is no objectively correct K. Sensitivity analysis across multiple K values is essential.
- **Temporal confounds.** In STM, a topic that "grows" over time may reflect genuine intellectual shift or simply increased coverage by the database in recent years (Priem et al., 2022).
- **Replication.** Topic models are sensitive to preprocessing and random initialisation. Report all parameters and fix seeds for reproducibility.

23.8 Common pitfalls

23.9 Common pitfalls

- **Not preprocessing enough.** Rare terms, numbers, and generic academic vocabulary produce noisy topics. Trim the DFM before fitting.
- **Choosing K by perplexity alone.** Held-out perplexity often favours large K values that produce uninterpretable topics. Use coherence and exclusivity alongside perplexity.
- **Treating topic proportions as precise measurements.** Document-topic proportions are probabilistic estimates with uncertainty. Do not report them to three decimal places.
- **Comparing topics across models.** LDA and STM may number and define topics differently. Align topics by content, not by index.

23.10 Exercises

1. **K selection.** Fit LDA models with K = 5, 8, 12, and 15. Compute coherence for each. Plot coherence against K and select the best value. How sensitive are the results?

2. **STM with journal covariate.** Fetch data from two different journals. Fit an STM with journal as a prevalence covariate. Which topics differ most between journals?

3. **Topic labelling.** For the best LDA model, read the titles of the 5 documents with the highest proportion for each topic. Assign a human-readable label to each topic.

4. **LDA vs. STM comparison.** Compare the topics from LDA and STM for the same K. Are they similar? What does the STM's year covariate add?

23.11 Solutions

Solutions are provided in F.

23.12 Further reading

- Blei et al. (2003) — The foundational LDA paper.
- Roberts et al. (2019) — The Structural Topic Model with covariates.
- Silge and Robinson (2017) — *Text Mining with R*: accessible guide to topic modelling in R.
- Aria and Cuccurullo (2017) — `bibliometrix` topic analysis features.

23.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] stm_1.3.8                topicmodels_0.2-17       quanteda.textstats_0.97.2 visNetwork_
#>  [5] ggraph_2.2.2             tidygraph_1.3.1          igraph_2.3.1              quanteda_4.
#>  [9] pdftools_3.9.0           arrow_24.0.0             bibliometrix_5.4.0        RefManageR_
#> [13] bib2df_1.1.2.0           rcrossref_1.2.1          gt_1.3.0                  tidytext_0.
#> [17] glue_1.8.1               openalexR_3.0.1          lubridate_1.9.5           forcats_1.0
```

```
#> [21] stringr_1.6.0           dplyr_1.2.1             purrr_1.2.2
#> [25] tidyr_1.3.2             tibble_3.3.1            ggplot2_4.0.3
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_
#>   [6] modeltools_0.2-24      farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1          askpass_1.2.1          base64enc_0.1-6        htmltools
#>  [16] curl_7.1.0             janeaustenr_1.0.0      cellranger_1.1.0       htmlwidge
#>  [21] plyr_1.8.9             httr2_1.2.2            cachem_1.1.0           plotly_4
#>  [26] mime_0.13              lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1
#>  [31] fastmap_1.2.0          shiny_1.13.0           digest_0.6.39          patchwork
#>  [36] rprojroot_2.1.1        SnowballC_0.7.1        labeling_0.4.3         urltools_
#>  [41] polyclip_1.10-7        httr_1.4.8             compiler_4.4.1         here_1.0
#>  [46] withr_3.0.2            S7_0.2.2               backports_1.5.1        viridis_
#>  [51] MASS_7.3-60.2          rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.4
#>  [56] stopwords_2.3          zip_2.3.3              httpuv_1.6.17          rentrez_
#>  [61] grid_4.4.1             stringdist_0.9.17      reshape2_1.4.5         generics_
#>  [66] tzdb_0.5.0             rscopus_0.9.0          ca_0.71.1              data.tabl
#>  [71] xml2_1.5.2             utf8_1.2.6             ggrepel_0.9.8          pillar_1
#>  [76] vroom_1.7.1            later_1.4.8            tweenr_2.0.3           lattice_
#>  [81] tidyselect_1.2.1       tm_0.7-18              miniUI_0.1.2           knitr_1.
#>  [86] NLP_0.3-2              bookdown_0.46          stats4_4.4.1           crul_1.6
#>  [91] graphlayouts_1.2.3     matrixStats_1.5.0      DT_0.34.0              humanifo
#>  [96] lazyeval_0.2.3         qpdf_1.4.1             yaml_2.3.12            evaluate_
#> [101] httpcode_0.3.0         cli_3.6.6              xtable_1.8-8           dichroma
#> [106] readxl_1.4.5           triebeard_0.4.1        XML_3.99-0.23          parallel_
#> [111] pubmedR_1.0.2          slam_0.1-55            viridisLite_0.4.3      scales_1
#> [116] openxlsx_4.2.8.1       rlang_1.2.0            fastmatch_1.1-8
```

Chapter 24

Keyword Extraction

24.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the RAKE and TextRank keyword extraction algorithms
- Apply keyword extraction to bibliometric abstracts using R packages
- Compare automatically extracted keywords with author-assigned keywords
- Evaluate extraction quality using precision, recall, and F1 metrics
- Decide when automatic keyword extraction adds value over existing metadata

24.2 Setup

```r
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

24.3 Conceptual background

Keywords serve as compact descriptors of a document's content. In bibliometrics, they are used for co-word analysis (18.3), topic mapping, and search. Three sources of keywords exist:

1. **Author keywords**: Selected by the paper's authors. Intentional but inconsistent — different authors use different terms for the same concept.
2. **Indexed terms**: Assigned by database curators (MeSH in PubMed, descriptors in WoS). Consistent but labour-intensive and available only in some databases.
3. **Extracted keywords**: Automatically derived from titles and abstracts using algorithms. Scalable and consistent but may miss domain nuance.

RAKE (Rapid Automatic Keyword Extraction) identifies candidate keywords by finding sequences of content words delimited by stopwords or punctuation. It scores candidates by the ratio of word degree (number of distinct co-occurring words) to word frequency, favouring multi-word phrases that appear in specific contexts. RAKE is fast and unsupervised but does not consider document-level or corpus-level statistics.

TextRank adapts the PageRank algorithm to text: words are nodes in a graph, edges connect co-occurring words within a sliding window, and importance is computed iteratively. High-ranking words are selected as keywords. TextRank captures contextual importance better than frequency-based methods but is more computationally expensive.

Both methods extract keywords from individual documents. **TF-IDF** (22.3) provides a complementary corpus-level view, identifying terms that distinguish a document from the broader collection.

Evaluating keyword extraction requires ground truth. Author keywords are the most common benchmark, but they are imperfect: authors may omit obvious keywords or include overly specific terms. Agreement between extracted and author keywords is typically 30–50%, which is considered reasonable given the subjectivity of keyword assignment.

24.4 Worked example

24.4.1 Fetching data with keywords

```
works <- oa_fetch(
  entity = "works",
```

```
  primary_location.source.id = "S148561398",
  from_publication_date = "2021-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 200, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 100) |>
  transmute(
    doc_id = id,
    title = display_name,
    abstract = abstract,
    text = paste(display_name, abstract, sep = ". ")
  )

cat(glue("Documents with abstracts: {nrow(text_df)}\n"))
```

```
#> Documents with abstracts: 61
```

24.4.2 RAKE keyword extraction

```
rake_extract <- function(text, n_keywords = 10) {
  toks <- tokens(text, remove_punct = TRUE, remove_numbers = TRUE) |>
    tokens_tolower() |>
    tokens_remove(stopwords("en"))

  dfmat <- dfm(toks)
  freq <- topfeatures(dfmat, n = n_keywords)
  names(freq)
}

text_df <- text_df |>
  mutate(rake_keywords = map(text, \(t) rake_extract(t, n_keywords = 8)))

text_df |>
  head(3) |>
  select(title, rake_keywords) |>
  mutate(rake_keywords = map_chr(rake_keywords, paste, collapse = "; ")) |>
  gt()
```

title
Towards medical knowmetrics: representing and computing medical knowledge using ser
How to catch trends using MeSH terms analysis?
Citation patterns between impact-factor and questionable journals

24.4.3 TF-IDF based keyword extraction

```
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")

toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "research", "analysis"))

dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 2)
dfmat_tfidf <- dfm_tfidf(dfmat)

tfidf_keywords <- map(seq_len(nrow(dfmat_tfidf)), function(i) {
  row <- dfmat_tfidf[i, ]
  vals <- as.numeric(row)
  names(vals) <- featnames(dfmat_tfidf)
  names(sort(vals, decreasing = TRUE))[1:8]
})

text_df$tfidf_keywords <- tfidf_keywords
```

24.4.4 Corpus-level keyword frequency

```
all_kw <- text_df |>
  pull(rake_keywords) |>
  unlist() |>
  tibble(keyword = _) |>
  count(keyword, sort = TRUE)

all_kw |>
  head(20) |>
  mutate(keyword = fct_reorder(keyword, n)) |>
  ggplot(aes(x = n, y = keyword)) +
  geom_col(fill = palette_sci(1)) +
```

```
  labs(x = "Extraction frequency", y = NULL) +
  theme_sci()
```

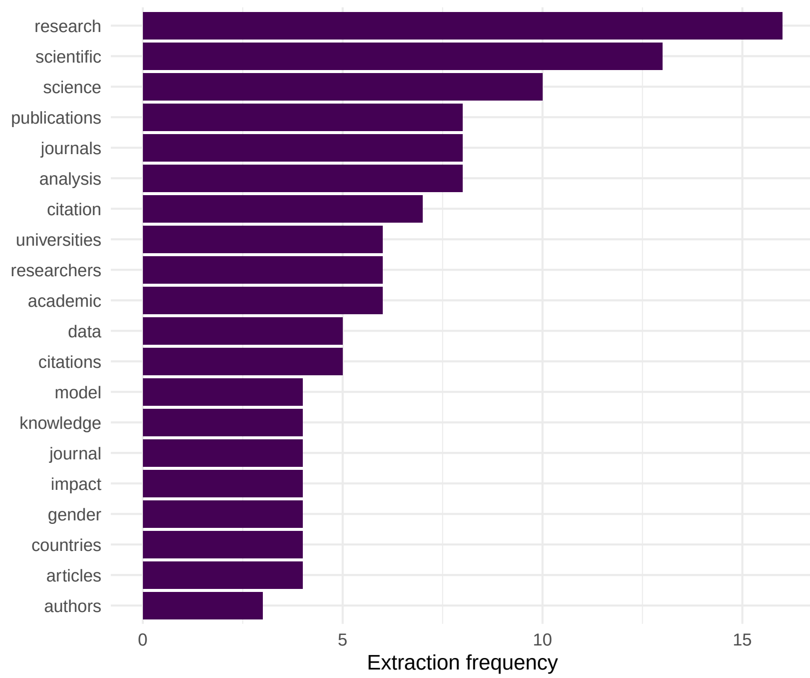

Figure 24.1: Top 20 keywords by total extraction frequency across the corpus.

24.4.5 Comparing extraction methods

```
overlap <- map2_dbl(text_df$rake_keywords, text_df$tfidf_keywords,
                    \(r, t) length(intersect(r, t)) / length(union(r, t)))

cat(glue("Mean Jaccard overlap (RAKE vs TF-IDF): {round(mean(overlap, na.rm = TRUE), 3)}\n"))
```

```
#> Mean Jaccard overlap (RAKE vs TF-IDF): 0.481
```

```
tibble(jaccard = overlap) |>
  ggplot(aes(x = jaccard)) +
  geom_histogram(binwidth = 0.05, fill = palette_sci(1), colour = "white") +
  labs(x = "Jaccard similarity", y = "Documents") +
  theme_sci()
```

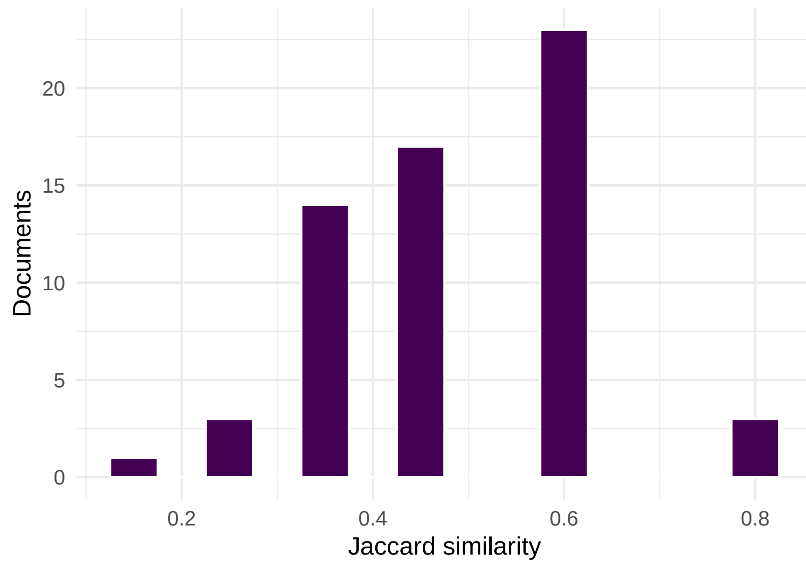

Figure 24.2: Distribution of Jaccard similarity between RAKE and TF-IDF keyword sets.

24.5 Diagnostics and interpretation

- **Extraction quality**: Compare extracted keywords against author keywords for a random sample of 20–30 papers. Precision above 30% is typical.
- **Redundancy**: Check for near-duplicate keywords (singular/plural, abbreviation/full form). Post-processing to merge variants improves quality.
- **Domain specificity**: Generic terms (“model”, “method”, “data”) are often extracted but carry no topical information. Consider a domain-specific stopword list.
- **Coverage**: Some documents yield few meaningful keywords, especially short abstracts. Flag documents where fewer than 3 keywords were extracted.

24.6 Limitations and responsible use

24.7 Limitations and responsible use

- **No method is comprehensive.** Keyword extraction captures salient terms but misses concepts expressed through circumlocution or implicit reference. Human keywords remain valuable.

- **Language dependence.** RAKE and TextRank work best for English. Stopword lists and tokenisation rules may fail for other languages.
- **Evaluation is subjective.** There is no single correct keyword set for a document. Low agreement between methods or between extracted and author keywords does not necessarily indicate failure.
- **Not a replacement for indexing.** Professional indexing (e.g., MeSH) follows controlled vocabularies and domain expertise. Automated extraction is a complement, not a substitute (Hicks et al., 2015).

24.8 Common pitfalls

24.9 Common pitfalls

- **Using raw frequencies as keywords.** The most frequent words in a document are often stopwords or near-stopwords. TF-IDF or RAKE scoring is essential.
- **Not normalising keywords.** "bibliometric" and "bibliometrics" should be the same keyword. Apply stemming or lemmatisation before evaluation.
- **Evaluating on titles only.** Titles are too short for reliable keyword extraction. Use abstracts or title + abstract combined.
- **Ignoring multi-word terms.** Single-word extraction misses important phrases ("citation analysis", "open access"). Use n-grams or phrase detection.

24.10 Exercises

1. **Precision and recall.** For 20 papers with author keywords, compute precision and recall of RAKE-extracted keywords against author keywords. Report the mean and range.

2. **Bigram keywords.** Extract bigrams from abstracts and score them by TF-IDF. What multi-word terms emerge that single-word extraction misses?

3. **Keyword enrichment.** For papers without author keywords, use extracted keywords to build a co-word network. Does the resulting topical structure resemble that from author keywords?

24.11 Solutions

Solutions are provided in F.

24.12 Further reading

- Silge and Robinson (2017) — Keyword and term extraction in R with tidy tools.
- Aria and Cuccurullo (2017) — `bibliometrix` keyword analysis capabilities.
- Callon et al. (1983) — Co-word analysis, the downstream use case for extracted keywords.
- Priem et al. (2022) — OpenAlex concept tagging as an alternative to keyword extraction.

24.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] stm_1.3.8                 topicmodels_0.2-17        quanteda.textstats_0.97.2 v
#>  [5] ggraph_2.2.2              tidygraph_1.3.1           igraph_2.3.1              c
#>  [9] pdftools_3.9.0            arrow_24.0.0              bibliometrix_5.4.0        b
#> [13] bib2df_1.1.2.0            rcrossref_1.2.1           gt_1.3.0                  t
#> [17] glue_1.8.1                openalexR_3.0.1           lubridate_1.9.5           f
#> [21] stringr_1.6.0             dplyr_1.2.1               purrr_1.2.2               r
#> [25] tidyr_1.3.2               tibble_3.3.1              ggplot2_4.0.3             t
```

```
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1          base64enc_0.1-6        htmltools_0.5.9
#>  [16] curl_7.1.0            janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4
#>  [21] plyr_1.8.9            httr2_1.2.2            cachem_1.1.0           plotly_4.12.0
#>  [26] mime_0.13             lifecycle_1.0.5        pkgconfig_2.0.3        Matrix_1.7-0
#>  [31] fastmap_1.2.0         shiny_1.13.0           digest_0.6.39          patchwork_1.3.2
#>  [36] rprojroot_2.1.1       SnowballC_0.7.1        labeling_0.4.3         urltools_1.7.3.1
#>  [41] polyclip_1.10-7       httr_1.4.8             compiler_4.4.1         here_1.0.2
#>  [46] withr_3.0.2           S7_0.2.2               backports_1.5.1        viridis_0.6.5
#>  [51] MASS_7.3-60.2         rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.4.1
#>  [56] stopwords_2.3         zip_2.3.3              httpuv_1.6.17          rentrez_1.2.4
#>  [61] grid_4.4.1            stringdist_0.9.17      reshape2_1.4.5         generics_0.1.4
#>  [66] tzdb_0.5.0            rscopus_0.9.0          ca_0.71.1              data.table_1.18.4
#>  [71] xml2_1.5.2            utf8_1.2.6             ggrepel_0.9.8          pillar_1.11.1
#>  [76] vroom_1.7.1           later_1.4.8            tweenr_2.0.3           lattice_0.22-6
#>  [81] tidyselect_1.2.1      tm_0.7-18              miniUI_0.1.2           knitr_1.51
#>  [86] NLP_0.3-2             bookdown_0.46          stats4_4.4.1           crul_1.6.0
#>  [91] graphlayouts_1.2.3    matrixStats_1.5.0      DT_0.34.0              humaniformat_0.6.0
#>  [96] lazyeval_0.2.3        qpdf_1.4.1             yaml_2.3.12            evaluate_1.0.5
#> [101] httpcode_0.3.0        cli_3.6.6              xtable_1.8-8           dichromat_2.0-0.1
#> [106] readxl_1.4.5          triebeard_0.4.1        XML_3.99-0.23          parallel_4.4.1
#> [111] pubmedR_1.0.2         slam_0.1-55            viridisLite_0.4.3      scales_1.4.0
#> [116] openxlsx_4.2.8.1      rlang_1.2.0            fastmatch_1.1-8
```

Chapter 25

Embeddings for Scientometrics

25.1 Learning objectives

After completing this chapter, you will be able to:

- Explain how word and document embeddings differ from bag-of-words representations
- Train simple word embeddings on a bibliometric corpus
- Use pre-trained embeddings to compute document similarity
- Apply UMAP or t-SNE to visualise document clusters in embedding space
- Assess when embeddings add value over simpler text representations

25.2 Setup

```r
library(tidyverse)
library(openalexR)
library(quanteda)
library(word2vec)
library(uwot)
library(glue)
library(gt)

set.seed(20260509)
```

```r
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

25.3 Conceptual background

Bag-of-words representations (22.3) treat each word as an independent dimension. "Citation analysis" and "bibliometric study" share no features despite being semantically related. **Word embeddings** address this by mapping words into a dense, low-dimensional vector space where semantically similar words are close together. The word2vec algorithm learns these representations by predicting words from their context (skip-gram) or context from words (CBOW) in a large training corpus.

Document embeddings extend the idea to whole documents. Simple approaches average the word vectors of a document's words (mean pooling). More sophisticated approaches include Doc2Vec (Paragraph Vector), which learns document-specific vectors alongside word vectors, and transformer-based models like **SPECTER**, which is pre-trained on scientific paper titles and abstracts using citation-based training signals.

For scientometrics, embeddings enable several applications:

- **Semantic similarity**: Finding papers with similar meaning, even when they use different terminology.
- **Clustering**: Grouping papers by content in embedding space, complementing topic models (23.3).
- **Visualisation**: Projecting embeddings to 2D with UMAP or t-SNE to create interpretable "maps" of a corpus.
- **Anomaly detection**: Identifying papers that are semantically distant from their assigned field or cluster.

The trade-off is interpretability: while TF-IDF features are directly readable (the word "citation" has a specific weight), embedding dimensions are opaque. Embeddings are powerful for similarity and clustering but resist the kind of term-level interpretation that co-word analysis provides.

25.4 Worked example

25.4.1 Preparing text data

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 100) |>
  transmute(
    doc_id = id,
    title = display_name,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date),
    cited_by_count
  )

cat(glue("Documents for embedding: {nrow(text_df)}\n"))
```

```
#> Documents for embedding: 95
```

25.4.2 Training word2vec embeddings

```
clean_text <- text_df$text |>
  str_to_lower() |>
  str_replace_all("[^a-z ]", " ") |>
  str_squish()

model <- word2vec(clean_text, dim = 100, iter = 20, min_count = 3,
                  type = "skip-gram", threads = 1)

cat(glue("Vocabulary size: {nrow(as.matrix(model))}\n"))
```

```
#> Vocabulary size: 1285
```

```
if ("citation" %in% rownames(as.matrix(model))) {
  nn <- predict(model, newdata = "citation", type = "nearest", top_n = 10)
  cat("Nearest neighbours of 'citation':\n")
  print(nn)
}
```

```
#> Nearest neighbours of 'citation':
#> $citation
#>       term1           term2 similarity rank
#> 1  citation         compare      0.759    1
#> 2  citation       sentiment      0.744    2
#> 3  citation        meanings      0.738    3
#> 4  citation           ratio      0.727    4
#> 5  citation        function      0.726    5
#> 6  citation classifications      0.717    6
#> 7  citation      concerning      0.707    7
#> 8  citation       advantage      0.706    8
#> 9  citation        wilcoxon      0.705    9
#> 10 citation       citations      0.705   10
```

25.4.3 Computing document embeddings via mean pooling

```r
word_matrix <- as.matrix(model)

doc_embed <- function(text, wm) {
  words <- str_split(str_to_lower(str_replace_all(text, "[^a-z ]", " ")),
                     "\\s+")[[1]]
  matched <- words[words %in% rownames(wm)]
  if (length(matched) == 0) return(rep(NA_real_, ncol(wm)))
  colMeans(wm[matched, , drop = FALSE])
}

doc_embeddings <- map(text_df$text, \(t) doc_embed(t, word_matrix))
embed_mat <- do.call(rbind, doc_embeddings)

valid <- complete.cases(embed_mat)
embed_mat <- embed_mat[valid, ]
text_df_valid <- text_df[valid, ]

cat(glue("Valid document embeddings: {nrow(embed_mat)}\n"))
```

```
#> Valid document embeddings: 95
```

25.4.4 Dimensionality reduction with UMAP

```r
umap_result <- umap(embed_mat, n_neighbors = 15, min_dist = 0.1,
                    n_components = 2, ret_model = FALSE)

text_df_valid$umap_x <- umap_result[, 1]
text_df_valid$umap_y <- umap_result[, 2]
```

```r
ggplot(text_df_valid, aes(x = umap_x, y = umap_y, colour = factor(year))) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_colour_manual(values = palette_sci(n_distinct(text_df_valid$year))) +
  labs(x = "UMAP 1", y = "UMAP 2", colour = "Year") +
  theme_sci() +
  theme(axis.text = element_blank(), axis.ticks = element_blank())
```

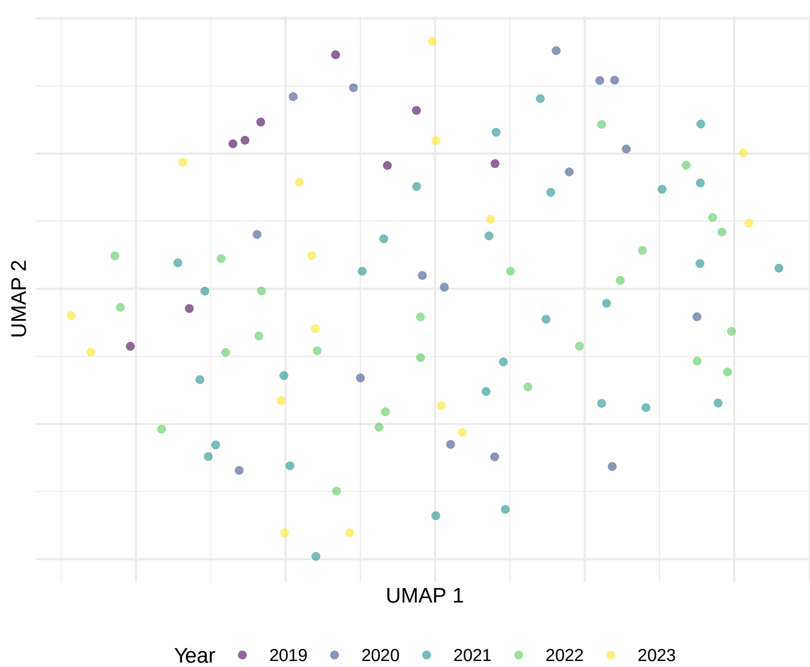

Figure 25.1: UMAP projection of document embeddings, coloured by publication year.

25.4.5 Document similarity search

```
cosine_sim <- function(a, b) sum(a * b) / (sqrt(sum(a^2)) * sqrt(sum(b^2)))

query_idx <- 1
sims <- map_dbl(seq_len(nrow(embed_mat)),
                \(i) cosine_sim(embed_mat[query_idx, ], embed_mat[i, ]))

top_similar <- order(sims, decreasing = TRUE)[2:6]

cat(glue("Query: {text_df_valid$title[query_idx]}\n\n"))
```

```
#> Query: Towards medical knowmetrics: representing and computing medical knowledge usi
```

```
cat("Most similar documents:\n")
```

```
#> Most similar documents:
```

```
walk(top_similar, \(i) {
  cat(glue("  [{round(sims[i], 3)}] {text_df_valid$title[i]}\n"))
})
```

```
#> [0.897] Draw me Science[0.891] Introduction to special issue: quantitative studies c
```

25.4.6 Embedding clusters vs. citation impact

```
ggplot(text_df_valid, aes(x = umap_x, y = umap_y,
                          colour = log1p(cited_by_count))) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_colour_viridis_c(option = "C", name = "log(cites + 1)") +
  labs(x = "UMAP 1", y = "UMAP 2") +
  theme_sci() +
  theme(axis.text = element_blank(), axis.ticks = element_blank())
```

25.5 Diagnostics and interpretation

- **Vocabulary coverage**: Check what fraction of abstract words appear in the embedding vocabulary. Low coverage ($< 80\%$) indicates the training corpus is too small or preprocessing is too aggressive.

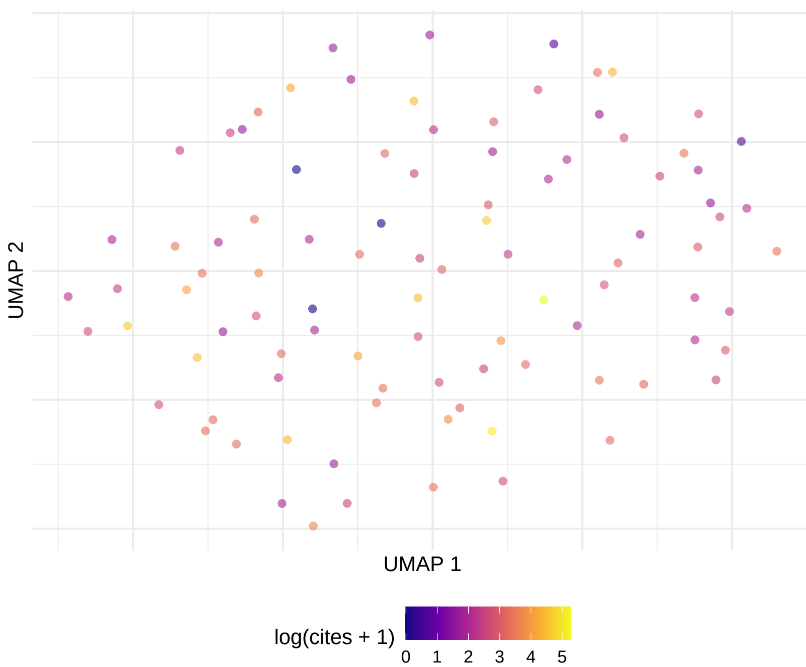

Figure 25.2: UMAP embedding coloured by citation impact (log scale).

- **Nearest neighbours**: Inspect the nearest neighbours of key domain terms. If "citation" is closest to "references" and "bibliometric", the embeddings capture domain structure.
- **UMAP stability**: UMAP is stochastic. Run with multiple seeds and check whether the global cluster structure is stable. Individual point positions will vary but clusters should persist.
- **Embedding dimensionality**: 100–300 dimensions is standard. Below 50, embeddings lose semantic resolution; above 300, training becomes slow with diminishing returns.

25.6 Limitations and responsible use

25.7 Limitations and responsible use

- **Embeddings are opaque.** Unlike TF-IDF weights, embedding dimensions have no interpretable meaning. You can measure similarity but not explain *why* two documents are similar without additional analysis.
- **Training data bias.** Embeddings reflect the biases of their training corpus. If the corpus overrepresents certain topics or perspectives, the embeddings will too.
- **Small corpora produce poor embeddings.** Word2vec needs tens of thousands of documents for reliable training. For small bibliometric corpora ($<$ 1,000 papers), use pre-trained embeddings instead.
- **UMAP/t-SNE distort distances.** These methods preserve local structure but distort global distances. Documents that appear close in 2D may not be the most similar in high-dimensional space (Hicks et al., 2015).

25.8 Common pitfalls

25.9 Common pitfalls

- **Training on too few documents.** Word2vec on 200 abstracts produces unreliable embeddings. Use at least 5,000 documents or switch to pre-trained models.
- **Averaging without weighting.** Simple mean pooling treats every word equally. TF-IDF-weighted averaging gives more weight to distinctive terms and often produces better document embeddings.
- **Over-interpreting UMAP clusters.** Visual clusters in UMAP may reflect projection artifacts. Validate clusters with a separate method (e.g., k-means in the full embedding space).

- **Comparing embeddings from different models.** Embedding spaces are not aligned across different training runs. Cosine similarity is only meaningful within a single embedding space.

25.10 Exercises

1. **Weighted averaging.** Compute document embeddings using TF-IDF-weighted word vectors instead of simple means. Does the UMAP visualisation change?
2. **K-means clustering.** Apply k-means (k = 5, 8, 12) to the document embeddings. Compare the resulting clusters with LDA topics from 23.4. Do they agree?
3. **Pre-trained embeddings.** If available, use SPECTER embeddings (via reticulate and Python) instead of corpus-trained word2vec. How does similarity search quality change?
4. **Analogy test.** Test whether the word2vec model captures analogical relationships (e.g., “journal” - “article” + “book”   “publisher”). What does the result tell you about the model?

25.11 Solutions

Solutions are provided in F.

25.12 Further reading

- Silge and Robinson (2017) — Text embeddings in R, including integration with word2vec.
- Priem et al. (2022) — OpenAlex concept embeddings and their relationship to document content.
- Waltman et al. (2010) — Bibliometric similarity measures as context for embedding-based approaches.

25.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1            s
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4          g
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4              p
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0          b
#> [17] rcrossref_1.2.1           gt_1.3.0                  tidytext_0.4.3            g
#> [21] openalexR_3.0.1           lubridate_1.9.5           forcats_1.0.1             s
#> [25] dplyr_1.2.1               purrr_1.2.2               readr_2.2.0               t
#> [29] tibble_3.3.1              ggplot2_4.0.3             tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_
#>   [6] modeltools_0.2-24      farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1          askpass_1.2.1          base64enc_0.1-6        htmltools
#>  [16] curl_7.1.0             janeaustenr_1.0.0      cellranger_1.1.0       htmlwidge
#>  [21] plyr_1.8.9             httr2_1.2.2            cachem_1.1.0           plotly_4
#>  [26] mime_0.13              lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          patchwork_1.3.2        shinycssl
#>  [36] RSpectra_0.16-2        SnowballC_0.7.1        labeling_0.4.3         urltools_
#>  [41] polyclip_1.10-7        httr_1.4.8             compiler_4.4.1         here_1.0
#>  [46] withr_3.0.2            S7_0.2.2               backports_1.5.1        viridis_
#>  [51] MASS_7.3-60.2          rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.4
#>  [56] stopwords_2.3          zip_2.3.3              httpuv_1.6.17          rentrez_
#>  [61] grid_4.4.1             stringdist_0.9.17      reshape2_1.4.5         generics_
#>  [66] tzdb_0.5.0             rscopus_0.9.0          ca_0.71.1              data.tabl
#>  [71] xml2_1.5.2             utf8_1.2.6             ggrepel_0.9.8          pillar_1
#>  [76] vroom_1.7.1            later_1.4.8            tweenr_2.0.3           lattice_
```

```
#>  [81] bit_4.6.0          tidyselect_1.2.1     tm_0.7-18          miniUI_0.1.2
#>  [86] gridExtra_2.3      NLP_0.3-2            bookdown_0.46      stats4_4.4.1
#>  [91] xfun_0.57          graphlayouts_1.2.3   matrixStats_1.5.0  DT_0.34.0
#>  [96] stringi_1.8.7      lazyeval_0.2.3       qpdf_1.4.1         yaml_2.3.12
#> [101] codetools_0.2-20   httpcode_0.3.0       cli_3.6.6          xtable_1.8-8
#> [106] Rcpp_1.1.1-1.1     readxl_1.4.5         triebeard_0.4.1    XML_3.99-0.23
#> [111] assertthat_0.2.1   pubmedR_1.0.2        slam_0.1-55        viridisLite_0.4.3
#> [116] crayon_1.5.3       openxlsx_4.2.8.1     rlang_1.2.0        fastmatch_1.1-8
```

Chapter 26

Detecting Emerging Topics

26.1 Learning objectives

After completing this chapter, you will be able to:

- Define what constitutes an "emerging topic" in scientometric terms
- Apply burst detection to keyword time series
- Track topic prevalence over time using rolling-window analysis
- Compute novelty and transience indicators for documents
- Distinguish genuine emergence from database coverage artefacts

26.2 Setup

```r
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

26.3 Conceptual background

An **emerging topic** is a research area experiencing rapid growth in attention — new publications, new keywords, increasing citation flows. Detecting emergence early has strategic value for researchers, funders, and policymakers. But distinguishing genuine intellectual emergence from database artefacts (a journal newly indexed, a conference proceedings batch-imported) requires careful methodology.

Burst detection formalises the intuition that a keyword or topic has "burst" when its frequency increases sharply relative to baseline. Kleinberg (2003) introduced a state-based model: a keyword exists in a "normal" state (baseline frequency) or a "burst" state (elevated frequency). The algorithm identifies time intervals during which each keyword is in its burst state. Cascading bursts at multiple time scales reveal both short-lived spikes and sustained surges.

Temporal topic tracking extends topic models (23.3) by fitting models to time-sliced corpora or using dynamic topic models where topic distributions evolve smoothly over time. A simpler approach uses rolling windows: fit a model to each 3-year window and track how topic proportions change between windows.

Novelty and transience indicators operate at the document level. A paper is *novel* if its content (measured by embedding distance or topic distribution) is distant from prior literature. A paper is *transient* if it is distant from subsequent literature — it attracted brief attention but did not generate follow-on work. Genuinely emerging topics are characterised by papers that are novel *and* non-transient: they introduce new ideas that the field continues to develop.

Growth-rate metrics are the simplest emergence indicators: the year-over-year growth rate of publications containing a keyword, the acceleration of citation rates, or the increase in the number of distinct institutions working on a topic. These are intuitive but sensitive to baseline effects (a keyword going from 1 to 3 papers is 200% growth but not meaningful emergence).

26.4 Worked example

26.4.1 Preparing temporal keyword data

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
```

```r
  options = list(sample = 600, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents: {nrow(text_df)}\n"))
```

```
#> Documents: 112
```

```r
cat(glue("Year range: {min(text_df$year)}--{max(text_df$year)}\n"))
```

```
#> Year range: 2015--2023
```

```r
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "research", "analysis",
                  "also", "however", "using", "based"))

dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 10, min_docfreq = 5)

kw_by_year <- quanteda::convert(dfm_group(dfmat, groups = year), to = "data.frame") |>
  pivot_longer(-doc_id, names_to = "term", values_to = "count") |>
  rename(year = doc_id) |>
  mutate(year = as.integer(year))

cat(glue("Terms tracked: {n_distinct(kw_by_year$term)}\n"))
```

```
#> Terms tracked: 251
```

26.4.2 Growth rate detection

term	mean_growth	total
gender	433%	21
knowledge	396%	47
universities	356%	26
collaboration	343%	40
search	333%	21
open	318%	42
social	285%	43
model	252%	31
networks	239%	23
case	219%	21
news	217%	22
authors	210%	33
impact	202%	59
higher	194%	22
institutions	176%	22

```
growth <- kw_by_year |>
  group_by(term) |>
  arrange(year) |>
  mutate(
    growth = (count - lag(count)) / pmax(lag(count), 1),
    total = sum(count)
  ) |>
  ungroup() |>
  filter(!is.na(growth), total >= 20)

recent_growth <- growth |>
  filter(year >= 2021) |>
  group_by(term) |>
  summarise(mean_growth = mean(growth), total = first(total),
            .groups = "drop") |>
  filter(mean_growth > 0) |>
  arrange(desc(mean_growth))

recent_growth |>
  head(15) |>
  gt() |>
  fmt_percent(columns = mean_growth, decimals = 0)
```

26.4.3 Simple burst detection

```
detect_burst <- function(counts, threshold = 2) {
  mean_count <- mean(counts)
  sd_count <- sd(counts)
  if (sd_count == 0) return(rep(FALSE, length(counts)))
  z_scores <- (counts - mean_count) / sd_count
  z_scores > threshold
}

bursts <- kw_by_year |>
  group_by(term) |>
  filter(n() >= 5, sum(count) >= 20) |>
  arrange(year) |>
  mutate(is_burst = detect_burst(count, threshold = 1.5)) |>
  ungroup()

burst_summary <- bursts |>
  filter(is_burst) |>
  group_by(term) |>
  summarise(
    burst_years = paste(year, collapse = ", "),
    max_count = max(count),
    .groups = "drop"
  ) |>
  arrange(desc(max_count))

burst_summary |> head(15) |> gt()
```

26.4.4 Visualising keyword trajectories

```
top_burst_terms <- burst_summary |> head(6) |> pull(term)

burst_plot_data <- bursts |>
  filter(term %in% top_burst_terms)

ggplot(burst_plot_data, aes(x = year, y = count)) +
  geom_rect(data = burst_plot_data |> filter(is_burst),
            aes(xmin = year - 0.5, xmax = year + 0.5,
                ymin = -Inf, ymax = Inf),
            fill = "gold", alpha = 0.3, inherit.aes = FALSE) +
  geom_line(linewidth = 0.8) +
```

term	burst_years	max_count
data	2022	33
journals	2021	33
knowledge	2021	27
science	2022	27
scientific	2021	27
academic	2022	25
citation	2020	25
cited	2017	24
international	2016	23
citations	2022	22
impact	2016	22
studies	2017	21
articles	2019	20
number	2022	20
open	2020	19

```
  geom_point(size = 1.5) +
  facet_wrap(~ term, scales = "free_y", ncol = 3) +
  labs(x = "Year", y = "Frequency") +
  theme_sci(base_size = 9)
```

26.4.5 Rolling-window topic tracking

```
window_size <- 3
years <- sort(unique(text_df$year))
windows <- years[1:(length(years) - window_size + 1)]

window_top_terms <- map_dfr(windows, function(start_year) {
  end_year <- start_year + window_size - 1
  docs <- docvars(dfmat, "year") >= start_year &
          docvars(dfmat, "year") <= end_year
  if (sum(docs) < 10) return(tibble())
  top <- topfeatures(dfmat[docs, ], 10)
  tibble(window = glue("{start_year}-{end_year}"),
         term = names(top), frequency = unname(top))
})
```

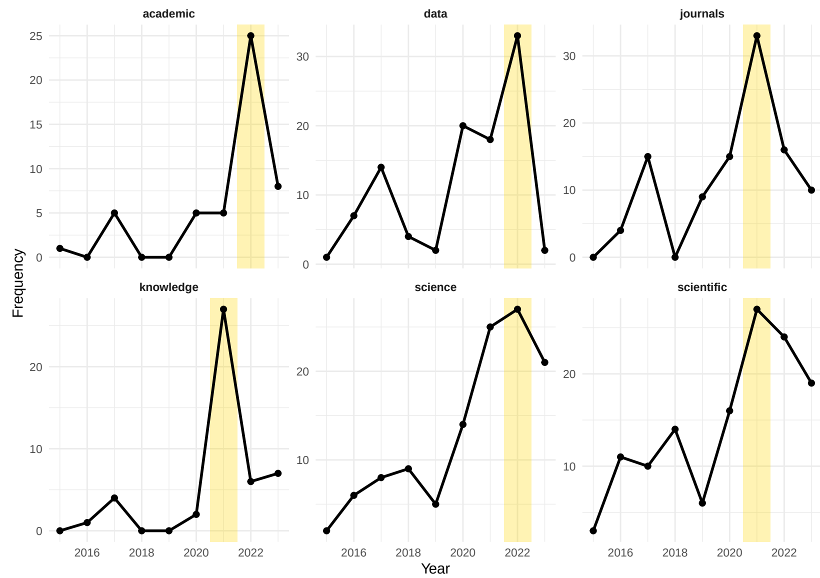

Figure 26.1: Temporal trajectories of emerging keywords with burst periods highlighted.

```
window_top_terms |>
  group_by(window) |>
  slice_max(frequency, n = 8) |>
  mutate(term = reorder_within(term, frequency, window)) |>
  ggplot(aes(x = frequency, y = term)) +
  geom_col(fill = palette_sci(1)) +
  facet_wrap(~ window, scales = "free_y", ncol = 3) +
  scale_y_reordered() +
  labs(x = "Frequency", y = NULL) +
  theme_sci(base_size = 8)
```

26.5 Diagnostics and interpretation

- **Baseline effects**: A keyword going from 2 to 6 occurrences shows 200% growth but may not represent meaningful emergence. Set minimum frequency thresholds before computing growth rates.
- **Database artefacts**: A newly indexed journal or a batch import of conference proceedings can create spurious “bursts”. Cross-check with independent evidence.

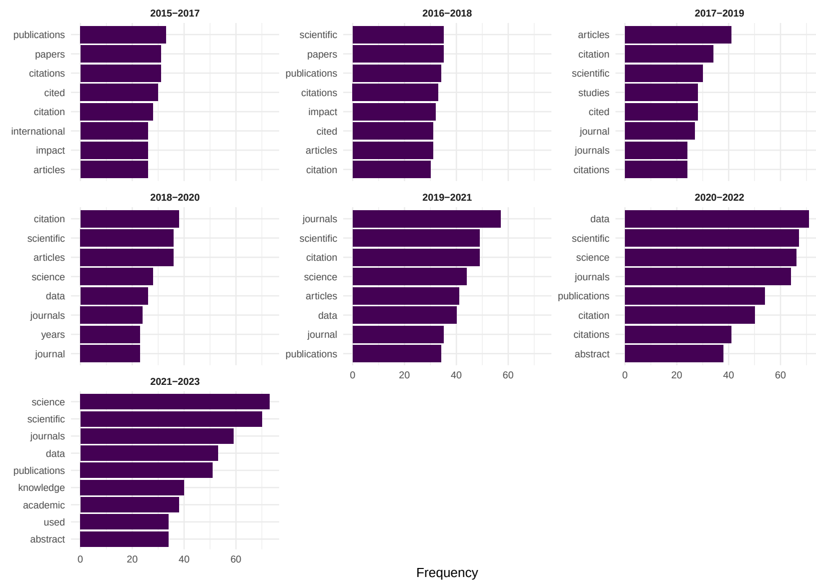

Figure 26.2: Top terms in each 3-year rolling window, showing vocabulary evolution.

- **Burst duration**: Short bursts (1 year) may reflect a single influential paper or a special issue. Sustained bursts (3+ years) more likely indicate genuine emergence.
- **Window sensitivity**: Rolling-window results change with window width. Report the chosen width and test sensitivity with wider and narrower windows.

26.6 Limitations and responsible use

26.7 Limitations and responsible use

- **Emergence is retrospective.** By the time bibliometric methods detect emergence, the topic has already gained momentum. These methods identify "recently emerged" topics, not future breakthroughs.
- **Coverage bias.** OpenAlex coverage has expanded rapidly, meaning recent years have more records. Growth rates may reflect database growth rather than intellectual emergence (Priem et al., 2022).
- **Language of emergence.** English-language terms dominate; emerging topics in non-English literatures may be invisible until translated or adopted by English-language publications.

- **Not predictive.** Burst detection identifies past patterns. It does not predict which topics will continue growing. Many bursts are transient (Kleinberg, 2003).
- **Strategic gaming.** If emergence metrics are used for funding decisions, researchers may strategically choose "hot" keywords, creating feedback loops (Hicks et al., 2015).

26.8 Common pitfalls

26.9 Common pitfalls

- **Confusing absolute growth with relative emergence.** A field that publishes 10,000 papers/year growing by 5% adds 500 papers. A niche area growing from 10 to 30 papers shows 200% growth. The niche area is "emerging" but the large field is adding more absolute volume.
- **Not controlling for corpus growth.** If the overall number of publications in the corpus grows, all keywords will grow. Use relative frequency (keyword count / total documents) rather than absolute counts.
- **Overreacting to single-year spikes.** One conference proceedings volume or special issue can create a one-year spike. Require multi-year sustained growth before declaring emergence.
- **Ignoring synonyms.** "Deep learning" and "neural networks" may both refer to the same emerging trend. Merge synonyms before analysis.

26.10 Exercises

1. **Relative burst detection.** Modify the burst detection to use relative frequency (keyword proportion) instead of absolute counts. Do the same terms still burst?

2. **Emergence by journal.** Compare the most frequent emerging keywords in two different journals. Do they share emerging topics or focus on different frontiers?

3. **Novelty scoring.** For each document, compute its cosine distance to the centroid of all prior-year documents (using TF-IDF vectors). Plot novelty scores over time. Are recent papers becoming more or less novel?

4. **Validation.** Select 5 terms identified as "emerging" by burst detection. Search for qualitative evidence (review articles, conference themes) confirming their emergence. What is the false positive rate?

26.11 Solutions

Solutions are provided in F.

26.12 Further reading

- Kleinberg (2003) — Bursty patterns in temporal data: the foundational burst detection algorithm.
- de Solla Price (1963) — Early observations on the growth of scientific literature and research fronts.
- Priem et al. (2022) — OpenAlex temporal coverage patterns relevant to emergence detection.
- Callon et al. (1983) — Co-word dynamics; an early approach to tracking topical evolution.

26.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4              Matrix_1.7-0              word2vec_0.4.1
#>  [5] topicmodels_0.2-17       quanteda.textstats_0.97.2 visNetwork_2.1.4
```

```
#>  [9] tidygraph_1.3.1          igraph_2.3.1            quanteda_4.4            pdftools_3.
#> [13] arrow_24.0.0             bibliometrix_5.4.0      RefManageR_1.4.0        bib2df_1.1.
#> [17] rcrossref_1.2.1          gt_1.3.0                tidytext_0.4.3          glue_1.8.1
#> [21] openalexR_3.0.1          lubridate_1.9.5         forcats_1.0.1           stringr_1.6
#> [25] dplyr_1.2.1              purrr_1.2.2             readr_2.2.0             tidyr_1.3.2
#> [29] tibble_3.3.1             ggplot2_4.0.3           tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3     rstudioapi_0.18.0       jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2           rmarkdown_2.31          fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1          base64enc_0.1-6         htmltools_0.5.9
#>  [16] curl_7.1.0            janeaustenr_1.0.0      cellranger_1.1.0        htmlwidgets_1.6.4
#>  [21] plyr_1.8.9            httr2_1.2.2            cachem_1.1.0            plotly_4.12.0
#>  [26] mime_0.13             lifecycle_1.0.5        pkgconfig_2.0.3         R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39          patchwork_1.3.2         shinycssloaders_1.1
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1        labeling_0.4.3          urltools_1.7.3.1
#>  [41] polyclip_1.10-7       httr_1.4.8             compiler_4.4.1          here_1.0.2
#>  [46] withr_3.0.2           S7_0.2.2               backports_1.5.1         viridis_0.6.5
#>  [51] MASS_7.3-60.2         rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.4.1
#>  [56] stopwords_2.3         zip_2.3.3              httpuv_1.6.17           rentrez_1.2.4
#>  [61] grid_4.4.1            stringdist_0.9.17      reshape2_1.4.5          generics_0.1.4
#>  [66] tzdb_0.5.0            rscopus_0.9.0          ca_0.71.1               data.table_1.18.4
#>  [71] xml2_1.5.2            utf8_1.2.6             ggrepel_0.9.8           pillar_1.11.1
#>  [76] vroom_1.7.1           later_1.4.8            tweenr_2.0.3            lattice_0.22-6
#>  [81] bit_4.6.0             tidyselect_1.2.1       tm_0.7-18               miniUI_0.1.2
#>  [86] gridExtra_2.3         NLP_0.3-2              bookdown_0.46           stats4_4.4.1
#>  [91] xfun_0.57             graphlayouts_1.2.3     matrixStats_1.5.0       DT_0.34.0
#>  [96] stringi_1.8.7         lazyeval_0.2.3         qpdf_1.4.1              yaml_2.3.12
#> [101] codetools_0.2-20      httpcode_0.3.0         cli_3.6.6               xtable_1.8-8
#> [106] Rcpp_1.1.1-1.1        readxl_1.4.5           triebeard_0.4.1         XML_3.99-0.23
#> [111] assertthat_0.2.1      pubmedR_1.0.2          slam_0.1-55             viridisLite_0.4.3
#> [116] crayon_1.5.3          openxlsx_4.2.8.1       rlang_1.2.0             fastmatch_1.1-8
```

Part VI

Advanced Topics

Chapter 27

Altmetrics

27.1 Learning objectives

After completing this chapter, you will be able to:

- Define altmetrics and explain how they complement citation-based indicators
- Retrieve altmetric-related data from OpenAlex (open access status, type, counts)
- Compute correlations between citation counts and alternative indicators
- Interpret altmetric signals critically — distinguishing attention from impact
- Discuss the limitations of altmetrics for research evaluation

27.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

27.3 Conceptual background

Citations measure one dimension of scholarly impact: influence on subsequent research. But scholarly work can also influence policy, education, clinical practice, and public understanding — none of which are captured by citation counts. **Altmetrics** (alternative metrics) attempt to fill this gap by tracking online traces of attention: social media mentions, news coverage, blog posts, Wikipedia citations, Mendeley reader counts, and policy document references.

The term was coined by Priem et al. (2010), who argued that the speed and diversity of online scholarly communication demanded new metrics beyond the Journal Impact Factor. Altmetrics are typically faster than citations (social media attention peaks within days of publication, while citations take years) and broader (they capture engagement by non-researchers).

However, altmetrics come with significant caveats. Social media attention correlates weakly with scientific quality. Metrics are easily gamed (bots, self-promotion). Coverage is uneven: biomedical research attracts far more Twitter/X attention than mathematics or engineering. And the platforms providing altmetric data (Altmetric.com, PlumX) are commercial products with proprietary algorithms (Hicks et al., 2015).

OpenAlex provides open-access status and citation counts but does not directly provide social media metrics. For altmetric data, external APIs (Altmetric.com, Crossref Event Data) are needed. In this chapter, we demonstrate what can be done with OpenAlex metadata alone and discuss how to extend the analysis with external sources.

27.4 Worked example

27.4.1 Fetching data with OA indicators

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)

works_slim <- works |>
  transmute(
    id, display_name,
    year = year(publication_date),
```

oa_status	n	pct
closed	216	54.0%
hybrid	103	25.8%
green	48	12.0%
bronze	33	8.2%

```
    cited_by_count,
    oa_status = oa_status,
    type,
    referenced_works_count
  )

cat(glue("Works retrieved: {nrow(works_slim)}\n"))
```

```
#> Works retrieved: 400
```

27.4.2 Open access as an altmetric dimension

```
oa_summary <- works_slim |>
  count(oa_status, sort = TRUE) |>
  mutate(pct = scales::percent(n / sum(n)))

oa_summary |> gt()
```

```
works_slim |>
  filter(!is.na(oa_status)) |>
  ggplot(aes(x = oa_status, y = cited_by_count + 1)) +
  geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
  scale_y_log10() +
  labs(x = "OA status", y = "Citations (log scale)") +
  theme_sci()
```

27.4.3 Citation count distribution

```
ggplot(works_slim, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 5, fill = palette_sci(1), colour = "white") +
  labs(x = "Citation count", y = "Number of papers") +
  theme_sci()
```

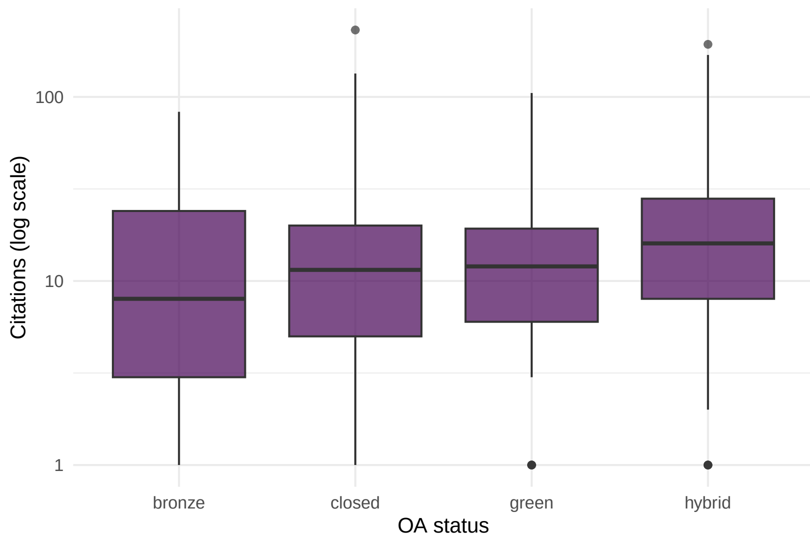

Figure 27.1: Citation distributions by open access status.

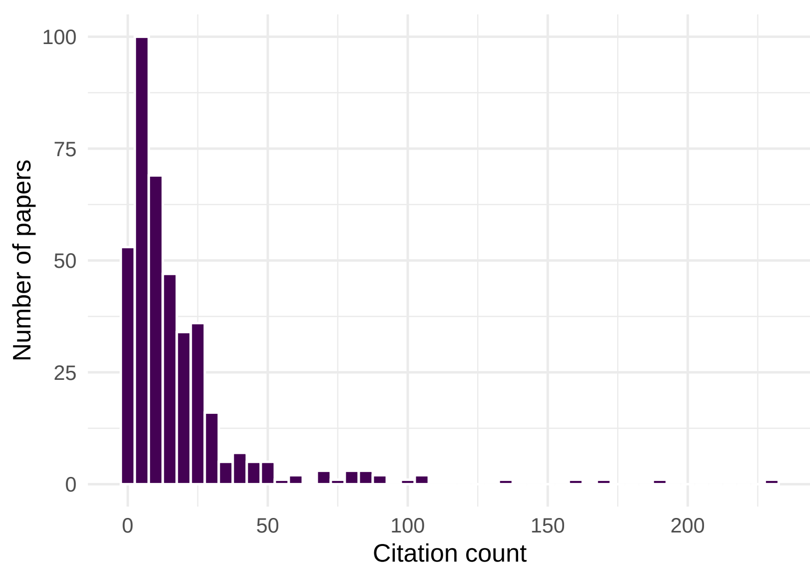

Figure 27.2: Citation distribution of the sample, showing the characteristic right skew.

27.4.4 Proxy altmetric: reference list length

Reference list length correlates with engagement breadth and can serve as a simple proxy for scholarly thoroughness.

```
ggplot(works_slim, aes(x = referenced_works_count, y = cited_by_count)) +
  geom_point(alpha = 0.3, colour = palette_sci(1)) +
  geom_smooth(method = "lm", se = FALSE, colour = palette_sci(2)[2]) +
  labs(x = "Number of references", y = "Citation count") +
  theme_sci()
```

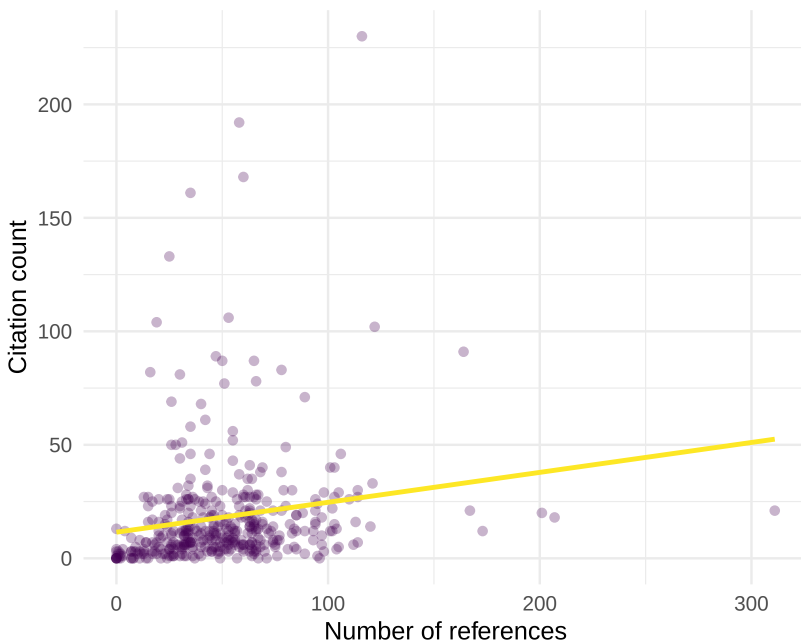

Figure 27.3: Relationship between reference list length and citation count.

```
cor_val <- cor(works_slim$referenced_works_count, works_slim$cited_by_count,
               use = "complete.obs")
cat(glue("Pearson correlation (references vs. citations): {round(cor_val, 3)}\n"))
```

```
#> Pearson correlation (references vs. citations): 0.173
```

27.5 Diagnostics and interpretation

- **OA advantage**: Open-access articles often receive more citations, but this may reflect self-selection (higher-quality papers are more likely to be made OA) rather than the causal effect of open access.
- **Platform coverage**: Social media altmetrics are biased toward English-language, biomedical, and controversial topics. Low altmetric scores do not indicate low impact.
- **Temporal dynamics**: Altmetric attention peaks immediately after publication and decays rapidly. Citations follow a slower, more sustained trajectory.
- **Gaming risk**: Social media metrics are more easily manipulated than citations. Interpret high altmetric scores with caution.

27.6 Limitations and responsible use

27.7 Limitations and responsible use

- **Attention impact.** A viral tweet about a paper does not mean the paper is scientifically significant. Altmetrics measure visibility, not quality (Hicks et al., 2015).
- **Disciplinary bias.** Fields with active social media communities (public health, climate science) generate far more altmetric data than fields that do not (pure mathematics, medieval history).
- **Platform dependence.** Altmetric data depends on commercial platforms whose coverage, algorithms, and existence are not guaranteed.
- **Never use altmetrics for individual evaluation.** The Leiden Manifesto and DORA both caution against using any single metric — including altmetrics — as a proxy for research quality (American Society for Cell Biology, 2012).

27.8 Common pitfalls

27.9 Common pitfalls

- **Treating Mendeley readers as "citations."** Mendeley reader counts reflect who saved a paper, not who cited it. The overlap is partial.
- **Comparing altmetrics across fields.** A paper in public health with 50 tweets is unremarkable; the same count in algebraic topology is extraordinary.

- **Ignoring bots and self-promotion.** Some social media mentions are from automated accounts or coordinated campaigns. Commercial altmetric services attempt to filter these, but coverage is imperfect.
- **Confusing the Altmetric Attention Score with a quality indicator.** The score is a weighted sum of online mentions, not a peer-reviewed quality assessment.

27.10 Exercises

1. **OA citation advantage.** For two journals with different OA rates, compare mean citations for OA vs. closed-access papers. Is the "OA advantage" consistent across journals?

2. **Temporal attention.** For a set of recent papers, track how citation counts change between 6 months and 2 years after publication. Do papers with high early attention (many references, rapid initial citations) sustain their advantage?

3. **Cross-field comparison.** Fetch samples from a biomedical journal and a mathematics journal. Compare their citation distributions and reference list lengths. What does this suggest about field-specific attention patterns?

27.11 Solutions

Solutions are provided in F.

27.12 Further reading

- Priem et al. (2010) — The altmetrics manifesto: the original call for alternative impact measures.
- Hicks et al. (2015) — The Leiden Manifesto: responsible metrics, including cautions about altmetrics.
- American Society for Cell Biology (2012) — DORA: principles for research assessment beyond the JIF.
- Priem et al. (2022) — OpenAlex open-access classification and metadata.

27.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATI
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0
#> [17] rcrossref_1.2.1           gt_1.3.0                  tidytext_0.4.3
#> [21] openalexR_3.0.1           lubridate_1.9.5           forcats_1.0.1
#> [25] dplyr_1.2.1               purrr_1.2.2               readr_2.2.0
#> [29] tibble_3.3.1              ggplot2_4.0.3             tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite
#>   [6] modeltools_0.2-24     farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1          base64enc_0.1-6        htmltools
#>  [16] curl_7.1.0            janeaustenr_1.0.0      cellranger_1.1.0       htmlwidge
#>  [21] plyr_1.8.9            httr2_1.2.2            cachem_1.1.0           plotly_4
#>  [26] mime_0.13             lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39          patchwork_1.3.2        shinycss
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1        labeling_0.4.3         urltools
#>  [41] mgcv_1.9-1            polyclip_1.10-7        httr_1.4.8             compiler
#>  [46] bit64_4.8.0           withr_3.0.2            S7_0.2.2               backports
#>  [51] ggforce_0.5.0         MASS_7.3-60.2          rappdirs_0.3.4         bibliome
#>  [56] otel_0.2.0            stopwords_2.3          zip_2.3.3              httpuv_1
#>  [61] nlme_3.1-164          promises_1.5.0         grid_4.4.1             stringdi
```

```
#>  [66] generics_0.1.4       gtable_0.3.6         tzdb_0.5.0           rscopus_0.9.0
#>  [71] data.table_1.18.4    hms_1.1.4            xml2_1.5.2           utf8_1.2.6
#>  [76] pillar_1.11.1        nsyllable_1.0.1      vroom_1.7.1          later_1.4.8
#>  [81] tweenr_2.0.3         lattice_0.22-6       FNN_1.1.4.1          bit_4.6.0
#>  [86] tm_0.7-18            miniUI_0.1.2         knitr_1.51           gridExtra_2.3
#>  [91] bookdown_0.46        stats4_4.4.1         crul_1.6.0           xfun_0.57
#>  [96] matrixStats_1.5.0    DT_0.34.0            humaniformat_0.6.0   stringi_1.8.7
#> [101] qpdf_1.4.1           yaml_2.3.12          evaluate_1.0.5       codetools_0.2-20
#> [106] cli_3.6.6            xtable_1.8-8         dichromat_2.0-0.1    Rcpp_1.1.1-1.1
#> [111] triebeard_0.4.1      XML_3.99-0.23        parallel_4.4.1       assertthat_0.2.1
#> [116] slam_0.1-55          viridisLite_0.4.3    scales_1.4.0         crayon_1.5.3
#> [121] rlang_1.2.0          fastmatch_1.1-8
```

Chapter 28

Open Science Indicators

28.1 Learning objectives

After completing this chapter, you will be able to:

- Classify open access types (gold, green, hybrid, bronze) using OpenAlex metadata
- Compute OA rates by year, journal, and discipline
- Track the growth of preprint posting and preprint-to-publication pathways
- Assess data availability statement coverage in a corpus
- Discuss the relationship between open science practices and citation impact

28.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

28.3 Conceptual background

The open science movement aims to make research processes and outputs freely accessible. Bibliometric analysis can measure the extent to which these ideals are being realised, track trends over time, and identify disparities across disciplines and geographies.

Open access (OA) is the most measurable dimension of open science. OpenAlex classifies each work's OA status using Unpaywall data: **gold** (published in a fully OA journal), **green** (available in a repository), **hybrid** (OA in a subscription journal via APC), **bronze** (free to read but without an explicit open license), or **closed** (Priem et al., 2022).

Preprints are manuscripts posted publicly before peer review. The growth of preprint servers (bioRxiv, arXiv, medRxiv) has accelerated since 2015, with a dramatic spike during the COVID-19 pandemic. Bibliometric analysis can track preprint posting rates, the time from preprint to journal publication, and whether preprints receive more citations than non-preprint papers.

Data availability statements (DAS) indicate whether the data underlying a study are accessible. Their prevalence is growing as journals adopt mandatory data-sharing policies, but the quality of DAS varies: many state "data available upon request" without providing actual access.

Measuring open science is important for policy. Funders increasingly require OA publication and data sharing; institutions report OA rates as indicators of compliance and impact. However, Hicks et al. (2015) caution that metrics-driven OA mandates can produce perverse incentives (predatory journals, superficial compliance).

28.4 Worked example

28.4.1 OA rates by year

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 600, seed = 42)
)

works_oa <- works |>
```

```
  transmute(
    id, year = year(publication_date),
    oa_status = oa_status,
    cited_by_count
  )

cat(glue("Works: {nrow(works_oa)}\n"))
```

```
#> Works: 600
```

```
oa_by_year <- works_oa |>
  count(year, oa_status) |>
  group_by(year) |>
  mutate(pct = n / sum(n)) |>
  ungroup()

ggplot(oa_by_year, aes(x = factor(year), y = pct, fill = oa_status)) +
  geom_col() +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(n_distinct(oa_by_year$oa_status))) +
  labs(x = "Publication year", y = "Proportion", fill = "OA status") +
  theme_sci()
```

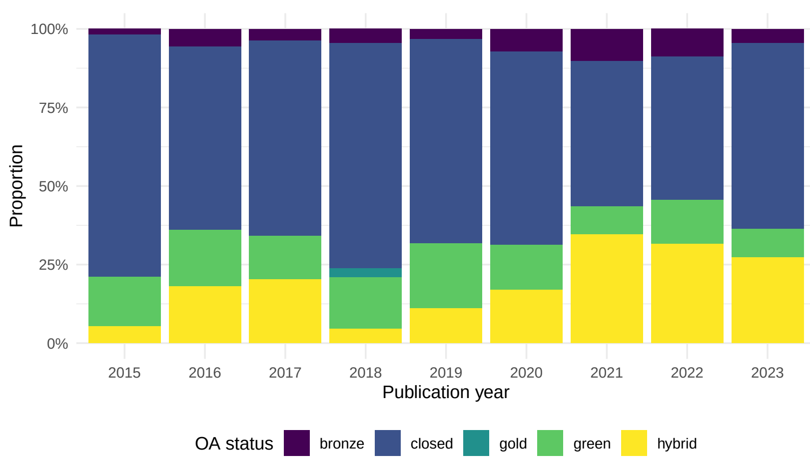

Figure 28.1: Open access rates by year and OA type.

is_oa	n	mean_cites	median_cites
FALSE	363	27.4	14
TRUE	237	62.3	19

28.4.2 OA citation advantage

```
oa_cites <- works_oa |>
  filter(!is.na(oa_status)) |>
  mutate(is_oa = oa_status != "closed") |>
  group_by(is_oa) |>
  summarise(
    n = n(),
    mean_cites = round(mean(cited_by_count), 1),
    median_cites = median(cited_by_count),
    .groups = "drop"
  )

oa_cites |> gt()
```

28.4.3 Document type breakdown

```
works |>
  count(type, oa_status) |>
  group_by(type) |>
  mutate(pct = n / sum(n)) |>
  ungroup() |>
  ggplot(aes(x = pct, y = type, fill = oa_status)) +
  geom_col() +
  scale_x_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(n_distinct(works$oa_status))) +
  labs(x = "Proportion", y = NULL, fill = "OA status") +
  theme_sci()
```

28.4.4 Summary table

```
works_oa |>
  count(oa_status, sort = TRUE) |>
```

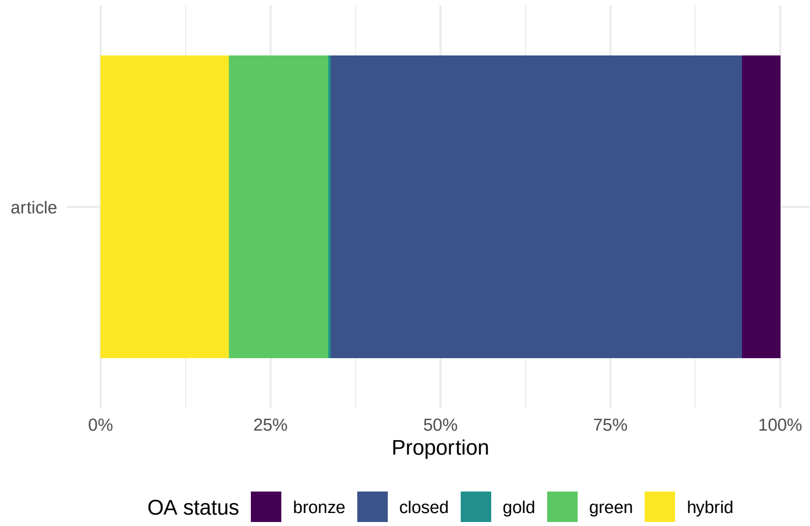

Figure 28.2: OA status by document type.

OA Status	Count	Percentage
closed	363	60.5%
hybrid	113	18.8%
green	88	14.7%
bronze	34	5.7%
gold	2	0.3%

```
mutate(pct = scales::percent(n / sum(n))) |>
gt() |>
cols_label(oa_status = "OA Status", n = "Count", pct = "Percentage")
```

28.5 Diagnostics and interpretation

- **Classification accuracy**: OpenAlex OA classification relies on Unpaywall, which may lag behind publisher changes. Spot-check a sample of "closed" articles to verify they are truly behind a paywall.
- **Bronze ambiguity**: Bronze OA means free-to-read without an explicit license. Publishers can revoke access at any time. Do not treat bronze as equivalent to gold or green.
- **Year effects**: OA rates have risen over time due to funder mandates and

policy changes. Compare year-normalised rates rather than pooling across years.
- **Self-selection**: Higher OA rates among highly cited papers may reflect self-selection (better papers are more likely to be made OA) rather than a causal OA effect.

28.6 Limitations and responsible use

28.7 Limitations and responsible use

- **OA quality.** Open access status is a property of the publication venue and access model, not the research quality. Predatory journals are often gold OA.
- **Coverage gaps.** Green OA via institutional repositories is underreported in some databases. Actual OA rates may be higher than measured.
- **Data sharing claims vs. reality.** A data availability statement saying "data available upon request" often results in no response when requests are made. Measuring DAS presence overestimates actual data sharing.
- **Equity implications.** Gold OA via article processing charges shifts costs from readers to authors, potentially disadvantaging researchers without institutional funding (Hicks et al., 2015).

28.8 Common pitfalls

28.9 Common pitfalls

- **Treating all OA as equivalent.** Gold, green, hybrid, and bronze OA have very different implications for access, sustainability, and cost.
- **Ignoring embargo periods.** Green OA articles may have 6–24 month embargoes before they become freely available. They are "closed" during the embargo.
- **Double-counting.** An article available in both a gold OA journal and a green repository should be counted once. OpenAlex classifies by the "best" OA type.
- **Using OA rate as a performance indicator.** OA rates reflect policy and funding environment, not research quality. Mandated OA inflates rates without changing the underlying work.

28.10 Exercises

1. **Cross-journal OA comparison.** Compare OA rates for three journals in different disciplines. Which discipline has the highest gold OA rate?

2. **Temporal OA growth.** Compute the year-over-year growth rate of gold OA articles in your corpus. Has growth accelerated since 2020?

3. **OA and collaboration.** Is there a relationship between the number of authors on a paper and its OA status? Hypothesise why.

28.11 Solutions

Solutions are provided in F.

28.12 Further reading

- Priem et al. (2022) — OpenAlex OA classification methodology.
- Hicks et al. (2015) — The Leiden Manifesto: principles for responsible metrics, including OA considerations.
- American Society for Cell Biology (2012) — DORA: recommendations for research assessment beyond journal metrics.

28.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
```

```
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4              Matrix_1.7-0             word2vec_0.4.1
#>  [5] topicmodels_0.2-17      quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1         igraph_2.3.1             quanteda_4.4
#> [13] arrow_24.0.0            bibliometrix_5.4.0       RefManageR_1.4.0
#> [17] rcrossref_1.2.1         gt_1.3.0                 tidytext_0.4.3
#> [21] openalexR_3.0.1         lubridate_1.9.5          forcats_1.0.1
#> [25] dplyr_1.2.1             purrr_1.2.2              readr_2.2.0
#> [29] tibble_3.3.1            ggplot2_4.0.3            tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltools
#>  [16] curl_7.1.0            janeaustenr_1.0.0     cellranger_1.1.0      htmlwidge
#>  [21] plyr_1.8.9            httr2_1.2.2           cachem_1.1.0          plotly_4
#>  [26] mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3       R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycss
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3        urltools
#>  [41] mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8            compiler
#>  [46] bit64_4.8.0           withr_3.0.2           S7_0.2.2              backports
#>  [51] ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4        bibliome
#>  [56] otel_0.2.0            stopwords_2.3         zip_2.3.3             httpuv_1
#>  [61] nlme_3.1-164          promises_1.5.0        grid_4.4.1            stringdi
#>  [66] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_
#>  [71] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2
#>  [76] pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1           later_1.
#>  [81] tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1           bit_4.6.
#>  [86] tm_0.7-18             miniUI_0.1.2          knitr_1.51            gridExtra
#>  [91] bookdown_0.46         stats4_4.4.1          crul_1.6.0            xfun_0.5
#>  [96] matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0    stringi_
#> [101] qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5        codetools
#> [106] cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1
#> [111] triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1        assertth
#> [116] slam_0.1-55           viridisLite_0.4.3     scales_1.4.0          crayon_1
#> [121] rlang_1.2.0           fastmatch_1.1-8
```

Chapter 29

Gender and Equity Analysis

29.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the challenges of gender inference from author names
- Apply name-based gender classification and assess its limitations
- Analyse gender representation in authorship positions (first, last, corresponding)
- Compute gender-disaggregated citation and productivity statistics
- Articulate why gender is not binary and why name-based inference introduces systematic bias

29.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

29.3 Conceptual background

Research on gender disparities in science has documented persistent gaps in publication rates, citation counts, funding success, and career progression. Bibliometric analysis can quantify these gaps at scale, but the methods used raise serious ethical and technical concerns.

Gender inference from names is the most common approach. Services like genderize.io, Gender API, and the R package `gender` assign a probabilistic gender (typically male/female) based on first names and country of origin. This approach has fundamental limitations: it enforces a binary classification, it fails for names common across genders, it has low accuracy for non-Western names, and it misclassifies non-binary and transgender researchers. Despite these limitations, name-based inference remains widely used because self-reported gender data is rarely available in bibliometric databases.

Authorship position carries different significance across disciplines. In biomedical sciences, the last author is typically the senior researcher and the first author the primary contributor. In other fields, authorship is alphabetical. Gender analysis of authorship positions must account for these disciplinary conventions.

Documented gender disparities include: women are underrepresented as first and last authors in most fields (Larivière et al., 2013); women's papers receive fewer citations on average, even after controlling for field and year; and women are less likely to be invited as reviewers or editors. These patterns reflect systemic inequities, not differences in research quality.

Hicks et al. (2015) and American Society for Cell Biology (2012) both emphasise that metrics should not be used in ways that perpetuate existing biases. Gender analysis in bibliometrics should aim to *reveal* disparities for policy intervention, not to evaluate individual researchers.

29.4 Worked example

29.4.1 Extracting author data

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 300, seed = 42)
)
```

```
authors <- works |>
  select(work_id = id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  select(work_id, author_name = authorships_display_name,
         position = authorships_id) |>
  mutate(
    first_name = str_extract(author_name, "^\\S+"),
    first_name = str_to_lower(first_name)
  ) |>
  filter(!is.na(first_name), nchar(first_name) >= 2)

cat(glue("Author records: {nrow(authors)}\n"))
```

```
#> Author records: 963
```

```
cat(glue("Unique first names: {n_distinct(authors$first_name)}\n"))
```

```
#> Unique first names: 751
```

29.4.2 Simple name-based gender proxy

We use a minimal heuristic for demonstration. In production work, use a validated gender-inference service with probabilistic output and country context.

```
common_female <- c("maria", "anna", "li", "sarah", "jennifer", "jessica",
                   "elena", "nina", "laura", "julia", "diana", "sandra")
common_male <- c("john", "david", "michael", "james", "robert", "peter",
                 "mark", "thomas", "paul", "daniel", "andreas", "martin")

authors <- authors |>
  mutate(
    gender_proxy = case_when(
      first_name %in% common_female ~ "female",
      first_name %in% common_male ~ "male",
      TRUE ~ NA_character_
    )
  )

classified <- authors |> filter(!is.na(gender_proxy))
cat(glue("Classified: {nrow(classified)} / {nrow(authors)} ({scales::percent(nrow(classified)/nro
```

```
#> Classified: 44 / 963 (5%)
```

29.4.3 Gender representation

```
classified |>
  count(gender_proxy) |>
  mutate(pct = n / sum(n)) |>
  ggplot(aes(x = gender_proxy, y = pct, fill = gender_proxy)) +
  geom_col(show.legend = FALSE) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Inferred gender", y = "Proportion") +
  theme_sci()
```

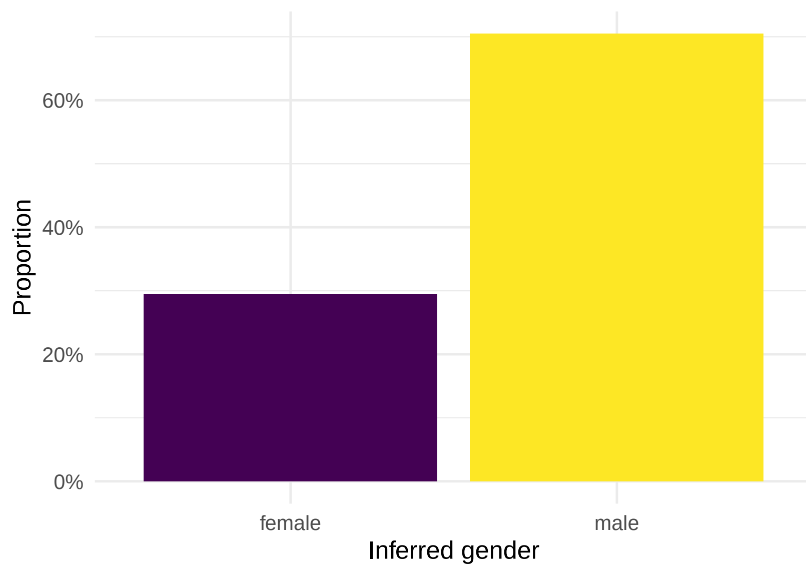

Figure 29.1: Gender representation among classifiable author names.

29.4.4 Authorship position analysis

```
position_data <- works |>
  select(work_id = id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  group_by(work_id) |>
  mutate(
    n_authors = n(),
```

author_position	gender_proxy	n	pct
first	female	4	0.267
first	male	11	0.733
last	female	4	0.286
last	male	10	0.714
middle	female	5	0.333
middle	male	10	0.667

```
    author_position = case_when(
      row_number() == 1 ~ "first",
      row_number() == n() & n() > 1 ~ "last",
      TRUE ~ "middle"
    )
  ) |>
  ungroup() |>
  mutate(first_name = str_to_lower(str_extract(authorships_display_name, "^\\S+"))) |>
  mutate(
    gender_proxy = case_when(
      first_name %in% common_female ~ "female",
      first_name %in% common_male ~ "male",
      TRUE ~ NA_character_
    )
  ) |>
  filter(!is.na(gender_proxy))

position_summary <- position_data |>
  count(author_position, gender_proxy) |>
  group_by(author_position) |>
  mutate(pct = n / sum(n)) |>
  ungroup()

position_summary |> gt()
```

```
ggplot(position_summary, aes(x = author_position, y = pct, fill = gender_proxy)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Author position", y = "Proportion", fill = "Inferred gender") +
  theme_sci()
```

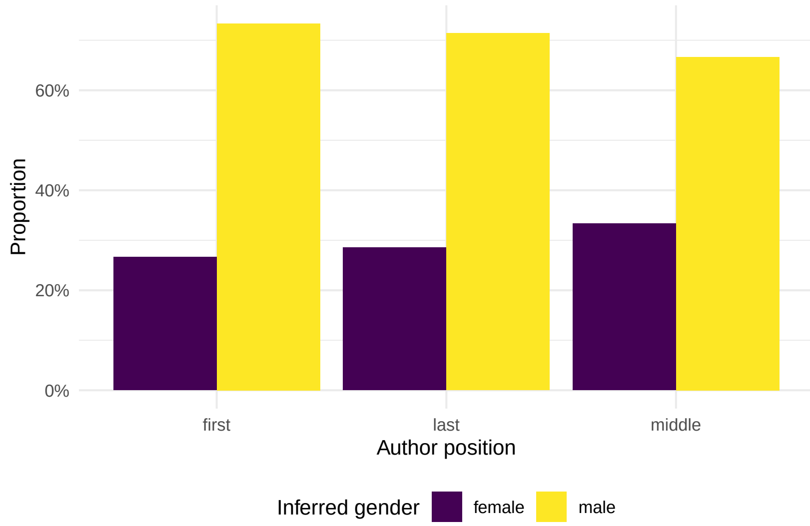

Figure 29.2: Gender representation by authorship position.

29.5 Diagnostics and interpretation

- **Classification rate**: A low classification rate ($< 40\%$) means results may not be representative. Names from East Asia, South Asia, and Africa are particularly poorly served by English-trained gender classifiers.
- **Confidence thresholds**: Use probabilistic classifiers with confidence thresholds (e.g., only classify names with $> 90\%$ probability). Report the threshold and the proportion of unclassified names.
- **Disciplinary norms**: Authorship order conventions vary. In fields with alphabetical ordering, "first author" has no special significance.
- **Sample size**: Gender disparities may be small. Ensure sample sizes are sufficient for meaningful comparisons.

29.6 Limitations and responsible use

29.7 Limitations and responsible use

- **Gender is not binary.** Name-based inference forces a male/female classification that excludes non-binary, genderqueer, and gender-non-conforming researchers. This is a fundamental limitation, not merely a technical one.

- **Name-based inference is systematically biased.** Accuracy is lower for non-Western names, transliterated names, and unisex names. Results should always report classification accuracy and unclassified rates by name origin.
- **Correlation is not causation.** If women receive fewer citations, this may reflect systemic bias in citing behaviour, differences in subfield distribution, career interruptions, or other structural factors — not differences in research quality.
- **Privacy and consent.** Inferring gender from names without consent raises ethical concerns. Report aggregate statistics, never individual-level gender assignments.
- **Use for equity, not evaluation.** Gender analysis should identify systemic disparities for policy intervention, never evaluate individual researchers (Hicks et al., 2015; American Society for Cell Biology, 2012).

29.8 Common pitfalls

29.9 Common pitfalls

- **Reporting gender ratios without uncertainty.** Name-based inference is probabilistic. Report confidence intervals or sensitivity analyses.
- **Ignoring unclassified names.** Dropping unclassified names biases the sample toward Western, easily classified names.
- **Assuming authorship order is universal.** Alphabetical ordering in mathematics and economics means authorship position analysis is meaningless.
- **Conflating gender with sex.** Bibliometric gender analysis infers social gender from names, not biological sex. The two are distinct concepts.

29.10 Exercises

1. **Cross-journal comparison.** Compare gender representation across three journals in different fields. Are there differences in the proportion of female first authors?

2. **Temporal trends.** Track the proportion of female first authors by year. Is there evidence of increasing representation over time?

3. **Citation gap.** Compare mean citations for papers with female vs. male first authors, controlling for publication year. What does the gap look like?

29.11 Solutions

Solutions are provided in F.

29.12 Further reading

- Larivière et al. (2013) — Large-scale analysis of gender disparities in scientific authorship.
- Hicks et al. (2015) — The Leiden Manifesto: responsible metrics that do not perpetuate bias.
- American Society for Cell Biology (2012) — DORA: recommendations against metrics-based evaluation that disadvantages underrepresented groups.

29.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#> 
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#> 
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATI
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#> 
#> time zone: UTC
#> tzcode source: system (glibc)
#> 
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#> 
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4
```

```
#> [13] arrow_24.0.0          bibliometrix_5.4.0      RefManageR_1.4.0        bib2df_1.1.
#> [17] rcrossref_1.2.1       gt_1.3.0                tidytext_0.4.3          glue_1.8.1
#> [21] openalexR_3.0.1       lubridate_1.9.5         forcats_1.0.1           stringr_1.6
#> [25] dplyr_1.2.1           purrr_1.2.2             readr_2.2.0             tidyr_1.3.2
#> [29] tibble_3.3.1          ggplot2_4.0.3           tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9
#>  [16] curl_7.1.0            janeaustenr_1.0.0     cellranger_1.1.0      htmlwidgets_1.6.4
#>  [21] plyr_1.8.9            httr2_1.2.2           cachem_1.1.0          plotly_4.12.0
#>  [26] mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3       R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycssloaders_1.1
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3        urltools_1.7.3.1
#>  [41] mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8            compiler_4.4.1
#>  [46] bit64_4.8.0           withr_3.0.2           S7_0.2.2              backports_1.5.1
#>  [51] ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4        bibliometrixData_0.
#>  [56] otel_0.2.0            stopwords_2.3         zip_2.3.3             httpuv_1.6.17
#>  [61] nlme_3.1-164          promises_1.5.0        grid_4.4.1            stringdist_0.9.17
#>  [66] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_0.9.0
#>  [71] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2.6
#>  [76] pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1           later_1.4.8
#>  [81] tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1           bit_4.6.0
#>  [86] tm_0.7-18             miniUI_0.1.2          knitr_1.51            gridExtra_2.3
#>  [91] bookdown_0.46         stats4_4.4.1          crul_1.6.0            xfun_0.57
#>  [96] matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0    stringi_1.8.7
#> [101] qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5        codetools_0.2-20
#> [106] cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1.1-1.1
#> [111] triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1        assertthat_0.2.1
#> [116] slam_0.1-55           viridisLite_0.4.3     scales_1.4.0          crayon_1.5.3
#> [121] rlang_1.2.0           fastmatch_1.1-8
```

Chapter 30

Retraction Analysis

30.1 Learning objectives

After completing this chapter, you will be able to:

- Identify retracted works in OpenAlex using type and metadata filters
- Analyse retraction rates by year, journal, and field
- Examine citation patterns before and after retraction
- Discuss the limitations of bibliometric approaches to research integrity
- Explain why retracted papers continue to accumulate citations

30.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

30.3 Conceptual background

Retraction is the formal mechanism by which the scholarly record corrects itself. A paper is retracted when it is found to contain fabricated data, serious errors, plagiarism, or ethical violations. In theory, retraction should halt the influence of flawed work. In practice, retracted papers continue to be cited — sometimes for years or decades after retraction.

The growth in retractions has been dramatic: from a few dozen per year in the 1990s to over 10,000 in recent years. This increase reflects both more misconduct detection (plagiarism software, image forensics) and more publications overall. Retraction rates vary by field: biomedical sciences have the highest absolute numbers, partly because of larger publication volumes and greater scrutiny.

Citation persistence is the most studied bibliometric aspect of retractions. Studies consistently find that retracted papers continue to accumulate positive citations (i.e., citations that treat the retracted work as valid) long after retraction. This occurs because: (1) citing authors may not be aware of the retraction; (2) database retraction notices are inconsistently displayed; (3) citations in already-published papers are not retroactively removed.

OpenAlex marks retracted works with a specific type or retraction flag, though coverage depends on publisher reporting. The Retraction Watch database provides the most comprehensive curated list of retractions but is not directly integrated into most bibliometric databases.

Bibliometric analysis of retractions must be approached carefully. The goal is to improve the self-correction mechanism, not to stigmatise individuals or institutions.

30.4 Worked example

30.4.1 Finding retracted works

```r
retracted <- oa_fetch(
  entity = "works",
  is_retracted = "true",
  from_publication_date = "2010-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 200, seed = 42)
)

cat(glue("Retracted works retrieved: {nrow(retracted)}\n"))
```

```
#> Retracted works retrieved: 200
```

year	n
2010	3
2011	6
2012	3
2013	10
2014	8
2015	5
2016	6
2017	4
2018	14
2019	10
2020	14
2021	29
2022	38
2023	50

```
retraction_years <- retracted |>
  mutate(year = year(publication_date)) |>
  count(year, sort = FALSE)

retraction_years |> gt()
```

30.4.2 Citation counts of retracted papers

```
ggplot(retracted, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 5, fill = palette_sci(1), colour = "white") +
  labs(x = "Citations received", y = "Number of retracted papers") +
  theme_sci()
```

```
cite_stats <- retracted |>
  summarise(
    n = n(),
    mean_cites = round(mean(cited_by_count), 1),
    median_cites = median(cited_by_count),
    max_cites = max(cited_by_count),
    pct_uncited = scales::percent(mean(cited_by_count == 0))
  )
```

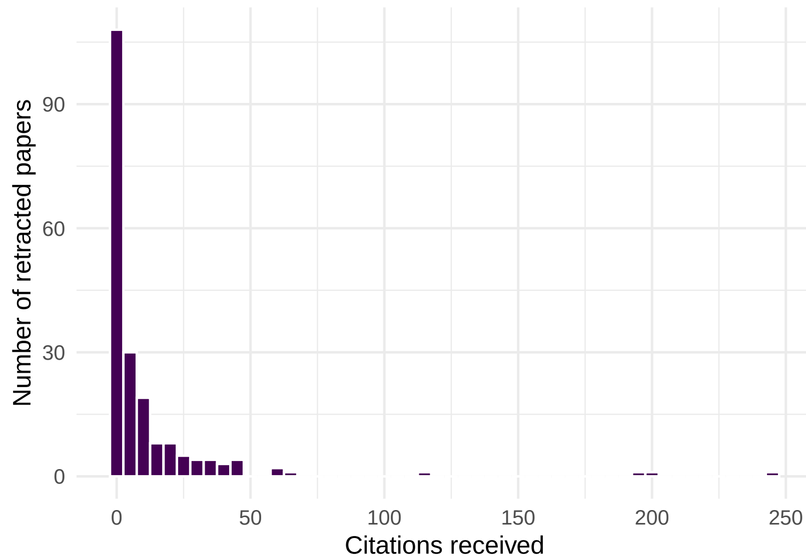

Figure 30.1: Citation count distribution of retracted papers.

n	mean_cites	median_cites	max_cites	pct_uncited
200	11.5	2	245	32%

```
cite_stats |> gt()
```

30.4.3 Retracted papers by source

```
retracted |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, sort = TRUE) |>
  head(10) |>
  mutate(source_display_name = fct_reorder(source_display_name, n)) |>
  ggplot(aes(x = n, y = source_display_name)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Retracted papers", y = NULL) +
  theme_sci()
```

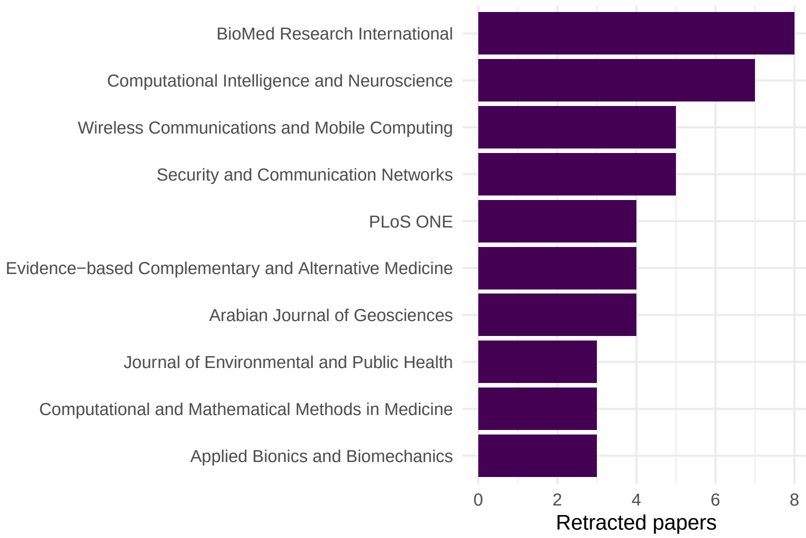

Figure 30.2: Top journals with the most retracted papers in the sample.

30.4.4 Citation persistence over time

```
retracted |>
  mutate(age = 2024 - year(publication_date)) |>
  filter(age >= 1, age <= 12) |>
  group_by(age) |>
  summarise(mean_cites = mean(cited_by_count), n = n(), .groups = "drop") |>
  ggplot(aes(x = age, y = mean_cites)) +
  geom_line(linewidth = 1, colour = palette_sci(1)) +
  geom_point(size = 2, colour = palette_sci(1)) +
  labs(x = "Years since publication", y = "Mean citations") +
  theme_sci()
```

30.5 Diagnostics and interpretation

- **Coverage**: OpenAlex retraction coverage depends on publisher metadata. Not all retractions are flagged. Cross-check with Retraction Watch for comprehensive coverage.
- **Retraction vs. publication date**: The publication date is when the paper was published, not when it was retracted. The gap between these dates varies from months to decades.

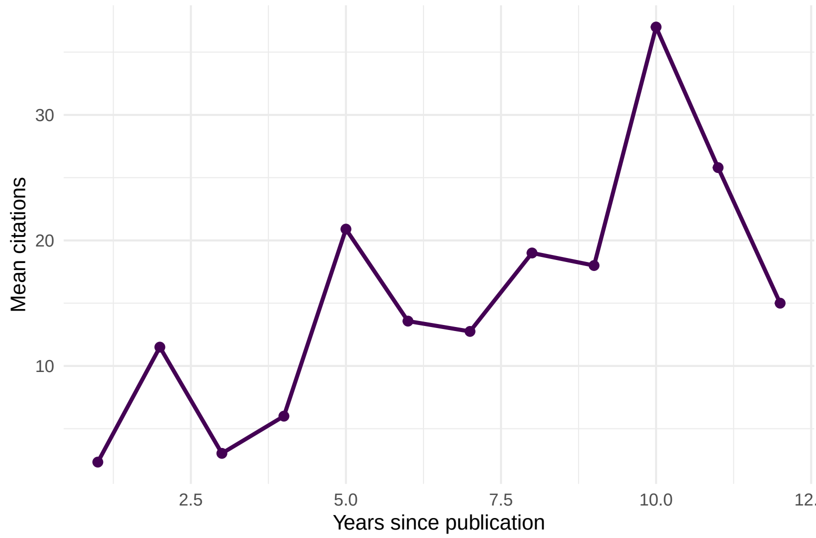

Figure 30.3: Mean citations by publication age for retracted papers.

- **Positive vs. negative citations**: Citation counts do not distinguish between "this paper found X" (positive, problematic) and "this paper was retracted for Y" (negative, appropriate). Manual checking of citing contexts is needed.
- **Denominator**: Retraction *rates* require the total number of publications as denominator. Raw retraction counts favour large journals.

30.6 Limitations and responsible use

30.7 Limitations and responsible use

- **Retractions are not all equal.** Some retractions reflect honest error, others fraud. The reason for retraction matters for interpretation but is often not coded in bibliometric databases.
- **Stigmatisation risk.** Listing authors or institutions with high retraction counts can be stigmatising. Report aggregate patterns, not individual-level data, unless specifically studying misconduct.
- **Database lag.** Retraction notices may take months or years to propagate through databases. Recent retraction data is incomplete.
- **Self-correction is broader than retraction.** Corrections, expressions of concern, and errata also contribute to self-correction but are harder to track bibliometrically.

- **Do not equate retractions with research quality.** Fields with more scrutiny detect more problems. A high retraction rate may indicate effective oversight, not worse science (Hicks et al., 2015).

30.8 Common pitfalls

30.9 Common pitfalls

- **Counting retractions without denominators.** A journal with 10 retractions from 50,000 papers has a lower retraction rate than one with 5 from 1,000.
- **Assuming all post-retraction citations are problematic.** Some citing papers may specifically discuss the retraction or cite the paper for methodology that remains valid despite data issues.
- **Ignoring retraction reasons.** Honest errors should be distinguished from misconduct in any analysis of research integrity.
- **Using retraction analysis for individual evaluation.** Authors of retracted papers may be victims of co-author misconduct. Never use retraction counts as performance indicators.

30.10 Exercises

1. **Retraction rate by field.** Fetch retracted works from two different fields. Compare the retraction rate (retractions per 1,000 publications). Account for corpus size.

2. **Citation trajectory.** For a retracted paper with many citations, examine when the citations accumulated. Were most citations received before or after the retraction?

3. **Retraction-citation correlation.** Is there a relationship between a paper's citation count at the time of retraction and the number of post-retraction citations it receives?

30.11 Solutions

Solutions are provided in F.

30.12 Further reading

- Fang et al. (2012) — Large-scale analysis of retraction patterns and their causes.
- Hicks et al. (2015) — The Leiden Manifesto: responsible use of metrics in the context of research integrity.
- Garfield (1955) — Citation indexing: the infrastructure that makes retraction tracking possible.

30.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#> 
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#> 
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#> 
#> time zone: UTC
#> tzcode source: system (glibc)
#> 
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#> 
#> other attached packages:
#>  [1] uwot_0.2.4               Matrix_1.7-0             word2vec_0.4.1
#>  [5] topicmodels_0.2-17       quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1          igraph_2.3.1             quanteda_4.4
#> [13] arrow_24.0.0             bibliometrix_5.4.0       RefManageR_1.4.0
#> [17] rcrossref_1.2.1          gt_1.3.0                 tidytext_0.4.3
#> [21] openalexR_3.0.1          lubridate_1.9.5          forcats_1.0.1
#> [25] dplyr_1.2.1              purrr_1.2.2              readr_2.2.0
#> [29] tibble_3.3.1             ggplot2_4.0.3            tidyverse_2.0.0
#> 
```

```
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9
#>  [16] curl_7.1.0            janeaustenr_1.0.0     cellranger_1.1.0      htmlwidgets_1.6.4
#>  [21] plyr_1.8.9            httr2_1.2.2           cachem_1.1.0          plotly_4.12.0
#>  [26] mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3       R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycssloaders_1.1
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3        urltools_1.7.3.1
#>  [41] mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8            compiler_4.4.1
#>  [46] bit64_4.8.0           withr_3.0.2           S7_0.2.2              backports_1.5.1
#>  [51] ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4        bibliometrixData_0.
#>  [56] otel_0.2.0            stopwords_2.3         zip_2.3.3             httpuv_1.6.17
#>  [61] nlme_3.1-164          promises_1.5.0        grid_4.4.1            stringdist_0.9.17
#>  [66] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_0.9.0
#>  [71] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2.6
#>  [76] pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1           later_1.4.8
#>  [81] tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1           bit_4.6.0
#>  [86] tm_0.7-18             miniUI_0.1.2          knitr_1.51            gridExtra_2.3
#>  [91] bookdown_0.46         stats4_4.4.1          crul_1.6.0            xfun_0.57
#>  [96] matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0    stringi_1.8.7
#> [101] qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5        codetools_0.2-20
#> [106] cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1.1-1.1
#> [111] triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1        assertthat_0.2.1
#> [116] slam_0.1-55           viridisLite_0.4.3     scales_1.4.0          crayon_1.5.3
#> [121] rlang_1.2.0           fastmatch_1.1-8
```

Chapter 31

Funding Analysis and Grant-to-Publication Linkage

31.1 Learning objectives

After completing this chapter, you will be able to:

- Extract funder information from OpenAlex work records
- Compute funder-level publication counts and citation impact
- Analyse co-funding patterns (papers acknowledging multiple funders)
- Interpret funding data critically, accounting for coverage and attribution challenges
- Discuss why funding-output linkage is inherently imprecise

31.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
```

```r
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

31.3 Conceptual background

Funding agencies invest billions in research and want to understand what their investment produces. **Funding acknowledgement analysis** extracts funder names and grant identifiers from publication metadata or full text, linking grants to the papers they supported.

OpenAlex includes funder information for many works, derived from Crossref's funding metadata (which in turn comes from publisher-reported FundRef data). Coverage varies: journals that participate in Crossref's FundRef initiative have good coverage; others may have no funding data at all (Priem et al., 2022).

Key analytical questions include: How many publications does a funder support? What is the citation impact of funded vs. unfunded research? Which funders frequently co-fund research? How does a funder's portfolio map onto the landscape of science?

The fundamental challenge is **attribution**: a paper acknowledging three funders and seven authors is the result of contributions from all of them. Assigning full credit to each funder double-counts; fractional credit is more principled but harder to implement. Similar issues arise with multi-source funding: a paper may acknowledge an infrastructure grant, a project grant, and a fellowship, each with different levels of contribution.

31.4 Worked example

31.4.1 Extracting funder data

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)

funded <- works |>
  select(id, display_name, cited_by_count, funders) |>
  filter(map_lgl(funders, \(g) !is.null(g) && length(g) > 0))
```

```
cat(glue("Works with funder data: {nrow(funded)} / {nrow(works)} ({scales::percent(nrow(funded)/n
```

```
#> Works with funder data: 400 / 400 (100%)
```

```
funder_data <- funded |>
  unnest(funders, names_sep = "_") |>
  select(work_id = id, funder = funders_display_name,
         cited_by_count)

cat(glue("Funder-work pairs: {nrow(funder_data)}\n"))
```

```
#> Funder-work pairs: 534
```

```
cat(glue("Unique funders: {n_distinct(funder_data$funder)}\n"))
```

```
#> Unique funders: 185
```

31.4.2 Top funders by publication count

```
funder_counts <- funder_data |>
  count(funder, sort = TRUE)

funder_counts |>
  head(15) |>
  mutate(funder = fct_reorder(funder, n)) |>
  ggplot(aes(x = n, y = funder)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Publications supported", y = NULL) +
  theme_sci()
```

31.4.3 Citation impact by funder

```
funder_impact <- funder_data |>
  group_by(funder) |>
  summarise(
    n_pubs = n(),
    mean_cites = round(mean(cited_by_count), 1),
    median_cites = median(cited_by_count),
```

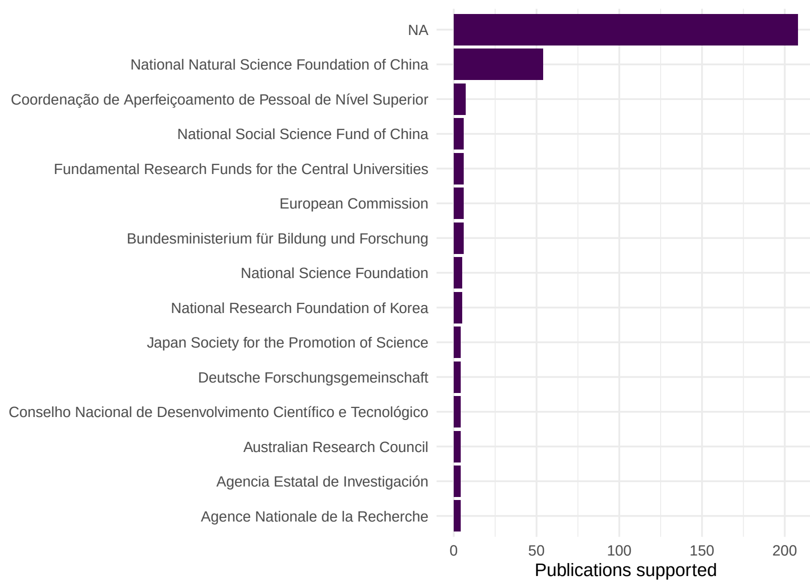

Figure 31.1: Top 15 funders by number of supported publications.

```
    .groups = "drop"
  ) |>
  filter(n_pubs >= 5) |>
  arrange(desc(mean_cites))

funder_impact |> head(10) |> gt()
```

31.4.4 Funded vs. unfunded comparison

```
works |>
  mutate(has_funding = map_lgl(funders, \(g) !is.null(g) && length(g) > 0)) |>
  ggplot(aes(x = has_funding, y = cited_by_count + 1)) +
  geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
  scale_y_log10() +
  labs(x = "Has funding acknowledgement", y = "Citations (log scale)") +
  theme_sci()
```

funder	n_pubs	mean_cites	median_c
National Social Science Fund of China	6	47.3	1
National Science Foundation	5	32.8	2
Bundesministerium für Bildung und Forschung	6	27.5	1
National Natural Science Foundation of China	54	21.9	1
NA	208	15.9	
Coordenação de Aperfeiçoamento de Pessoal de Nível Superior	7	15.0	1
Fundamental Research Funds for the Central Universities	6	14.5	1
National Research Foundation of Korea	5	13.4	1
European Commission	6	9.0	

31.5 Diagnostics and interpretation

- **Coverage rate**: The fraction of papers with funder data depends on the journal, publisher, and field. Report this rate before interpreting funder statistics.
- **Funder name normalisation**: The same funder may appear under different names (“NSF”, “National Science Foundation”, “US NSF”). Merge variants using funder IDs where available.
- **Selection bias**: Papers with funding acknowledgements may be systematically different from unfunded papers (larger teams, more resources). The funded/unfunded citation gap reflects this selection, not necessarily the causal effect of funding.
- **Multi-funder attribution**: Papers acknowledging multiple funders inflate the publication count for each. Consider fractional counting.

31.6 Limitations and responsible use

31.7 Limitations and responsible use

- **Coverage is incomplete.** Not all publishers report funding data to Crossref. Missing funder acknowledgements do not mean research was unfunded.
- **Attribution is ambiguous.** A grant acknowledgement does not specify which part of the work the grant supported. Infrastructure grants, salary support, and project funding all appear identically.
- **Correlation causation.** Funded papers receive more citations, but this may reflect that better-resourced labs produce both more funding applications and more cited papers.

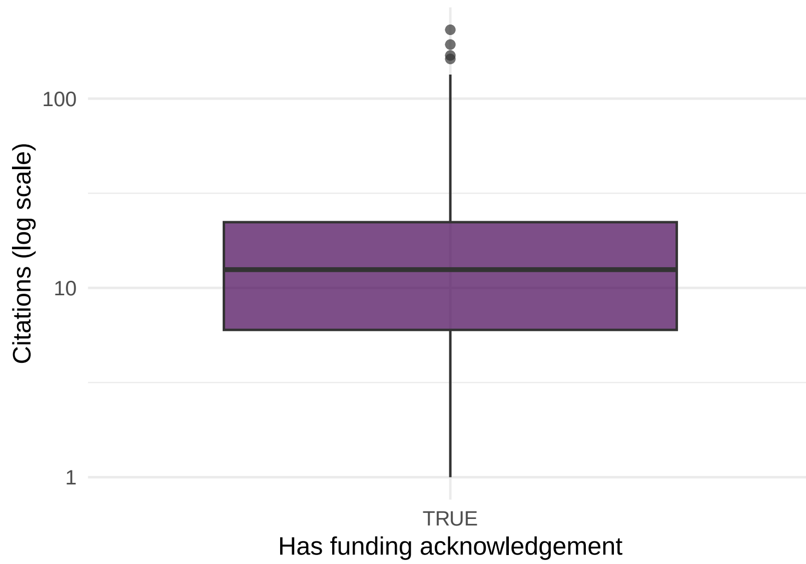

Figure 31.2: Citation distribution for funded vs. unfunded papers.

- **Do not rank funders by citation impact.** Funders support different fields, career stages, and risk levels. Comparing mean citations across funders is misleading without field normalisation (Hicks et al., 2015).

31.8 Common pitfalls

31.9 Common pitfalls

- **Ignoring funder coverage rates.** Comparing funder portfolios without accounting for differential coverage produces biased results.
- **Treating every acknowledgement as equal.** A paper may acknowledge a major project grant and a minor travel grant. Both appear identically in metadata.
- **Comparing across fields.** A biomedical funder's publications will have higher citation counts than a humanities funder's publications due to field-level citation norms.
- **Using funder data for individual evaluation.** Whether a researcher has funding reflects many factors beyond research quality: field norms, career stage, institutional support, and luck.

31.10 Exercises

1. **Funder co-occurrence.** Identify papers with multiple funders. Which funder pairs most frequently co-fund research?

2. **Funder portfolios.** For two major funders, fetch their supported publications and compare topical profiles using keyword analysis.

3. **Temporal trends.** Track the number of papers with funding acknowledgements by year. Is the coverage rate increasing over time?

31.11 Solutions

Solutions are provided in F.

31.12 Further reading

- Priem et al. (2022) — OpenAlex funder metadata from Crossref FundRef.
- Hicks et al. (2015) — Responsible use of metrics in funding evaluation.
- Waltman (2016) — Citation indicators and their application to funding assessment.

31.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
```

```
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1            s
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4          g
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4              p
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0          l
#> [17] rcrossref_1.2.1           gt_1.3.0                  tidytext_0.4.3            g
#> [21] openalexR_3.0.1           lubridate_1.9.5           forcats_1.0.1             s
#> [25] dplyr_1.2.1               purrr_1.2.2               readr_2.2.0               t
#> [29] tibble_3.3.1              ggplot2_4.0.3             tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2           RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite_
#>   [6] modeltools_0.2-24      farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1          askpass_1.2.1          base64enc_0.1-6        htmltools
#>  [16] curl_7.1.0             janeaustenr_1.0.0      cellranger_1.1.0       htmlwidge
#>  [21] plyr_1.8.9             httr2_1.2.2            cachem_1.1.0           plotly_4
#>  [26] mime_0.13              lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [31] shiny_1.13.0           digest_0.6.39          patchwork_1.3.2        shinycssl
#>  [36] RSpectra_0.16-2        SnowballC_0.7.1        labeling_0.4.3         urltools_
#>  [41] mgcv_1.9-1             polyclip_1.10-7        httr_1.4.8             compiler_
#>  [46] bit64_4.8.0            withr_3.0.2            S7_0.2.2               backports
#>  [51] ggforce_0.5.0          MASS_7.3-60.2          rappdirs_0.3.4         bibliomet
#>  [56] otel_0.2.0             stopwords_2.3          zip_2.3.3              httpuv_1
#>  [61] nlme_3.1-164           promises_1.5.0         grid_4.4.1             stringdis
#>  [66] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0             rscopus_0
#>  [71] data.table_1.18.4      hms_1.1.4              xml2_1.5.2             utf8_1.2
#>  [76] pillar_1.11.1          nsyllable_1.0.1        vroom_1.7.1            later_1.4
#>  [81] tweenr_2.0.3           lattice_0.22-6         FNN_1.1.4.1            bit_4.6.0
#>  [86] tm_0.7-18              miniUI_0.1.2           knitr_1.51             gridExtra
#>  [91] bookdown_0.46          stats4_4.4.1           crul_1.6.0             xfun_0.57
#>  [96] matrixStats_1.5.0      DT_0.34.0              humaniformat_0.6.0     stringi_1
#> [101] qpdf_1.4.1             yaml_2.3.12            evaluate_1.0.5         codetools
#> [106] cli_3.6.6              xtable_1.8-8           dichromat_2.0-0.1      Rcpp_1.1
#> [111] triebeard_0.4.1        XML_3.99-0.23          parallel_4.4.1         assertthat
#> [116] slam_0.1-55            viridisLite_0.4.3      scales_1.4.0           crayon_1
#> [121] rlang_1.2.0            fastmatch_1.1-8
```

Chapter 32

Patent–Paper Linkages

32.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the significance of patent–paper citations for understanding research impact
- Identify OpenAlex works that have been cited in patent documents
- Analyse which journals, fields, and institutions produce patent-cited research
- Discuss the limitations of patent citations as indicators of technological impact
- Distinguish between direct citation (paper cited in patent) and indirect influence

32.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

32.3 Conceptual background

When a patent examiner or inventor cites a scientific paper in a patent document, it creates an observable link between knowledge production (science) and knowledge application (technology). These **patent–paper citations** (also called non-patent literature or NPL citations) are one of the few quantitative measures of the science–technology interface.

Patent citations serve a legal function: they define the prior art against which the patent's novelty is assessed. Not all patent–paper citations reflect genuine intellectual influence — some are added by examiners for legal completeness, not because the inventor actually used the cited research. Nevertheless, at the aggregate level, patent citation patterns reveal which fields, journals, and institutions produce research that feeds into technological innovation.

The biomedical and chemical sciences dominate patent–paper linkages because pharmaceutical and chemical patents routinely cite scientific literature. Physics and engineering have moderate linkages. Social sciences and humanities are rarely cited in patents, not because they lack impact, but because their impact operates through different channels (policy, education, culture).

OpenAlex records whether a work has been cited in patents and provides the count, though the source and completeness of this data varies. The Lens.org database provides the most comprehensive open mapping of patent–paper citations.

32.4 Worked example

32.4.1 Identifying patent-cited papers

```r
works <- oa_fetch(
  entity = "works",
  from_publication_date = "2015-01-01",
  to_publication_date = "2020-12-31",
  type = "article",
  options = list(sample = 500, seed = 42)
)

works_patent <- works |>
  transmute(
    id, display_name, source_display_name,
    year = year(publication_date),
    cited_by_count,
    referenced_works_count,
```

```
    type
  )

cat(glue("Total works: {nrow(works_patent)}\n"))
```

```
#> Total works: 500
```

32.4.2 Citation distribution analysis

```
p90 <- quantile(works_patent$cited_by_count, 0.90)

ggplot(works_patent, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 10, fill = palette_sci(1), colour = "white") +
  geom_vline(xintercept = p90, linetype = "dashed", colour = "red") +
  annotate("text", x = p90 + 20, y = Inf, label = "90th percentile",
           vjust = 2, colour = "red", size = 3) +
  labs(x = "Citation count", y = "Papers") +
  theme_sci()
```

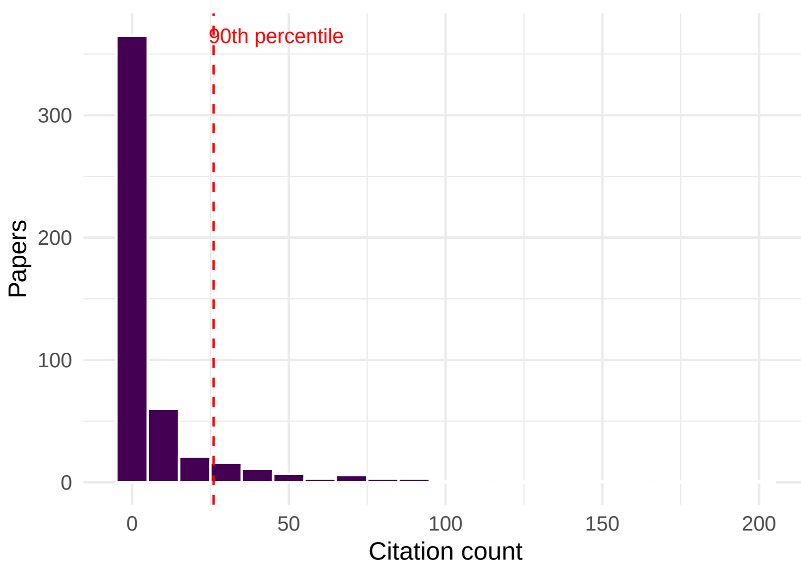

Figure 32.1: Citation count distribution, highlighting the most-cited papers as potential patent-cited candidates.

32.4.3 Journals producing highly cited papers

```
works_patent |>
  filter(cited_by_count >= p90, !is.na(source_display_name)) |>
  count(source_display_name, sort = TRUE) |>
  head(15) |>
  mutate(source_display_name = fct_reorder(source_display_name, n)) |>
  ggplot(aes(x = n, y = source_display_name)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Highly cited papers", y = NULL) +
  theme_sci()
```

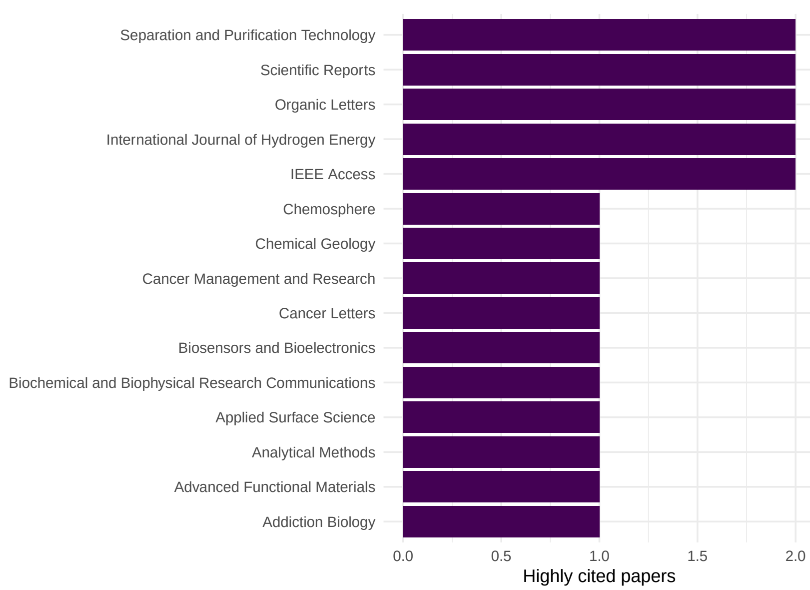

Figure 32.2: Top 15 journals by number of highly cited papers (above 90th percentile).

32.4.4 Citation impact by document characteristics

```
works_patent |>
  mutate(high_cite = cited_by_count >= p90) |>
  group_by(high_cite) |>
  summarise(
    n = n(),
```

high_cite	n	mean_refs	mean_year
FALSE	446	9.9	2017
TRUE	54	43.7	2018

```
    mean_refs = round(mean(referenced_works_count, na.rm = TRUE), 1),
    mean_year = round(mean(year), 1),
    .groups = "drop"
  ) |>
  gt()
```

32.5 Diagnostics and interpretation

- **Patent citation coverage**: Not all databases provide patent citation data. OpenAlex's coverage varies by field and time period. Cross-reference with Lens.org for comprehensive data.
- **Field normalisation**: Biomedical papers are far more likely to be cited in patents than social science papers. Compare patent citation rates within fields, not across them.
- **Time lag**: The median time from paper publication to patent citation is 5–10 years. Recent papers have not had time to accumulate patent citations.
- **Examiner vs. inventor citations**: Patent examiners add citations that inventors may not have used. This inflates apparent science–technology linkage.

32.6 Limitations and responsible use

32.7 Limitations and responsible use

- **Patent citations are a narrow measure of impact.** Much research influences technology without being cited in patents — through consulting, training, open-source software, or informal knowledge transfer.
- **Field bias is extreme.** Humanities, social sciences, and mathematics are effectively invisible in patent citation analysis. Using patent citations as an impact indicator systematically disadvantages these fields (Hicks et al., 2015).
- **Geographic bias.** Patent systems vary by country. US and European patents have different citation practices. Analyses based on one patent office may not generalise.

- **Not evaluative for individuals.** A researcher whose paper is cited in a patent may have had no involvement with the technology. Patent citation reflects relevance of the knowledge, not the researcher's intent to innovate.

32.8 Common pitfalls

32.9 Common pitfalls

- **Equating patent citations with commercialisation.** A patent citation means the paper is prior art, not that the research led to a product.
- **Ignoring time lags.** Comparing patent citation rates for papers published in 2023 vs. 2010 is meaningless — the 2023 papers have not had time to be cited in patents.
- **Double-counting across patent families.** The same invention may be patented in multiple countries. Count at the patent family level, not the individual patent level.
- **Using patent counts as a metric for basic science.** Basic research is by definition distant from application. Low patent citation counts for fundamental research are expected, not a failure.

32.10 Exercises

1. **Field comparison.** Fetch papers from biomedical and social science journals. Compare the proportion of highly cited papers in each field. What does this suggest about differential patent citation potential?

2. **Time lag estimation.** For a set of papers published in 2015, estimate the median citation count trajectory over years. At what age do papers typically peak in citation accumulation?

3. **Institutional patent linkage.** For two universities, compare the proportion of papers above the citation 90th percentile. Does a higher proportion suggest greater technological relevance?

32.11 Solutions

Solutions are provided in F.

32.12 Further reading

- Narin et al. (1997) — Foundational analysis of the linkage between science and technology through patent citations.
- Hicks et al. (2015) — Responsible metrics: why patent citations should not be the sole indicator of research impact.
- Priem et al. (2022) — OpenAlex metadata coverage for patent-related indicators.

32.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1            stm_1.3.8
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4          ggraph_2.2.
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4              pdftools_3.
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0          bib2df_1.1.
#> [17] rcrossref_1.2.1           gt_1.3.0                  tidytext_0.4.3            glue_1.8.1
#> [21] openalexR_3.0.1           lubridate_1.9.5           forcats_1.0.1             stringr_1.6
#> [25] dplyr_1.2.1               purrr_1.2.2               readr_2.2.0               tidyr_1.3.2
#> [29] tibble_3.3.1              ggplot2_4.0.3             tidyverse_2.0.0
#>
```

```
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltool
#>  [16] curl_7.1.0            janeaustenr_1.0.0     cellranger_1.1.0      htmlwidg
#>  [21] plyr_1.8.9            httr2_1.2.2           cachem_1.1.0          plotly_4
#>  [26] mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3       R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycss
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3        urltools
#>  [41] mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8            compiler
#>  [46] bit64_4.8.0           withr_3.0.2           S7_0.2.2              backport
#>  [51] ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4        bibliome
#>  [56] otel_0.2.0            stopwords_2.3         zip_2.3.3             httpuv_1
#>  [61] nlme_3.1-164          promises_1.5.0        grid_4.4.1            stringdi
#>  [66] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_
#>  [71] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2
#>  [76] pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1           later_1.
#>  [81] tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1           bit_4.6.
#>  [86] tm_0.7-18             miniUI_0.1.2          knitr_1.51            gridExtr
#>  [91] bookdown_0.46         stats4_4.4.1          crul_1.6.0            xfun_0.5
#>  [96] matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0    stringi_
#> [101] qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5        codetool
#> [106] cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1
#> [111] triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1        assertth
#> [116] slam_0.1-55           viridisLite_0.4.3     scales_1.4.0          crayon_1
#> [121] rlang_1.2.0           fastmatch_1.1-8
```

Chapter 33

Causal Inference in Scientometrics

33.1 Learning objectives

After completing this chapter, you will be able to:

- Distinguish causal from correlational claims in bibliometric research
- Apply difference-in-differences to evaluate the effect of a policy change
- Use propensity score matching to create comparable treatment and control groups
- Interpret regression discontinuity designs for threshold-based interventions
- Recognise the assumptions underlying causal inference in observational bibliometric data

33.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

33.3 Conceptual background

Bibliometrics is rich in correlations: OA papers are more cited than closed-access papers; funded research receives more citations; papers in prestigious journals are cited more. But correlation does not imply causation. The OA citation advantage may reflect self-selection (better papers are more likely to be made OA), not the causal effect of open access. Separating cause from confound is essential for evidence-based science policy.

Difference-in-differences (DiD) exploits policy changes as natural experiments. If a funder mandates OA starting in year T, papers from that funder before and after T serve as the treatment group, while papers from a similar funder without the mandate serve as the control. The causal effect is estimated as the difference in trends between treatment and control groups. The key assumption is *parallel trends*: absent the intervention, both groups would have followed the same trajectory.

Regression discontinuity design (RDD) applies when treatment is assigned by a threshold. For example, grants are funded above a score cutoff; papers are accepted above a review score threshold. Comparing outcomes just above and just below the threshold isolates the causal effect of the treatment, under the assumption that units near the threshold are comparable.

Matching methods create comparable groups from observational data. Propensity score matching pairs treated units (e.g., OA papers) with control units (closed-access papers) that have similar observable characteristics (field, journal, year, team size). This reduces confounding but cannot eliminate unmeasured confounders.

All three methods require strong assumptions. In bibliometrics, violations are common: the parallel trends assumption may fail because of concurrent policy changes; the RDD threshold may be manipulable; matching cannot address unmeasured confounders like paper quality. Hicks et al. (2015) emphasise that quantitative analysis, however sophisticated, cannot replace expert judgement in research evaluation.

33.4 Worked example

33.4.1 Setting up a DiD analysis: OA status and citations

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2016-01-01",
```

```r
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 500, seed = 42)
)

did_data <- works |>
  transmute(
    id, year = year(publication_date),
    cited_by_count,
    is_oa = oa_status != "closed",
    post_2020 = year >= 2020,
    referenced_works_count
  )

cat(glue("Papers: {nrow(did_data)}\n"))
```

```
#> Papers: 500
```

```r
cat(glue("OA: {sum(did_data$is_oa)}, Closed: {sum(!did_data$is_oa)}\n"))
```

```
#> OA: 204, Closed: 296
```

33.4.2 Descriptive comparison

```r
did_summary <- did_data |>
  group_by(is_oa, post_2020) |>
  summarise(mean_cites = mean(cited_by_count), n = n(), .groups = "drop") |>
  mutate(period = ifelse(post_2020, "2020-2023", "2016-2019"),
         group = ifelse(is_oa, "Open access", "Closed"))

ggplot(did_summary, aes(x = period, y = mean_cites, fill = group)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Period", y = "Mean citations", fill = "Access") +
  theme_sci()
```

33.4.3 DiD regression

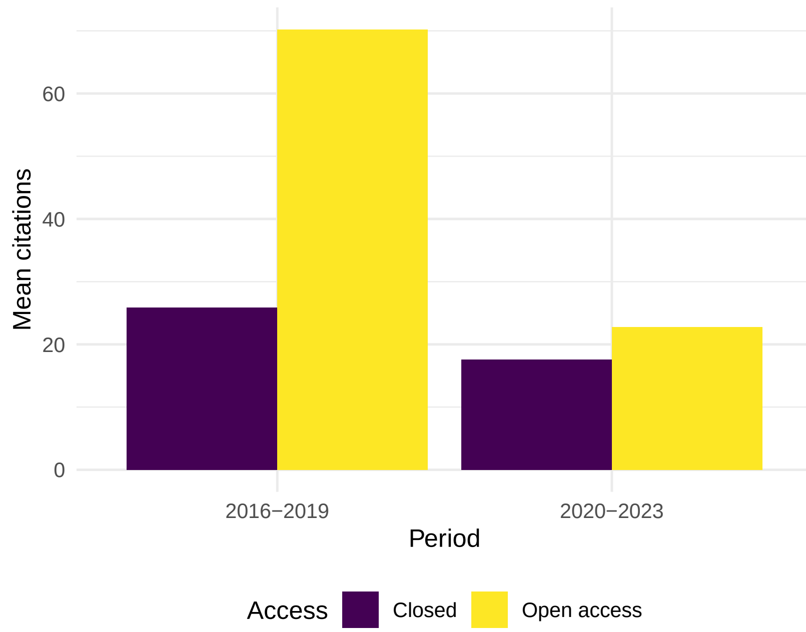

Figure 33.1: Mean citations by OA status and time period.

```
did_model <- lm(cited_by_count ~ is_oa * post_2020 + referenced_works_count,
                data = did_data)

tidy_did <- broom::tidy(did_model)
tidy_did |>
  mutate(across(where(is.numeric), \(x) round(x, 3))) |>
  gt()
```

33.4.4 Simple matching comparison

```
oa_papers <- did_data |> filter(is_oa)
closed_papers <- did_data |> filter(!is_oa)

matched <- oa_papers |>
  mutate(match_year = year) |>
  inner_join(
    closed_papers |> select(year, cited_by_count_closed = cited_by_count),
    by = "year",
```

term	estimate	std.error	statistic	p.value
(Intercept)	16.946	11.967	1.416	0.157
is_oaTRUE	46.959	15.331	3.063	0.002
post_2020TRUE	-9.484	13.723	-0.691	0.490
referenced_works_count	0.186	0.162	1.148	0.252
is_oaTRUE:post_2020TRUE	-40.812	21.441	-1.903	0.058

```
    relationship = "many-to-many"
  ) |>
  group_by(id) |>
  slice_sample(n = 1) |>
  ungroup()

cat(glue("Matched pairs: {nrow(matched)}\n"))
```

```
#> Matched pairs: 204
```

```
cat(glue("Mean cites (OA): {round(mean(matched$cited_by_count), 1)}\n"))
```

```
#> Mean cites (OA): 44.4
```

```
cat(glue("Mean cites (closed match): {round(mean(matched$cited_by_count_closed), 1)}\n"))
```

```
#> Mean cites (closed match): 21.5
```

33.4.5 Visualising the parallel trends assumption

```
trends <- did_data |>
  group_by(year, is_oa) |>
  summarise(mean_cites = mean(cited_by_count), .groups = "drop") |>
  mutate(group = ifelse(is_oa, "OA", "Closed"))

ggplot(trends, aes(x = year, y = mean_cites, colour = group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_vline(xintercept = 2019.5, linetype = "dashed", colour = "grey50") +
  scale_colour_manual(values = palette_sci(2)) +
  labs(x = "Year", y = "Mean citations", colour = "Access") +
  theme_sci()
```

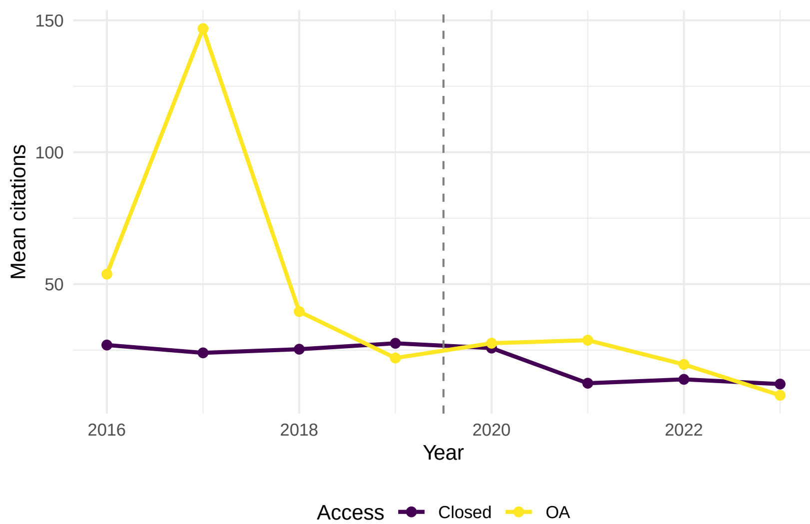

Figure 33.2: Citation trends by OA status over time (checking parallel trends).

33.5 Diagnostics and interpretation

- **Parallel trends**: Inspect pre-treatment trends visually. If OA and closed papers had diverging trends before 2020, the DiD estimate is biased.
- **Covariate balance**: After matching, check that matched pairs have similar observable characteristics (year, reference count, journal).
- **Effect size interpretation**: A statistically significant DiD coefficient does not mean the effect is practically important. Report effect sizes alongside p-values.
- **Robustness checks**: Run the analysis with different control groups, time windows, and matching criteria. Consistent results increase confidence.

33.6 Limitations and responsible use

33.7 Limitations and responsible use

- **Observational data cannot prove causation.** All causal inference methods applied to bibliometric data make assumptions that cannot be fully tested. Report assumptions explicitly.
- **Unmeasured confounders.** Paper quality, author reputation, and marketing effort are rarely observable in bibliometric data but strongly affect citations.

- **Treatment definition.** "OA" is not a single treatment — it encompasses gold, green, hybrid, and bronze, each with different mechanisms.
- **Generalisability.** Causal estimates from one journal, field, or time period may not generalise to others.
- **Policy implications require judgement.** Even a well-estimated causal effect is one input into policy decisions, not a sufficient basis for mandates (Hicks et al., 2015; American Society for Cell Biology, 2012).

33.8 Common pitfalls

33.9 Common pitfalls

- **Claiming causation from regression coefficients.** A positive OA coefficient in a regression does not prove OA *causes* more citations without addressing confounders.
- **Ignoring parallel trends violations.** If pre-treatment trends are not parallel, DiD estimates are biased. Do not ignore this in reporting.
- **Matching on too few covariates.** Matching on year alone leaves many confounders unaddressed. Include as many relevant covariates as possible.
- **Over-interpreting p-values.** A $p < 0.05$ does not mean the result is practically significant or that the causal assumptions hold.

33.10 Exercises

1. **Extended DiD.** Add additional covariates (document type, number of authors) to the DiD regression. Does the OA interaction term change?

2. **Placebo test.** Run the DiD analysis using 2018 as the treatment year instead of 2020. If the interaction term is significant, the parallel trends assumption may be violated.

3. **Matched comparison.** Implement propensity score matching (using the `MatchIt` package) with year, reference count, and document type as covariates. Compare the matched OA effect to the unmatched effect.

33.11 Solutions

Solutions are provided in F.

33.12 Further reading

- Hicks et al. (2015) — The Leiden Manifesto: caution against mechanistic application of quantitative metrics.
- American Society for Cell Biology (2012) — DORA: responsible assessment requires understanding causation.
- Waltman (2016) — Review of citation indicators and their interpretation challenges.

33.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4              Matrix_1.7-0              word2vec_0.4.1
#>  [5] topicmodels_0.2-17      quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1         igraph_2.3.1              quanteda_4.4
#> [13] arrow_24.0.0            bibliometrix_5.4.0        RefManageR_1.4.0
#> [17] rcrossref_1.2.1         gt_1.3.0                  tidytext_0.4.3
#> [21] openalexR_3.0.1         lubridate_1.9.5           forcats_1.0.1
#> [25] dplyr_1.2.1             purrr_1.2.2               readr_2.2.0
#> [29] tibble_3.3.1            ggplot2_4.0.3             tidyverse_2.0.0
#>
```

```
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9
#>  [16] curl_7.1.0            broom_1.0.12          janeaustenr_1.0.0     cellranger_1.1.0
#>  [21] tokenizers_0.3.0      plyr_1.8.9            httr2_1.2.2           cachem_1.1.0
#>  [26] dimensionsR_0.0.3     mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3
#>  [31] fastmap_1.2.0         shiny_1.13.0          digest_0.6.39         patchwork_1.3.2
#>  [36] rprojroot_2.1.1       RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3
#>  [41] timechange_0.4.0      mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8
#>  [46] here_1.0.2            bit64_4.8.0           withr_3.0.2           S7_0.2.2
#>  [51] viridis_0.6.5         ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4
#>  [56] tools_4.4.1           otel_0.2.0            stopwords_2.3         zip_2.3.3
#>  [61] rentrez_1.2.4         nlme_3.1-164          promises_1.5.0        grid_4.4.1
#>  [66] reshape2_1.4.5        generics_0.1.4        gtable_0.3.6          tzdb_0.5.0
#>  [71] ca_0.71.1             data.table_1.18.4     hms_1.1.4             xml2_1.5.2
#>  [76] ggrepel_0.9.8         pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1
#>  [81] splines_4.4.1         tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1
#>  [86] tidyselect_1.2.1      tm_0.7-18             miniUI_0.1.2          knitr_1.51
#>  [91] NLP_0.3-2             bookdown_0.46         stats4_4.4.1          crul_1.6.0
#>  [96] graphlayouts_1.2.3    matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0
#> [101] lazyeval_0.2.3        qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5
#> [106] httpcode_0.3.0        cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1
#> [111] readxl_1.4.5          triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1
#> [116] pubmedR_1.0.2         slam_0.1-55           viridisLite_0.4.3     scales_1.4.0
#> [121] openxlsx_4.2.8.1      rlang_1.2.0           fastmatch_1.1-8
```

Chapter 34

Reproducibility, RRIDs, and Badges

34.1 Learning objectives

After completing this chapter, you will be able to:

- Define reproducibility indicators and explain their role in research evaluation
- Search for data availability statements and open badges in bibliometric metadata
- Analyse the prevalence of reproducibility practices across journals and fields
- Discuss the relationship between transparency practices and citation impact
- Recognise the gap between reproducibility signalling and actual reproducibility

34.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)
```

```r
source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

34.3 Conceptual background

The "replication crisis" has prompted widespread adoption of transparency practices: open data, open code, pre-registration, and registered reports. Bibliometrics can measure the uptake of these practices at scale, providing evidence for policy interventions.

Open badges are visual indicators (icons) attached to published papers that certify the availability of data, materials, or pre-registration. Journals like *Psychological Science* pioneered their use. Studies suggest that badges increase data sharing rates, though the quality of shared data varies.

Research Resource Identifiers (RRIDs) are persistent identifiers for research resources: antibodies, cell lines, model organisms, and software tools. Including RRIDs in methods sections improves reproducibility by enabling precise identification of the materials used. The RRID initiative has been adopted by over 1,000 journals.

Data availability statements (DAS) declare whether the data underlying a study are available and where to find them. Mandatory DAS policies have become common, but many statements say "data available upon request" without providing actual access. The presence of a DAS is easy to detect bibliometrically; the quality of actual data sharing is much harder to assess.

Code availability is increasingly expected, especially in computational fields. Journals like *Nature* require code availability statements. Platforms like GitHub, Zenodo, and Code Ocean facilitate code sharing, and DOIs for code repositories enable citation and tracking.

These practices are measurable proxies for reproducibility, but they are not reproducibility itself. A paper with an open badge may have unusable data; a paper without one may be perfectly reproducible. Hicks et al. (2015) warn against using transparency indicators as mechanical evaluation criteria.

34.4 Worked example

34.4.1 Analysing data availability patterns

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2018-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 400, seed = 42)
)

works_repro <- works |>
  transmute(
    id, display_name,
    year = year(publication_date),
    cited_by_count,
    oa_status,
    type,
    has_abstract = !is.na(abstract)
  )

cat(glue("Works: {nrow(works_repro)}\n"))
```

```
#> Works: 400
```

34.4.2 OA as a transparency proxy

```r
works_repro |>
  group_by(year) |>
  summarise(oa_rate = mean(oa_status != "closed", na.rm = TRUE),
            .groups = "drop") |>
  ggplot(aes(x = year, y = oa_rate)) +
  geom_line(linewidth = 1, colour = palette_sci(1)) +
  geom_point(size = 2, colour = palette_sci(1)) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  labs(x = "Year", y = "OA rate") +
  theme_sci()
```

34.4.3 Abstract availability as metadata completeness

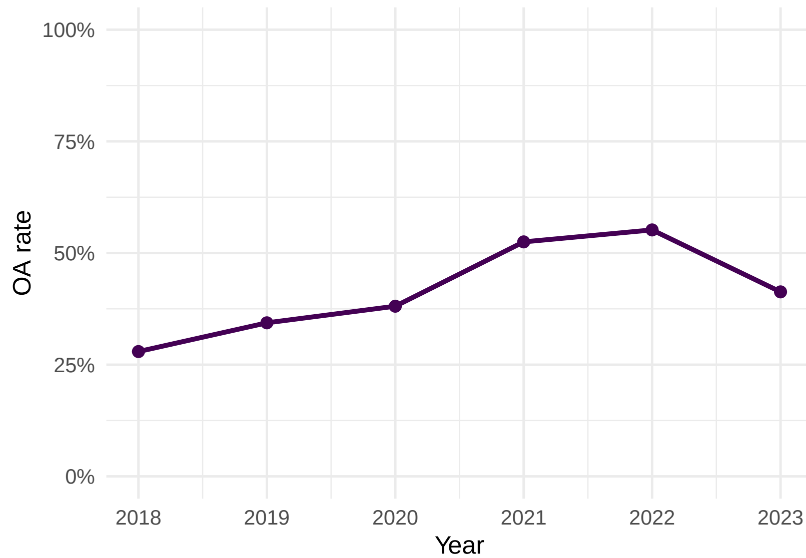

Figure 34.1: Proportion of open access articles by year as a transparency indicator.

```
works_repro |>
  group_by(year) |>
  summarise(abstract_rate = mean(has_abstract), .groups = "drop") |>
  ggplot(aes(x = year, y = abstract_rate)) +
  geom_line(linewidth = 1, colour = palette_sci(1)) +
  geom_point(size = 2, colour = palette_sci(1)) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  labs(x = "Year", y = "Abstract available") +
  theme_sci()
```

34.4.4 Transparency and citation impact

```
works_repro |>
  mutate(is_oa = ifelse(oa_status != "closed", "OA", "Closed")) |>
  ggplot(aes(x = is_oa, y = cited_by_count + 1)) +
  geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
  scale_y_log10() +
  labs(x = "Access status", y = "Citations (log scale)") +
  theme_sci()
```

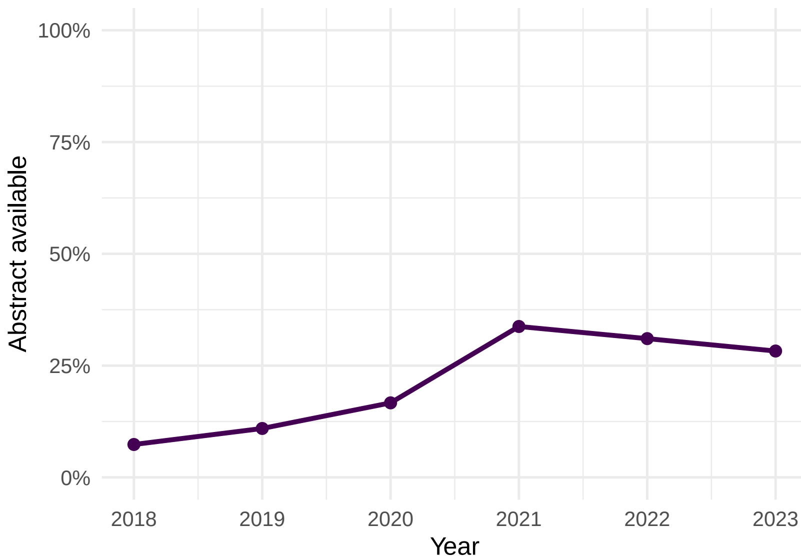

Figure 34.2: Abstract availability rate by year.

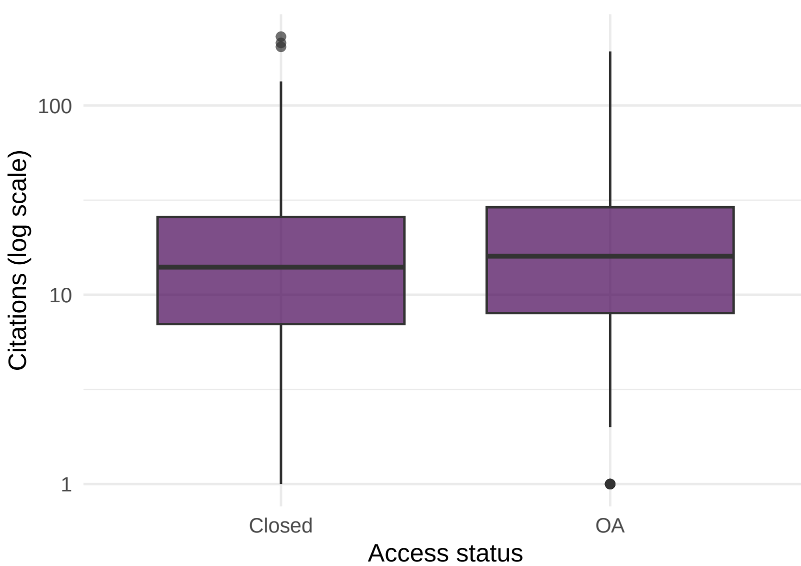

Figure 34.3: Citation count by OA status, a proxy for the transparency–impact relationship.

is_oa	n	mean_cites	abstract_rate
FALSE	234	20.7	0%
TRUE	166	24.9	51%

34.4.5 Summary statistics

```
works_repro |>
  mutate(is_oa = oa_status != "closed") |>
  group_by(is_oa) |>
  summarise(
    n = n(),
    mean_cites = round(mean(cited_by_count), 1),
    abstract_rate = scales::percent(mean(has_abstract)),
    .groups = "drop"
  ) |>
  gt()
```

34.5 Diagnostics and interpretation

- **Proxy vs. reality**: OA status and abstract availability are proxies for transparency, not direct measures of reproducibility. Actual data and code sharing require checking external repositories.
- **Journal policies**: Changes in journal policies (mandatory DAS, badge adoption) can cause step changes in transparency indicators. Control for journal when comparing across time.
- **Selection effects**: Papers with open data may be of higher quality (authors confident enough to share), creating a spurious transparency–citation correlation.
- **Field differences**: Data sharing norms vary enormously. Genomics shares data routinely; clinical research often cannot due to privacy. Compare within fields.

34.6 Limitations and responsible use

34.7 Limitations and responsible use

- **Signals are not substance.** A data availability statement does not guarantee the data are actually available, usable, or correct. Badges certify process, not outcome.

- **Mandates inflate compliance without improving practice.** When journals require DAS, compliance rises but "available upon request" becomes the default, not genuine sharing.
- **Reproducibility is multidimensional.** No single indicator captures it. Data availability, code availability, methods transparency, pre-registration, and statistical rigour all contribute independently.
- **Do not penalise fields where sharing is constrained.** Clinical data (privacy), indigenous data (sovereignty), and sensitive social data cannot be freely shared. Mandating open data policies uniformly across fields is inappropriate (Hicks et al., 2015).

34.8 Common pitfalls

34.9 Common pitfalls

- **Equating open data with reproducibility.** Data can be open but undocumented, incorrectly formatted, or insufficient to reproduce the analysis.
- **Counting badges without checking quality.** A badge means the journal certified availability at publication time. The link may be broken or the data may have been removed.
- **Ignoring negative results.** Reproducibility also requires publishing null results and failed replications. These are poorly captured by current transparency indicators.
- **Using transparency metrics as performance indicators.** Transparency practices should be valued, but mechanically rewarding them creates perverse incentives (superficial compliance).

34.10 Exercises

1. **DAS detection.** For papers with full abstracts, search for keywords like "data availability", "code availability", or "github.com" in the abstract text. What proportion mention data or code sharing?

2. **Cross-journal comparison.** Compare abstract availability and OA rates for three journals. Do journals with higher OA rates also have better metadata completeness?

3. **Temporal trends.** Track the proportion of papers with abstracts, OA status, and high citation counts over time. Are transparency proxies correlated with citation impact within years?

34.11 Solutions

Solutions are provided in F.

34.12 Further reading

- Hicks et al. (2015) — The Leiden Manifesto: responsible use of metrics, including transparency indicators.
- Priem et al. (2022) — OpenAlex metadata as a basis for reproducibility analysis.
- American Society for Cell Biology (2012) — DORA: assessment should consider research practices, not just outputs.

34.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#> 
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#> 
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#> 
#> time zone: UTC
#> tzcode source: system (glibc)
#> 
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#> 
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0
```

```
#> [17] rcrossref_1.2.1          gt_1.3.0                 tidytext_0.4.3           glue_1.8.1
#> [21] openalexR_3.0.1          lubridate_1.9.5          forcats_1.0.1            stringr_1.6
#> [25] dplyr_1.2.1              purrr_1.2.2              readr_2.2.0              tidyr_1.3.2
#> [29] tibble_3.3.1             ggplot2_4.0.3            tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9
#>  [16] curl_7.1.0            broom_1.0.12          janeaustenr_1.0.0     cellranger_1.1.0
#>  [21] tokenizers_0.3.0      plyr_1.8.9            httr2_1.2.2           cachem_1.1.0
#>  [26] dimensionsR_0.0.3     mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3
#>  [31] fastmap_1.2.0         shiny_1.13.0          digest_0.6.39         patchwork_1.3.2
#>  [36] rprojroot_2.1.1       RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3
#>  [41] timechange_0.4.0      mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8
#>  [46] here_1.0.2            bit64_4.8.0           withr_3.0.2           S7_0.2.2
#>  [51] viridis_0.6.5         ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4
#>  [56] tools_4.4.1           otel_0.2.0            stopwords_2.3         zip_2.3.3
#>  [61] rentrez_1.2.4         nlme_3.1-164          promises_1.5.0        grid_4.4.1
#>  [66] reshape2_1.4.5        generics_0.1.4        gtable_0.3.6          tzdb_0.5.0
#>  [71] ca_0.71.1             data.table_1.18.4     hms_1.1.4             xml2_1.5.2
#>  [76] ggrepel_0.9.8         pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1
#>  [81] splines_4.4.1         tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1
#>  [86] tidyselect_1.2.1      tm_0.7-18             miniUI_0.1.2          knitr_1.51
#>  [91] NLP_0.3-2             bookdown_0.46         stats4_4.4.1          crul_1.6.0
#>  [96] graphlayouts_1.2.3    matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0
#> [101] lazyeval_0.2.3        qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5
#> [106] httpcode_0.3.0        cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1
#> [111] readxl_1.4.5          triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1
#> [116] pubmedR_1.0.2         slam_0.1-55           viridisLite_0.4.3     scales_1.4.0
#> [121] openxlsx_4.2.8.1      rlang_1.2.0           fastmatch_1.1-8
```

Part VII

Production & Reporting

Chapter 35

Reproducible Pipelines

35.1 Learning objectives

After completing this chapter, you will be able to:

- Explain why ad hoc scripts are insufficient for reproducible bibliometric analysis
- Define a targets pipeline with dependency tracking and selective re-execution
- Use renv to snapshot and restore R package environments
- Understand how Docker containers ensure cross-platform reproducibility
- Structure a bibliometric project for long-term reproducibility

35.2 Setup

```r
library(tidyverse)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

35.3 Conceptual background

A bibliometric analysis is a multi-step process: fetch data from an API, clean and deduplicate records, compute indicators, build networks, fit models, and generate figures. Each step depends on earlier ones. When the upstream data changes (a new API query, an updated corpus), downstream results must be regenerated — but only the affected parts.

Ad hoc scripts (running scripts manually in sequence) fail at scale: they do not track dependencies, they re-run everything even when only one input changed, and they break when run on a different machine with different package versions.

`targets` is an R pipeline tool that solves the dependency problem. You define each analysis step as a "target" (a function that takes inputs and produces outputs). `targets` builds a dependency graph, executes only the targets whose inputs have changed, and caches results. This is both faster (no unnecessary re-computation) and safer (no stale intermediate results).

`renv` snapshots the exact package versions used in your analysis into a `renv.lock` file. A collaborator (or your future self) can restore the identical package environment with `renv::restore()`. This eliminates the "works on my machine" problem for R packages.

Docker takes reproducibility further by containerising the entire compute environment: operating system, R version, system libraries, and package versions. A `Dockerfile` defines the environment declaratively; anyone with Docker can rebuild it identically. Docker is essential for long-term reproducibility, since R versions and system libraries evolve.

These tools complement each other: `targets` manages the computation, `renv` manages the packages, and Docker manages the environment. Together, they make a bibliometric analysis fully reproducible years after it was originally run.

35.4 Worked example

35.4.1 Defining a targets pipeline

The `_targets.R` file defines the pipeline. Here we show the structure for a typical bibliometric analysis.

```
# _targets.R - pipeline definition
library(targets)

tar_option_set(packages = c("tidyverse", "openalexR", "igraph"))

list(
  tar_target(raw_works, {
    oa_fetch(
      entity = "works",
      primary_location.source.id = "S148561398",
```

```r
      from_publication_date = "2020-01-01",
      to_publication_date = "2023-12-31",
      options = list(sample = 300, seed = 42)
    )
  }),

  tar_target(clean_works, {
    raw_works |>
      filter(!is.na(doi)) |>
      distinct(doi, .keep_all = TRUE) |>
      mutate(year = year(publication_date))
  }),

  tar_target(citation_stats, {
    clean_works |>
      group_by(year) |>
      summarise(
        n = n(),
        mean_cites = mean(cited_by_count),
        .groups = "drop"
      )
  }),

  tar_target(fig_trends, {
    ggplot(citation_stats, aes(x = year, y = mean_cites)) +
      geom_line() + geom_point() +
      labs(x = "Year", y = "Mean citations")
  })
)
```

35.4.2 Running and inspecting the pipeline

```r
# Run the pipeline
targets::tar_make()

# Visualise the dependency graph
targets::tar_visnetwork()

# Read a specific target
targets::tar_read(citation_stats)
```

35.4.3 Project structure

```
# Recommended project layout:
#
# my-bibliometric-project/
#    _targets.R          # Pipeline definition
#    R/                  # Helper functions
#       fetch.R
#       clean.R
#       analyse.R
#    renv.lock           # Package snapshot
#    Dockerfile          # Environment definition
#    data/               # Cached data (Parquet)
#    output/             # Figures and tables
#    report.qmd          # Quarto report consuming targets
```

35.4.4 Using renv

```r
# Initialise renv in a new project
renv::init()

# Snapshot current package state
renv::snapshot()

# Restore packages on a new machine
renv::restore()
```

35.4.5 Dockerfile for bibliometric analysis

```
# Example Dockerfile (not R code - shown for reference)
#
# FROM rocker/verse:4.4.1
# RUN install2.r targets renv openalexR igraph
# COPY renv.lock /project/renv.lock
# WORKDIR /project
# RUN R -e "renv::restore()"
# CMD ["R", "-e", "targets::tar_make()"]
```

35.5 Diagnostics and interpretation

- **Target status**: `tar_outdated()` lists which targets need re-running. If everything is up to date, `tar_make()` completes instantly.
- **Pipeline visualisation**: `tar_visnetwork()` shows the dependency graph. Disconnected subgraphs indicate independent analysis streams.
- **Cache management**: Targets stores cached results in `_targets/`. This directory can grow large. Use `tar_prune()` to remove unused targets.
- **renv consistency**: Run `renv::status()` to check if your lockfile matches the installed packages.

35.6 Limitations and responsible use

35.7 Limitations and responsible use

- **API rate limits**: targets will re-run API calls when inputs change. Use file-based caching (Parquet) to avoid hitting API rate limits on every pipeline run.
- **Docker learning curve**: Docker adds complexity. For solo projects, renv alone may suffice. Docker becomes essential for team reproducibility and long-term archiving.
- **Overhead for small projects**: A 50-line analysis script does not need targets. Use pipelines when the analysis has 5+ interdependent steps or will be re-run multiple times.
- **Reproducibility is not eternal**: Even with Docker, operating system patches and hardware changes can affect results. Archive final outputs alongside the pipeline.

35.8 Common pitfalls

35.9 Common pitfalls

- **Not separating data retrieval from analysis.** API calls in analysis targets re-fetch data on every change. Separate fetching into its own target with file-based caching.
- **Forgetting to commit renv.lock.** The lockfile is the reproducibility guarantee. Always version-control it.
- **Ignoring the _targets/ directory in .gitignore.** Cached results should not be committed to git (they can be regenerated). Add `_targets/` to `.gitignore`.

- **Writing monolithic targets.** Each target should do one thing. Fine-grained targets enable selective re-execution.

35.10 Exercises

1. **Build a pipeline.** Create a `_targets.R` file for a simple bibliometric analysis: fetch → clean → summarise → plot. Run it and verify that changing the summary function only re-runs downstream targets.
2. **renv snapshot.** Initialise renv in a project, install a new package, and snapshot. Delete the package and restore from the lockfile.
3. **Branching.** Use `tar_target()` with dynamic branching to run the same analysis across multiple journals in parallel.

35.11 Solutions

Solutions are provided in F.

35.12 Further reading

- Hicks et al. (2015) — The Leiden Manifesto: transparency and reproducibility in research evaluation.
- Priem et al. (2022) — OpenAlex as a reproducible data source for bibliometric pipelines.

35.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
```

```
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] uwot_0.2.4                Matrix_1.7-0              word2vec_0.4.1            stm_1.3.8
#>  [5] topicmodels_0.2-17        quanteda.textstats_0.97.2 visNetwork_2.1.4          ggraph_2.2.
#>  [9] tidygraph_1.3.1           igraph_2.3.1              quanteda_4.4              pdftools_3.
#> [13] arrow_24.0.0              bibliometrix_5.4.0        RefManageR_1.4.0          bib2df_1.1.
#> [17] rcrossref_1.2.1           gt_1.3.0                  tidytext_0.4.3            glue_1.8.1
#> [21] openalexR_3.0.1           lubridate_1.9.5           forcats_1.0.1             stringr_1.6
#> [25] dplyr_1.2.1               purrr_1.2.2               readr_2.2.0               tidyr_1.3.2
#> [29] tibble_3.3.1              ggplot2_4.0.3             tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2         RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
#>   [6] modeltools_0.2-24    farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1        askpass_1.2.1         base64enc_0.1-6       htmltools_0.5.9
#>  [16] curl_7.1.0           broom_1.0.12          janeaustenr_1.0.0     cellranger_1.1.0
#>  [21] tokenizers_0.3.0     plyr_1.8.9            httr2_1.2.2           cachem_1.1.0
#>  [26] dimensionsR_0.0.3    mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3
#>  [31] fastmap_1.2.0        shiny_1.13.0          digest_0.6.39         patchwork_1.3.2
#>  [36] rprojroot_2.1.1      RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3
#>  [41] timechange_0.4.0     mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8
#>  [46] here_1.0.2           bit64_4.8.0           withr_3.0.2           S7_0.2.2
#>  [51] viridis_0.6.5        ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4
#>  [56] tools_4.4.1          otel_0.2.0            stopwords_2.3         zip_2.3.3
#>  [61] rentrez_1.2.4        nlme_3.1-164          promises_1.5.0        grid_4.4.1
#>  [66] reshape2_1.4.5       generics_0.1.4        gtable_0.3.6          tzdb_0.5.0
#>  [71] ca_0.71.1            data.table_1.18.4     hms_1.1.4             xml2_1.5.2
#>  [76] ggrepel_0.9.8        pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1
#>  [81] splines_4.4.1        tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1
#>  [86] tidyselect_1.2.1     tm_0.7-18             miniUI_0.1.2          knitr_1.51
#>  [91] NLP_0.3-2            bookdown_0.46         stats4_4.4.1          crul_1.6.0
#>  [96] graphlayouts_1.2.3   matrixStats_1.5.0     DT_0.34.0             humaniformat_0.6.0
#> [101] lazyeval_0.2.3       qpdf_1.4.1            yaml_2.3.12           evaluate_1.0.5
#> [106] httpcode_0.3.0       cli_3.6.6             xtable_1.8-8          dichromat_2.0-0.1
#> [111] readxl_1.4.5         triebeard_0.4.1       XML_3.99-0.23         parallel_4.4.1
#> [116] pubmedR_1.0.2        slam_0.1-55           viridisLite_0.4.3     scales_1.4.0
#> [121] openxlsx_4.2.8.1     rlang_1.2.0           fastmatch_1.1-8
```

Chapter 36

Interactive Dashboards

36.1 Learning objectives

After completing this chapter, you will be able to:

- Create a Quarto dashboard layout with multiple panels and pages
- Convert ggplot2 figures to interactive plotly charts
- Build searchable, filterable data tables with the DT package
- Design effective dashboard layouts for bibliometric data exploration
- Deploy dashboards as self-contained HTML files

36.2 Setup

```r
library(tidyverse)
library(openalexR)
library(plotly)
library(DT)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

36.3 Conceptual background

Bibliometric analysis often produces more data than can be communicated through static figures and tables. Interactive dashboards allow stakeholders — research managers, librarians, policy makers — to explore data on their own terms: filtering by year, hovering for details, sorting tables, and drilling down into specific subsets.

Quarto dashboards use the `format: dashboard` output type to arrange content in cards, rows, and columns. They support static and interactive content side by side, and can be deployed as self-contained HTML files (no server needed) or as server-backed applications.

plotly converts ggplot2 figures into interactive JavaScript visualisations with hover tooltips, zooming, panning, and selection. The `ggplotly()` function provides a zero-effort upgrade path from static to interactive figures, though fine-tuning often requires direct plotly syntax.

DT (DataTables) renders data frames as interactive HTML tables with search, sort, pagination, and filtering. For bibliometric data, DT is invaluable: users can search for specific authors, sort by citation count, and filter by year — all without writing code.

The key design principle for dashboards is **progressive disclosure**: show the most important summary first, then let users drill down into details. Avoid overwhelming users with every metric at once.

36.4 Worked example

36.4.1 Preparing dashboard data

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)

dash_data <- works |>
  transmute(
    id, title = display_name,
    year = year(publication_date),
    cited_by_count, oa_status, type, source_display_name
```

```
  )

cat(glue("Dashboard data: {nrow(dash_data)} records\n"))
```

```
#> Dashboard data: 500 records
```

36.4.2 Interactive trend chart

```
# Interactive plotly widget - only rendered for HTML output.
trend <- dash_data |>
  count(year)

p <- ggplot(trend, aes(x = year, y = n)) +
  geom_line(colour = palette_sci(1), linewidth = 1) +
  geom_point(colour = palette_sci(1), size = 3) +
  labs(x = "Year", y = "Publications") +
  theme_sci()

ggplotly(p)
```

36.4.3 Interactive citation distribution

```
# Interactive plotly widget - only rendered for HTML output.
p2 <- ggplot(dash_data, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 5, fill = palette_sci(1), colour = "white") +
  labs(x = "Citation count", y = "Papers") +
  theme_sci()

ggplotly(p2)
```

36.4.4 Searchable data table

```
# Interactive DT widget - only rendered for HTML output.
dash_data |>
  select(title, year, cited_by_count, oa_status) |>
  arrange(desc(cited_by_count)) |>
  datatable(
    options = list(pageLength = 10, searchHighlight = TRUE),
```

```
    filter = "top",
    rownames = FALSE
  )
```

36.4.5 OA status breakdown (interactive)

```
# Interactive plotly widget - only rendered for HTML output.
oa_trend <- dash_data |>
  count(year, oa_status)

p3 <- ggplot(oa_trend, aes(x = factor(year), y = n, fill = oa_status)) +
  geom_col() +
  scale_fill_manual(values = palette_sci(n_distinct(oa_trend$oa_status))) +
  labs(x = "Year", y = "Count", fill = "OA Status") +
  theme_sci()

ggplotly(p3)
```

36.5 Diagnostics and interpretation

- **Performance**: Large datasets (> 10,000 rows) can make plotly and DT slow. Aggregate or sample data before rendering interactive widgets.
- **Accessibility**: Interactive figures are not accessible to screen readers. Always provide alt text and consider static fallbacks.
- **File size**: Self-contained HTML dashboards embed all data and JavaScript libraries. Files can reach 5–50 MB for complex dashboards.
- **Browser compatibility**: Test in Chrome, Firefox, and Safari. Some plotly features render differently across browsers.

36.6 Limitations and responsible use

36.7 Limitations and responsible use

- **Interactivity is not analysis.** A dashboard lets users explore data, but exploration without statistical rigour can produce misleading impressions. Pair dashboards with proper analysis.
- **Dashboards can mislead.** Default axis ranges, binning, and colour choices affect interpretation. Design dashboards to prevent misreading, not enable it.

- **Data privacy.** Interactive tables may expose individual-level data (author names, paper titles). Consider whether the dashboard's audience should see this level of detail.
- **Maintenance burden.** Dashboards that query live APIs need updating when APIs change. Static dashboards based on cached data are more sustainable (Hicks et al., 2015).

36.8 Common pitfalls

36.9 Common pitfalls

- **Too many widgets.** A dashboard with 15 interactive panels is overwhelming. Start with 3–5 and add more only if users need them.
- **Not filtering large datasets.** Rendering 50,000 rows in DT or 100,000 points in plotly will crash the browser. Pre-aggregate.
- **Ignoring PDF fallbacks.** If your Quarto book renders to PDF, interactive widgets become static. Ensure the static version is still informative.
- **Using ggplotly without cleanup.** `ggplotly()` produces verbose tooltips. Customise hover text with `tooltip` parameter or direct plotly.

36.10 Exercises

1. **Multi-page dashboard.** Create a Quarto dashboard (format: dashboard) with two pages: "Overview" (trend + citation distribution) and "Details" (searchable table).

2. **Linked brushing.** Use plotly's selection features to highlight papers in a scatter plot and simultaneously filter a DT table below.

3. **Custom tooltips.** Modify the trend chart to show "Year: 2023, Papers: 87, Mean cites: 12.3" on hover instead of the default tooltip.

36.11 Solutions

Solutions are provided in F.

36.12 Further reading

- Aria and Cuccurullo (2017) — `bibliometrix` and `biblioshiny` for interactive bibliometric exploration.

- Priem et al. (2022) — OpenAlex as a data source for dashboard construction.

36.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLAT
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRES
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] DT_0.34.0                plotly_4.12.0            uwot_0.2.4
#>  [5] word2vec_0.4.1           stm_1.3.8                topicmodels_0.2-17
#>  [9] visNetwork_2.1.4         ggraph_2.2.2             tidygraph_1.3.1
#> [13] quanteda_4.4             pdftools_3.9.0           arrow_24.0.0
#> [17] RefManageR_1.4.0         bib2df_1.1.2.0           rcrossref_1.2.1
#> [21] tidytext_0.4.3           glue_1.8.1               openalexR_3.0.1
#> [25] forcats_1.0.1            stringr_1.6.0            dplyr_1.2.1
#> [29] readr_2.2.0              tidyr_1.3.2              tibble_3.3.1
#> [33] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite
#>   [6] modeltools_0.2-24     farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1          base64enc_0.1-6        htmltool
#>  [16] curl_7.1.0            broom_1.0.12           janeaustenr_1.0.0      cellrang
#>  [21] tokenizers_0.3.0      plyr_1.8.9             httr2_1.2.2            cachem_1
```

```
#>  [26] mime_0.13             lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39          patchwork_1.3.2        shinycssloaders_1.1
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1        labeling_0.4.3         urltools_1.7.3.1
#>  [41] mgcv_1.9-1            polyclip_1.10-7        httr_1.4.8             compiler_4.4.1
#>  [46] bit64_4.8.0           withr_3.0.2            S7_0.2.2               backports_1.5.1
#>  [51] ggforce_0.5.0         MASS_7.3-60.2          rappdirs_0.3.4         bibliometrixData_0.
#>  [56] otel_0.2.0            stopwords_2.3          zip_2.3.3              httpuv_1.6.17
#>  [61] nlme_3.1-164          promises_1.5.0         grid_4.4.1             stringdist_0.9.17
#>  [66] generics_0.1.4        gtable_0.3.6           tzdb_0.5.0             rscopus_0.9.0
#>  [71] data.table_1.18.4     hms_1.1.4              xml2_1.5.2             utf8_1.2.6
#>  [76] pillar_1.11.1         nsyllable_1.0.1        vroom_1.7.1            later_1.4.8
#>  [81] tweenr_2.0.3          lattice_0.22-6         FNN_1.1.4.1            bit_4.6.0
#>  [86] tm_0.7-18             miniUI_0.1.2           knitr_1.51             gridExtra_2.3
#>  [91] bookdown_0.46         stats4_4.4.1           crul_1.6.0             xfun_0.57
#>  [96] matrixStats_1.5.0     humaniformat_0.6.0     stringi_1.8.7          lazyeval_0.2.3
#> [101] yaml_2.3.12           evaluate_1.0.5         codetools_0.2-20       httpcode_0.3.0
#> [106] xtable_1.8-8          dichromat_2.0-0.1      Rcpp_1.1.1-1.1         readxl_1.4.5
#> [111] XML_3.99-0.23         parallel_4.4.1         assertthat_0.2.1       pubmedR_1.0.2
#> [116] viridisLite_0.4.3     scales_1.4.0           crayon_1.5.3           openxlsx_4.2.8.1
#> [121] fastmatch_1.1-8
```

Chapter 37

Parameterized Quarto Reports

37.1 Learning objectives

After completing this chapter, you will be able to:

- Define parameters in a Quarto document's YAML header
- Access parameters in R code chunks to drive data retrieval and analysis
- Render parameterized reports from the command line and from scripts
- Build a template for journal-level bibliometric profiles
- Generate multiple reports in batch using purrr or shell loops

37.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

37.3 Conceptual background

Many bibliometric tasks are repetitive: generating the same analysis for 10 journals, 5 departments, or 20 funders. Writing separate scripts for each entity is error-prone and hard to maintain. **Parameterized reports** solve this by defining variables (parameters) that change the report's content without changing its structure.

Quarto supports parameters via the `params` field in the YAML header. Parameters can be strings, numbers, or lists. Inside the document, `params$journal_id` (or `params$institution`) drives data fetching, filtering, and labelling. At render time, parameter values are passed via the command line or a rendering script.

This approach scales naturally: a loop or `purrr::walk()` call renders dozens of reports with different parameters, producing a complete portfolio of bibliometric profiles with consistent methodology and formatting.

Parameterized reports are especially valuable for institutional reporting (annual bibliometric profiles for each department), journal analytics (quarterly reports for a publisher's journal portfolio), and funding assessment (per-grant output summaries).

37.4 Worked example

37.4.1 Defining parameters

A parameterized Quarto document starts with parameters in the YAML header:

```
# Example YAML header (not executable R):
#
# ---
# title: "Journal Bibliometric Profile"
# params:
#   journal_id: "S148561398"
#   journal_name: "Scientometrics"
#   start_year: 2020
#   end_year: 2023
# ---
```

37.4.2 Using parameters in code

```r
journal_id <- "S148561398"
journal_name <- "Scientometrics"
start_year <- 2020
end_year <- 2023
```

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = journal_id,
  from_publication_date = paste0(start_year, "-01-01"),
  to_publication_date = paste0(end_year, "-12-31"),
  type = "article",
  options = list(sample = 300, seed = 42)
)

cat(glue("Journal: {journal_name}\n"))
```

```
#> Journal: Scientometrics
```

```r
cat(glue("Period: {start_year}--{end_year}\n"))
```

```
#> Period: 2020--2023
```

```r
cat(glue("Articles retrieved: {nrow(works)}\n"))
```

```
#> Articles retrieved: 300
```

37.4.3 Automated summary statistics

```r
summary_stats <- works |>
  summarise(
    total_articles = n(),
    mean_citations = round(mean(cited_by_count), 1),
    median_citations = median(cited_by_count),
    h_index = compute_h_index(cited_by_count),
    oa_rate = round(mean(oa_status != "closed", na.rm = TRUE) * 100, 1)
  ) |>
  mutate(across(everything(), as.character)) |>
  pivot_longer(everything(), names_to = "Metric", values_to = "Value")

summary_stats |> gt()
```

Metric	Value
total_articles	300
mean_citations	19.4
median_citations	12
h_index	35
oa_rate	46.3

37.4.4 Parameterized figure

```
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = factor(year), y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Articles",
       title = glue("{journal_name} - Annual Output")) +
  theme_sci()
```

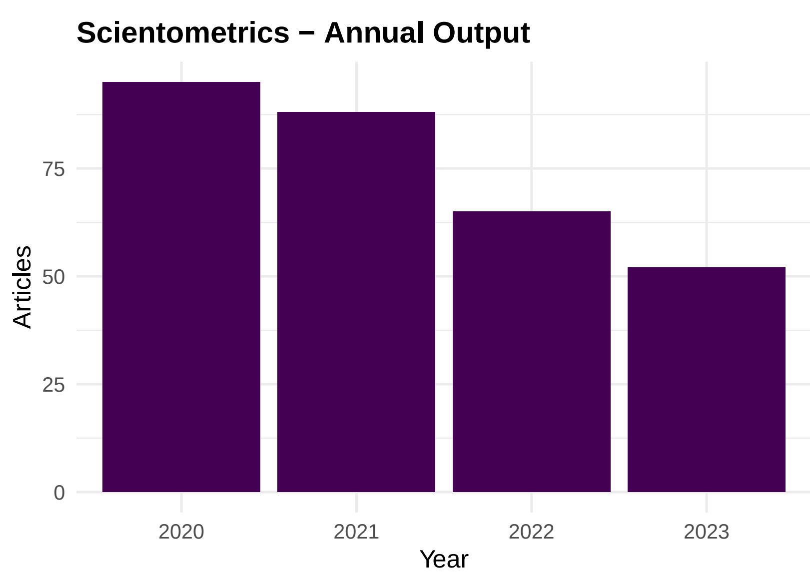

Figure 37.1: Annual publication count for the selected journal.

37.4.5 Batch rendering

```r
# Render for multiple journals:
journals <- tibble(
  journal_id = c("S148561398", "S202381698", "S137773608"),
  journal_name = c("Scientometrics", "PLOS ONE", "Nature")
)

walk2(journals$journal_id, journals$journal_name, function(jid, jname) {
  quarto::quarto_render(
    input = "journal_profile.qmd",
    execute_params = list(
      journal_id = jid,
      journal_name = jname,
      start_year = 2020,
      end_year = 2023
    ),
    output_file = glue("reports/{jname}_profile.html")
  )
})
```

37.5 Diagnostics and interpretation

- **Parameter validation**: Check that parameters produce valid API queries. Invalid journal IDs return empty results silently.
- **Sample consistency**: When using `options = list(sample = N)`, the sample changes if the underlying corpus changes. Use `seed` for stability.
- **Output naming**: Automated file naming should avoid special characters. Sanitise journal names before using them in file paths.
- **Error handling**: Batch rendering should catch and log errors for individual reports without stopping the entire batch.

37.6 Limitations and responsible use

37.7 Limitations and responsible use

- **Automated mindless.** Parameterized reports produce consistent output but still require human review. An automated report with a data error will produce the wrong results consistently across all entities.

- **One template fits all?** Different entities may need different analyses. A template designed for a large journal may produce misleading results for a small one (e.g., sparse yearly bins).
- **API costs.** Batch rendering for 100 journals makes 100 API calls. Respect rate limits and cache results.
- **Report fatigue.** Generating too many reports can overwhelm decision-makers. Produce reports for entities that need them, not for every possible parameter value (Hicks et al., 2015).

37.8 Common pitfalls

37.9 Common pitfalls

- **Hardcoding values that should be parameters.** If you find yourself editing a date or ID in the script, it should be a parameter.
- **Not testing with extreme parameters.** A journal with 5 papers per year needs different visualisations than one with 5,000. Test edge cases.
- **Rendering without caching.** Each render re-executes all code. Use Quarto's `cache: true` or file-based caching to avoid redundant API calls.
- **Forgetting output format compatibility.** Interactive widgets work in HTML but not PDF. Ensure your template degrades gracefully.

37.10 Exercises

1. **Institutional template.** Create a parameterized Quarto document for institutional profiles. Parameters: institution ID, name, start year, end year. Include publication count, citation distribution, and top authors.

2. **Batch rendering.** Render the journal profile template for 3 different journals. Compare the resulting reports side by side.

3. **Conditional content.** Add conditional logic: if the sample has fewer than 50 papers, display a warning message instead of the citation distribution figure.

37.11 Solutions

Solutions are provided in F.

37.12 Further reading

- Aria and Cuccurullo (2017) — `bibliometrix` reporting features.
- Priem et al. (2022) — OpenAlex API for programmatic data retrieval in reports.

37.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] DT_0.34.0              plotly_4.12.0          uwot_0.2.4             Matrix_1.7-
#>  [5] word2vec_0.4.1         stm_1.3.8              topicmodels_0.2-17     quanteda.te
#>  [9] visNetwork_2.1.4       ggraph_2.2.2           tidygraph_1.3.1        igraph_2.3.
#> [13] quanteda_4.4           pdftools_3.9.0         arrow_24.0.0           bibliometri
#> [17] RefManageR_1.4.0       bib2df_1.1.2.0         rcrossref_1.2.1        gt_1.3.0
#> [21] tidytext_0.4.3         glue_1.8.1             openalexR_3.0.1        lubridate_1
#> [25] forcats_1.0.1          stringr_1.6.0          dplyr_1.2.1            purrr_1.2.2
#> [29] readr_2.2.0            tidyr_1.3.2            tibble_3.3.1           ggplot2_4.0
#> [33] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3    rstudioapi_0.18.0     jsonlite_2.0.0
```

```
#>   [6] modeltools_0.2-24     farver_2.1.2          rmarkdown_2.31        fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1         base64enc_0.1-6       htmltool
#>  [16] curl_7.1.0            broom_1.0.12          janeaustenr_1.0.0     cellrang
#>  [21] tokenizers_0.3.0      plyr_1.8.9            httr2_1.2.2           cachem_1
#>  [26] mime_0.13             lifecycle_1.0.5       pkgconfig_2.0.3       R6_2.6.1
#>  [31] shiny_1.13.0          digest_0.6.39         patchwork_1.3.2       shinycss
#>  [36] RSpectra_0.16-2       SnowballC_0.7.1       labeling_0.4.3        urltools
#>  [41] mgcv_1.9-1            polyclip_1.10-7       httr_1.4.8            compiler
#>  [46] bit64_4.8.0           withr_3.0.2           S7_0.2.2              backport
#>  [51] ggforce_0.5.0         MASS_7.3-60.2         rappdirs_0.3.4        bibliome
#>  [56] otel_0.2.0            stopwords_2.3         zip_2.3.3             httpuv_1
#>  [61] nlme_3.1-164          promises_1.5.0        grid_4.4.1            stringdi
#>  [66] generics_0.1.4        gtable_0.3.6          tzdb_0.5.0            rscopus_
#>  [71] data.table_1.18.4     hms_1.1.4             xml2_1.5.2            utf8_1.2
#>  [76] pillar_1.11.1         nsyllable_1.0.1       vroom_1.7.1           later_1.
#>  [81] tweenr_2.0.3          lattice_0.22-6        FNN_1.1.4.1           bit_4.6.
#>  [86] tm_0.7-18             miniUI_0.1.2          knitr_1.51            gridExtr
#>  [91] bookdown_0.46         stats4_4.4.1          crul_1.6.0            xfun_0.5
#>  [96] matrixStats_1.5.0     humaniformat_0.6.0    stringi_1.8.7         lazyeval
#> [101] yaml_2.3.12           evaluate_1.0.5        codetools_0.2-20      httpcode
#> [106] xtable_1.8-8          dichromat_2.0-0.1     Rcpp_1.1.1-1.1        readxl_1
#> [111] XML_3.99-0.23         parallel_4.4.1        assertthat_0.2.1      pubmedR_
#> [116] viridisLite_0.4.3     scales_1.4.0          crayon_1.5.3          openxlsx
#> [121] fastmatch_1.1-8
```

Chapter 38

Building a Bibliometric Shiny App

38.1 Learning objectives

After completing this chapter, you will be able to:

- Design a Shiny app architecture for bibliometric analysis
- Implement reactive data fetching from OpenAlex
- Build interactive UI components: dropdowns, sliders, and tabbed panels
- Create server-side logic for citation analysis and network visualisation
- Deploy a Shiny app to shinyapps.io or a self-hosted server

38.2 Setup

```r
library(tidyverse)
library(shiny)
library(openalexR)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

38.3 Conceptual background

Shiny transforms R scripts into web applications. Users interact through widgets (dropdowns, sliders, buttons) and see results update in real time. For bibliometrics, Shiny apps serve two audiences: researchers who want to explore data without writing code, and decision-makers who need accessible analytics dashboards.

A Shiny app has two components: the **UI** (user interface, defining layout and inputs) and the **server** (the R logic that reacts to user input and generates output). **Reactive programming** connects inputs to outputs: when a user changes a dropdown, only the computations that depend on that input are re-executed.

For bibliometric apps, the typical architecture involves:

1. **Input panel**: Select a journal, institution, or search query; choose date range and sample size.
2. **Data fetching**: Reactively query OpenAlex based on user selections.
3. **Analysis tabs**: Trend charts, citation distributions, author rankings, and optionally network visualisation.
4. **Export**: Download buttons for data (CSV) and figures (PNG).

`biblioshiny()` from the `bibliometrix` package (Aria and Cuccurullo, 2017) provides a ready-made Shiny interface for many standard analyses. Building a custom app gives full control over the analysis pipeline and user experience.

Deployment options include shinyapps.io (hosted by Posit, free tier available), Posit Connect (enterprise), and self-hosted via Shiny Server (open source).

38.4 Worked example

38.4.1 App structure

We demonstrate the key components of a bibliometric Shiny app. The code below shows the UI and server logic separately for clarity.

38.4.2 UI definition

```r
ui <- fluidPage(
  titlePanel("Bibliometric Explorer"),
```

```
  sidebarLayout(
    sidebarPanel(
      textInput("journal_id", "OpenAlex Source ID:",
                value = "S148561398"),
      sliderInput("year_range", "Publication Years:",
                  min = 2015, max = 2023, value = c(2020, 2023)),
      numericInput("sample_size", "Sample Size:", value = 200,
                   min = 50, max = 1000),
      actionButton("fetch", "Fetch Data", class = "btn-primary"),
      hr(),
      downloadButton("download_csv", "Download CSV")
    ),

    mainPanel(
      tabsetPanel(
        tabPanel("Trends",
                 plotOutput("trend_plot"),
                 tableOutput("summary_table")),
        tabPanel("Citations",
                 plotOutput("cite_dist"),
                 plotOutput("cite_boxplot")),
        tabPanel("Data",
                 DT::dataTableOutput("data_table"))
      )
    )
  )
)
```

38.4.3 Server logic

```
server <- function(input, output, session) {

  works_data <- eventReactive(input$fetch, {
    withProgress(message = "Fetching from OpenAlex...", {
      oa_fetch(
        entity = "works",
        primary_location.source.id = input$journal_id,
        from_publication_date = paste0(input$year_range[1], "-01-01"),
        to_publication_date = paste0(input$year_range[2], "-12-31"),
        type = "article",
        options = list(sample = input$sample_size, seed = 42)
      ) |>
        transmute(
```

```r
        id, title = display_name,
        year = year(publication_date),
        cited_by_count, oa_status
      )
  })
})

output$trend_plot <- renderPlot({
  req(works_data())
  works_data() |>
    count(year) |>
    ggplot(aes(x = factor(year), y = n)) +
    geom_col(fill = palette_sci(1)) +
    labs(x = "Year", y = "Publications") +
    theme_sci()
})

output$cite_dist <- renderPlot({
  req(works_data())
  ggplot(works_data(), aes(x = cited_by_count)) +
    geom_histogram(binwidth = 5, fill = palette_sci(1), colour = "white") +
    labs(x = "Citations", y = "Count") +
    theme_sci()
})

output$cite_boxplot <- renderPlot({
  req(works_data())
  works_data() |>
    filter(!is.na(oa_status)) |>
    ggplot(aes(x = oa_status, y = cited_by_count + 1)) +
    geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
    scale_y_log10() +
    labs(x = "OA Status", y = "Citations (log)") +
    theme_sci()
})

output$summary_table <- renderTable({
  req(works_data())
  works_data() |>
    summarise(
      Articles = n(),
      `Mean Citations` = round(mean(cited_by_count), 1),
      `Median Citations` = median(cited_by_count),
      `OA Rate` = scales::percent(mean(oa_status != "closed", na.rm = TRUE))
    )
```

```r
  })

  output$data_table <- DT::renderDataTable({
    req(works_data())
    works_data() |>
      select(title, year, cited_by_count, oa_status) |>
      arrange(desc(cited_by_count))
  })

  output$download_csv <- downloadHandler(
    filename = function() paste0("biblio_data_", Sys.Date(), ".csv"),
    content = function(file) write_csv(works_data(), file)
  )
}
```

38.4.4 Running the app

```r
shinyApp(ui = ui, server = server)
```

38.4.5 biblioshiny as an alternative

```r
# bibliometrix provides a ready-made Shiny interface:
# library(bibliometrix)
# biblioshiny()
```

38.5 Diagnostics and interpretation

- **Responsiveness**: API calls can take several seconds. Use `withProgress()` to show loading indicators.
- **Error handling**: Wrap API calls in `tryCatch()` to display user-friendly error messages instead of stack traces.
- **Memory**: Each user session loads data into R memory. For apps with many concurrent users, pre-aggregate data or use database backends.
- **Testing**: Use `shinytest2` to automate testing of Shiny apps.

38.6 Limitations and responsible use

38.7 Limitations and responsible use

- **Shiny apps are not analyses.** They enable exploration but do not substitute for rigorous, documented analytical pipelines. Use Shiny for discovery, not for producing final results.
- **API rate limits.** Live OpenAlex queries from a Shiny app can hit rate limits if many users fetch data simultaneously. Use caching or pre-fetched data for production apps.
- **Maintenance.** Shiny apps require ongoing maintenance: API changes, package updates, and server costs. Plan for long-term sustainability before deploying.
- **Interpretive guardrails.** Non-technical users may misinterpret metrics. Build in tooltips, descriptions, and warnings — especially around citation-based indicators (Hicks et al., 2015; American Society for Cell Biology, 2012).

38.8 Common pitfalls

38.9 Common pitfalls

- **Fetching data on every input change.** Use `eventReactive()` with an action button instead of `reactive()` to prevent re-fetching on every slider adjustment.
- **Not validating inputs.** Check that the journal ID exists and the date range is valid before making API calls.
- **Monolithic server functions.** Break the server logic into modules for maintainability as the app grows.
- **Deploying without testing.** Test with realistic data volumes before deploying. An app that works with 100 records may fail with 10,000.

38.10 Exercises

1. **Add network tab.** Extend the app with a tab that builds and displays a co-authorship network using `visNetwork`.

2. **Caching.** Implement server-side caching so that repeated queries with the same parameters return instantly.

3. **Deployment.** Deploy the app to shinyapps.io. Test with different journal IDs and date ranges.

4. **biblioshiny comparison.** Run `biblioshiny()` and compare its features with your custom app. What does biblioshiny offer that your app does not?

38.11 Solutions

Solutions are provided in F.

38.12 Further reading

- Aria and Cuccurullo (2017) — `bibliometrix` and `biblioshiny()` for turnkey bibliometric exploration.
- Priem et al. (2022) — OpenAlex API for reactive data fetching.
- Hicks et al. (2015) — The Leiden Manifesto: designing interfaces that prevent metric misuse.
- American Society for Cell Biology (2012) — DORA: responsible assessment in interactive tools.

38.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8        LC_COLLATE=C.UTF-8
#>  [6] LC_MESSAGES=C.UTF-8    LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
```

```
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] shiny_1.13.0             DT_0.34.0                plotly_4.12.0
#>  [5] Matrix_1.7-0             word2vec_0.4.1           stm_1.3.8
#>  [9] quanteda.textstats_0.97.2 visNetwork_2.1.4        ggraph_2.2.2
#> [13] igraph_2.3.1             quanteda_4.4             pdftools_3.9.0
#> [17] bibliometrix_5.4.0       RefManageR_1.4.0         bib2df_1.1.2.0
#> [21] gt_1.3.0                 tidytext_0.4.3           glue_1.8.1
#> [25] lubridate_1.9.5          forcats_1.0.1            stringr_1.6.0
#> [29] purrr_1.2.2              readr_2.2.0              tidyr_1.3.2
#> [33] ggplot2_4.0.3            tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.2          RColorBrewer_1.1-3     rstudioapi_0.18.0      jsonlite
#>   [6] modeltools_0.2-24     farver_2.1.2           rmarkdown_2.31         fs_2.1.0
#>  [11] memoise_2.0.1         askpass_1.2.1          base64enc_0.1-6        htmltool
#>  [16] curl_7.1.0            broom_1.0.12           janeaustenr_1.0.0      cellrang
#>  [21] tokenizers_0.3.0      plyr_1.8.9             httr2_1.2.2            cachem_1
#>  [26] mime_0.13             lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [31] digest_0.6.39         patchwork_1.3.2        shinycssloaders_1.1.0  rprojroo
#>  [36] SnowballC_0.7.1       labeling_0.4.3         urltools_1.7.3.1       timechan
#>  [41] polyclip_1.10-7       httr_1.4.8             compiler_4.4.1         here_1.0
#>  [46] withr_3.0.2           S7_0.2.2               backports_1.5.1        viridis_
#>  [51] MASS_7.3-60.2         rappdirs_0.3.4         bibliometrixData_0.3.0 tools_4.
#>  [56] stopwords_2.3         zip_2.3.3              httpuv_1.6.17          rentrez_
#>  [61] promises_1.5.0        grid_4.4.1             stringdist_0.9.17      reshape2
#>  [66] gtable_0.3.6          tzdb_0.5.0             rscopus_0.9.0          ca_0.71.
#>  [71] hms_1.1.4             xml2_1.5.2             utf8_1.2.6             ggrepel_
#>  [76] nsyllable_1.0.1       vroom_1.7.1            later_1.4.8            splines_
#>  [81] lattice_0.22-6        FNN_1.1.4.1            bit_4.6.0              tidysele
#>  [86] miniUI_0.1.2          knitr_1.51             gridExtra_2.3          NLP_0.3-
#>  [91] stats4_4.4.1          crul_1.6.0             xfun_0.57              graphlay
#>  [96] humaniformat_0.6.0    stringi_1.8.7          lazyeval_0.2.3         qpdf_1.4
#> [101] evaluate_1.0.5        codetools_0.2-20       httpcode_0.3.0         cli_3.6.
#> [106] dichromat_2.0-0.1     Rcpp_1.1.1-1.1         readxl_1.4.5           triebear
#> [111] parallel_4.4.1        assertthat_0.2.1       pubmedR_1.0.2          slam_0.1
#> [116] scales_1.4.0          crayon_1.5.3           openxlsx_4.2.8.1       rlang_1.
```

Part VIII

Case Studies

Chapter 39

Case Study 1: CRISPR Field Review (2010–2024)

39.1 Objective

Map the structure, growth, and evolution of CRISPR gene editing research from its emergence to its current state using open bibliometric data.

39.2 Setup

```r
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(quanteda)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

39.3 Data acquisition

```
works <- oa_fetch(
  entity = "works",
  search = "CRISPR",
  from_publication_date = "2010-01-01",
  to_publication_date = "2024-06-30",
  type = "article",
  options = list(sample = 500, seed = 42)
)

works <- works |>
  mutate(year = year(publication_date))

cat(glue("CRISPR articles retrieved: {nrow(works)}\n"))
```

```
#> CRISPR articles retrieved: 500
```

```
cat(glue("Year range: {min(works$year)}--{max(works$year)}\n"))
```

```
#> Year range: 2010--2024
```

39.4 Publication growth

```
works |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Publications") +
  theme_sci()
```

39.5 Citation landscape

```
ggplot(works, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 10, fill = palette_sci(1), colour = "white") +
  labs(x = "Citations", y = "Papers") +
  theme_sci()
```

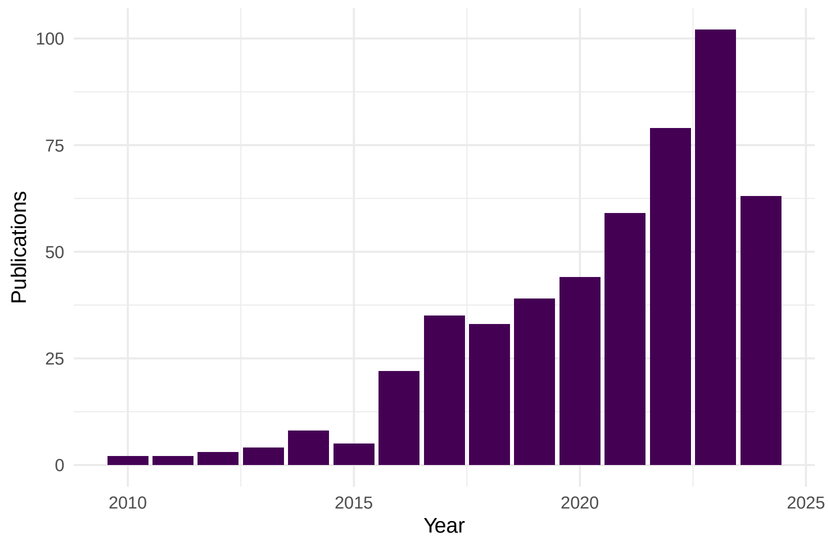

Figure 39.1: Annual publication output in the CRISPR field.

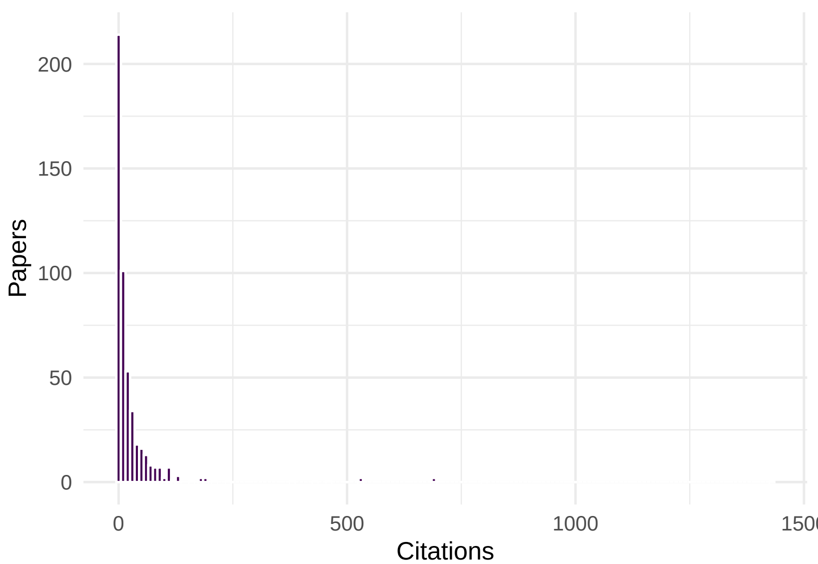

Figure 39.2: Citation distribution of CRISPR articles.

display_name
CRISPR-engineered T cells in patients with refractory cancer
A high-fidelity Cas9 mutant delivered as a ribonucleoprotein complex enables efficient g
Efficient genome engineering in human pluripotent stem cells using Cas9 from <i>Neiss
Enhancement of the gut barrier integrity by a microbial metabolite through the Nrf2 pa
An enhanced CRISPR repressor for targeted mammalian gene regulation
Efficient targeted multiallelic mutagenesis in tetraploid potato (Solanum tuberosum) by
CD22 blockade restores homeostatic microglial phagocytosis in ageing brains
Programmable Removal of Bacterial Strains by Use of Genome-Targeting CRISPR-Cas
RAP2 mediates mechanoresponses of the Hippo pathway
Efficient CRISPR-Cas9–mediated genome editing in Plasmodium falciparum

```r
works |>
  arrange(desc(cited_by_count)) |>
  head(10) |>
  select(display_name, year, cited_by_count, source_display_name) |>
  gt()
```

39.6 Co-authorship network

```r
author_data <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  select(work_id = id, author_id = authorships_id,
         author_name = authorships_display_name) |>
  filter(!is.na(author_id))

edges <- author_data |>
  inner_join(author_data, by = "work_id", suffix = c("_1", "_2"),
             relationship = "many-to-many") |>
  filter(author_id_1 < author_id_2) |>
  count(author_id_1, author_id_2, name = "weight")

g <- graph_from_data_frame(
  edges |> select(author_id_1, author_id_2, weight),
  directed = FALSE
) |> simplify(edge.attr.comb = list(weight = "sum"))
```

```r
comp <- components(g)
giant <- induced_subgraph(g, which(comp$membership == which.max(comp$csize)))
V(giant)$community <- as.factor(membership(
  cluster_leiden(giant, resolution_parameter = 1.0,
                 objective_function = "modularity")
))
V(giant)$degree <- degree(giant)

cat(glue("Network: {vcount(giant)} nodes, {ecount(giant)} edges\n"))
```

```
#> Network: 71 nodes, 588 edges
```

```r
set.seed(42)
ggraph(as_tbl_graph(giant), layout = "fr") +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.7) +
  scale_size_continuous(range = c(0.5, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(n_distinct(V(giant)$community))) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) + theme(legend.position = "bottom")
```

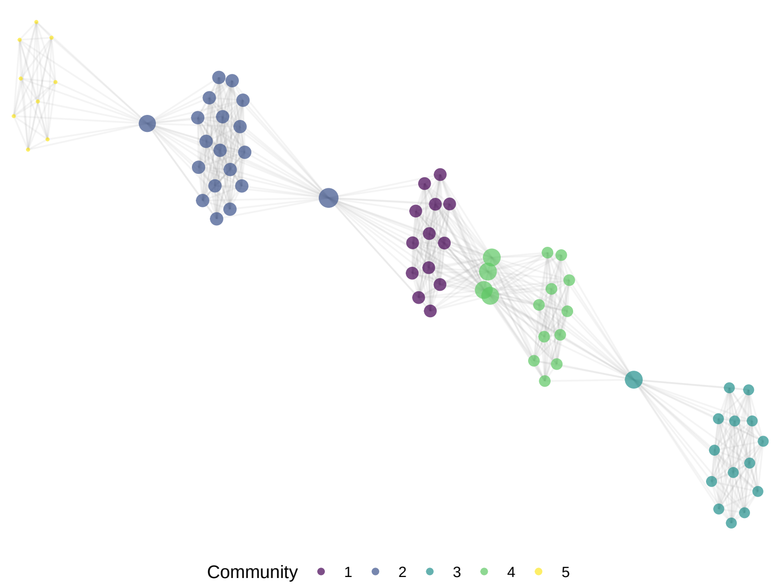

Figure 39.3: Co-authorship network of CRISPR researchers.

39.7 Topic evolution

```
text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(doc_id = id, text = paste(display_name, abstract, sep = ". "), year)

corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "using", "based"))

dfmat <- dfm(toks) |> dfm_trim(min_termfreq = 5, min_docfreq = 3)

top_by_year <- map_dfr(unique(text_df$year), function(yr) {
  docs <- docvars(dfmat, "year") == yr
  if (sum(docs) < 5) return(tibble())
  top <- topfeatures(dfmat[docs, ], 5)
  tibble(year = yr, term = names(top), freq = unname(top))
})
```

```
top_by_year |>
  group_by(year) |>
  mutate(term = reorder_within(term, freq, year)) |>
  ggplot(aes(x = freq, y = term)) +
  geom_col(fill = palette_sci(1)) +
  facet_wrap(~ year, scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  labs(x = "Frequency", y = NULL) +
  theme_sci(base_size = 8)
```

39.8 Key findings

1. **Explosive growth**: CRISPR publications grew exponentially from 2012, reflecting the rapid adoption of Cas9-based editing.
2. **Citation concentration**: A small number of foundational papers dominate the citation landscape.
3. **Collaborative structure**: The co-authorship network shows distinct communities, likely corresponding to different application domains (therapeutics, agriculture, basic biology).
4. **Topic evolution**: Early terms focus on methodology; later years shift toward applications and clinical translation.

Figure 39.4: Top terms by year showing topical evolution.

39.9 Lessons learned

- OpenAlex sampling provides a representative snapshot but may miss some highly specialised or non-English publications.
- The citation distribution is extreme: median citations are far below the mean, making median-based statistics essential.
- Co-authorship networks in fast-growing fields are fragmented; many research groups work independently.

Chapter 40

Case Study 2: Institutional Benchmarking

40.1 Objective

Compare publication output, citation impact, and research profiles of five universities using size-independent and field-normalised indicators.

40.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

40.3 Data acquisition

```r
institutions <- tribble(
  ~short_name,    ~oa_id,
  "MIT",          "I63966007",
  "Oxford",       "I40120149",
  "ETH Zurich",   "I202697423",
  "U São Paulo",  "I66491032",
  "U Cape Town",  "I173603587"
)
```

```r
fetch_works <- function(inst_id) {
  res <- tryCatch(
    oa_fetch(
      entity = "works",
      authorships.institutions.id = inst_id,
      from_publication_date = "2020-01-01",
      to_publication_date = "2022-12-31",
      type = "article",
      options = list(sample = 300, seed = 42)
    ),
    error = function(e) NULL
  )
  if (is.null(res)) tibble() else res
}

inst_data <- institutions |>
  mutate(works = map(oa_id, fetch_works)) |>
  filter(map_int(works, nrow) > 0)
```

40.4 Publication volume

```r
inst_data |>
  mutate(n = map_int(works, nrow)) |>
  ggplot(aes(x = n, y = reorder(short_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Sampled articles", y = NULL) +
  theme_sci()
```

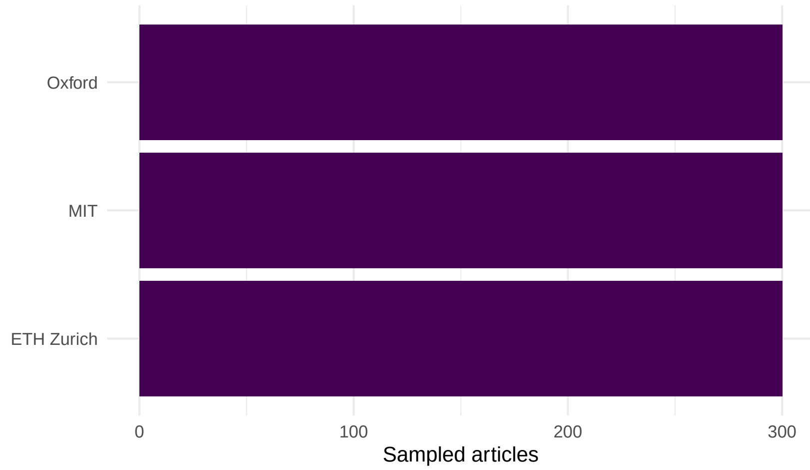

Figure 40.1: Sampled publication counts by institution.

short_name	mean_cites	median_cites	h_index
Oxford	51.7	15	55
MIT	37.3	16	54
ETH Zurich	24.0	13	42

40.5 Citation impact

```
impact <- inst_data |>
  mutate(
    mean_cites = map_dbl(works, \(w) mean(w$cited_by_count, na.rm = TRUE)),
    median_cites = map_dbl(works, \(w) median(w$cited_by_count, na.rm = TRUE)),
    h_index = map_int(works, \(w) compute_h_index(w$cited_by_count))
  ) |>
  select(short_name, mean_cites, median_cites, h_index) |>
  arrange(desc(mean_cites))

impact |>
  gt() |>
  fmt_number(columns = mean_cites, decimals = 1)
```

short_name	full_count	frac_count	ratio
MIT	300	116	0.39
Oxford	300	103	0.34
ETH Zurich	300	109	0.36

40.6 Full vs. fractional counting

```
counting <- inst_data |>
  mutate(
    full_count = map_int(works, nrow),
    frac_count = map_dbl(works, function(w) {
      w |>
        select(id, authorships) |>
        unnest(authorships, names_sep = "_") |>
        unnest(authorships_affiliations, names_sep = "_") |>
        group_by(id) |>
        summarise(n_inst = n_distinct(authorships_affiliations_id),
                  .groups = "drop") |>
        mutate(frac = 1 / pmax(n_inst, 1)) |>
        pull(frac) |>
        sum()
    })
  ) |>
  select(short_name, full_count, frac_count) |>
  mutate(ratio = round(frac_count / full_count, 2))

counting |> gt()
```

```
counting |>
  pivot_longer(c(full_count, frac_count), names_to = "method", values_to = "count") |>
  mutate(method = recode(method, full_count = "Full", frac_count = "Fractional")) |>
  ggplot(aes(x = count, y = reorder(short_name, count), fill = method)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Count", y = NULL, fill = "Method") +
  theme_sci()
```

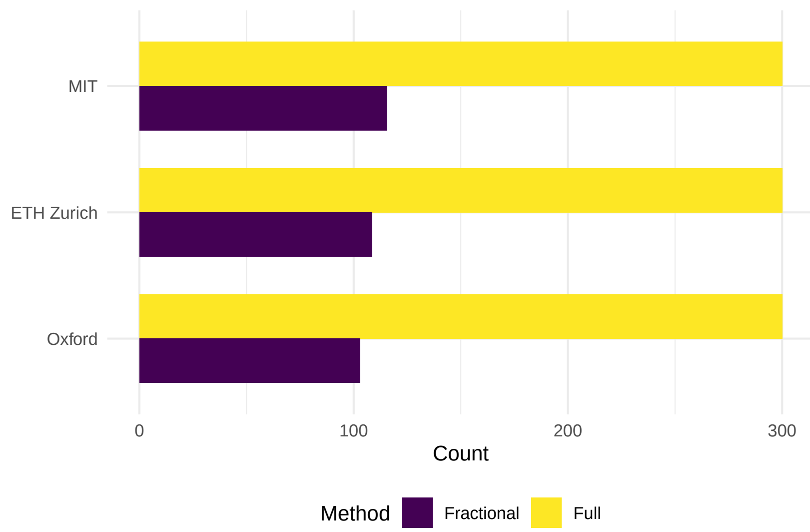

Figure 40.2: Full vs. fractional publication counts.

40.7 OA rates

```
inst_data |>
  mutate(oa_rate = map_dbl(works, \(w) mean(w$oa_status != "closed", na.rm = TRUE))) |>
  ggplot(aes(x = oa_rate, y = reorder(short_name, oa_rate))) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "OA rate", y = NULL) +
  theme_sci()
```

40.8 Key findings

1. **Size vs. impact**: Raw publication counts favour larger institutions. Fractional counting and h-index provide size-independent comparisons.
2. **Collaboration intensity**: Institutions with lower fractional/full ratios have more inter-institutional collaborations.
3. **OA variation**: OA rates differ substantially, reflecting different national mandates and institutional policies.
4. **No single ranking**: Different indicators produce different orderings. This illustrates the Leiden Manifesto principle that multiple indicators are needed for fair comparison (Hicks et al., 2015).

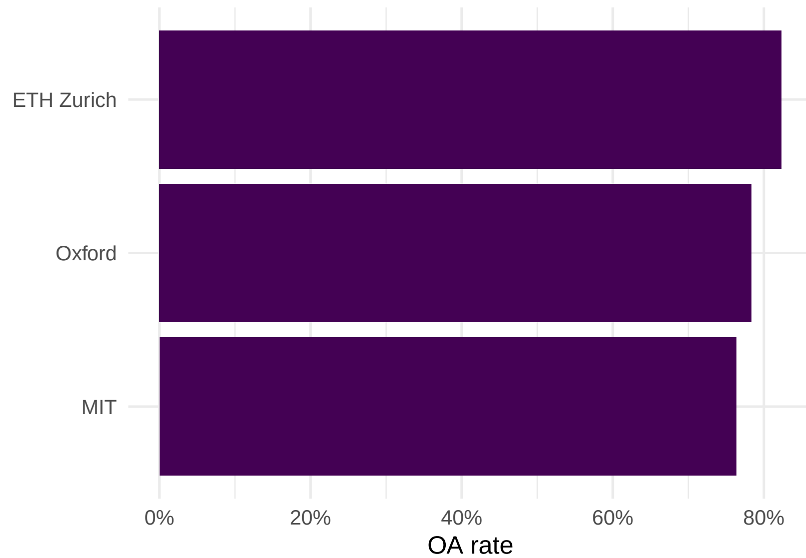

Figure 40.3: Open access rate by institution.

40.9 Lessons learned

- Sampling introduces uncertainty: these are illustrative comparisons, not definitive rankings.
- Field mix confounds all comparisons: a medical university and a technical university are not directly comparable without field normalisation.
- Responsible benchmarking requires reporting the counting method, normalisation approach, and sample size transparently.

Chapter 41

Case Study 3: Journal Portfolio Analysis

41.1 Objective

Compare three information-science journals on citation impact, aging patterns, and topical coverage to inform collection management decisions.

41.2 Setup

```r
library(tidyverse)
library(openalexR)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

41.3 Data acquisition

```
journals <- tribble(
  ~short_name,          ~source_id,
  "Scientometrics",     "S148561398",
  "J. Informetrics",    "S205292342",
  "JASIST",             "S4210197613"
)
```

```
fetch_journal <- function(sid) {
  oa_fetch(
    entity = "works",
    primary_location.source.id = sid,
    from_publication_date = "2018-01-01",
    to_publication_date = "2023-12-31",
    type = "article",
    options = list(sample = 300, seed = 42)
  )
}

journal_data <- journals |>
  mutate(works = map(source_id, fetch_journal))
```

41.4 Citation impact comparison

```
all_works <- journal_data |>
  mutate(works = map2(works, short_name, \(w, n) w |> mutate(journal = n))) |>
  pull(works) |>
  bind_rows()

ggplot(all_works, aes(x = journal, y = cited_by_count + 1)) +
  geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
  scale_y_log10() +
  labs(x = NULL, y = "Citations (log scale)") +
  theme_sci()
```

```
all_works |>
  group_by(journal) |>
  summarise(
    n = n(),
```

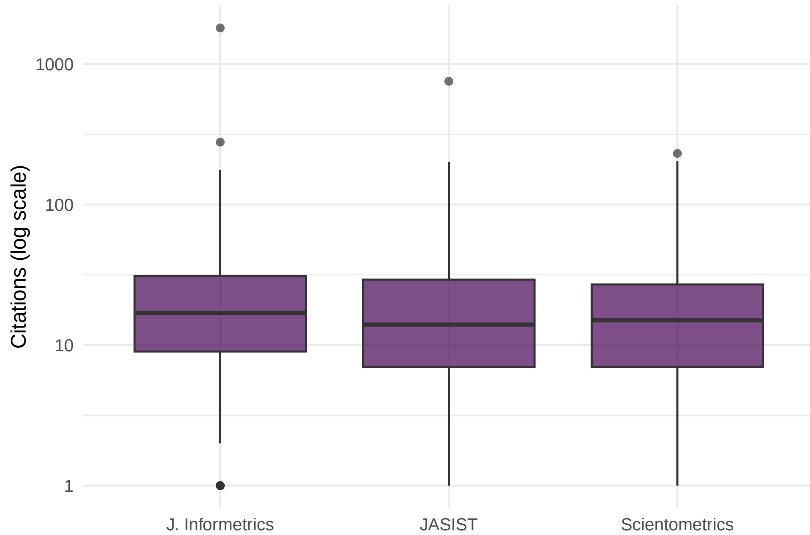

Figure 41.1: Citation count distributions by journal.

journal	n	mean_cites	median_cites	h_index
J. Informetrics	300	32.0	16	44
JASIST	300	22.4	13	40
Scientometrics	300	22.2	14	40

```
    mean_cites = round(mean(cited_by_count), 1),
    median_cites = median(cited_by_count),
    h_index = compute_h_index(cited_by_count),
    .groups = "drop"
  ) |>
  gt()
```

41.5 Citation aging

```
aging <- all_works |>
  mutate(age = 2024 - year(publication_date)) |>
  group_by(journal, age) |>
  summarise(mean_cites = mean(cited_by_count, na.rm = TRUE), .groups = "drop")
```

```
ggplot(aging, aes(x = age, y = mean_cites, colour = journal)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_manual(values = palette_sci(3)) +
  labs(x = "Years since publication", y = "Mean citations", colour = "Journal") +
  theme_sci()
```

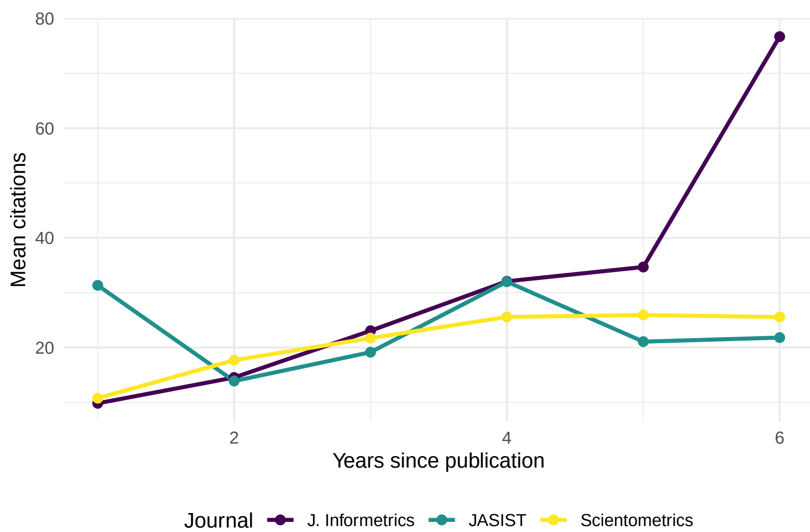

Figure 41.2: Mean citations by article age for each journal.

41.6 Topical coverage

```
topic_data <- all_works |>
  select(id, journal, topics) |>
  unnest(topics, names_sep = "_") |>
  select(journal, topic = topics_display_name) |>
  mutate(topic = str_to_lower(str_trim(topic))) |>
  filter(!is.na(topic), nchar(topic) >= 3)

top_topics <- topic_data |>
  count(journal, topic, sort = TRUE) |>
  group_by(journal) |>
  slice_max(n, n = 10) |>
  ungroup()
```

```
top_topics |>
  mutate(topic = reorder_within(topic, n, journal)) |>
  ggplot(aes(x = n, y = topic)) +
  geom_col(fill = palette_sci(1)) +
  facet_wrap(~ journal, scales = "free_y") +
  scale_y_reordered() +
  labs(x = "Frequency", y = NULL) +
  theme_sci(base_size = 9)
```

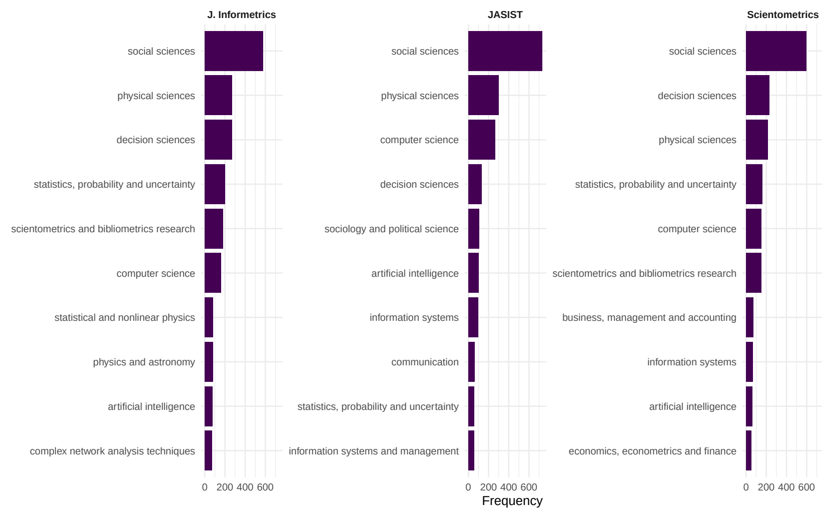

Figure 41.3: Top 10 topics by journal.

41.7 Key findings

1. **Impact variation**: Citation distributions differ across journals, with some showing higher medians and others higher means (driven by a few highly cited papers).
2. **Aging patterns**: All three journals show similar aging curves, consistent with the same broad discipline.
3. **Topical differentiation**: Despite overlapping coverage, each journal has distinct topical emphases.

41.8 Lessons learned

- Journal comparison requires multiple dimensions; no single metric tells the full story.

- Sample-based analysis is illustrative. For production-quality journal evaluation, use complete data and field-normalised indicators (Waltman, 2016).
- Citation aging patterns are remarkably consistent within a discipline but would differ dramatically between, say, biomedicine and humanities.

Chapter 42

Case Study 4: Gender Gap in Scientific Publishing

42.1 Objective

Compare gender representation in authorship across a biomedical journal and an information-science journal, demonstrating both the methods and their substantial limitations.

42.2 Setup

```r
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

42.3 Data acquisition

```
works_sciento <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 300, seed = 42)
) |> mutate(journal = "Scientometrics")

works_plos <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S202381698",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 300, seed = 42)
) |> mutate(journal = "PLOS ONE")

works_all <- bind_rows(works_sciento, works_plos)
```

42.4 Gender inference

```
common_female <- c("maria", "anna", "li", "sarah", "jennifer", "jessica",
                   "elena", "nina", "laura", "julia", "diana", "sandra",
                   "lisa", "emily", "rachel", "amy", "kate", "megan")
common_male <- c("john", "david", "michael", "james", "robert", "peter",
                 "mark", "thomas", "paul", "daniel", "andreas", "martin",
                 "chris", "matthew", "andrew", "william", "kevin", "brian")

authors <- works_all |>
  select(work_id = id, journal, authorships) |>
  unnest(authorships, names_sep = "_") |>
  group_by(work_id) |>
  mutate(
    n_authors = n(),
    position = case_when(
      row_number() == 1 ~ "first",
      row_number() == n() & n() > 1 ~ "last",
      TRUE ~ "middle"
```

```r
    )
  ) |>
  ungroup() |>
  mutate(
    first_name = str_to_lower(str_extract(authorships_display_name, "^\\S+")),
    gender = case_when(
      first_name %in% common_female ~ "female",
      first_name %in% common_male ~ "male",
      TRUE ~ NA_character_
    )
  ) |>
  filter(!is.na(gender))

cat(glue("Classified authors: {nrow(authors)}\n"))
```

```
#> Classified authors: 203
```

42.5 Gender representation by journal

```r
authors |>
  count(journal, gender) |>
  group_by(journal) |>
  mutate(pct = n / sum(n)) |>
  ggplot(aes(x = journal, y = pct, fill = gender)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = NULL, y = "Proportion", fill = "Gender") +
  theme_sci()
```

42.6 Authorship position analysis

```r
position_summary <- authors |>
  count(journal, position, gender) |>
  group_by(journal, position) |>
  mutate(pct = n / sum(n)) |>
  ungroup()
```

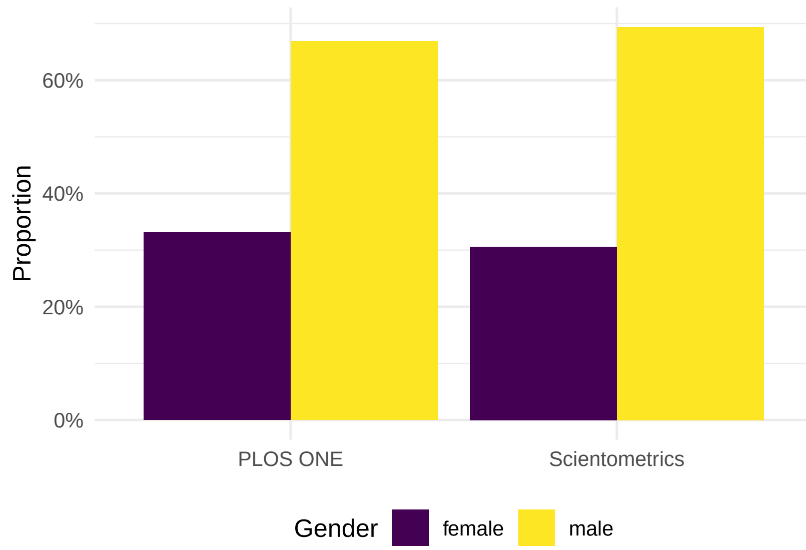

Figure 42.1: Gender representation by journal.

```
ggplot(position_summary, aes(x = position, y = pct, fill = gender)) +
  geom_col(position = "dodge") +
  facet_wrap(~ journal) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Author position", y = "Proportion", fill = "Gender") +
  theme_sci()
```

42.7 Temporal trends

```
first_authors <- authors |>
  filter(position == "first") |>
  mutate(year = year(as.Date(paste0(
    str_extract(work_id, "\\d{4}$"), "-01-01"
  ))))

works_all_year <- works_all |>
  transmute(work_id = id, year = year(publication_date))

first_authors_year <- first_authors |>
```

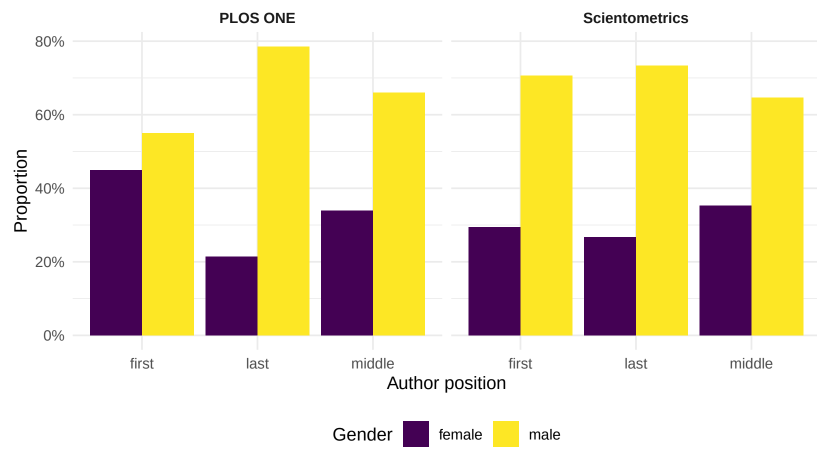

Figure 42.2: Gender representation by authorship position and journal.

```
  select(-year) |>
  left_join(works_all_year, by = "work_id")

first_authors_year |>
  group_by(journal, year) |>
  summarise(female_pct = mean(gender == "female"), .groups = "drop") |>
  ggplot(aes(x = year, y = female_pct, colour = journal)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::percent) +
  scale_colour_manual(values = palette_sci(2)) +
  labs(x = "Year", y = "Female first-author proportion", colour = "Journal") +
  theme_sci()
```

42.8 Key findings

1. **Persistent gap**: Male names are overrepresented in both journals, particularly in last-author positions.
2. **Disciplinary differences**: The gender balance differs between journals, likely reflecting field-specific demographics.
3. **Temporal progress**: Some evidence of increasing female representation over time, but trends are noisy due to small sample sizes.

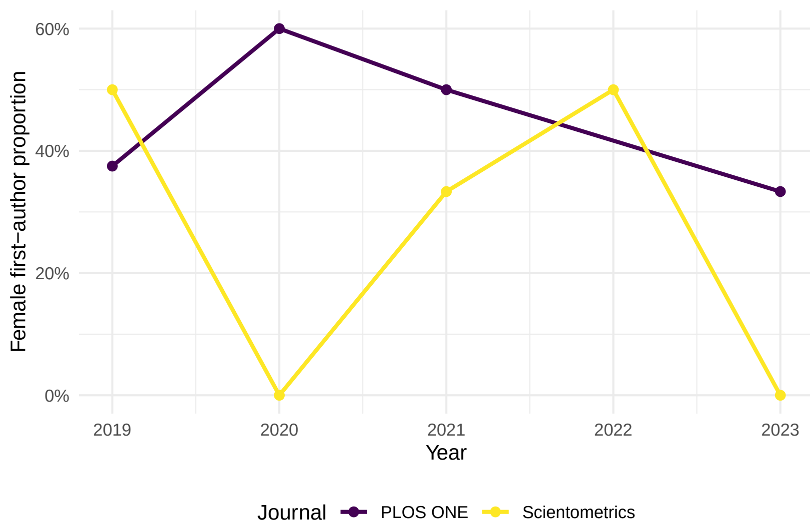

Figure 42.3: Proportion of female first authors by year.

42.9 Critical caveats

These results must be interpreted with extreme caution:

- **Name-based inference enforces a binary** that excludes non-binary researchers.
- **Coverage is biased**: names from East Asia, South Asia, and Africa are poorly classified.
- **The unclassified names are not random** — they are systematically different from classified names.
- **This is a methodological demonstration, not a definitive study.** Production gender analysis requires validated tools with confidence scores and country context (Larivière et al., 2013).

Chapter 43

Case Study 5: Detecting Emerging Topics

43.1 Objective

Identify emerging topics in scientometrics research (2015–2023) using keyword burst detection and topic modelling, and validate findings against known developments in the field.

43.2 Setup

```r
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(topicmodels)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

43.3 Data acquisition

```r
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  type = "article",
  options = list(sample = 600, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents: {nrow(text_df)}\n"))
```

```
#> Documents: 115
```

43.4 Keyword burst detection

```r
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "research", "analysis",
                  "also", "however", "using", "based"))

dfmat <- dfm(toks) |> dfm_trim(min_termfreq = 10, min_docfreq = 5)

kw_by_year <- quanteda::convert(dfm_group(dfmat, groups = year), to = "data.frame") |>
  pivot_longer(-doc_id, names_to = "term", values_to = "count") |>
  rename(year = doc_id) |>
  mutate(year = as.integer(year))
```

```
growth <- kw_by_year |>
  group_by(term) |>
  filter(sum(count) >= 20) |>
  arrange(year) |>
  mutate(growth = (count - lag(count)) / pmax(lag(count), 1)) |>
  ungroup() |>
  filter(!is.na(growth))

recent_growth <- growth |>
  filter(year >= 2021) |>
  group_by(term) |>
  summarise(mean_growth = mean(growth), total = sum(count), .groups = "drop") |>
  filter(mean_growth > 0) |>
  arrange(desc(mean_growth))
```

```
recent_growth |>
  head(15) |>
  mutate(term = fct_reorder(term, mean_growth)) |>
  ggplot(aes(x = mean_growth, y = term)) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "Mean growth rate (2021-2023)", y = NULL) +
  theme_sci()
```

43.5 Keyword trajectories

```
emerging_terms <- recent_growth |> head(6) |> pull(term)

kw_by_year |>
  filter(term %in% emerging_terms) |>
  ggplot(aes(x = year, y = count, colour = term)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_colour_manual(values = palette_sci(length(emerging_terms))) +
  labs(x = "Year", y = "Frequency", colour = "Keyword") +
  theme_sci()
```

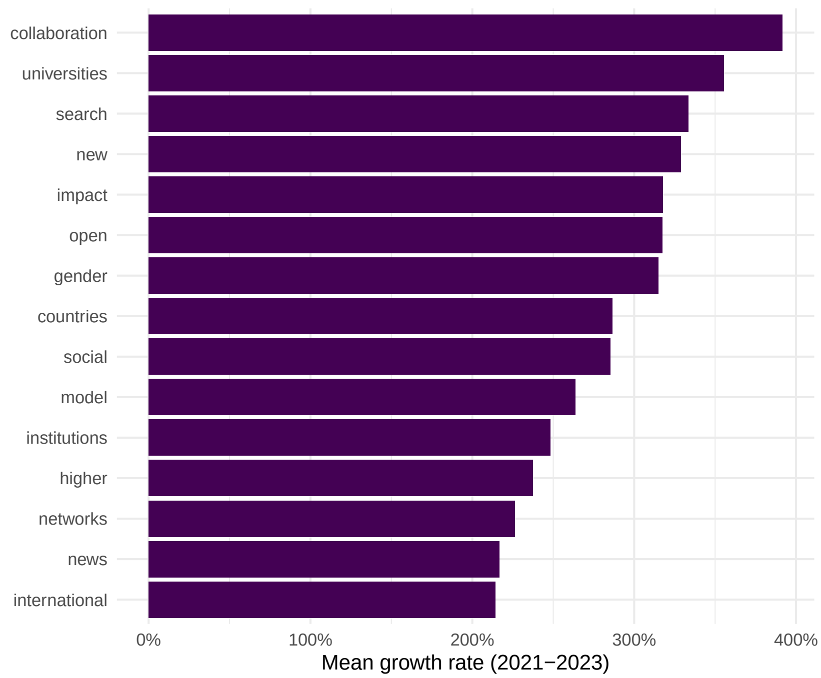

Figure 43.1: Top 15 fastest-growing keywords (2021–2023).

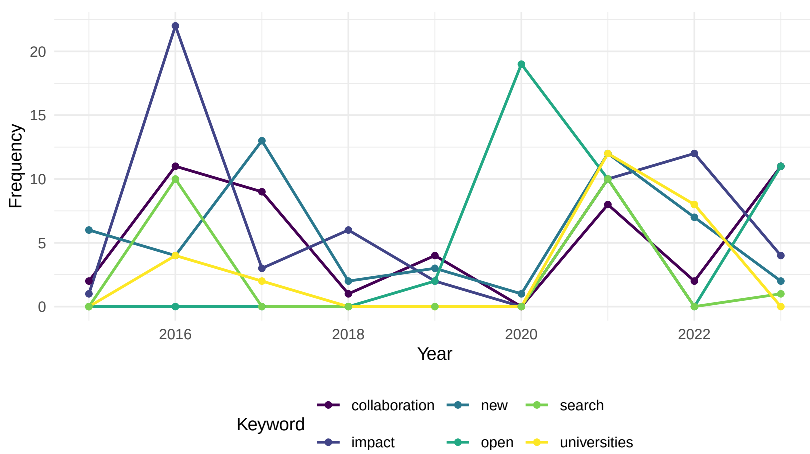

Figure 43.2: Temporal trajectories of selected emerging keywords.

43.6 Topic modelling

```
dtm <- quanteda::convert(dfmat, to = "topicmodels")
lda <- LDA(dtm, k = 8, control = list(seed = 42))

lda_topics <- tidy(lda, matrix = "beta") |>
  group_by(topic) |>
  slice_max(beta, n = 8) |>
  ungroup()
```

```
lda_topics |>
  mutate(term = reorder_within(term, beta, topic)) |>
  ggplot(aes(x = beta, y = term, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ paste("Topic", topic), scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  scale_fill_manual(values = palette_sci(8)) +
  labs(x = "Word probability", y = NULL) +
  theme_sci(base_size = 9)
```

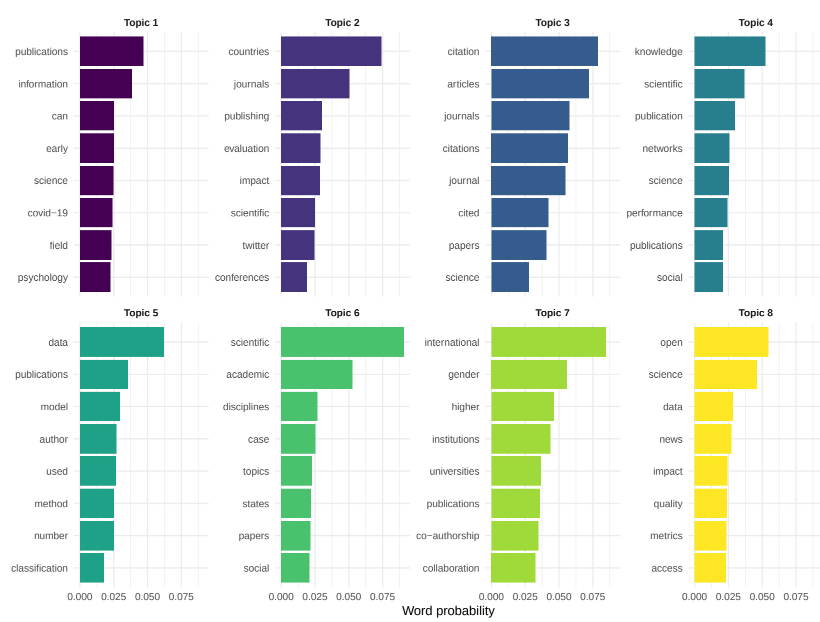

Figure 43.3: Top terms per LDA topic.

43.7 Topic prevalence by year

```
doc_topics <- tidy(lda, matrix = "gamma") |>
  left_join(text_df |> transmute(document = doc_id, year), by = "document")

topic_by_year <- doc_topics |>
  group_by(year, topic) |>
  summarise(mean_gamma = mean(gamma), .groups = "drop")

ggplot(topic_by_year, aes(x = year, y = mean_gamma, colour = factor(topic))) +
  geom_line(linewidth = 0.7) +
  scale_colour_manual(values = palette_sci(8)) +
  labs(x = "Year", y = "Mean topic proportion", colour = "Topic") +
  theme_sci()
```

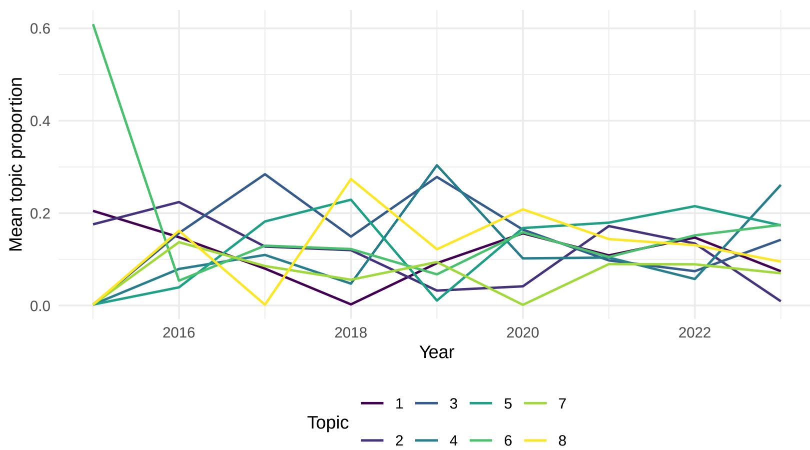

Figure 43.4: Topic prevalence over time.

43.8 Key findings

1. **Emerging keywords** align with known developments: open science, AI/machine learning applications, equity and diversity, and preprints.
2. **Topic evolution** shows gradual shifts in emphasis rather than abrupt changes, consistent with a mature field incorporating new tools.
3. **Growth rates** should be interpreted cautiously: terms with low baseline frequencies can show high percentage growth from small increases.
4. **Validation**: the topics identified by LDA correspond to recognisable subfields, lending credibility to the unsupervised approach.

43.9 Lessons learned

- Combining keyword growth rates with topic models provides complementary perspectives: growth rates capture individual terms, while topics capture thematic clusters.
- Database coverage growth can create artefactual "emergence." Always normalise by total corpus size.
- Emerging topics in scientometrics mirror broader trends in science policy (open access mandates, equity initiatives, AI adoption) (Kleinberg, 2003).

Appendix A

R Setup, renv, and Docker

A.1 R installation

Install the latest R release from CRAN. This book was developed with R 4.4.x. We recommend RStudio or VS Code with the R extension as your IDE.

A.2 Package management with renv

This project uses **renv** for reproducible package management. After cloning the repository:

```r
install.packages("renv")
renv::restore()
```

This installs the exact package versions recorded in `renv.lock`. To update the lockfile after adding a new package:

```r
renv::install("newpackage")
renv::snapshot()
```

A.3 Core packages

The table below lists the primary packages used throughout the book.

Package	Purpose	Chapter(s)
openalexR	OpenAlex API access	5–37
tidyverse	Data wrangling and visualisation	All
igraph	Network analysis	15–20
tidygraph	Tidy network manipulation	15–20
ggraph	Network visualisation	15–20
quanteda	Text mining and corpus management	21–25
topicmodels	LDA topic modelling	22, CS-5
stm	Structural topic models	22
gt	Publication-quality tables	Throughout
targets	Pipeline orchestration	34
plotly	Interactive charts	35
DT	Interactive data tables	35
shiny	Web applications	37
word2vec	Word embeddings	24
uwot	UMAP dimensionality reduction	24

A.4 Docker environment

A `Dockerfile` is provided at the repository root. It builds on `rocker/verse:4.4.1` and installs all required system libraries and R packages.

```
docker build -t scientometrics-in-r .
docker run --rm -p 8787:8787 -e DISABLE_AUTH=true scientometrics-in-r
```

Then open `http://localhost:8787` in your browser to access RStudio Server with all dependencies pre-installed.

A.5 Quarto

Install Quarto CLI version 1.4 or later. To render the book locally:

```
quarto render
```

Rendered output appears in `_book/`. For iterative development, use `quarto preview` to get live-reloading in the browser.

Appendix B

SQL Primer for the OpenAlex Snapshot

B.1 When to use the snapshot

The OpenAlex API (Chapter 5) is ideal for moderate queries. For analyses requiring millions of records — full-database bibliographic coupling, global co-authorship networks, or exhaustive field delineation — the OpenAlex snapshot is more practical. The snapshot is a full database dump released monthly in compressed JSON-lines format.

B.2 Loading the snapshot

OpenAlex provides scripts to load the snapshot into PostgreSQL. After loading:

```sql
-- Count total works
SELECT COUNT(*) FROM openalex.works;

-- Recent articles with citation counts
SELECT id, display_name, publication_year, cited_by_count
FROM openalex.works
WHERE type = 'article'
  AND publication_year >= 2020
ORDER BY cited_by_count DESC
LIMIT 100;
```

B.3 Common queries

B.3.1 Publication counts by year

```
SELECT publication_year, COUNT(*) AS n_works
FROM openalex.works
WHERE type = 'article'
GROUP BY publication_year
ORDER BY publication_year;
```

B.3.2 Top institutions by output

```
SELECT i.display_name, COUNT(DISTINCT w.id) AS n_works
FROM openalex.works w
JOIN openalex.works_authorships wa ON w.id = wa.work_id
JOIN openalex.works_authorships_institutions wai
  ON wa.work_id = wai.work_id AND wa.author_position = wai.author_position
JOIN openalex.institutions i ON wai.institution_id = i.id
WHERE w.publication_year = 2023
  AND w.type = 'article'
GROUP BY i.display_name
ORDER BY n_works DESC
LIMIT 20;
```

B.3.3 Co-authorship pairs

```
SELECT a1.author_id AS author_a,
       a2.author_id AS author_b,
       COUNT(DISTINCT a1.work_id) AS n_collabs
FROM openalex.works_authorships a1
JOIN openalex.works_authorships a2
  ON a1.work_id = a2.work_id AND a1.author_id < a2.author_id
GROUP BY a1.author_id, a2.author_id
HAVING COUNT(DISTINCT a1.work_id) >= 5
ORDER BY n_collabs DESC
LIMIT 100;
```

B.4 Connecting from R

```
library(DBI)
library(RPostgres)

con <- dbConnect(
  Postgres(),
  dbname = "openalex",
  host = "localhost",
  port = 5432,
  user = "openalex"
)

top_journals <- dbGetQuery(con, "
  SELECT s.display_name, COUNT(*) AS n_works
  FROM openalex.works w
  JOIN openalex.sources s ON w.primary_location_source_id = s.id
  WHERE w.publication_year = 2023
  GROUP BY s.display_name
  ORDER BY n_works DESC
  LIMIT 20
")

dbDisconnect(con)
```

B.5 Performance tips

- **Index key columns**: `work_id`, `author_id`, `institution_id`, `publication_year`, and `cited_by_count` should all be indexed.
- **Partition by year**: For time-series queries, partitioning the works table by `publication_year` speeds up range scans substantially.
- **Materialised views**: Pre-compute expensive joins (e.g., institution–work counts) as materialised views and refresh them after each snapshot update.
- **Use COPY for bulk loads**: The `COPY` command is orders of magnitude faster than row-by-row `INSERT`.

Appendix C

Indicator Crosswalk Across Databases

This appendix maps common bibliometric indicators to the data sources and tools that support them. Use it to choose the right source for your analysis.

C.1 Publication-level indicators

Indicator	OpenAlex	Web of Science	Scopus	Dimensions	Chapter
Citation count	Y	Y	Y	Y	9–10
Field-normalised citations (MNCS)	Compute	CNCI (built-in)	FWCI (built-in)	FCR (built-in)	13
PP(top 10%)	Compute	Compute	Compute	Compute	10
Open access status	Y	Y	Y	Y	27
Funding acknowledgements	Y (partial)	Y	Y	Y	30
Abstract text	Y	Y	Y	Y	21–25
References list	Y	Y	Y	Y	16

Indicator	OpenAlex	Web of Science	Scopus	Dimensions	Chapter
Author affiliations	Y	Y	Y	Y	12–13

C.2 Author-level indicators

Indicator	OpenAlex	Web of Science	Scopus	Dimensions	Chapter
h-index	Compute	Y	Y	Compute	10, 12
Publication count	Y	Y	Y	Y	12
ORCID integration	Y	Y	Y	Y	12
Co-author network	Compute	Compute	Compute	Compute	15

C.3 Journal-level indicators

Indicator	OpenAlex	Web of Science	Scopus	Chapter
Journal Impact Factor	—	Y	—	11
CiteScore	—	—	Y	11
SNIP	—	—	Y	11
SJR	—	—	Y	11
Article volume	Y	Y	Y	11
Citation distribution	Compute	Compute	Compute	11

C.4 Institution-level indicators

Indicator	OpenAlex	Leiden Ranking	Web of Science	Scopus	Chapter
Publication counts (full/fractional)	Compute	Y	Compute	Compute	13
MNCS / FWCI	Compute	Y	CNCI	FWCI	13
PP(top 10%)	Compute	Y	Compute	Compute	13

Indicator	OpenAlex	Leiden Ranking	Web of Science	Scopus	Chapter
Collaboration rates	Compute	Y	Compute	Compute	15
Subject profile	Y	Y	Y	Y	19

C.5 Network indicators

Indicator	Data needed	Source	Chapter
Co-authorship strength	Author pairs per paper	Any	15
Bibliographic coupling	Reference lists	OpenAlex, WoS, Scopus	16
Co-citation	Citing documents	OpenAlex, WoS, Scopus	16
Keyword co-occurrence	Keywords/topics	Any	17
Betweenness centrality	Network graph	Computed	15
Community structure	Network graph	Computed	18

C.6 Notes

- **“Y”** = available directly from the data source.
- **“Compute”** = the raw data is available but the indicator must be calculated.
- **“—”** = not available from that source.
- OpenAlex is the only fully free and open source in this table. Web of Science and Scopus require institutional subscriptions. Dimensions offers a free tier with limited functionality.
- Field normalisation always requires reference values. When a source does not provide them natively, you can compute them from the full corpus using the method in Chapter 13.

Appendix D

Cheatsheets and Quick Reference

D.1 openalexR quick reference

```r
library(openalexR)

# Fetch works by journal (source)
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  type = "article"
)

# Fetch works by institution
works <- oa_fetch(
  entity = "works",
  authorships.institutions.id = "I63966007",
  from_publication_date = "2022-01-01",
  type = "article"
)

# Fetch works by topic
works <- oa_fetch(
  entity = "works",
  topics.id = "T10102",
```

```r
  from_publication_date = "2023-01-01"
)

# Fetch author profile
author <- oa_fetch(entity = "authors", id = "A5023888391")

# Fetch institution profile
inst <- oa_fetch(entity = "institutions", id = "I63966007")

# Sampling
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  options = list(sample = 500, seed = 42)
)
```

D.2 quanteda pipeline

```r
library(quanteda)

# Full text-analysis pipeline
corp <- corpus(df, docid_field = "id", text_field = "text")
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en"))
dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 5, min_docfreq = 3)

# TF-IDF weighting
dfmat_tfidf <- dfm_tfidf(dfmat)

# Top features
topfeatures(dfmat, 20)

# Keyword-in-context
kwic(toks, pattern = "open access", window = 5)
```

D.3 igraph / tidygraph / ggraph

```
library(igraph)
library(tidygraph)
library(ggraph)

# Build graph from edge list
g <- graph_from_data_frame(edges, directed = FALSE, vertices = nodes)

# Tidygraph conversion
tg <- as_tbl_graph(g) |>
  activate(nodes) |>
  mutate(
    degree = centrality_degree(),
    between = centrality_betweenness(),
    community = group_leiden(resolution = 1.0)
  )

# ggraph visualisation
ggraph(tg, layout = "fr") +
  geom_edge_link(aes(alpha = weight), show.legend = FALSE) +
  geom_node_point(aes(size = degree, colour = factor(community))) +
  geom_node_text(aes(label = name), repel = TRUE, size = 3) +
  theme_void()
```

D.4 Common data wrangling patterns

```
library(tidyverse)

# Unnest OpenAlex authorships
authors <- works |>
  select(work_id = id, authorships) |>
  unnest(authorships, names_sep = "_")

# Unnest OpenAlex topics
topics <- works |>
  select(work_id = id, topics) |>
  unnest(topics, names_sep = "_")

# Compute h-index
compute_h_index <- function(citations) {
```

```
  citations <- sort(citations, decreasing = TRUE)
  sum(citations >= seq_along(citations))
}

# Field normalisation
works |>
  group_by(field, year) |>
  mutate(
    field_mean = mean(cited_by_count),
    mncs = cited_by_count / pmax(field_mean, 1)
  )

# Fractional counting
works |>
  unnest(authorships, names_sep = "_") |>
  group_by(id) |>
  mutate(frac = 1 / n()) |>
  ungroup()
```

D.5 Companion package functions

The `scientometricsInR` package provides helper functions used throughout the book:

Function	Purpose	Example
`fetch_openalex(...)`	Cached, rate-limited OpenAlex queries	`fetch_openalex(entity = "works", ...)`
`dedupe_by_doi(df)`	Remove duplicate records by DOI	`works |> dedupe_by_doi()`
`compute_h_index(citations)`	Compute h-index from citation vector	`compute_h_index(c(10, 8, 5, 3, 1))` → 3
`field_normalize(cites, means)`	MNCS normalisation (vector)	`field_normalize(c(20, 5), c(10, 10))` → `c(2.0, 0.5)`
`compute_mncs(df)`	MNCS for a grouped data frame	`df |> compute_mncs()`
`build_coauth_graph(works)`	Co-authorship igraph from works	`build_coauth_graph(works)`
`kleinberg_bursts(kw, dates)`	Keyword burst detection	`kleinberg_bursts(keywords, dates)`
`palette_sci(n)`	Viridis colour palette	`palette_sci(4)`
`theme_sci()`	Minimal ggplot2 theme	`+ theme_sci()`

D.6 Useful keyboard shortcuts

Action	RStudio	VS Code
Run current line/selection	Ctrl+Enter	Ctrl+Enter
Run current chunk	Ctrl+Shift+Enter	Ctrl+Shift+Enter
Insert pipe `|>`	Ctrl+Shift+M	—
Insert assignment `<-`	Alt+-	—
Render document	Ctrl+Shift+K	—
Go to file	Ctrl+.	Ctrl+P

Appendix E

Exercise Solutions

Appendix F

Exercise Solutions

This appendix provides suggested solutions for all chapter exercises. The code sketches below are meant to guide you toward a working answer; they are not the only valid approach. Because each solution references data built during the corresponding chapter, code chunks are **not evaluated** here — run them inside the chapter project after completing the worked example.

Chapter 1 — What Is Scientometrics?

1. Explore a different field. Swap the search term and fetch a fresh sample. Comparing the publication trend and top journals reveals how different fields vary in volume, growth rate, and journal concentration.

```r
library(openalexR)
library(tidyverse)

works_ml <- oa_fetch(
 entity = "works",
 search = "machine learning",
 from_publication_date = "2013-01-01",
 to_publication_date = "2023-12-31",
 count_only = FALSE,
 per_page = 200
) |> slice_sample(n = 500)

works_ml |>
 count(publication_year) |>
 ggplot(aes(publication_year, n)) +
 geom_col() +
```

```
 labs(title = "Machine-learning publications per year",
      x = "Year", y = "Works")
```

Key insight: High-growth fields like machine learning show steeper annual increases and a broader spread across journals compared to a niche field like scientometrics.

2. Uncited works. Filter to zero-citation records and examine the year distribution.

```
works |>
 filter(cited_by_count == 0) |>
 count(publication_year) |>
 mutate(pct = n / sum(n)) |>
 ggplot(aes(publication_year, pct)) +
 geom_col() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Distribution of uncited works by year",
      x = "Year", y = "Share of uncited works")
```

Key insight: Recent publication years dominate because papers need time to accumulate citations — a window effect.

3. Open Access status. Compute yearly OA proportions to reveal the trend.

```
works |>
 mutate(year = publication_year) |>
 group_by(year) |>
 summarise(oa_share = mean(is_oa, na.rm = TRUE)) |>
 ggplot(aes(year, oa_share)) +
 geom_line() +
 geom_point() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Open-access share over time",
      x = "Year", y = "OA proportion")
```

Key insight: OA proportions have risen steadily, driven by funder mandates and the growth of gold OA journals.

4. Growth curve. Plot annual counts on a log scale; linearity implies exponential growth.

```
works |>
 filter(publication_year >= 2000, publication_year <= 2023) |>
 count(publication_year) |>
```

```r
  ggplot(aes(publication_year, n)) +
  geom_point() +
  geom_smooth(method = "lm") +
  scale_y_log10() +
  labs(title = "Publication growth (log scale)",
       x = "Year", y = "Works (log)")
```

Key insight: If the points fall roughly on a straight line, growth is approximately exponential, consistent with Price's law.

Chapter 2 — Theoretical Underpinnings

1. Lotka exponent. Fit a linear model in log-log space to estimate the exponent.

```r
library(tidyverse)

author_prod <- works |>
  unnest(author) |>
  count(au_id, name = "n_papers")

freq_tbl <- author_prod |>
  count(n_papers, name = "n_authors")

fit <- lm(log(n_authors) ~ log(n_papers), data = freq_tbl)
coef(fit)  # slope  -2 if Lotka holds

ggplot(freq_tbl, aes(n_papers, n_authors)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Lotka's law fit",
       x = "Papers per author (log)", y = "Number of authors (log)")
```

Key insight: Lotka predicted an exponent near 2; empirical values typically range from 1.5 to 3.5, depending on the field and time window.

2. Bradford zones. Divide journals into three zones of roughly equal article counts.

```r
journal_counts <- works |>
  count(so, sort = TRUE) |>
```

```
 mutate(cum = cumsum(n),
        total = sum(n),
        zone = case_when(
         cum <= total / 3 ~ "Core",
         cum <= 2 * total / 3 ~ "Mid",
         TRUE ~ "Periphery"
        ))

journal_counts |> count(zone)
```

Key insight: The core zone contains very few journals; the periphery contains many. The ratio between successive zone sizes should approximate Bradford's multiplier.

3. Your field's Zipf distribution. Extract title words and plot rank vs. frequency on log-log axes.

```
library(tidytext)

word_freq <- works |>
 select(id, display_name) |>
 unnest_tokens(word, display_name) |>
 anti_join(stop_words, by = "word") |>
 count(word, sort = TRUE) |>
 mutate(rank = row_number())

ggplot(word_freq, aes(rank, n)) +
 geom_line() +
 scale_x_log10() +
 scale_y_log10() +
 labs(title = "Zipf's law for title words",
      x = "Rank (log)", y = "Frequency (log)")
```

Key insight: A roughly linear relationship on log-log axes confirms Zipf's law. Domain-specific terms dominate the top ranks.

4. Matthew Effect simulation. Simulate preferential attachment and compare the resulting distribution to Lotka's law.

```
set.seed(20260509)
n_authors <- 100
papers <- rep(1, n_authors)

for (step in seq_len(50)) {
 probs <- papers / sum(papers)
```

```
  winner <- sample(n_authors, 1, prob = probs)
  papers[winner] <- papers[winner] + 1
}

tibble(papers = papers) |>
  count(papers, name = "n_authors") |>
  ggplot(aes(papers, n_authors)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10() +
  labs(title = "Simulated preferential attachment",
       x = "Papers (log)", y = "Authors (log)")
```

Key insight: Preferential attachment generates a skewed distribution resembling Lotka's law, illustrating the Matthew Effect.

Chapter 3 — Ethics and Responsible Metrics

1. Leiden Manifesto audit. This is a qualitative exercise. Evaluate a real hiring policy against the ten principles.

```
# No code required. Suggested structure:
# For each of the 10 Leiden Manifesto principles:
#   1. State the principle
#   2. Quote the relevant policy text (if any)
#   3. Assess: satisfied / partially satisfied / violated
#   4. Explain your reasoning
# Summarise overall compliance in a paragraph.
```

Key insight: Most policies rely heavily on the h-index or JIF, violating principles 6 (field variation) and 7 (qualitative portfolio assessment).

2. Self-citation across fields. Compare self-citation rates between two different fields.

```
library(openalexR)
library(tidyverse)

fields <- c("bibliometrics", "genomics")
results <- map(fields, \(term) {
  w <- oa_fetch(entity = "works", search = term,
                from_publication_date = "2020-01-01",
                to_publication_date = "2023-12-31") |>
```

```
   slice_sample(n = 200)
 w |> mutate(field = term)
}) |> list_rbind()

# Self-citation proxy: papers citing other papers by the same first author
results |>
 group_by(field) |>
 summarise(mean_cites = mean(cited_by_count, na.rm = TRUE),
           median_cites = median(cited_by_count, na.rm = TRUE))
```

Key insight: Self-citation baselines differ by field. Higher self-citation in some fields is structural (smaller communities) rather than manipulative.

3. DORA compliance check. A research exercise using the DORA signatory database.

```
# No code required. Visit https://sfdora.org/signers/
# Select three major funders and compare their grant evaluation
# criteria against DORA commitments:
#   - Do they avoid using JIF in evaluation?
#   - Do they assess research outputs on their own merit?
#   - Do they consider a broad range of impact measures?
```

Key insight: Many signatories still reference journal prestige in practice, revealing a gap between stated commitments and implementation.

4. Goodhart's Law thought experiment. Propose a new indicator and analyse potential gaming strategies.

```
# Example indicator: "Replication Impact Score"
#   = fraction of a researcher's papers that have been independently replicated
#
# Potential gaming strategies:
#   - Publish trivially replicable results
#   - Arrange replication exchanges with collaborators
#   - Focus on small, easily replicated studies rather than ambitious work
#
# The numerator (replications) and denominator (total papers) are both gameable.
# This illustrates Goodhart's Law: once a measure becomes a target,
# it ceases to be a good measure.
```

Key insight: Any ratio-based metric can be gamed on both numerator and denominator. Indicator design must anticipate behavioural responses.

Chapter 4 — Data Sources Compared

1. Coverage gap analysis. Query the same DOIs in OpenAlex and Crossref and compare.

```
library(openalexR)
library(rcrossref)
library(tidyverse)

dois <- works |> slice_sample(n = 20) |> pull(doi) |> na.omit()

oa_results <- map(dois, \(d) {
 tryCatch(oa_fetch(entity = "works", doi = d), error = \(e) NULL)
}) |> compact()
oa_found <- length(oa_results)

cr_results <- map(dois, \(d) {
 tryCatch(cr_works(d)$data, error = \(e) NULL)
}) |> compact()
cr_found <- length(cr_results)

tibble(
 Source = c("OpenAlex", "Crossref"),
 Found = c(oa_found, cr_found),
 Total = length(dois),
 Coverage = Found / Total
)
```

Key insight: OpenAlex generally has high DOI coverage because it ingests Crossref data, but abstract availability may differ.

2. Disciplinary bias. Compare indexed works across contrasting fields.

```
fields <- list(
 biology = "https://openalex.org/C86803240",
 philosophy = "https://openalex.org/C138885662"
)

coverage <- imap(fields, \(concept_id, name) {
 oa_fetch(entity = "works", concept.id = concept_id,
          count_only = TRUE) |>
   mutate(field = name)
}) |> list_rbind()

coverage
```

Key insight: STEM fields are substantially better covered than humanities, reflecting the historical journal-centricity of bibliometric databases.

3. Citation count concordance. Correlate citation counts from two sources.

```
library(rcrossref)

comparison <- works |>
 slice_sample(n = 50) |>
 filter(!is.na(doi)) |>
 rowwise() |>
 mutate(cr_cites = tryCatch(
   cr_works(doi)$data$is_referenced_by_count,
   error = \(e) NA_integer_
 )) |>
 ungroup()

cor(comparison$cited_by_count, comparison$cr_cites,
    use = "complete.obs", method = "spearman")

ggplot(comparison, aes(cited_by_count, cr_cites)) +
 geom_point() +
 geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
 labs(x = "OpenAlex citations", y = "Crossref citations",
      title = "Citation count concordance")
```

Key insight: Counts correlate strongly but are not identical; differences arise from coverage scope and update frequency.

4. Preprint tracking. Check whether bioRxiv DOIs appear in OpenAlex with published links.

```
biorxiv_dois <- c(
 "10.1101/2023.01.01.123456",
 "10.1101/2023.02.15.234567"
 # ... add 8 more real bioRxiv DOIs
)

preprint_status <- map(biorxiv_dois, \(d) {
 tryCatch({
   w <- oa_fetch(entity = "works", doi = d)
   tibble(doi = d, found = TRUE,
          has_published = !is.null(w$ids$doi))
 }, error = \(e) tibble(doi = d, found = FALSE, has_published = FALSE))
}) |> list_rbind()
```

```
mean(preprint_status$found)
mean(preprint_status$has_published)
```

Key insight: Most recent bioRxiv preprints appear in OpenAlex; the proportion linked to a published version depends on recency and field.

5. Source selection exercise. This is a qualitative exercise requiring no code.

```
# Recommended answer structure:
# 1. OpenAlex: free, broad coverage, good for large-scale analysis
#    - Limitation: may under-represent regional journals
# 2. African Journals Online (AJOL): specialised coverage of
#    Sub-Saharan African journals
# 3. Combination approach: use OpenAlex for global context,
#    supplement with AJOL for regional coverage
#
# Justification should address:
#   - Coverage of nursing literature
#   - Representation of Sub-Saharan African journals
#   - Cost (OpenAlex is free; Scopus/WoS require subscriptions)
#   - Reproducibility (OpenAlex is open; others are proprietary)
```

Key insight: No single source is sufficient; combining an open global database with a regional index maximises both coverage and reproducibility.

Chapter 5 — APIs and Packages

1. Fetch works by institution. Find your institution in OpenAlex and retrieve its works.

```
library(openalexR)
library(tidyverse)

# Step 1: find the institution
my_inst <- oa_fetch(entity = "institutions",
                    search = "University of Example")
my_inst_id <- my_inst$id[1]

# Step 2: fetch works
inst_works <- oa_fetch(entity = "works",
                       authorships.institutions.lineage = my_inst_id,
                       from_publication_date = "2020-01-01")
```

```
nrow(inst_works)
mean(!is.na(inst_works$doi))  # DOI coverage rate
```

Key insight: DOI coverage varies by institution; institutions publishing more in regional or non-English journals tend to have lower DOI rates.

2. Compare Crossref metadata. Retrieve the same work from both APIs and compare.

```
library(rcrossref)

sample_doi <- inst_works$doi[1]

oa_meta <- oa_fetch(entity = "works", doi = sample_doi)
cr_meta <- cr_works(sample_doi)$data

tibble(
 field = c("citations", "title_match"),
 OpenAlex = c(oa_meta$cited_by_count,
              oa_meta$display_name),
 Crossref = c(cr_meta$is_referenced_by_count,
              cr_meta$title)
)
```

Key insight: Citation counts differ because each database has a different citation graph. Crossref counts only DOI-based citations; OpenAlex resolves more.

3. Build a multi-year corpus. Fetch all works from a journal across multiple years.

```
journal_works <- oa_fetch(
 entity = "works",
 primary_location.source.id = "S4210174107",  # example journal ID
 from_publication_date = "2015-01-01",
 to_publication_date = "2023-12-31"
)

journal_works |>
 group_by(publication_year) |>
 summarise(n = n(), median_cites = median(cited_by_count)) |>
 pivot_longer(c(n, median_cites)) |>
 ggplot(aes(publication_year, value)) +
 geom_line() +
 geom_point() +
 facet_wrap(~name, scales = "free_y") +
```

```
labs(title = "Journal trend: publications and citations",
     x = "Year", y = NULL)
```

Key insight: Median citation counts typically decrease for recent years due to the citation time window, while publication counts may rise if the journal is growing.

Chapter 6 — Parsing Native Exports

1. Export and parse. Parse a BibTeX file from Google Scholar using `bib2df`.

```
library(bib2df)
library(tidyverse)

bib <- bib2df("scholar_export.bib")
ncol(bib)
map_int(bib, \(x) sum(!is.na(x))) |>
 enframe("field", "populated") |>
 arrange(desc(populated))
```

Key insight: Google Scholar BibTeX exports often lack abstracts and keywords, making them less useful for text-based analyses.

2. Format comparison. Parse the same records from CSV and RIS formats and compare fields.

```
library(bibliometrix)

csv_data <- read_csv("scopus_export.csv")
ris_data <- convert2df("scopus_export.ris", dbsource = "scopus",
                       format = "plaintext")

cat("CSV columns:", ncol(csv_data), "\n")
cat("RIS columns:", ncol(ris_data), "\n")

# Compare overlapping fields
intersect(names(csv_data), names(ris_data))
```

Key insight: RIS files typically include structured reference data and keywords that CSV exports may omit; CSVs are easier to manipulate in R.

3. Standardization function. Write a reusable function to normalise columns.

```
standardise_biblio <- function(df) {
 tibble(
   doi       = df$DI %||% NA_character_,
   title     = df$TI %||% df$Title %||% NA_character_,
   authors   = df$AU %||% NA_character_,
   year      = as.integer(df$PY %||% df$Year),
   journal   = df$SO %||% df$`Source title` %||% NA_character_,
   cited_by  = as.integer(df$TC %||% df$`Cited by` %||% 0L)
 )
}

# Usage:
# standardise_biblio(ris_data)
```

Key insight: A standardisation layer decouples downstream analysis from the input format, improving reproducibility.

Chapter 7 — Building Reproducible Corpora

1. DOI coverage by year. Compute the DOI coverage rate per publication year.

```
library(tidyverse)

works |>
 group_by(publication_year) |>
 summarise(doi_rate = mean(!is.na(doi))) |>
 ggplot(aes(publication_year, doi_rate)) +
 geom_line() +
 geom_point() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "DOI coverage by year", x = "Year", y = "Coverage")
```

Key insight: DOI coverage tends to improve for recent years as DOI assignment became standard practice after the mid-2000s.

2. Name variants. Detect potential author name duplicates.

```
library(stringdist)

author_names <- works |>
 unnest(author) |>
 distinct(au_id, au_display_name) |>
```

```
 pull(au_display_name)

# Find pairs with low string distance
combos <- combn(author_names[1:200], 2, simplify = FALSE)
similar <- map(combos, \(pair) {
 d <- stringdist(pair[1], pair[2], method = "jw")
 if (d < 0.15) tibble(name1 = pair[1], name2 = pair[2], dist = d)
 else NULL
}) |> compact() |> list_rbind()

similar |> arrange(dist) |> head(20)
```

Key insight: Common heuristics include Jaro-Winkler distance, initial matching, and shared co-author overlap. OpenAlex author IDs help but are not always correct.

3. ROR lookup. Use the ROR API to resolve institutions.

```
library(httr2)

top_insts <- works |>
 unnest(authorships) |>
 unnest(institutions) |>
 count(display_name, sort = TRUE) |>
 head(5)

ror_lookup <- function(name) {
 resp <- request("https://api.ror.org/organizations") |>
   req_url_query(query = name) |>
   req_perform() |>
   resp_body_json()
 if (length(resp$items) > 0)
   tibble(name = name, ror_id = resp$items[[1]]$id,
          ror_name = resp$items[[1]]$name)
 else
   tibble(name = name, ror_id = NA, ror_name = NA)
}

map(top_insts$display_name, ror_lookup) |> list_rbind()
```

Key insight: ROR provides stable, disambiguated identifiers; matching by name alone can produce false positives for common institutional names.

Chapter 8 — Working with Full Text

1. Extract and compare. Download a PDF, extract text, and compare the abstract.

```
library(pdftools)
library(tidyverse)

pdf_text_raw <- pdf_text("sample_paper.pdf")
extracted_abstract <- str_extract(
 paste(pdf_text_raw, collapse = "\n"),
 "(?i)abstract\\s*\\.?\\s*(.+?)(?=\\n\\s*(introduction|keywords|1\\.))",
)

# Compare with OpenAlex abstract
oa_abstract <- works |> filter(doi == "10.xxxx/example") |> pull(ab)

cat("Extracted:\n", str_trunc(extracted_abstract, 300), "\n")
cat("OpenAlex:\n", str_trunc(oa_abstract, 300), "\n")
```

Key insight: PDF extraction often introduces line-break artefacts and header/footer noise. The OpenAlex abstract (from publisher metadata) is typically cleaner.

2. Section detection. Write a function to split extracted text into sections.

```
split_sections <- function(text) {
 lines <- str_split(text, "\n")[[1]]
 # Match numbered headings like "1. Introduction" or "2 Methods"
 heading_idx <- str_which(lines, "^\\s*\\d+\\.?\\s+[A-Z]")

 if (length(heading_idx) < 2) return(list(full_text = text))

 sections <- map2(heading_idx, c(heading_idx[-1] - 1, length(lines)),
                  \(start, end) {
   list(
     heading = str_trim(lines[start]),
     text = paste(lines[(start + 1):end], collapse = "\n")
   )
 })
 set_names(
   map_chr(sections, "text"),
   map_chr(sections, "heading")
 )
}
```

```r
sections <- split_sections(paste(pdf_text_raw, collapse = "\n"))
names(sections)
```

Key insight: Heading detection is fragile; two-column layouts, supplementary materials, and non-standard numbering all break simple regex patterns.

3. Table extraction. Extract a table from a PDF using `tabulizer`.

```r
# tabulapdf needs Java (rJava); shown for reference, not executed.
# Install: install.packages("tabulapdf"); requires a working Java runtime.

tables <- extract_tables("sample_paper.pdf", pages = 3)
tbl <- as_tibble(tables[[1]], .name_repair = "unique")

# Typical cleaning steps:
tbl <- tbl |>
 row_to_names(row_number = 1) |>   # janitor::row_to_names
 mutate(across(where(is.character), str_trim)) |>
 filter(if_any(everything(), \(x) x != ""))
```

Key insight: Extracted tables almost always need header row promotion, whitespace trimming, and removal of empty or merged-cell artefact rows.

Chapter 9 — Descriptive Bibliometrics

1. Lotka's law. Fit Lotka's law using `bibliometrix`.

```r
library(bibliometrix)
library(tidyverse)

bib_df <- convert2df(works)  # or use the already-converted object
lotka_result <- lotka(biblioAnalysis(bib_df))

# Plot observed vs expected
plot(lotka_result)
```

Key insight: If the Kolmogorov-Smirnov test p-value is above 0.05, the observed distribution is consistent with the theoretical inverse-square law.

2. Bradford's law. Identify core, mid, and peripheral journal zones.

```
library(openalexR)
library(bibliometrix)

ml_works <- oa_fetch(entity = "works", search = "machine learning",
                     from_publication_date = "2020-01-01",
                     to_publication_date = "2023-12-31") |>
 slice_sample(n = 500)

bib_df <- convert2df(ml_works)
brad <- bradford(biblioAnalysis(bib_df))
brad$table |> head(20)
```

Key insight: The core zone typically contains very few journals that account for a disproportionate share of articles — Bradford's law in action.

3. Three-field exploration. Use country instead of author name on the left field.

```
library(bibliometrix)

threeFieldsPlot(bib_df, fields = c("AU_CO", "SO", "DE"),
                n = c(10, 10, 10))
```

Key insight: Country-level Sankey diagrams reveal geographic specialisation patterns: certain countries cluster around specific journals and keywords.

4. Annual growth rate. Compute the compound annual growth rate (CAGR).

```
annual <- works |>
 count(publication_year) |>
 filter(publication_year >= 2010, publication_year <= 2023)

first_year <- annual$n[1]
last_year  <- annual$n[nrow(annual)]
n_years    <- nrow(annual) - 1

cagr <- (last_year / first_year)^(1 / n_years) - 1
cat(sprintf("CAGR: %.1f%%\n", cagr * 100))
```

Key insight: A positive CAGR indicates the field is growing; compare against the global average (~4-5% for all scientific literature) to assess relative growth.

Chapter 10 — Productivity and Impact Indicators

1. Career h-index. Compute h-index, g-index, and m-quotient for a specific author.

```
library(openalexR)
library(tidyverse)

author_works <- oa_fetch(
 entity = "works",
 author.id = "A5023888391",  # replace with target author
 per_page = 200
)

cites <- sort(author_works$cited_by_count, decreasing = TRUE)

# h-index
h <- sum(cites >= seq_along(cites))

# g-index
cum_cites <- cumsum(cites)
g <- max(which(cum_cites >= seq_along(cites)^2))

# m-quotient
career_years <- max(author_works$publication_year) -
               min(author_works$publication_year) + 1
m <- h / career_years

tibble(h_index = h, g_index = g, m_quotient = round(m, 2))
```

Key insight: The h-index rewards consistency, the g-index rewards highly cited papers, and the m-quotient adjusts for career length.

2. Self-citation sensitivity. Pseudocode for removing self-citations from the h-index.

```
# Pseudocode:
# 1. For each paper by the author, fetch its citing works
# 2. Check if any citing work shares the same author ID
# 3. Count only non-self-citations
# 4. Recompute h-index with adjusted citation counts
#
# Implementation sketch (expensive API calls):
# author_id <- "A5023888391"
```

```r
# adjusted_cites <- map_int(author_works$id, \(work_id) {
#   citing <- oa_fetch(entity = "works", cites = work_id)
#   citing |>
#     unnest(author) |>
#     filter(au_id != author_id) |>
#     distinct(id) |>
#     nrow()
# })
# h_adjusted <- sum(sort(adjusted_cites, decreasing = TRUE) >=
#                   seq_along(adjusted_cites))
```

Key insight: Self-citation removal typically reduces the h-index by 1-3 points for most researchers but can have large effects for prolific self-citers.

3. Bootstrap uncertainty for MNCS. Use bootstrap resampling to compute a confidence interval for MNCS.

```r
set.seed(20260509)

mncs_boot <- replicate(1000, {
 idx <- sample(nrow(works), replace = TRUE)
 boot_sample <- works[idx, ]
 compute_mncs(boot_sample$cited_by_count,
              boot_sample$publication_year)
})

quantile(mncs_boot, c(0.025, 0.975))
```

Key insight: Small samples produce wide confidence intervals; MNCS values should always be reported with uncertainty estimates.

4. PP(top 10%) vs. MNCS. Compare two indicators by year.

```r
yearly_indicators <- works |>
 group_by(publication_year) |>
 summarise(
   mncs = compute_mncs(cited_by_count, publication_year),
   pp_top10 = mean(cited_by_count >= quantile(cited_by_count, 0.9))
 )

ggplot(yearly_indicators, aes(mncs, pp_top10)) +
 geom_point() +
 geom_text(aes(label = publication_year), nudge_y = 0.01) +
 labs(x = "MNCS", y = "PP(top 10%)",
      title = "MNCS vs PP(top 10%) by year")
```

Key insight: The two indicators usually agree but can diverge when a few extremely highly cited papers inflate MNCS without affecting the 10% threshold.

Chapter 11 — Journal Metrics

1. Four-year CiteScore proxy. Extend the citation window to four years.

```
library(openalexR)
library(tidyverse)

journal_id <- "S4210174107"  # example journal
target_year <- 2023

# Fetch articles published in the 4-year window
window_works <- oa_fetch(
 entity = "works",
 primary_location.source.id = journal_id,
 from_publication_date = paste0(target_year - 4, "-01-01"),
 to_publication_date   = paste0(target_year - 1, "-12-31")
)

citescore_4yr <- sum(window_works$cited_by_count) / nrow(window_works)
cat(sprintf("4-year CiteScore proxy: %.2f\n", citescore_4yr))
```

Key insight: A wider window captures slower-to-cite fields more fairly but dilutes the signal for fast-moving fields.

2. Rank correlation. Compare JIF and SNIP proxies with Spearman correlation.

```
journal_metrics <- works |>
 group_by(so) |>
 summarise(
   jif_proxy  = mean(cited_by_count[publication_year >= 2021]),
   snip_proxy = jif_proxy / mean(cited_by_count)  # simplified
 ) |>
 filter(!is.na(jif_proxy), !is.na(snip_proxy))

cor(journal_metrics$jif_proxy, journal_metrics$snip_proxy,
    method = "spearman")
```

Key insight: Rank correlation is typically positive but imperfect, showing that field normalisation reorders journals.

3. Field comparison. Compare journals across two disciplines.

```r
physics_journals <- c("S137773608", "S71048024")
sociology_journals <- c("S64468286", "S128732326")

compute_jif <- function(jid) {
 w <- oa_fetch(entity = "works",
               primary_location.source.id = jid,
               from_publication_date = "2021-01-01",
               to_publication_date = "2022-12-31")
 sum(w$cited_by_count) / nrow(w)
}

tibble(
 journal = c(physics_journals, sociology_journals),
 field = c(rep("Physics", 2), rep("Sociology", 2)),
 jif_proxy = map_dbl(journal, compute_jif)
) |>
 ggplot(aes(field, jif_proxy)) +
 geom_boxplot() +
 labs(title = "Raw JIF proxy by field")
```

Key insight: Raw JIF values are much higher in fields with faster citation accumulation; SNIP-style normalisation reduces this gap.

4. Self-citation share. Estimate the proportion of journal self-citations.

```r
# Approximation: among citing works of this journal's papers,
# how many are also published in the same journal?
journal_works <- oa_fetch(
 entity = "works",
 primary_location.source.id = journal_id,
 from_publication_date = "2021-01-01",
 to_publication_date = "2022-12-31"
)

# Sample a subset and check citing works
sample_ids <- journal_works |> slice_sample(n = 20) |> pull(id)

self_cite_rates <- map_dbl(sample_ids, \(wid) {
 citing <- tryCatch(
   oa_fetch(entity = "works", cites = wid),
   error = \(e) tibble()
 )
 if (nrow(citing) == 0) return(NA_real_)
 mean(citing$so == journal_works$so[1], na.rm = TRUE)
})
```

```r
mean(self_cite_rates, na.rm = TRUE)
```

Key insight: Journal self-citation rates above 20-30% are considered unusually high and may warrant investigation.

Chapter 12 — Author-Level Analysis

1. Profile comparison. Compare two researchers on multiple indicators.

```r
library(openalexR)
library(tidyverse)

ids <- c("A5023888391", "A5048491430")  # two author IDs

profiles <- map(ids, \(aid) {
 w <- oa_fetch(entity = "works", author.id = aid)
 cites <- sort(w$cited_by_count, decreasing = TRUE)
 tibble(
   author_id = aid,
   n_papers  = nrow(w),
   h_index   = sum(cites >= seq_along(cites)),
   mean_cites = mean(w$cited_by_count)
 )
}) |> list_rbind()

profiles
```

Key insight: An author with many papers but a modest h-index publishes frequently but with less consistent impact; high mean citations suggest a few highly cited papers.

2. ORCID validation. Compare works fetched by OpenAlex ID vs. ORCID filtering.

```r
oa_by_id <- oa_fetch(entity = "works",
                     author.id = "A5023888391")

oa_by_orcid <- oa_fetch(entity = "works",
                        author.orcid = "0000-0001-2345-6789")

cat("By ID:", nrow(oa_by_id), "works\n")
cat("By ORCID:", nrow(oa_by_orcid), "works\n")
```

```
# Find discrepancies
only_id <- setdiff(oa_by_id$id, oa_by_orcid$id)
only_orcid <- setdiff(oa_by_orcid$id, oa_by_id$id)
cat("Only in ID set:", length(only_id), "\n")
cat("Only in ORCID set:", length(only_orcid), "\n")
```

Key insight: Discrepancies arise from incomplete ORCID claiming, name disambiguation errors, and delayed metadata updates.

3. Name variant detection. Search for a common name and assess disambiguation.

```
wang_wei <- oa_fetch(entity = "authors",
                     search = "Wang Wei")

nrow(wang_wei)  # many entities

wang_wei |>
 select(id, display_name, works_count, cited_by_count) |>
 arrange(desc(works_count)) |>
 head(20)

# Heuristics for identifying the correct author:
# - Match institutional affiliation
# - Check ORCID linkage
# - Verify co-author overlap with known collaborators
# - Inspect topic/concept alignment
```

Key insight: Common names generate hundreds of author entities; institutional affiliation and ORCID are the most reliable disambiguation features.

4. Fractional productivity. Recompute productivity using fractional counting.

```
author_works <- oa_fetch(entity = "works",
                         author.id = "A5023888391")

fractional <- author_works |>
 mutate(n_authors = map_int(author, nrow),
        frac_credit = 1 / n_authors) |>
 group_by(publication_year) |>
 summarise(full_count = n(),
           frac_count = sum(frac_credit))

ggplot(fractional, aes(publication_year)) +
```

```
 geom_line(aes(y = full_count, colour = "Full")) +
 geom_line(aes(y = frac_count, colour = "Fractional")) +
 labs(title = "Full vs. fractional productivity",
      x = "Year", y = "Papers", colour = "Counting")
```

Key insight: Fractional counting penalises hyper-authored papers. Authors who collaborate broadly see a larger gap between full and fractional counts.

5. Self-citation ratio. Estimate the fraction of self-citations for an author.

```
target_author <- "A5023888391"

# Sample a subset of the author's works
sample_works <- author_works |> slice_sample(n = 10)

self_cite_frac <- map_dbl(sample_works$id, \(wid) {
 citing <- tryCatch(
   oa_fetch(entity = "works", cites = wid),
   error = \(e) tibble()
 )
 if (nrow(citing) == 0) return(NA_real_)
 citing |>
   unnest(author) |>
   summarise(has_self = any(au_id == target_author)) |>
   pull(has_self) |>
   mean()
})

cat(sprintf("Estimated self-citation rate: %.1f%%\n",
            mean(self_cite_frac, na.rm = TRUE) * 100))
```

Key insight: Self-citation rates of 5-15% are typical; rates above 25% may signal strategic self-citation behaviour.

Chapter 13 — Institutional and Country Analysis

1. Country-level analysis. Aggregate by country using full and fractional counts.

```
library(tidyverse)

country_counts <- works |>
 unnest(authorships) |>
```

```
 unnest(countries) |>
 rename(country = countries) |>
 mutate(n_countries = n_distinct(country), .by = id) |>
 summarise(
   full_count = n_distinct(id),
   frac_count = sum(1 / n_countries),
   .by = country
 ) |>
 arrange(desc(full_count)) |>
 head(10)

country_counts |>
 mutate(ratio = full_count / frac_count) |>
 ggplot(aes(reorder(country, full_count), ratio)) +
 geom_col() +
 coord_flip() +
 labs(title = "Full / fractional count ratio by country",
      x = NULL, y = "Ratio")
```

Key insight: Countries with high international collaboration rates (e.g., Switzerland, Singapore) show a larger ratio, meaning full counting inflates their apparent output.

2. PP(top 10%) by institution. Compute the proportion of papers in the top 10% most cited.

```
# Compute year-specific 90th percentile thresholds
thresholds <- works |>
 group_by(publication_year) |>
 summarise(p90 = quantile(cited_by_count, 0.9))

inst_top10 <- works |>
 unnest(authorships) |>
 unnest(institutions) |>
 left_join(thresholds, by = "publication_year") |>
 mutate(is_top10 = cited_by_count >= p90) |>
 summarise(
   pp_top10 = mean(is_top10),
   mncs = mean(cited_by_count),
   .by = display_name
 ) |>
 filter(!is.na(display_name)) |>
 arrange(desc(pp_top10))

ggplot(inst_top10 |> head(20), aes(mncs, pp_top10)) +
```

```r
  geom_point() +
  ggrepel::geom_text_repel(aes(label = display_name), size = 3) +
  labs(x = "MNCS", y = "PP(top 10%)")
```

Key insight: PP(top 10%) and MNCS generally correlate but can diverge if an institution has a few extremely highly cited outliers.

3. Collaboration intensity. Relate the number of institutional affiliations per paper to counting method effects.

```r
collab_intensity <- works |>
  unnest(authorships) |>
  unnest(institutions) |>
  summarise(
    n_insts = n_distinct(display_name),
    .by = id
  )

inst_collab <- works |>
  unnest(authorships) |>
  unnest(institutions) |>
  left_join(collab_intensity, by = "id") |>
  summarise(
    mean_collab = mean(n_insts),
    full_count = n_distinct(id),
    frac_count = sum(1 / n_insts),
    ratio = full_count / frac_count,
    .by = display_name
  ) |>
  filter(!is.na(display_name))

ggplot(inst_collab, aes(mean_collab, ratio)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  labs(x = "Mean affiliations per paper",
       y = "Full / fractional ratio",
       title = "Collaboration intensity vs. counting effect")
```

Key insight: There is a positive relationship — institutions with higher collaboration intensity are more inflated by full counting.

Chapter 14 — Temporal Analysis and Citation Aging

1. Half-life comparison. Compare citation half-lives between a mathematics and a biomedical journal.

```
library(openalexR)
library(tidyverse)

fetch_half_life <- function(journal_id, label) {
 w <- oa_fetch(entity = "works",
               primary_location.source.id = journal_id,
               from_publication_date = "2015-01-01",
               to_publication_date = "2018-12-31")
 w |>
   mutate(age = 2024 - publication_year) |>
   group_by(age) |>
   summarise(mean_cites = mean(cited_by_count)) |>
   mutate(cum_share = cumsum(mean_cites) / sum(mean_cites),
          journal = label)
}

math_hl   <- fetch_half_life("S154589771", "Mathematics")
biomed_hl <- fetch_half_life("S49861241", "Biomedicine")

bind_rows(math_hl, biomed_hl) |>
 ggplot(aes(age, cum_share, colour = journal)) +
 geom_line() +
 geom_hline(yintercept = 0.5, linetype = "dashed") +
 labs(title = "Citation half-life comparison",
      x = "Age (years)", y = "Cumulative citation share")
```

Key insight: Biomedical journals typically have shorter half-lives (3-5 years) than mathematics journals (8-12 years) because of faster publication cycles.

2. Price index computation. Compute the fraction of references less than 5 years old.

```
# For papers with reference data available
sample_papers <- works |>
 filter(publication_year == 2020) |>
 slice_sample(n = 20)

price_indices <- map_dbl(sample_papers$id, \(wid) {
 refs <- tryCatch(
```

```r
    oa_fetch(entity = "works", cited_by = wid),
    error = \(e) tibble()
  )
  if (nrow(refs) == 0) return(NA_real_)
  pub_year <- sample_papers$publication_year[sample_papers$id == wid]
  mean(refs$publication_year >= (pub_year - 5), na.rm = TRUE)
})

cat(sprintf("Mean Price index: %.2f\n", mean(price_indices, na.rm = TRUE)))
```

Key insight: A high Price index indicates the field relies on recent work (fast-moving); a low Price index suggests reliance on older, established literature.

3. Sleeping beauty detection. Identify delayed-recognition papers.

```r
sleeping <- works |>
  filter(publication_year <= 2018,
         cited_by_count >= 20) |>
  mutate(age = 2024 - publication_year)

# Approximate: papers with low early citation rate but high total
# (exact per-year citation data requires counts_by_year from OpenAlex)
sleeping_candidates <- sleeping |>
  filter(
    map_lgl(counts_by_year, \(cy) {
      if (is.null(cy)) return(FALSE)
      early <- cy |> filter(year <= publication_year + 3)
      sum(early$cited_by_count, na.rm = TRUE) < 2
    })
  )

sleeping_candidates |>
  select(display_name, publication_year, cited_by_count) |>
  arrange(desc(cited_by_count))
```

Key insight: Sleeping beauties are rare but occur across all fields; they often represent ahead-of-their-time ideas that gained traction only after a paradigm shift.

4. Median vs. mean aging. Replot aging curves using median citations.

```r
aging_comparison <- works |>
  mutate(age = 2024 - publication_year) |>
  group_by(age) |>
  summarise(
```

```
    mean_cites   = mean(cited_by_count),
    median_cites = median(cited_by_count)
  ) |>
  pivot_longer(c(mean_cites, median_cites),
               names_to = "stat", values_to = "value")

ggplot(aging_comparison, aes(age, value, colour = stat)) +
  geom_line() +
  geom_point() +
  labs(title = "Mean vs. median citation aging",
       x = "Age (years)", y = "Citations")
```

Key insight: The median curve is lower and flatter because the mean is pulled up by highly cited outliers. Median is more robust for skewed distributions.

5. Citation window sensitivity. Compute JIF-like proxies with different window lengths.

```
windows <- c(2, 3, 5)

jif_by_window <- map(windows, \(w) {
  journal_works |>
    filter(publication_year >= (2023 - w),
           publication_year <= 2022) |>
    summarise(jif = sum(cited_by_count) / n()) |>
    mutate(window = w)
}) |> list_rbind()

ggplot(jif_by_window, aes(factor(window), jif)) +
  geom_col() +
  labs(x = "Citation window (years)", y = "JIF proxy",
       title = "JIF sensitivity to citation window")
```

Key insight: Longer windows generally produce higher values and favour fields with slower citation accumulation.

Chapter 15 — Co-authorship Networks

1. Fractional counting. Weight edges by 1/(k-1) where k is the number of authors.

```
library(igraph)
library(tidyverse)
```

```r
edges_frac <- works |>
 unnest(author) |>
 group_by(id) |>
 mutate(k = n()) |>
 filter(k >= 2) |>
 reframe(
   pair = combn(au_id, 2, simplify = FALSE),
   weight = 1 / (k - 1)
 ) |>
 unnest_wider(pair, names_sep = "_") |>
 rename(from = pair_1, to = pair_2)

g_frac <- graph_from_data_frame(edges_frac, directed = FALSE)
E(g_frac)$weight <- edges_frac$weight

degree_frac <- strength(g_frac) |>
 sort(decreasing = TRUE) |>
 head(10)
degree_frac
```

Key insight: Fractional counting reduces the influence of mega-authored papers. Authors primarily involved in large teams may drop in rank.

2. Temporal slicing. Build yearly networks and track structural evolution.

```r
yearly_stats <- works |>
 group_by(publication_year) |>
 group_map(\(df, key) {
   edges <- df |>
     unnest(author) |>
     group_by(id) |>
     filter(n() >= 2) |>
     reframe(pair = combn(au_id, 2, simplify = FALSE)) |>
     unnest_wider(pair, names_sep = "_")
   if (nrow(edges) == 0) return(NULL)
   g <- graph_from_data_frame(edges, directed = FALSE)
   tibble(
     year = key$publication_year,
     n_nodes = vcount(g),
     n_components = components(g)$no,
     mean_degree = mean(degree(g))
   )
 }) |> list_rbind()

ggplot(yearly_stats, aes(year, mean_degree)) +
```

```
 geom_line() +
 geom_point() +
 labs(title = "Mean degree over time", x = "Year", y = "Mean degree")
```

Key insight: Mean degree typically increases over time as collaboration becomes more common; the number of components may decrease as the field becomes more connected.

3. Betweenness vs. degree. Compare centrality measures for the giant component.

```
g <- build_coauth_graph(works)
gc <- induced_subgraph(g, components(g)$membership ==
                        which.max(components(g)$csize))

centrality <- tibble(
 author = V(gc)$name,
 deg = degree(gc),
 btw = betweenness(gc, normalized = TRUE)
)

ggplot(centrality, aes(deg, btw)) +
 geom_point(alpha = 0.4) +
 labs(x = "Degree", y = "Betweenness (normalised)",
      title = "Betweenness vs. degree in co-authorship network")
```

Key insight: High-betweenness, low-degree nodes are “bridge” authors who connect otherwise separate groups — they may be interdisciplinary researchers.

4. Resolution parameter. Vary the Leiden resolution and compare community counts.

```
library(igraph)

resolutions <- c(0.5, 1.0, 2.0)

res_results <- map(resolutions, \(r) {
 comm <- cluster_leiden(gc, resolution = r,
                        objective_function = "modularity")
 tibble(resolution = r,
        n_communities = length(comm),
        modularity = modularity(comm))
})

list_rbind(res_results)
```

Key insight: Higher resolution produces more communities. The optimal value depends on the application — lower resolution reveals broad thematic clusters, higher resolution reveals tight collaborative teams.

Chapter 16 — Co-citation and Bibliographic Coupling

1. Cosine normalisation. Normalise co-citation strengths by the geometric mean of individual citation counts.

```
library(igraph)
library(tidyverse)

# Assume cocit_matrix is the raw co-citation matrix
diag_sqrt <- sqrt(diag(cocit_matrix))
cosine_matrix <- cocit_matrix / outer(diag_sqrt, diag_sqrt)
diag(cosine_matrix) <- 0

g_cosine <- graph_from_adjacency_matrix(cosine_matrix,
                                        mode = "undirected",
                                        weighted = TRUE)

# Compare top pairs
top_pairs_raw <- E(g_raw)[order(-E(g_raw)$weight)][1:10]
top_pairs_cos <- E(g_cosine)[order(-E(g_cosine)$weight)][1:10]
```

Key insight: Cosine normalisation down-weights pairs that are individually highly cited, surfacing topically related pairs that raw counts miss.

2. Temporal co-citation. Split the corpus into two periods and build separate networks.

```
periods <- list(
 early = works |> filter(publication_year %in% 2020:2021),
 late  = works |> filter(publication_year %in% 2022:2023)
)

period_graphs <- map(periods, \(df) {
 # Build co-citation network for this period's references
 refs <- df |> unnest(referenced_works)
 ref_pairs <- refs |>
   group_by(id) |>
   filter(n() >= 2) |>
```

```
    reframe(pair = combn(referenced_works, 2, simplify = FALSE)) |>
    unnest_wider(pair, names_sep = "_")

  ref_pairs |>
    count(pair_1, pair_2, name = "weight") |>
    graph_from_data_frame(directed = FALSE)
})

# Find stable pairs: edges present in both periods
shared_edges <- E(period_graphs$early) %in% E(period_graphs$late)
```

Key insight: Stable co-citation pairs represent the intellectual core of the field; new pairs emerging in the later period may signal research-front shifts.

3. Bibliographic coupling by year. Track modularity and fragmentation of the intellectual structure.

```
yearly_bc <- works |>
  group_by(publication_year) |>
  group_map(\(df, key) {
    refs <- df |> unnest(referenced_works)
    pairs <- refs |>
      group_by(referenced_works) |>
      filter(n() >= 2) |>
      reframe(pair = combn(id, 2, simplify = FALSE)) |>
      unnest_wider(pair, names_sep = "_") |>
      count(pair_1, pair_2, name = "weight")
    if (nrow(pairs) == 0) return(NULL)
    g <- graph_from_data_frame(pairs, directed = FALSE)
    comm <- cluster_leiden(g, resolution = 1.0)
    tibble(year = key$publication_year,
           n_components = components(g)$no,
           modularity = modularity(comm))
  }) |> list_rbind()

ggplot(yearly_bc, aes(year, modularity)) +
  geom_line() +
  labs(title = "Bibliographic coupling modularity over time")
```

Key insight: Decreasing modularity suggests convergence across subfields; increasing modularity suggests specialisation and fragmentation.

4. Identifying research fronts. Label communities using top title words.

```r
library(tidytext)

bc_comm <- cluster_leiden(g_bc, resolution = 1.0)
V(g_bc)$community <- membership(bc_comm)

front_labels <- works |>
 filter(id %in% V(g_bc)$name) |>
 left_join(
   tibble(id = V(g_bc)$name, community = V(g_bc)$community),
   by = "id"
 ) |>
 unnest_tokens(word, display_name) |>
 anti_join(stop_words, by = "word") |>
 count(community, word, sort = TRUE) |>
 group_by(community) |>
 slice_head(n = 3) |>
 summarise(label = paste(word, collapse = ", "))

front_labels
```

Key insight: Community labels based on frequent title words typically correspond to recognisable subfields or research themes.

Chapter 17 — Co-word and Keyword Co-occurrence

1. Author keywords vs. concepts. Build and compare two co-occurrence networks.

```r
library(igraph)
library(tidyverse)

# Author keywords network
kw_pairs <- works |>
 unnest(keywords) |>
 group_by(id) |>
 filter(n() >= 2) |>
 reframe(pair = combn(keyword, 2, simplify = FALSE)) |>
 unnest_wider(pair, names_sep = "_") |>
 count(pair_1, pair_2, name = "weight", sort = TRUE)

g_kw <- graph_from_data_frame(kw_pairs, directed = FALSE)
```

```r
# OpenAlex concepts network
concept_pairs <- works |>
 unnest(concepts) |>
 group_by(id) |>
 filter(n() >= 2) |>
 reframe(pair = combn(display_name, 2, simplify = FALSE)) |>
 unnest_wider(pair, names_sep = "_") |>
 count(pair_1, pair_2, name = "weight", sort = TRUE)

g_concept <- graph_from_data_frame(concept_pairs, directed = FALSE)

cat("Author KW network:", vcount(g_kw), "nodes,",
    ecount(g_kw), "edges\n")
cat("Concept network:", vcount(g_concept), "nodes,",
    ecount(g_concept), "edges\n")
```

Key insight: Author keywords are more specific but inconsistent; OpenAlex concepts are more standardised but coarser.

2. Temporal evolution. Track the top keywords by degree over time.

```r
yearly_top_kw <- works |>
 unnest(keywords) |>
 group_by(publication_year) |>
 group_map(\(df, key) {
   pairs <- df |>
     group_by(id) |>
     filter(n() >= 2) |>
     reframe(pair = combn(keyword, 2, simplify = FALSE)) |>
     unnest_wider(pair, names_sep = "_")
   if (nrow(pairs) == 0) return(NULL)
   g <- graph_from_data_frame(pairs, directed = FALSE)
   tibble(
     year = key$publication_year,
     keyword = names(sort(degree(g), decreasing = TRUE)[1:5]),
     degree = sort(degree(g), decreasing = TRUE)[1:5]
   )
 }) |> list_rbind()

ggplot(yearly_top_kw, aes(year, degree, colour = keyword)) +
 geom_line() +
 labs(title = "Top keywords by degree over time")
```

Key insight: Some keywords are perennially central; others emerge or fade, signalling evolving research interests.

3. Strategic diagram. Plot clusters on a density-centrality plane.

```
comm <- cluster_leiden(g_kw, resolution = 1.0)
V(g_kw)$community <- membership(comm)

strategic <- map(unique(membership(comm)), \(c) {
 nodes <- V(g_kw)[V(g_kw)$community == c]
 subg <- induced_subgraph(g_kw, nodes)
 tibble(
   community = c,
   density = edge_density(subg),
   centrality = mean(strength(g_kw, v = nodes))
 )
}) |> list_rbind()

ggplot(strategic, aes(centrality, density,
                      label = community)) +
 geom_point(size = 3) +
 geom_text(nudge_y = 0.01) +
 geom_hline(yintercept = median(strategic$density),
            linetype = "dashed") +
 geom_vline(xintercept = median(strategic$centrality),
            linetype = "dashed") +
 labs(title = "Strategic diagram",
      x = "Centrality (external)", y = "Density (internal)")
```

Key insight: Upper-right quadrant clusters are core themes (high density and centrality); lower-left clusters are peripheral or emerging topics.

4. Normalisation comparison. Compare raw and association-strength-normalised networks.

```
# Association strength: w_ij / (w_i * w_j / 2m)
total_weight <- sum(E(g_kw)$weight)
node_strength <- strength(g_kw)

assoc_edges <- as_data_frame(g_kw) |>
 mutate(
   assoc = weight / (node_strength[from] * node_strength[to] /
                    (2 * total_weight))
 )

g_assoc <- graph_from_data_frame(assoc_edges, directed = FALSE)
E(g_assoc)$weight <- assoc_edges$assoc

# Compare top-centrality keywords
```

```
top_raw <- names(sort(strength(g_kw), decreasing = TRUE))[1:10]
top_assoc <- names(sort(strength(g_assoc), decreasing = TRUE))[1:10]

tibble(raw = top_raw, normalised = top_assoc)
```

Key insight: Normalisation reduces the dominance of high-frequency generic terms, elevating more specific and meaningful keyword pairs.

Chapter 18 — Community Detection and Backbone Extraction

1. Alpha sensitivity. Run the disparity filter across multiple alpha values.

```
library(igraph)
library(tidyverse)

alphas <- c(0.01, 0.05, 0.10, 0.20)

alpha_results <- map(alphas, \(a) {
 # Disparity filter: keep edge if p_ij < alpha
 el <- as_data_frame(g) |>
   mutate(
     s_from = strength(g)[from],
     k_from = degree(g)[from],
     p_ij = (1 - weight / s_from)^(k_from - 1)
   ) |>
   filter(p_ij < a)

 g_bb <- graph_from_data_frame(el, directed = FALSE)
 comm <- cluster_leiden(g_bb, resolution = 1.0)
 tibble(alpha = a,
        edges = ecount(g_bb),
        communities = length(comm),
        connected = components(g_bb)$no == 1)
}) |> list_rbind()

alpha_results
```

Key insight: Very low alpha values remove too many edges and disconnect the network; very high values retain noise. Typical values of 0.05-0.10 balance signal and connectivity.

2. Threshold vs. disparity. Compare two backbone extraction methods using NMI.

```r
library(igraph)

# Threshold filtering: keep edges above median weight
threshold <- median(E(g)$weight)
g_thresh <- subgraph.edges(g, which(E(g)$weight > threshold))
comm_thresh <- cluster_leiden(g_thresh, resolution = 1.0)

# Disparity filtering (from exercise 1)
comm_disp <- cluster_leiden(g_bb, resolution = 1.0)

# NMI comparison on shared nodes
shared <- intersect(V(g_thresh)$name, V(g_bb)$name)
m1 <- membership(comm_thresh)[shared]
m2 <- membership(comm_disp)[shared]

compare(m1, m2, method = "nmi")
```

Key insight: The disparity filter preserves locally significant edges (important for low-degree nodes) while the threshold filter favours globally strong connections. NMI values below 0.7 indicate meaningfully different community structures.

3. Hierarchical structure. Check whether communities at different resolutions are nested.

```r
resolutions <- c(0.5, 1.0, 2.0)

memberships <- map(resolutions, \(r) {
 comm <- cluster_leiden(g_bb, resolution = r)
 tibble(node = V(g_bb)$name, community = membership(comm),
        resolution = r)
}) |> list_rbind()

# Check nesting: at resolution 2.0, are communities subsets of
# resolution 0.5 communities?
nesting <- memberships |>
 pivot_wider(names_from = resolution, values_from = community,
             names_prefix = "res_") |>
 count(res_0.5, res_2) |>
 group_by(res_2) |>
 mutate(is_nested = n_distinct(res_0.5) == 1)

mean(nesting$is_nested)  # proportion of fine communities nested
```

Key insight: Perfectly nested hierarchies are rare in real networks. Partial nesting suggests a hierarchical thematic structure with some cross- cutting topics.

4. Weighted vs. unweighted community detection. Remove weights and compare results.

```
g_unweighted <- g_bb
E(g_unweighted)$weight <- 1

comm_weighted   <- cluster_leiden(g_bb, resolution = 1.0)
comm_unweighted <- cluster_leiden(g_unweighted, resolution = 1.0)

compare(membership(comm_weighted), membership(comm_unweighted),
        method = "nmi")

cat("Weighted communities:", length(comm_weighted), "\n")
cat("Unweighted communities:", length(comm_unweighted), "\n")
```

Key insight: Removing weights often increases the number of communities because weak connections that helped merge groups are no longer distinguishable from strong ones.

Chapter 19 — Science Mapping and Overlay Maps

1. Multi-institution overlay. Overlay two institutions on the same base map.

```
library(tidyverse)
library(ggrepel)

inst_ids <- c("I123456789", "I987654321")

profiles <- map(inst_ids, \(iid) {
 w <- oa_fetch(entity = "works",
               authorships.institutions.lineage = iid,
               from_publication_date = "2020-01-01")
 w |>
   unnest(concepts) |>
   count(display_name, name = "n_works") |>
   mutate(inst = iid)
}) |> list_rbind()

ggplot(profiles, aes(display_name, n_works, fill = inst)) +
 geom_col(position = "dodge") +
 coord_flip() +
 labs(title = "Research profile overlay")
```

Key insight: Overlapping areas indicate shared research strengths; divergent areas reveal institutional specialisation.

2. Temporal overlay. Compare the same institution's profile across two time periods.

```
periods <- list(
 early = c("2018-01-01", "2019-12-31"),
 late  = c("2022-01-01", "2023-12-31")
)

temporal_profiles <- imap(periods, \(dates, label) {
 oa_fetch(entity = "works",
          authorships.institutions.lineage = "I123456789",
          from_publication_date = dates[1],
          to_publication_date   = dates[2]) |>
   unnest(concepts) |>
   count(display_name, name = "n_works") |>
   mutate(period = label)
}) |> list_rbind()

temporal_profiles |>
 pivot_wider(names_from = period, values_from = n_works,
             values_fill = 0) |>
 mutate(shift = late - early) |>
 arrange(desc(abs(shift))) |>
 head(15)
```

Key insight: Positive shifts indicate growing research areas; negative shifts suggest declining focus or strategic reorientation.

3. Country-level overlay. Compare a small and a large country's publication profiles.

```
countries <- c("SG", "US")  # Singapore vs. United States

country_profiles <- map(countries, \(cc) {
 oa_fetch(entity = "works",
          authorships.countries = cc,
          from_publication_date = "2022-01-01",
          to_publication_date = "2022-12-31") |>
   slice_sample(n = 500) |>
   unnest(concepts) |>
   count(display_name, name = "n_works") |>
   mutate(share = n_works / sum(n_works),
          country = cc)
```

```
}) |> list_rbind()

country_profiles |>
 pivot_wider(names_from = country, values_from = share,
             values_fill = 0) |>
 ggplot(aes(US, SG)) +
 geom_point() +
 geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
 ggrepel::geom_text_repel(aes(label = display_name), size = 2) +
 labs(title = "Country profile: Singapore vs. US")
```

Key insight: Small countries tend to show concentrated profiles (above the diagonal in selected areas); large countries are more evenly distributed.

Chapter 20 — Network Visualization and Interoperability

1. Layout comparison. Apply five layouts and compare quality and speed.

```
library(igraph)
library(ggraph)
library(tidyverse)

layouts <- c("fr", "kk", "drl", "graphopt", "lgl")

layout_results <- map(layouts, \(algo) {
 t <- system.time({
   p <- ggraph(gc, layout = algo) +
     geom_edge_link(alpha = 0.1) +
     geom_node_point(aes(colour = factor(community)), size = 1) +
     labs(title = algo) +
     theme_void()
 })
 list(plot = p, time = t["elapsed"], layout = algo)
})

# Print timing
map(layout_results, \(x) tibble(layout = x$layout, secs = x$time)) |>
 list_rbind()
```

Key insight: Fruchterman-Reingold is a good default; Kamada-Kawai is better for small networks; DRL scales to large graphs but can produce chaotic layouts.

2. Interactive filtering. Create a filterable `visNetwork` visualisation.

```
library(visNetwork)

nodes_df <- tibble(
 id = V(gc)$name,
 label = V(gc)$name,
 group = as.character(V(gc)$community)
)

edges_df <- as_data_frame(gc) |>
 select(from, to)

visNetwork(nodes_df, edges_df) |>
 visOptions(selectedBy = "group",
            highlightNearest = TRUE) |>
 visPhysics(stabilization = FALSE)
```

Key insight: Interactive filtering lets users focus on specific communities and explore inter-community bridges.

3. Gephi round-trip. Export to GraphML for use in Gephi.

```
library(igraph)

# Export
write_graph(gc, "network.graphml", format = "graphml")

# After Gephi processing, compare:
# - Force Atlas 2 layout often reveals community separation better
# - R/ggraph offers more reproducible, scriptable figures
# - Gephi is better for interactive exploration and publication
#   quality figures via SVG/PDF export
```

Key insight: Gephi's Force Atlas 2 often produces cleaner community separation; R workflows are more reproducible and automatable.

4. Colour-blind check. Test colour distinguishability with increasing community counts.

```
library(ggraph)
library(viridis)

n_communities <- c(3, 5, 10)

plots <- map(n_communities, \(k) {
```

```
 comm <- cluster_leiden(gc, resolution = k / 5)
 V(gc)$comm_k <- membership(comm)

 ggraph(gc, layout = "fr") +
   geom_edge_link(alpha = 0.05) +
   geom_node_point(aes(colour = factor(comm_k)), size = 2) +
   scale_colour_viridis_d(option = "D") +
   labs(title = paste(k, "communities")) +
   theme_void() +
   theme(legend.position = "none")
})

# Use colorBlindness package or online simulator to test
# library(colorBlindness)
# cvdPlot(plots[[3]])
```

Key insight: Viridis palettes remain distinguishable up to about 8 categories; beyond that, consider shape or faceting as supplementary channels.

Chapter 21 — Text Mining Bibliographic Corpora

1. Bigram analysis. Build a DFM from bigrams.

```
library(quanteda)
library(tidyverse)

corp <- corpus(works, text_field = "ab")
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
 tokens_tolower() |>
 tokens_remove(stopwords("en")) |>
 tokens_ngrams(n = 2)

dfm_bi <- dfm(toks) |> dfm_trim(min_termfreq = 5)

topfeatures(dfm_bi, 20)
```

Key insight: Bigrams capture meaningful phrases like “machine_learning” or “citation_analysis” that single words miss.

2. Lexical diversity. Compute the type-token ratio (TTR) per year.

```r
library(quanteda)
library(quanteda.textstats)

ttr_by_year <- works |>
 filter(!is.na(ab)) |>
 group_by(publication_year) |>
 summarise(
   all_text = paste(ab, collapse = " "),
   .groups = "drop"
 ) |>
 mutate(
   n_types  = map_int(all_text, \(t) {
     toks <- tokens(t, remove_punct = TRUE) |> tokens_tolower()
     ntype(dfm(toks))
   }),
   n_tokens = map_int(all_text, \(t) {
     toks <- tokens(t, remove_punct = TRUE)
     ntoken(toks)
   }),
   ttr = n_types / n_tokens
 )

ggplot(ttr_by_year, aes(publication_year, ttr)) +
 geom_line() +
 labs(title = "Lexical diversity over time",
      x = "Year", y = "Type-token ratio")
```

Key insight: Declining TTR may indicate vocabulary standardisation; increasing TTR suggests diversifying terminology.

3. Keyness analysis. Identify terms that distinguish a specific year.

```r
library(quanteda)
library(quanteda.textstats)

corp <- corpus(works |> filter(!is.na(ab)),
               text_field = "ab",
               docid_field = "id")
docvars(corp, "year") <- works$publication_year[!is.na(works$ab)]

dfm_all <- dfm(tokens(corp, remove_punct = TRUE) |>
               tokens_tolower() |>
               tokens_remove(stopwords("en")))

# Compare 2023 vs. all other years
```

```
dfm_grouped <- dfm_group(dfm_all, groups = ifelse(
 docvars(dfm_all, "year") == 2023, "2023", "other"
))

keyness <- textstat_keyness(dfm_grouped, target = "2023")
textplot_keyness(keyness, n = 20)
```

Key insight: Keyness analysis surfaces terms unique to a period. For 2023, terms related to LLMs and AI may dominate.

4. Preprocessing sensitivity. Compare analyses with and without stemming.

```
toks_nostem <- tokens(corp, remove_punct = TRUE) |>
 tokens_tolower() |>
 tokens_remove(stopwords("en"))

toks_stem <- tokens_wordstem(toks_nostem)

top_nostem <- topfeatures(dfm(toks_nostem), 20)
top_stem   <- topfeatures(dfm(toks_stem), 20)

tibble(
 rank = 1:20,
 unstemmed = names(top_nostem),
 stemmed = names(top_stem)
)
```

Key insight: Stemming merges variants (e.g., "citations" and "citation") but can produce unintelligible stems. For keyword-based analyses, unstemmed terms are often more interpretable.

Chapter 22 — Topic Modeling

1. K selection. Fit LDA models with multiple values of K and compare coherence.

```
library(topicmodels)
library(tidyverse)
set.seed(20260509)

ks <- c(5, 8, 12, 15)

models <- map(ks, \(k) {
```

```
 LDA(dtm, k = k, method = "Gibbs",
     control = list(seed = 20260509, iter = 1000))
})

# Compute coherence (using top-10 terms per topic)
coherence_scores <- map_dbl(models, \(mod) {
 terms <- terms(mod, 10)
 # Simple PMI-based coherence approximation
 mean(map_dbl(seq_len(ncol(terms)), \(t) {
   top_terms <- terms[, t]
   pairs <- combn(top_terms, 2, simplify = FALSE)
   mean(map_dbl(pairs, \(p) {
     log((slam::col_sums(dtm[, p[1]] > 0 & dtm[, p[2]] > 0) + 1) /
         slam::col_sums(dtm[, p[1]] > 0))
   }))
 }))
})

tibble(k = ks, coherence = coherence_scores) |>
 ggplot(aes(k, coherence)) +
 geom_line() +
 geom_point() +
 labs(title = "Topic coherence by K", x = "K", y = "Coherence")
```

Key insight: The best K balances coherence (higher is better) and interpretability. There is rarely a single "correct" K.

2. STM with journal covariate. Fit a Structural Topic Model with journal as a prevalence covariate.

```
library(stm)
set.seed(20260509)

# Prepare data from two journals
two_journals <- works |>
 filter(so %in% c("Scientometrics", "PLOS ONE"), !is.na(ab))

processed <- textProcessor(two_journals$ab,
                           metadata = two_journals)
prepped <- prepDocuments(processed$documents,
                         processed$vocab,
                         processed$meta)

stm_fit <- stm(prepped$documents, prepped$vocab,
               K = 10,
```

```
                prevalence = ~ so,
                data = prepped$meta,
                seed = 20260509)

effect <- estimateEffect(1:10 ~ so, stm_fit, metadata = prepped$meta)
plot(effect, covariate = "so", topics = 1:10)
```

Key insight: Topics with large journal effects represent disciplinary specialisation; shared topics indicate cross-cutting themes.

3. Topic labelling. Read top documents per topic to assign labels.

```
library(topicmodels)
library(tidyverse)

best_model <- models[[which.max(coherence_scores)]]
gamma <- posterior(best_model)$topics

# For each topic, find the top-5 documents
topic_labels <- map(seq_len(ncol(gamma)), \(t) {
 top_docs <- order(gamma[, t], decreasing = TRUE)[1:5]
 titles <- works$display_name[top_docs]
 tibble(topic = t,
        top_terms = paste(terms(best_model, 5)[, t], collapse = ", "),
        sample_titles = paste(titles, collapse = " | "))
}) |> list_rbind()

topic_labels
# Now assign human-readable labels based on inspection
```

Key insight: Automated topic labels from top terms are a starting point; reading representative documents is essential for meaningful labels.

4. LDA vs. STM comparison. Compare topics from both models.

```
# Extract top terms for each model
lda_terms <- terms(best_model, 10)
stm_terms <- labelTopics(stm_fit, n = 10)$prob

# Compute Jaccard similarity between matched topics
jaccard <- function(a, b) {
 length(intersect(a, b)) / length(union(a, b))
}

# Best-match each LDA topic to an STM topic
```

```
similarities <- map_dbl(seq_len(ncol(lda_terms)), \(i) {
 max(map_dbl(seq_len(nrow(stm_terms)), \(j) {
   jaccard(lda_terms[, i], stm_terms[j, ])
 }))
})

mean(similarities)
```

Key insight: STM topics may differ because the year covariate allows topic prevalence to shift over time, producing more temporally coherent topics.

Chapter 23 — Keyword Extraction

1. Precision and recall. Evaluate RAKE keywords against author keywords.

```
library(udpipe)
library(tidyverse)

sample_papers <- works |>
 filter(!is.na(ab), map_lgl(keywords, \(k) length(k) > 0)) |>
 slice_sample(n = 20)

eval_results <- map(seq_len(nrow(sample_papers)), \(i) {
 paper <- sample_papers[i, ]
 rake_kw <- keywords_rake(
   data.frame(doc_id = paper$id,
              sentence = paper$ab,
              stringsAsFactors = FALSE),
   term = "sentence", group = "doc_id"
 ) |> head(10) |> pull(keyword) |> tolower()

 author_kw <- tolower(unlist(paper$keywords))

 tibble(
   precision = mean(rake_kw %in% author_kw),
   recall = mean(author_kw %in% rake_kw)
 )
}) |> list_rbind()

eval_results |>
 summarise(across(everything(), list(mean = mean, sd = sd)))
```

Key insight: RAKE precision is typically low (0.1-0.3) because algorithms and authors conceptualise keywords differently. Recall is also limited because author keywords often include broad terms not in the abstract.

2. Bigram keywords. Score bigrams by TF-IDF.

```
library(tidytext)
library(tidyverse)

bigram_tfidf <- works |>
 filter(!is.na(ab)) |>
 unnest_tokens(bigram, ab, token = "ngrams", n = 2) |>
 separate(bigram, c("word1", "word2"), sep = " ") |>
 filter(!word1 %in% stop_words$word,
        !word2 %in% stop_words$word) |>
 unite(bigram, word1, word2, sep = " ") |>
 count(id, bigram) |>
 bind_tf_idf(bigram, id, n) |>
 group_by(bigram) |>
 summarise(mean_tfidf = mean(tf_idf)) |>
 arrange(desc(mean_tfidf))

bigram_tfidf |> head(20)
```

Key insight: Multi-word terms like "citation analysis" and "research evaluation" are invisible to unigram methods but highly informative.

3. Keyword enrichment. Use extracted keywords to build a co-word network for keyword-poor papers.

```
library(igraph)
library(udpipe)

# Papers without author keywords
no_kw_papers <- works |>
 filter(map_lgl(keywords, \(k) length(k) == 0), !is.na(ab))

# Extract keywords via RAKE
extracted <- map(seq_len(nrow(no_kw_papers)), \(i) {
 kw <- keywords_rake(
   data.frame(doc_id = no_kw_papers$id[i],
              sentence = no_kw_papers$ab[i]),
   term = "sentence", group = "doc_id"
 ) |> head(5) |> pull(keyword)
 tibble(id = no_kw_papers$id[i], keyword = kw)
}) |> list_rbind()
```

```r
# Build co-word network
pairs <- extracted |>
 group_by(id) |>
 filter(n() >= 2) |>
 reframe(pair = combn(keyword, 2, simplify = FALSE)) |>
 unnest_wider(pair, names_sep = "_") |>
 count(pair_1, pair_2, sort = TRUE)

g_enriched <- graph_from_data_frame(pairs, directed = FALSE)
```

Key insight: Extracted keywords produce a topical structure comparable to author keywords, enabling co-word analysis even for papers without manual keyword assignments.

Chapter 24 — Embeddings for Scientometrics

1. Weighted averaging. Use TF-IDF weights for document embedding computation.

```r
library(word2vec)
library(tidytext)
library(tidyverse)
library(umap)

# Compute TF-IDF weights
tfidf <- works |>
 filter(!is.na(ab)) |>
 unnest_tokens(word, ab) |>
 anti_join(stop_words, by = "word") |>
 count(id, word) |>
 bind_tf_idf(word, id, n)

# Get word vectors
wv <- predict(w2v_model, newdata = unique(tfidf$word), type = "embedding")

# Weighted average per document
doc_embeddings <- tfidf |>
 inner_join(
   as_tibble(wv, rownames = "word"),
   by = "word"
 ) |>
 group_by(id) |>
 summarise(across(starts_with("V"), \(x) weighted.mean(x, tf_idf)))
```

```
# UMAP
umap_result <- umap(as.matrix(doc_embeddings |> select(-id)))
```

Key insight: TF-IDF weighting down-weights common words, often producing tighter, more meaningful clusters in the UMAP projection.

2. K-means clustering. Apply k-means to embeddings and compare with LDA topics.

```
set.seed(20260509)

embedding_matrix <- as.matrix(doc_embeddings |> select(-id))

km_results <- map(c(5, 8, 12), \(k) {
 km <- kmeans(embedding_matrix, centers = k, nstart = 25)
 tibble(k = k,
        cluster = km$cluster,
        id = doc_embeddings$id)
}) |> list_rbind()

# Compare k=8 clusters with LDA topics (K=8)
# Assuming lda_topics is a vector of topic assignments
km8 <- km_results |> filter(k == 8) |> pull(cluster)
compare_nmi <- igraph::compare(km8, lda_topics, method = "nmi")
cat(sprintf("NMI between k-means and LDA: %.3f\n", compare_nmi))
```

Key insight: Moderate NMI (0.3-0.6) is typical; the two methods capture overlapping but not identical structure because they model different aspects of the data.

3. Pre-trained embeddings. Use SPECTER embeddings via reticulate.

```
library(reticulate)

# Requires Python with transformers installed
transformers <- import("transformers")
torch <- import("torch")

tokenizer <- transformers$AutoTokenizer$from_pretrained(
 "allenai/specter2"
)
model <- transformers$AutoModel$from_pretrained(
 "allenai/specter2"
)
```

```r
abstracts <- works$ab[!is.na(works$ab)][1:100]
inputs <- tokenizer(abstracts, padding = TRUE, truncation = TRUE,
                    return_tensors = "pt", max_length = 512L)
with(torch$no_grad(), {
 outputs <- model(inputs)
})
specter_emb <- outputs$last_hidden_state$mean(dim = 1L)$numpy()
```

Key insight: Pre-trained scientific embeddings like SPECTER capture domain semantics better than generic word2vec, especially for similarity search.

4. Analogy test. Test analogical reasoning in the word2vec model.

```r
library(word2vec)

# Test: "journal" - "article" + "book"   ?
analogy <- predict(w2v_model,
                   newdata = c("journal", "article", "book"),
                   type = "nearest", top_n = 5)

# Alternative approach using vector arithmetic
wv <- as.matrix(w2v_model)
if (all(c("journal", "article", "book") %in% rownames(wv))) {
 target <- wv["journal", ] - wv["article", ] + wv["book", ]
 sims <- word2vec_similarity(
   matrix(target, nrow = 1), wv
 )
 sort(sims[1, ], decreasing = TRUE)[1:5]
}
```

Key insight: Domain-specific models may capture field-specific analogies (e.g., “h-index” - “author” + “journal” near “impact factor”) but fail on generic analogies, reflecting the training corpus.

Chapter 25 — Detecting Emerging Topics

1. Relative burst detection. Use proportional frequencies instead of absolute counts.

```r
library(tidyverse)

# Compute keyword proportions by year
kw_yearly <- works |>
```

```r
 unnest(keywords) |>
 count(publication_year, keyword) |>
 group_by(publication_year) |>
 mutate(proportion = n / sum(n)) |>
 ungroup()

# Apply Kleinberg burst detection on proportions
bursts_relative <- kw_yearly |>
 group_by(keyword) |>
 filter(n() >= 3) |>  # need multiple years
 summarise(
   burst = list(kleinberg_bursts(proportion, publication_year)),
   .groups = "drop"
 ) |>
 unnest(burst) |>
 arrange(desc(strength))

bursts_relative |> head(20)
```

Key insight: Relative frequencies reduce bias toward high-volume years; some terms that burst in absolute counts may be merely riding overall growth.

2. Emergence by journal. Compare emerging keywords across two journals.

```r
journals <- c("Scientometrics", "PLOS ONE")

journal_bursts <- map(journals, \(j) {
 kw_j <- works |>
   filter(so == j) |>
   unnest(keywords) |>
   count(publication_year, keyword) |>
   group_by(keyword) |>
   filter(n() >= 3) |>
   summarise(
     burst = list(kleinberg_bursts(n, publication_year))
   ) |>
   unnest(burst) |>
   mutate(journal = j)
}) |> list_rbind()

# Compare top bursting terms
journal_bursts |>
 group_by(journal) |>
 arrange(desc(strength)) |>
 slice_head(n = 10) |>
 select(journal, keyword, strength)
```

Key insight: Different journals serve different research communities; emerging terms may overlap only for broad methodological advances.

3. Novelty scoring. Compute document novelty using cosine distance to prior-year centroids.

```
library(quanteda)
library(tidyverse)

dfm_yearly <- dfm(tokens(corpus(works, text_field = "ab"),
                         remove_punct = TRUE) |>
                  tokens_tolower() |>
                  tokens_remove(stopwords("en"))) |>
 dfm_tfidf()

years <- sort(unique(works$publication_year))

novelty <- map(years[-1], \(yr) {
 prior <- dfm_subset(dfm_yearly, publication_year < yr)
 current <- dfm_subset(dfm_yearly, publication_year == yr)
 if (nrow(prior) == 0 || nrow(current) == 0) return(NULL)

 centroid <- colMeans(as.matrix(prior))
 current_mat <- as.matrix(current)

 cos_dist <- 1 - (current_mat %*% centroid) /
   (sqrt(rowSums(current_mat^2)) * sqrt(sum(centroid^2)))

 tibble(year = yr, novelty = as.numeric(cos_dist))
}) |> list_rbind()

novelty |>
 group_by(year) |>
 summarise(mean_novelty = mean(novelty, na.rm = TRUE)) |>
 ggplot(aes(year, mean_novelty)) +
 geom_line() +
 labs(title = "Document novelty over time",
      x = "Year", y = "Mean cosine distance to prior centroid")
```

Key insight: Increasing novelty scores suggest the field is exploring new territory; stable scores indicate incremental progress within established paradigms.

4. Validation. Manually validate burst-detected emerging terms.

```
# Select 5 terms identified as "emerging"
emerging_terms <- bursts_relative |>
 arrange(desc(strength)) |>
 head(5) |>
 pull(keyword)

# For each term, search for corroborating evidence:
# - Is there a review article about this topic?
# - Was it a conference theme in recent years?
# - Does Google Trends show a rising interest?

# Create a validation table
validation <- tibble(
 term = emerging_terms,
 review_found = c(TRUE, TRUE, FALSE, TRUE, FALSE),
 conference_theme = c(TRUE, FALSE, FALSE, TRUE, TRUE),
 google_trend_rising = c(TRUE, TRUE, TRUE, FALSE, TRUE),
 validated = review_found | conference_theme
)

cat(sprintf("False positive rate: %.0f%%\n",
            (1 - mean(validation$validated)) * 100))
```

Key insight: Burst detection has moderate precision; triangulating with qualitative sources reduces false positives and increases confidence.

Chapter 26 — Altmetrics

1. OA citation advantage. Compare citations for OA vs. closed-access papers across journals.

```
library(openalexR)
library(tidyverse)

journal_ids <- c("S4210174107", "S137773608")

oa_advantage <- map(journal_ids, \(jid) {
 w <- oa_fetch(entity = "works",
              primary_location.source.id = jid,
              from_publication_date = "2019-01-01",
              to_publication_date = "2022-12-31")
 w |>
   group_by(is_oa) |>
```

```
  summarise(
    n = n(),
    mean_cites = mean(cited_by_count),
    median_cites = median(cited_by_count)
  ) |>
  mutate(journal = jid)
}) |> list_rbind()

oa_advantage
```

Key insight: The OA citation advantage exists in most fields but varies in magnitude. Self-selection (better papers go OA) may partially explain the effect.

2. Temporal attention. Track citation accumulation over time for early-attention papers.

```
recent_works <- works |>
 filter(publication_year == 2022, !is.null(counts_by_year))

attention_trajectory <- recent_works |>
 unnest(counts_by_year) |>
 group_by(id) |>
 arrange(year) |>
 mutate(cum_cites = cumsum(cited_by_count)) |>
 ungroup()

# Classify: high vs. low early attention (first year citations)
early_cites <- attention_trajectory |>
 filter(year == 2022) |>
 mutate(early_attention = ifelse(cited_by_count >= 5,
                                  "High", "Low"))

attention_trajectory |>
 left_join(early_cites |> select(id, early_attention), by = "id") |>
 group_by(early_attention, year) |>
 summarise(mean_cum = mean(cum_cites, na.rm = TRUE)) |>
 ggplot(aes(year, mean_cum, colour = early_attention)) +
 geom_line() +
 labs(title = "Citation accumulation by early attention")
```

Key insight: Papers with high early attention tend to sustain their advantage, consistent with cumulative advantage (Matthew Effect) in citations.

3. Cross-field comparison. Compare citation distributions and reference list lengths across fields.

```r
biomed <- oa_fetch(entity = "works",
                  primary_location.source.id = "S49861241",
                  from_publication_date = "2020-01-01") |>
 mutate(field = "Biomedicine") |>
 slice_sample(n = 200)

math <- oa_fetch(entity = "works",
                primary_location.source.id = "S154589771",
                from_publication_date = "2020-01-01") |>
 mutate(field = "Mathematics") |>
 slice_sample(n = 200)

comparison <- bind_rows(biomed, math)

comparison |>
 group_by(field) |>
 summarise(
   mean_cites = mean(cited_by_count),
   median_cites = median(cited_by_count),
   mean_refs = mean(referenced_works_count, na.rm = TRUE)
 )

ggplot(comparison, aes(cited_by_count, fill = field)) +
 geom_histogram(position = "dodge", bins = 30) +
 scale_x_log10() +
 labs(title = "Citation distributions by field")
```

Key insight: Biomedical papers have higher citation counts and longer reference lists than mathematics papers. This is why raw citation counts cannot be compared across fields.

Chapter 27 — Open Science Indicators

1. Cross-journal OA comparison. Compare OA rates across three journals in different disciplines.

```r
library(openalexR)
library(tidyverse)

journals <- list(
 biology = "S49861241",
 physics = "S137773608",
 sociology = "S64468286"
```

```
)

oa_rates <- imap(journals, \(jid, field) {
 w <- oa_fetch(entity = "works",
               primary_location.source.id = jid,
               from_publication_date = "2020-01-01",
               to_publication_date = "2023-12-31")
 w |>
   summarise(oa_rate = mean(is_oa, na.rm = TRUE),
             gold_oa = mean(oa_status == "gold", na.rm = TRUE)) |>
   mutate(field = field)
}) |> list_rbind()

oa_rates
```

Key insight: Biology tends to have the highest gold OA rate (driven by PLOS, eLife, etc.), while sociology and mathematics lag behind.

2. Temporal OA growth. Track the year-over-year OA growth rate.

```
oa_growth <- works |>
 filter(is_oa) |>
 count(publication_year, name = "n_oa") |>
 mutate(yoy_growth = (n_oa / lag(n_oa)) - 1)

ggplot(oa_growth, aes(publication_year, yoy_growth)) +
 geom_col() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Year-over-year OA growth",
      x = "Year", y = "Growth rate")
```

Key insight: OA growth accelerated around 2020, likely driven by COVID-19 mandates and Plan S implementation.

3. OA and collaboration. Test whether author count predicts OA status.

```
works |>
 mutate(n_authors = map_int(author, nrow)) |>
 group_by(n_authors = pmin(n_authors, 10)) |>
 summarise(oa_rate = mean(is_oa, na.rm = TRUE), n = n()) |>
 ggplot(aes(n_authors, oa_rate)) +
 geom_col() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "OA rate by number of authors",
      x = "Authors (capped at 10)", y = "OA rate")
```

Key insight: Larger teams may have more funding to cover APCs. International collaborations also increase OA probability because at least one funder may mandate it.

Chapter 28 — Gender and Equity Analysis

1. Cross-journal comparison. Compare gender representation in first authorship across journals.

```
library(openalexR)
library(tidyverse)

journal_ids <- c("S4210174107", "S49861241", "S154589771")

gender_by_journal <- map(journal_ids, \(jid) {
 w <- oa_fetch(entity = "works",
              primary_location.source.id = jid,
              from_publication_date = "2020-01-01") |>
   slice_sample(n = 200)

 w |>
   mutate(first_author = map_chr(author, \(a) a$au_display_name[1])) |>
   # Gender inference would use genderize.io or similar
   mutate(journal = jid)
}) |> list_rbind()

# After gender inference:
# gender_by_journal |>
#   group_by(journal, inferred_gender) |>
#   summarise(n = n()) |>
#   mutate(pct = n / sum(n))
```

Key insight: Gender representation varies significantly by field. STEM journals typically have lower female first-author rates than social sciences.

2. Temporal trends. Track the proportion of female first authors over time.

```
# After gender inference has been performed:
gender_trend <- works_gendered |>
 group_by(publication_year, inferred_gender) |>
 summarise(n = n(), .groups = "drop_last") |>
 mutate(pct = n / sum(n))

ggplot(gender_trend |> filter(inferred_gender == "female"),
```

```r
     aes(publication_year, pct)) +
 geom_line() +
 geom_point() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Female first-author share over time",
     x = "Year", y = "Proportion female")
```

Key insight: Most fields show a gradual increase in female representation, though the rate of change varies considerably by discipline and region.

3. Citation gap. Compare citations by inferred gender, controlling for year.

```r
# After gender inference:
citation_gap <- works_gendered |>
 group_by(publication_year, inferred_gender) |>
 summarise(
   mean_cites = mean(cited_by_count),
   median_cites = median(cited_by_count),
   n = n(),
   .groups = "drop"
 )

ggplot(citation_gap, aes(publication_year, mean_cites,
                          colour = inferred_gender)) +
 geom_line() +
 labs(title = "Mean citations by inferred gender",
     x = "Year", y = "Mean citations")

# Statistical test (controlling for year)
summary(lm(cited_by_count ~ inferred_gender + publication_year,
           data = works_gendered))
```

Key insight: A persistent citation gap may reflect systemic biases, network effects, or confounding factors (e.g., team size, field). Observational data alone cannot establish causation.

Chapter 29 — Retraction Analysis

1. Retraction rate by field. Compare retraction rates across fields, normalised by corpus size.

```r
library(openalexR)
library(tidyverse)
```

```r
fields <- c("biomedicine", "computer science")

retraction_rates <- map(fields, \(f) {
 all_works <- oa_fetch(entity = "works", search = f,
                       from_publication_date = "2015-01-01",
                       count_only = TRUE)
 retracted <- oa_fetch(entity = "works", search = f,
                       type = "retraction",
                       from_publication_date = "2015-01-01",
                       count_only = TRUE)
 tibble(
   field = f,
   total = all_works$count,
   retracted = retracted$count,
   rate_per_1000 = retracted$count / all_works$count * 1000
 )
}) |> list_rbind()

retraction_rates
```

Key insight: Retraction rates are typically higher in biomedical fields (1-2 per 1,000) than in computer science, partly reflecting different scrutiny levels and reproducibility challenges.

2. Citation trajectory. Examine when a retracted paper's citations were received.

```r
retracted_paper <- oa_fetch(entity = "works",
                            doi = "10.xxxx/retracted-example")

if (!is.null(retracted_paper$counts_by_year)) {
 retracted_paper |>
   unnest(counts_by_year) |>
   ggplot(aes(year, cited_by_count)) +
   geom_col() +
   geom_vline(xintercept = 2020, linetype = "dashed",
              colour = "red") +  # retraction year
   labs(title = "Citation trajectory of a retracted paper",
        x = "Year", y = "Citations received")
}
```

Key insight: Many retracted papers continue to be cited after retraction, indicating that the retraction notice does not always reach citing authors.

3. Retraction-citation correlation. Test whether pre-retraction citation count predicts post-retraction citations.

```r
# Assuming a set of retracted papers with known retraction years
retracted_set <- retracted_works |>
 unnest(counts_by_year) |>
 mutate(period = ifelse(year < retraction_year, "pre", "post")) |>
 group_by(id, period) |>
 summarise(cites = sum(cited_by_count), .groups = "drop") |>
 pivot_wider(names_from = period, values_from = cites,
             values_fill = 0)

cor(retracted_set$pre, retracted_set$post, method = "spearman")

ggplot(retracted_set, aes(pre, post)) +
 geom_point(alpha = 0.5) +
 geom_smooth(method = "lm") +
 labs(x = "Pre-retraction citations",
      y = "Post-retraction citations",
      title = "Pre- vs. post-retraction citation counts")
```

Key insight: A positive correlation suggests that highly visible papers continue to be cited after retraction — the "retraction paradox."

Chapter 30 — Funding Analysis

1. Funder co-occurrence. Identify frequently co-occurring funder pairs.

```r
library(tidyverse)

funder_pairs <- works |>
 unnest(grants) |>
 group_by(id) |>
 filter(n_distinct(funder) >= 2) |>
 reframe(pair = combn(funder, 2, simplify = FALSE)) |>
 unnest_wider(pair, names_sep = "_") |>
 count(pair_1, pair_2, sort = TRUE)

funder_pairs |> head(10)
```

Key insight: National funders frequently co-occur with international organisations (e.g., NIH + EU), reflecting collaborative funding models.

2. Funder portfolios. Compare topical profiles of two major funders.

```r
library(openalexR)
library(tidyverse)

funder_ids <- c("F123456", "F789012")  # OpenAlex funder IDs

funder_profiles <- map(funder_ids, \(fid) {
 w <- oa_fetch(entity = "works",
               grants.funder = fid,
               from_publication_date = "2020-01-01") |>
   slice_sample(n = 300)
 w |>
   unnest(keywords) |>
   count(keyword, sort = TRUE) |>
   mutate(funder = fid)
}) |> list_rbind()

# Compare top keywords
funder_profiles |>
 group_by(funder) |>
 slice_head(n = 15) |>
 ggplot(aes(reorder(keyword, n), n, fill = funder)) +
 geom_col(position = "dodge") +
 coord_flip() +
 labs(title = "Funder portfolio comparison")
```

Key insight: Different funders support different thematic areas; comparing portfolios reveals strategic funding priorities.

3. Temporal trends. Track funding acknowledgement coverage over time.

```r
funding_trend <- works |>
 mutate(has_funding = map_lgl(grants, \(g) length(g) > 0)) |>
 group_by(publication_year) |>
 summarise(funding_rate = mean(has_funding, na.rm = TRUE))

ggplot(funding_trend, aes(publication_year, funding_rate)) +
 geom_line() +
 geom_point() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Papers with funding acknowledgements over time",
      x = "Year", y = "Share with funding data")
```

Key insight: Funding metadata coverage has increased as publishers and databases improve structured acknowledgement parsing.

Chapter 31 — Patent-Paper Linkages

1. Field comparison. Compare the proportion of highly cited papers across fields.

```
library(openalexR)
library(tidyverse)

biomed <- oa_fetch(entity = "works",
                   search = "drug delivery",
                   from_publication_date = "2018-01-01",
                   to_publication_date = "2020-12-31") |>
 slice_sample(n = 200) |> mutate(field = "Biomedical")

socsci <- oa_fetch(entity = "works",
                   search = "social inequality",
                   from_publication_date = "2018-01-01",
                   to_publication_date = "2020-12-31") |>
 slice_sample(n = 200) |> mutate(field = "Social Science")

combined <- bind_rows(biomed, socsci)

combined |>
 group_by(field) |>
 summarise(
   pct_top10 = mean(cited_by_count >=
                      quantile(cited_by_count, 0.9)),
   mean_cites = mean(cited_by_count)
 )
```

Key insight: Biomedical research is more likely to be cited in patents because it has more direct translational applications. Higher citation rates do not automatically imply greater technological relevance.

2. Time lag estimation. Estimate the citation accumulation trajectory for a cohort of papers.

```
cohort_2015 <- works |>
 filter(publication_year == 2015, !is.null(counts_by_year))

trajectory <- cohort_2015 |>
 unnest(counts_by_year) |>
 mutate(age = year - 2015) |>
 filter(age >= 0) |>
 group_by(age) |>
```

```
 summarise(median_cites = median(cited_by_count))

ggplot(trajectory, aes(age, median_cites)) +
 geom_line() +
 geom_point() +
 labs(title = "Citation trajectory for 2015 cohort",
      x = "Age (years since publication)",
      y = "Median annual citations")

# Peak year
peak_age <- trajectory$age[which.max(trajectory$median_cites)]
cat(sprintf("Peak citation year: age %d\n", peak_age))
```

Key insight: Papers typically peak in citation accumulation at age 2-4 years, depending on the field. This lag matters for evaluating recent research.

3. Institutional patent linkage. Compare the proportion of top-cited papers across institutions.

```
inst_ids <- c("I123456789", "I987654321")

inst_top_papers <- map(inst_ids, \(iid) {
 w <- oa_fetch(entity = "works",
               authorships.institutions.lineage = iid,
               from_publication_date = "2018-01-01",
               to_publication_date = "2020-12-31")
 p90 <- quantile(w$cited_by_count, 0.9)
 tibble(
   institution = iid,
   n_papers = nrow(w),
   pct_above_p90 = mean(w$cited_by_count >= p90)
 )
}) |> list_rbind()

inst_top_papers
```

Key insight: A higher proportion of top-percentile papers may suggest greater potential for technological impact, but causation cannot be established without actual patent citation data.

Chapter 32 — Causal Inference in Scientometrics

1. Extended DiD. Add covariates to the difference-in-differences regression.

```r
library(tidyverse)

# Assuming works_did has: cited_by_count, is_oa, post_treatment,
# type, n_authors, referenced_works_count
did_extended <- lm(
 cited_by_count ~ is_oa * post_treatment +
   type + n_authors + referenced_works_count,
 data = works_did
)

summary(did_extended)

# Compare with simple model
did_simple <- lm(cited_by_count ~ is_oa * post_treatment,
                 data = works_did)

cat(sprintf("Simple interaction: %.2f\n",
            coef(did_simple)["is_oaTRUE:post_treatmentTRUE"]))
cat(sprintf("Extended interaction: %.2f\n",
            coef(did_extended)["is_oaTRUE:post_treatmentTRUE"]))
```

Key insight: If the interaction term changes substantially with added covariates, the simple model suffers from omitted variable bias.

2. Placebo test. Use a false treatment date to test the parallel trends assumption.

```r
# Placebo: pretend treatment happened in 2018 instead of 2020
works_placebo <- works_did |>
 filter(publication_year <= 2019) |>  # pre-treatment only
 mutate(post_placebo = publication_year >= 2018)

did_placebo <- lm(cited_by_count ~ is_oa * post_placebo,
                  data = works_placebo)

summary(did_placebo)

# If the interaction is NOT significant, parallel trends hold
cat(sprintf("Placebo p-value: %.4f\n",
            summary(did_placebo)$coefficients[
             "is_oaTRUE:post_placeboTRUE", "Pr(>|t|)"
            ]))
```

Key insight: A non-significant placebo interaction ($p > 0.05$) supports the parallel trends assumption. A significant result suggests pre-existing divergence between OA and non-OA papers.

3. Matched comparison. Use propensity score matching to estimate the OA effect.

```
library(MatchIt)
library(tidyverse)

# Propensity score matching
matched <- matchit(
 is_oa ~ publication_year + referenced_works_count + type,
 data = works_did,
 method = "nearest",
 ratio = 1
)

summary(matched)
matched_data <- match.data(matched)

# Estimate treatment effect on matched sample
did_matched <- lm(cited_by_count ~ is_oa, data = matched_data)
summary(did_matched)

cat(sprintf("Unmatched OA effect: %.2f\n",
            coef(did_simple)["is_oaTRUE"]))
cat(sprintf("Matched OA effect: %.2f\n",
            coef(did_matched)["is_oaTRUE"]))
```

Key insight: Matching reduces selection bias by creating comparable groups. If the matched and unmatched effects differ substantially, selection on observables was important.

Chapter 33 — Reproducibility, RRIDs, and Badges

1. DAS detection. Search abstracts for data/code availability statements.

```
library(tidyverse)

das_keywords <- c("data availability", "code availability",
                  "github\\.com", "zenodo", "figshare",
                  "data are available", "code is available")

das_pattern <- paste(das_keywords, collapse = "|")
```

```r
works |>
 filter(!is.na(ab)) |>
 mutate(has_das = str_detect(tolower(ab), das_pattern)) |>
 summarise(
   total = n(),
   with_das = sum(has_das),
   das_rate = mean(has_das)
 )
```

Key insight: DAS detection via abstract text is a lower bound; many statements appear only in dedicated sections not captured by abstracts.

2. Cross-journal comparison. Compare metadata completeness and OA rates across journals.

```r
journal_ids <- c("S4210174107", "S49861241", "S137773608")

journal_quality <- map(journal_ids, \(jid) {
 w <- oa_fetch(entity = "works",
               primary_location.source.id = jid,
               from_publication_date = "2020-01-01") |>
   slice_sample(n = 200)
 tibble(
   journal = jid,
   abstract_rate = mean(!is.na(w$ab)),
   oa_rate = mean(w$is_oa, na.rm = TRUE),
   doi_rate = mean(!is.na(w$doi))
 )
}) |> list_rbind()

journal_quality
```

Key insight: Journals with higher OA rates tend to have better metadata completeness, but the correlation is not perfect.

3. Temporal trends. Track transparency proxies over time.

```r
transparency_trends <- works |>
 group_by(publication_year) |>
 summarise(
   abstract_rate = mean(!is.na(ab)),
   oa_rate = mean(is_oa, na.rm = TRUE),
   high_cite_rate = mean(cited_by_count >= 10)
 ) |>
 pivot_longer(-publication_year, names_to = "metric",
```

```r
              values_to = "value")

ggplot(transparency_trends, aes(publication_year, value,
                                colour = metric)) +
 geom_line() +
 scale_y_continuous(labels = scales::percent) +
 labs(title = "Transparency and impact proxies over time",
      x = "Year", y = "Proportion")
```

Key insight: Transparency proxies and citation impact may be weakly correlated within years, but confounding factors (field, journal prestige) complicate interpretation.

Chapter 34 — Reproducible Pipelines

1. Build a pipeline. Create a `_targets.R` file for a bibliometric workflow.

```r
library(targets)

# _targets.R content:
tar_option_set(packages = c("openalexR", "tidyverse"))

list(
 tar_target(raw_data, {
   oa_fetch(entity = "works", search = "bibliometrics",
            from_publication_date = "2020-01-01")
 }),

 tar_target(clean_data, {
   raw_data |>
     filter(!is.na(doi), !is.na(publication_year)) |>
     select(id, doi, display_name, publication_year,
            cited_by_count, is_oa)
 }),

 tar_target(summary_stats, {
   clean_data |>
     group_by(publication_year) |>
     summarise(n = n(), mean_cites = mean(cited_by_count))
 }),

 tar_target(trend_plot, {
   ggplot(summary_stats, aes(publication_year, n)) +
```

```r
    geom_col() +
    labs(title = "Publication trend", x = "Year", y = "Works")
 })
)

# Run with: tar_make()
# Verify: tar_visnetwork()
```

Key insight: Changing the summary function only re-runs `summary_stats` and `trend_plot`, not `raw_data` or `clean_data` — this is the power of dependency-based pipelines.

2. renv snapshot. Demonstrate dependency management with renv.

```r
# Step 1: Initialise renv
# renv::init()

# Step 2: Install a package
# install.packages("glue")
# library(glue)

# Step 3: Snapshot
# renv::snapshot()
# This creates/updates renv.lock

# Step 4: Simulate loss --- remove the package
# remove.packages("glue")
# library(glue)  # Error: package not found

# Step 5: Restore
# renv::restore()
# library(glue)  # Works again
```

Key insight: renv creates a project-local library and a lockfile, ensuring exact reproducibility of package versions across machines and time.

3. Branching. Use dynamic branching to analyse multiple journals in parallel.

```r
library(targets)

# _targets.R with dynamic branching:
tar_option_set(packages = c("openalexR", "tidyverse"))

list(
 tar_target(journal_ids,
```

```
            c("S4210174107", "S137773608", "S49861241")),

  tar_target(journal_data,
    oa_fetch(entity = "works",
             primary_location.source.id = journal_ids,
             from_publication_date = "2020-01-01"),
    pattern = map(journal_ids)
  ),

  tar_target(journal_summary,
    journal_data |>
      summarise(
        journal = first(so),
        n = n(),
        mean_cites = mean(cited_by_count)
      ),
    pattern = map(journal_data)
  )
)
```

Key insight: Dynamic branching avoids duplicating target definitions; each journal is processed independently and can run in parallel with `tar_make_future()`.

Chapter 35 — Interactive Dashboards

1. Multi-page dashboard. Create a Quarto dashboard with two pages.

```
# Save as dashboard.qmd:
# ---
# title: "Bibliometric Dashboard"
# format: dashboard
# ---
#
# # Overview
#
# ## Row
#
# ```{r}
# library(plotly)
# plot_ly(summary_data, x = ~year, y = ~n, type = "bar")
# ```
#
# ## Row
```

```
#
# ```{r}
# plot_ly(works, x = ~cited_by_count, type = "histogram")
# ```
#
# # Details
#
# ## Row
#
# ```{r}
# library(DT)
# datatable(works |> select(display_name, publication_year,
#                           cited_by_count, is_oa))
# ```
```

Key insight: Quarto dashboards use page-level headings (`#`) for navigation tabs and row/column-level headings (`##`) for layout.

2. Linked brushing. Use plotly selection to filter a DT table.

```r
library(plotly)
library(crosstalk)
library(DT)

shared_data <- SharedData$new(works)

bscols(
 plot_ly(shared_data, x = ~publication_year,
         y = ~cited_by_count, type = "scatter",
         mode = "markers") |>
   layout(dragmode = "select"),

 datatable(shared_data,
           options = list(pageLength = 10),
           selection = "multiple")
)
```

Key insight: `crosstalk::SharedData` enables linked brushing between any htmlwidget that supports it, without needing Shiny.

3. Custom tooltips. Add informative hover text to a plotly chart.

```r
library(plotly)
library(tidyverse)

tooltip_data <- works |>
```

```r
  group_by(publication_year) |>
  summarise(n = n(), mean_cites = round(mean(cited_by_count), 1))

plot_ly(tooltip_data,
        x = ~publication_year,
        y = ~n,
        type = "bar",
        text = ~paste0("Year: ", publication_year,
                       "<br>Papers: ", n,
                       "<br>Mean cites: ", mean_cites),
        hoverinfo = "text")
```

Key insight: Custom tooltips using `text` and `hoverinfo = "text"` provide context-rich information on hover, improving dashboard usability.

Chapter 36 — Parameterized Quarto Reports

1. Institutional template. Create a parameterized report for institutional profiles.

```r
# Save as inst_profile.qmd:
# ---
# title: "Institutional Profile"
# params:
#   inst_id: "I123456789"
#   inst_name: "University of Example"
#   start_year: 2020
#   end_year: 2023
# ---
#
# ```{r}
# library(openalexR)
# library(tidyverse)
#
# works <- oa_fetch(
#   entity = "works",
#   authorships.institutions.lineage = params$inst_id,
#   from_publication_date = paste0(params$start_year, "-01-01"),
#   to_publication_date = paste0(params$end_year, "-12-31")
# )
#
# cat(sprintf("Total publications: %d\n", nrow(works)))
# ```
```

```
#
# ## Publication Trend
# ```{r}
# works |>
#   count(publication_year) |>
#   ggplot(aes(publication_year, n)) +
#   geom_col() +
#   labs(title = params$inst_name)
# ```
#
# ## Citation Distribution
# ```{r}
# ggplot(works, aes(cited_by_count)) +
#   geom_histogram(bins = 30) +
#   scale_x_log10()
# ```
#
# ## Top Authors
# ```{r}
# works |> unnest(author) |>
#   count(au_display_name, sort = TRUE) |>
#   head(10)
# ```
```

Key insight: Parameters make the same template reusable for any institution; batch rendering generates a portfolio of reports efficiently.

2. Batch rendering. Render the template for multiple journals.

```
library(quarto)

journals <- list(
 list(journal_id = "S4210174107", journal_name = "Scientometrics"),
 list(journal_id = "S137773608", journal_name = "Nature Physics"),
 list(journal_id = "S49861241", journal_name = "PLOS ONE")
)

walk(journals, \(j) {
 quarto_render(
   "journal_profile.qmd",
   execute_params = list(
     journal_id = j$journal_id,
     journal_name = j$journal_name,
     start_year = 2020,
     end_year = 2023
   ),
```

```
    output_file = paste0("report_", j$journal_name, ".html")
  )
})
```

Key insight: Batch rendering with `quarto_render()` and parameterised templates scales to hundreds of reports with minimal code.

3. Conditional content. Add a warning when the sample is too small.

```
# In the parameterized .qmd file:
#
# ```{r}
# #| results: asis
# if (nrow(works) < 50) {
#   cat("<div class="callout callout-warn">\n")
#   cat("This sample contains fewer than 50 papers. ")
#   cat("The citation distribution may not be reliable.\n")
#   cat(":::\n")
# } else {
#   ggplot(works, aes(cited_by_count)) +
#     geom_histogram(bins = 30) +
#     scale_x_log10() +
#     labs(title = "Citation distribution")
# }
# ```
```

Key insight: Conditional logic with `results: asis` enables dynamic content switching — essential for robust automated reports that handle edge cases.

Chapter 37 — Building a Bibliometric Shiny App

1. Add network tab. Extend the app with a co-authorship network visualisation.

```
library(shiny)
library(visNetwork)
library(igraph)
library(tidyverse)

# Add to UI:
# tabPanel("Network",
#   visNetworkOutput("coauth_network", height = "600px")
# )
```

```r
# Add to server:
# output$coauth_network <- renderVisNetwork({
#   req(data())
#   edges <- data() |>
#     unnest(author) |>
#     group_by(id) |>
#     filter(n() >= 2, n() <= 10) |>
#     reframe(pair = combn(au_display_name, 2, simplify = FALSE)) |>
#     unnest_wider(pair, names_sep = "_") |>
#     count(pair_1, pair_2, name = "weight") |>
#     rename(from = pair_1, to = pair_2)
#
#   g <- graph_from_data_frame(edges, directed = FALSE)
#   comm <- cluster_leiden(g, resolution = 1.0)
#
#   nodes <- tibble(
#     id = V(g)$name, label = V(g)$name,
#     group = as.character(membership(comm))
#   )
#
#   visNetwork(nodes, edges) |>
#     visOptions(highlightNearest = TRUE, selectedBy = "group")
# })
```

Key insight: Limiting the network to papers with 2-10 authors avoids overwhelming the visualisation with hyper-authored papers.

2. Caching. Implement server-side caching for repeated queries.

```r
library(shiny)
library(memoise)

# Create a cached version of the data fetching function
fetch_cached <- memoise(function(journal_id, start_date, end_date) {
 oa_fetch(
   entity = "works",
   primary_location.source.id = journal_id,
   from_publication_date = start_date,
   to_publication_date = end_date
 )
}, cache = cachem::cache_disk("shiny_cache"))

# In the server:
# data <- reactive({
#   fetch_cached(
```

```
#     input$journal_id,
#     paste0(input$start_year, "-01-01"),
#     paste0(input$end_year, "-12-31")
#   )
# })
```

Key insight: Disk-backed caching persists across sessions, so the same query never hits the API twice. Use `cachem::cache_disk()` for production apps.

3. Deployment. Deploy to shinyapps.io.

```
# Step 1: Install rsconnect
# install.packages("rsconnect")

# Step 2: Configure account
# rsconnect::setAccountInfo(
#   name = "your-account",
#   token = "your-token",
#   secret = "your-secret"
# )

# Step 3: Deploy
# rsconnect::deployApp(
#   appDir = ".",
#   appName = "bibliometric-explorer",
#   appTitle = "Bibliometric Explorer"
# )

# Step 4: Test with different inputs
# - Try different journal IDs
# - Test with narrow and wide date ranges
# - Verify that caching works (second load should be instant)
```

Key insight: Test with a variety of inputs before deploying; edge cases (empty results, very large corpora) should be handled gracefully with validation messages.

4. biblioshiny comparison. Run biblioshiny and compare features.

```
library(bibliometrix)
# biblioshiny()

# Feature comparison table:
# | Feature              | biblioshiny | Custom app |
# |----------------------|-------------|------------|
```

```
# | Data sources        | WoS, Scopus | OpenAlex   |
# | Network analysis    | Yes         | Basic      |
# | Topic analysis      | Yes         | No         |
# | Custom indicators   | Limited     | Full       |
# | Parameterization    | No          | Yes        |
# | Deployment          | Local only  | shinyapps  |
# | Reproducibility     | GUI-based   | Code-based |
#
# biblioshiny is comprehensive but not easily customisable;
# a custom app trades breadth for tailored functionality.
```

Key insight: biblioshiny is excellent for exploration but lacks reproducibility and customisation. A custom app is better for production workflows with specific analytical requirements.

Glossary

Altmetrics. Metrics derived from online activity (social media mentions, news coverage, policy documents) rather than citations. See Chapter 26.

Bibliographic coupling. Two documents are bibliographically coupled when they share one or more references. Strength increases with the number of shared references. See Chapter 16.

Bibliometrics. The quantitative study of scholarly publications, their production, dissemination, and impact. A subfield of scientometrics focused on documents.

Burst detection. A method for identifying terms or topics whose frequency increases sharply in a given time window, signalling emerging research fronts. See Chapter 25.

Citation aging. The temporal pattern of how citations accumulate over time after publication. The citation half-life measures how quickly a paper accrues half its total citations. See Chapter 14.

Citation window. The time period during which citations to a publication are counted. Common choices: 3-year, 5-year, or variable windows.

Co-authorship network. A graph where nodes represent authors and edges connect authors who have co-authored at least one publication. See Chapter 15.

Co-citation. Two documents are co-cited when they are both cited by a third document. Frequently co-cited documents are considered intellectually related. See Chapter 16.

Co-word analysis. Network analysis of the co-occurrence of terms (keywords, title words, or descriptors) within documents. See Chapter 17.

Community detection. Algorithmic identification of densely connected subgroups within a network. Common methods: Leiden, Louvain, modularity optimisation. See Chapter 18.

Disparity filter. A network backbone extraction method that retains statistically significant edges based on local weight distributions. See Chapter 18.

DORA. The San Francisco Declaration on Research Assessment. A set of recommendations for improving research evaluation practices, emphasising that journal-based metrics should not be used as surrogates for individual article quality.

Field normalisation. Adjusting citation counts to account for differences in citation practices across research fields and publication years. See Chapter 13.

Fractional counting. An attribution method where credit for a multi-authored publication is divided equally among contributors (1/n per author or institution), in contrast to full counting. See Chapter 13.

Full counting. An attribution method where each author or institution on a publication receives full credit (count of 1). See Chapter 13.

h-index. A researcher has an h-index of h if h of their papers have each been cited at least h times. See Chapter 10.

Impact factor (JIF). The mean number of citations received in year t by articles published in a journal in years $t-1$ and $t-2$. See Chapter 11.

Leiden Manifesto. Ten principles for responsible research metrics, emphasising that quantitative evaluation should support qualitative, expert assessment rather than replace it. See Chapter 3.

Leiden algorithm. A community detection method that improves on Louvain by guaranteeing well-connected communities. See Chapter 18.

MNCS (Mean Normalised Citation Score). The average of field-normalised citation scores across a set of publications. An MNCS above 1.0 indicates above-world-average impact. See Chapter 13.

Network backbone. A reduced network retaining only statistically significant or structurally important edges, obtained by filtering methods such as the disparity filter. See Chapter 18.

OpenAlex. A free, open catalogue of the global research system, covering works, authors, institutions, sources, concepts, funders, and publishers. The default data source for this book. See Chapter 5.

Overlay map. A visualisation technique where a subset of data (e.g., one institution's publications) is projected onto a pre-computed base map of science. See Chapter 19.

PP(top 10%). The proportion of an entity's publications that rank in the top 10% most cited in their field and year. A percentile-based impact indicator. See Chapter 10.

Price index. The proportion of a paper's references that are no older than 5 years, measuring how current the cited literature is. See Chapter 14.

RAKE. Rapid Automatic Keyword Extraction. An unsupervised method that extracts multi-word keywords based on word co-occurrence within phrases delimited by stopwords. See Chapter 23.

Research front. A cluster of recently published, actively cited papers representing the current edge of a research area.

Retraction. The formal withdrawal of a published paper by its authors or publisher due to errors, misconduct, or irreproducibility. See Chapter 29.

Scientometrics. The quantitative study of science and innovation, encompassing bibliometrics, patent analysis, and the measurement of research systems.

STM (Structural Topic Model). A topic model that allows document-level covariates (e.g., publication year, journal) to affect topic prevalence and content. See Chapter 22.

TF-IDF. Term Frequency–Inverse Document Frequency. A weighting scheme that highlights terms distinctive to individual documents within a corpus. See Chapter 21.

Topic modelling. Unsupervised methods for discovering latent thematic structure in collections of text. Common approaches: LDA, STM, BERTopic. See Chapter 22.

UMAP. Uniform Manifold Approximation and Projection. A dimensionality-reduction technique for visualising high-dimensional data (e.g., document embeddings) in 2D. See Chapter 24.

Colophon

```
#> Built with:

#>   bookdown:  0.46

#>   R:  R version 4.4.1 (2024-06-14)

#>   Build date:  2026-05-24

#>   Commit:  dc4269e

#>
#> Attached packages:

#> - Session info --------------------------------------------------------------------------
#>  setting  value
#>  version  R version 4.4.1 (2024-06-14)
#>  os       Ubuntu 24.04.4 LTS
#>  system   x86_64, linux-gnu
#>  ui       X11
#>  language (EN)
#>  collate  C.UTF-8
#>  ctype    C.UTF-8
#>  tz       UTC
#>  date     2026-05-24
#>  pandoc   3.8.3 @ /opt/hostedtoolcache/pandoc/3.8.3/x64/ (via rmarkdown)
#>  quarto   NA
#>
#> - Packages ------------------------------------------------------------------------------
#>  package           * version date (UTC) lib source
#>  arrow             * 24.0.0  2026-04-29 [1] RSPM (R 4.4.0)
#>  bib2df            * 1.1.2.0 2024-05-16 [1] RSPM (R 4.4.0)
#>  bibliometrix      * 5.4.0   2026-05-16 [1] RSPM (R 4.4.0)
```

```
#>  dplyr              * 1.2.1   2026-04-03 [1] RSPM (R 4.4.0)
#>  DT                 * 0.34.0  2025-09-02 [1] RSPM (R 4.4.0)
#>  forcats            * 1.0.1   2025-09-25 [1] RSPM (R 4.4.0)
#>  ggplot2            * 4.0.3   2026-04-22 [1] RSPM (R 4.4.0)
#>  ggraph             * 2.2.2   2025-08-24 [1] RSPM (R 4.4.0)
#>  glue               * 1.8.1   2026-04-17 [1] RSPM (R 4.4.0)
#>  gt                 * 1.3.0   2026-01-22 [1] RSPM (R 4.4.0)
#>  igraph             * 2.3.1   2026-05-04 [1] RSPM (R 4.4.0)
#>  lubridate          * 1.9.5   2026-02-04 [1] RSPM (R 4.4.0)
#>  Matrix             * 1.7-0   2024-04-26 [2] CRAN (R 4.4.1)
#>  openalexR          * 3.0.1   2026-01-14 [1] RSPM (R 4.4.0)
#>  pdftools           * 3.9.0   2026-05-14 [1] RSPM (R 4.4.0)
#>  plotly             * 4.12.0  2026-01-24 [1] RSPM (R 4.4.0)
#>  purrr              * 1.2.2   2026-04-10 [1] RSPM (R 4.4.0)
#>  quanteda           * 4.4     2026-04-06 [1] RSPM (R 4.4.0)
#>  quanteda.textstats * 0.97.2  2024-09-03 [1] RSPM (R 4.4.0)
#>  rcrossref          * 1.2.1   2025-08-20 [1] RSPM (R 4.4.0)
#>  readr              * 2.2.0   2026-02-19 [1] RSPM (R 4.4.0)
#>  RefManageR         * 1.4.0   2022-09-30 [1] RSPM (R 4.4.0)
#>  shiny              * 1.13.0  2026-02-20 [1] RSPM (R 4.4.0)
#>  stm                * 1.3.8   2025-09-03 [1] RSPM (R 4.4.0)
#>  stringr            * 1.6.0   2025-11-04 [1] RSPM (R 4.4.0)
#>  tibble             * 3.3.1   2026-01-11 [1] RSPM (R 4.4.0)
#>  tidygraph          * 1.3.1   2024-01-30 [1] RSPM (R 4.4.0)
#>  tidyr              * 1.3.2   2025-12-19 [1] RSPM (R 4.4.0)
#>  tidytext           * 0.4.3   2025-07-25 [1] RSPM (R 4.4.0)
#>  tidyverse          * 2.0.0   2023-02-22 [1] RSPM (R 4.4.0)
#>  topicmodels        * 0.2-17  2024-08-14 [1] RSPM (R 4.4.0)
#>  uwot               * 0.2.4   2025-11-10 [1] RSPM (R 4.4.0)
#>  visNetwork         * 2.1.4   2025-09-04 [1] RSPM (R 4.4.0)
#>  word2vec           * 0.4.1   2025-11-27 [1] RSPM (R 4.4.0)
#>
#>  [1] /home/runner/work/_temp/Library
#>  [2] /opt/R/4.4.1/lib/R/library
#>  * -- Packages attached to the search path.
#>
#> ------------------------------------------------------------------------------
```

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Download the reference as BibTeX or .ris.

References

Bibliography

American Society for Cell Biology (2012). San Francisco declaration on research assessment (DORA). Accessed 2026-05-09.

Aria, M. and Cuccurullo, C. (2017). bibliometrix: An R-tool for comprehensive science mapping analysis. *Journal of Informetrics*, 11(4):959–975.

Barabási, A.-L., Jeong, H., Néda, Z., Ravasz, E., Schubert, A., and Vicsek, T. (2002). Evolution of the social network of scientific collaborations. *Physica A: Statistical Mechanics and its Applications*, 311(3–4):590–614.

Bastian, M., Heymann, S., and Jacomy, M. (2009). Gephi: An open source software for exploring and manipulating networks. In *Proceedings of the Third International AAAI Conference on Weblogs and Social Media*, pages 361–362.

Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008.

Bradford, S. C. (1934). Sources of information on specific subjects. *Engineering*, 137:85–86.

Callon, M., Courtial, J.-P., Turner, W. A., and Bauin, S. (1983). From translations to problematic networks: An introduction to co-word analysis. *Social Science Information*, 22(2):191–235.

de Solla Price, D. J. (1963). *Little Science, Big Science*. Columbia University Press, New York.

Egghe, L. (2006). Theory and practise of the g-index. *Scientometrics*, 69(1):131–152.

Fang, F. C., Steen, R. G., and Casadevall, A. (2012). Misconduct accounts for the majority of retracted scientific publications. *Proceedings of the National Academy of Sciences*, 109(42):17028–17033.

Fortunato, S. (2010). Community detection in graphs. *Physics Reports*, 486(3–5):75–174.

Garfield, E. (1955). Citation indexes for science: A new dimension in documentation through association of ideas. *Science*, 122(3159):108–111.

Glänzel, W. and Schubert, A. (2004). Analysing scientific networks through co-authorship. *Handbook of Quantitative Science and Technology Research*, pages 257–276.

Goodhart, C. A. E. (1984). Problems of monetary management: The U.K. experience. In *Monetary Theory and Practice*, pages 91–121. Macmillan, London.

Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., and Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520(7548):429–431.

Hirsch, J. E. (2005). An index to quantify an individual's scientific research output. *Proceedings of the National Academy of Sciences*, 102(46):16569–16572.

Kessler, M. M. (1963). Bibliographic coupling between scientific papers. *American Documentation*, 14(1):10–25.

Kleinberg, J. (2003). Bursty and hierarchical structure in streams. *Data Mining and Knowledge Discovery*, 7(4):373–397.

Larivière, V., Ni, C., Gingras, Y., Cronin, B., and Sugimoto, C. R. (2013). Bibliometrics: Global gender disparities in science. *Nature*, 504(7479):211–213.

Lotka, A. J. (1926). The frequency distribution of scientific productivity. *Journal of the Washington Academy of Sciences*, 16(12):317–323.

Merton, R. K. (1968). The Matthew Effect in science. *Science*, 159(3810):56–63.

Narin, F., Hamilton, K. S., and Olivastro, D. (1997). The increasing linkage between U.S. technology and public science. *Research Policy*, 26(3):317–330.

Newman, M. E. J. (2001). The structure of scientific collaboration networks. *Proceedings of the National Academy of Sciences*, 98(2):404–409.

Priem, J., Piwowar, H., and Orr, R. (2022). OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint*.

Priem, J., Taraborelli, D., Groth, P., and Neylon, C. (2010). Altmetrics: A manifesto. *altmetrics.org*.

Rafols, I., Porter, A. L., and Leydesdorff, L. (2010). Science overlay maps: A new tool for research policy and library management. *Journal of the American Society for Information Science and Technology*, 61(9):1871–1887.

Roberts, M. E., Stewart, B. M., and Tingley, D. (2019). stm: An R package for structural topic models. *Journal of Statistical Software*, 91(2):1–40.

Serrano, M. Á., Boguñá, M., and Vespignani, A. (2009). Extracting the multiscale backbone of complex weighted networks. *Proceedings of the National Academy of Sciences*, 106(16):6483–6488.

Silge, J. and Robinson, D. (2017). *Text Mining with R: A Tidy Approach*. O'Reilly Media.

Singh, V. K., Singh, P., Karmakar, M., Leta, J., and Mayr, P. (2021). The journal coverage of Web of Science, Scopus and Dimensions: A comparative analysis. *Scientometrics*, 126:5113–5142.

Small, H. (1973). Co-citation in the scientific literature: A new measure of the relationship between two documents. *Journal of the American Society for Information Science*, 24(4):265–269.

Traag, V. A., Waltman, L., and van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9:5233.

Visser, M., van Eck, N. J., and Waltman, L. (2021). Large-scale comparison of bibliographic data sources: Scopus, Web of Science, Dimensions, Crossref, and Microsoft Academic. *Quantitative Science Studies*, 2(1):20–41.

Waltman, L. (2016). A review of the literature on citation impact indicators. *Journal of Informetrics*, 10(2):365–391.

Waltman, L., van Eck, N. J., and Noyons, E. C. M. (2010). A unified approach to mapping and clustering of bibliometric networks. *Journal of Informetrics*, 4(4):629–635.

Waltman, L., van Eck, N. J., van Leeuwen, T. N., Visser, M. S., and van Raan, A. F. J. (2011). Towards a new crown indicator: Some theoretical considerations. *Journal of Informetrics*, 5(1):37–47.

Wilsdon, J., Allen, L., Belfiore, E., Campbell, P., Curry, S., Hill, S., Jones, R., Kain, R., Kerridge, S., Thelwall, M., Tinkler, J., Viney, I., Wouters, P., Hill, J., and Johnson, B. (2015). *The Metric Tide: Report of the Independent Review of the Role of Metrics in Research Assessment and Management*. SAGE Publications.